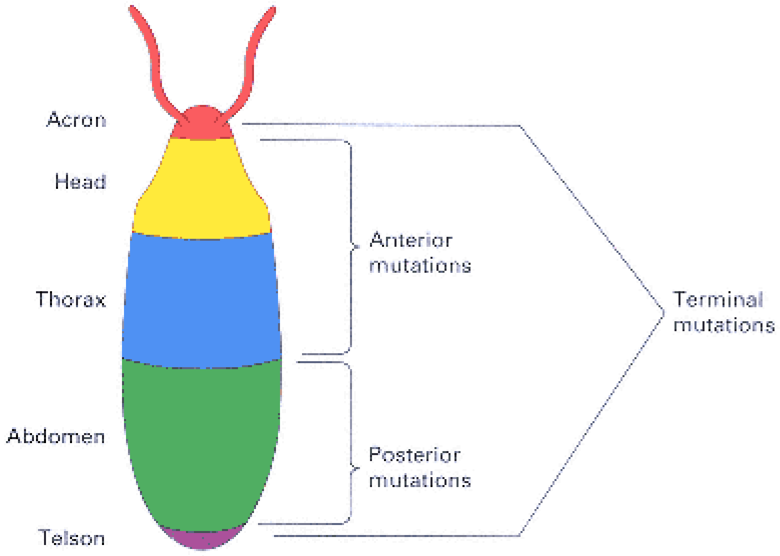

Figure 12.25
Regional differentiation of the early *Drosophila* embryo along the anterior-posterior axis, Mutations in any of the classes of genes shown result in elimination of the corresponding region of the embryo.

the next along the bicoid concentration gradient. One of the important genes activated by *bicoid* is the gap gene *hunchback*. Five other genes in the anterior class are known, and they code for cellular components necessary for *bicoid* localization.

2. The second group of coordinate genes includes the *posterior genes*, which affect the abdominal segments (Figure 12.25). Some of the mutants also lack pole cells. One of

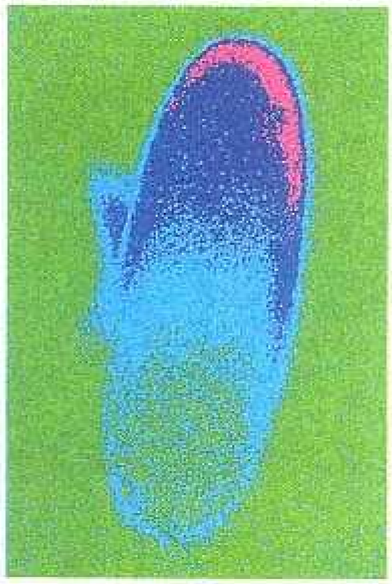

Figure 12.26
A gradient of gene expression resembling that of *bicoid* in the early *Drosophila* embryo. In this photograph, the intensity of the fluorescent signal has been pseudocolored so that the region of highest expression is pink and the region of lowest expression is green.
[Courtesy of James Langeland, Stephen Paddock, and Sean Carroll.]

the posterior mutations, *nanos*, yields embryos with defective abdominal segmentation but normal pole cells, abnormalities that resemble those produced by surgical removal of the posterior cytoplasm. The phenotype does not result merely from a generalized disruption of development at the posterior end, because the pole cells—as well as a posterior structure called the telson, which normally develops between the pole cells and the abdomen—are not affected in either *nanos* or the surgically manipulated embryos. The *nanos* mRNA is localized tightly to the posterior pole of the oocyte, and the gene product is a repressor of translation. Among the genes whose mRNA is not translated in the presence of nanos protein is the gene *hunchback.* Hence *hunchback* expression is controlled jointly by the bicoid and nanos proteins; bicoid protein activates transcription in an anterior-posterior gradient, and nanos protein represses translation in the posterior region.

3. The third group of coordinate genes includes the *terminal genes,* which simultaneously affect the most anterior structure (the acron) and the most posterior structure (the telson) (Figure 12.25). The key gene in this class is *torso,* which codes for a transmembrane receptor that is uniformly distributed throughout the embryo in the early developmental stages. The torso receptor is activated by a signal released only at the poles of the egg by the nurse cells in that location (Figure 12.7). The torso receptor is a tyrosine kinase that initiates cellular differentiation by means of phosphorylation of specific tyrosine residues in one or more target proteins, among them a *Drosophila* homolog of the vertebrate oncogene *D-raf.*

Apart from the three sets of genes that determine the anterior-posterior axis of the embryo, a fourth set of genes determines the dorsal-ventral axis. The morphogen for dorsal-ventral determination is the present in a pronounced ventral-to-dorsal gradient in the late syncytial blastderm. The dorsal protein is a transcription factor related to the avian oncogene *v-ref.* An additional 16 other genes are known to affect dorsalventral determination. Mutations in these genes eliminate ventral and lateral pattern elements. In many cases, the mutant embryos can be rescued by the injection of wildtype cytoplasm, no matter where the wildtype cytoplasm is taken from or where it is injected. Examples include the genes called *snake, gastrulation-defective,* and *easter.* All three genes code for proteins called *serine proteases.* Serine proteases are synthesized as inactive precursors that require a specific cleavage for activation. They often act in a temporal sequence, which means that activation of one enzyme in the pathway is necessary for activation of the next enzyme in line (Figure 12.27). About half the clotting factors in human blood are serine proteases. The serial activation of the enzymes results in a **cascade effect** that greatly amplifies an initial signal. Each step in the cascade multiplies the signal produced in the preceding step.

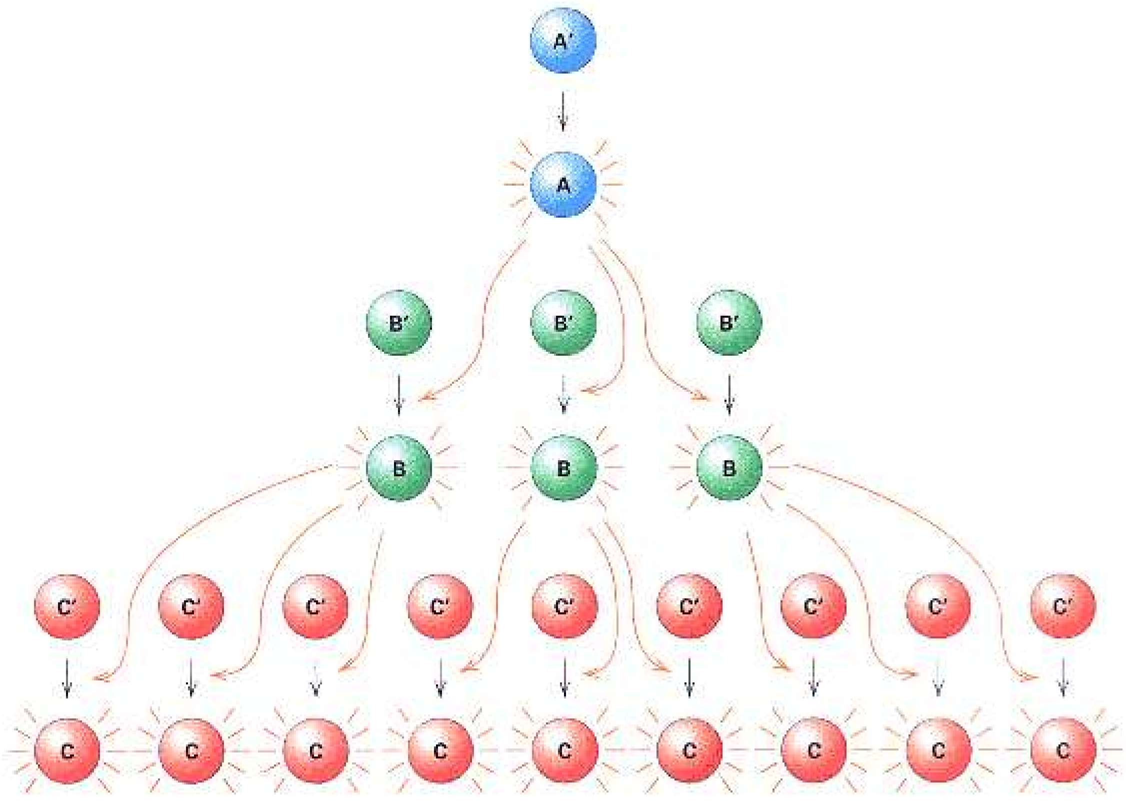

Figure 12.27
Amplification of a signal by a cascade of activation. The number of activated components at each step increases exponentially. This is a simplified example with a threefold amplification at each step. The primed symbols denote inactive enzyme forms; the unprimed symbols denote active forms.

Gap Genes

The main role of the coordinate genes is to regulate the expression of a small group of approximately six gap genes along the anterior-posterior axis. The genes are called **gap genes** because mutations in them result in the absence of pattern elements derived from a group of contiguous segments (Figure 12.24A). Gap genes are zygotic genes. The gene *hunchback* serves as an example of the class because *hunchback* expression is controlled by offsetting effects of *bicoid* and *nanos.* Transcription of *hunchback* is stimulated in an anterior-to-posterior gradient by the bicoid transcription factor, but posterior *hunchback* expression is prevented by translational repression because of the posteriorly localized nanos protein. In the early *Drosophila* embryo in Figure 12.28, the gradient of *hunchback* expression is indicated by the green fluorescence of an antibody specific to the *hunchback* gene product. The superimposed red fluorescence results from antibody specific to the product of *Krüppel,* another gap gene. The region of overlapping gene expression appears in yellow. The products of both *hunchback* and *Krüppel* are transcription factors of the zinc fingre type (Chapter 11). Other gap genes also are transcription factors. Together, the gap genes have a pattern of regional specificity and partly overlapping domains of expression that enable them to act in combinatorial fashion to control the next set of genes in the segmentation hierarchy, the pairrule genes.

Pair-Rule Genes

The coordinate and gap genes determine the polarity of the embryo and establish broad regions within which subsequent development takes place. As development proceeds, the progressively more refined organization of the embryo is correlated with the patterns of expression of the segmentation genes. Among these are the **pair-rule genes,** in which the mutant phenotype has alternating segments absent or malformed (Figure 12.24B). Approximately eight pair-rule genes have been identified. For example, mutations of the pair-rule gene *even-skipped* affect even- numbered segments, and those of another pair-rule gene, *add-skipped,* affect odd-numbered segments. The function of the pair-rule genes is to give the early *Drosophila* larva a segmented body pattern with both repetitiveness and individuality of segments. For example, there are eight abdominal segments that are repetitive in that they are regularly spaced and share several common features, but they differ in the details of their differentiation.

One of the earliest pair-rule genes expressed is *hairy,* whose pattern of expression is under both positive and negative regulation by the products of *hunchback, Krüppel,* and other gap genes. Expression of *hairy* yields seven stripes (Figure 12.29). The striped pattern of pair-rule gene expression is typical, but the stripes of expression of one gene are usually slightly out of register with those of another. Together with the continued regional expression of the gap genes, the combinatorial patterns of gene expression in the embryo are already complex and linearly differentiated. Figure 12.30 shows an embryo stained for the products of three genes: *hairy* (green), *Krüppel* (red), and *giant* (blue). The regions of overlapping expression appear as color mixtures—orange, yellow, light green, or purple. Even at the early stage in Figure 12.30, there is a unique combinatorial pattern of gene expression in every segment and parasegment. The complexity of combinatorial control can be appreciated by considering that the expression of the *hairy* gene in stripe 7 depends on a promoter element smaller than 1.5 kb that contains a series of binding sites for the protein products of the genes *caudal, hunchback, knirps, Krüppel, tailless, huckbein, bicoid,* and perhaps still other proteins yet to be identified. The combinatorial patterns of gene expression of the pair-rule genes define the boundaries of expression of the segment-polarity genes, which function next in the hierarchy.

Segment-Polarity Genes

Whereas the pair-rule genes determine the body plan at the level of segments and parasegments, the **segment-polarity**

genes create a spatial differentiation within each segment. Approximately 14 segment-polarity genes have been identified. The mutant phenotype has repetitive deletion of pattern along the embryo (Figure 12.24C) and usually a mirrorimage duplication of the part that remains. Among the earliest segment-polarity genes expressed is *engrailed*, whose stripes of expression approximately coincide with the boundaries of the parasegments and so divide each segment into anterior and posterior domains (Figure 12.31).

Expression of the segment-polarity genes finally establishes the early polarity and linear differentiation of the embryo into segments and parasegments. The regulatory interactions within the hierarchy of segmentation genes are illustrated in Figure 12.32. These interactions govern the activities of the second set of developmental genes, the *homeotic genes*, which control the pathways of differentiation in each segment or parasegment.

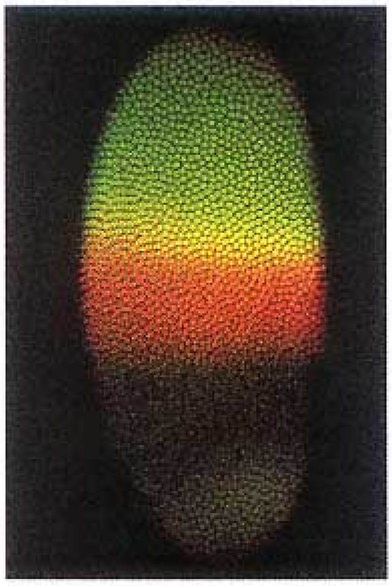

Figure 12.28
An embryo of *Drosophila*, approximately 2.5 hours after fertilization, showing the regional localization of the *hunchback* gene product (green) the Krüppel gene product (red), and their overlap (yellow). [Courtesy of James Langeland, Stephen Paddock, and Sean Carroll.

Figure 12.29
Characteristic seven stripes of expression of the gene *hairy* in a *Drosophila* embryo approximately 3 hours after

fertilization.
[Courtesy of James Langeland, Stephen Paddock, and Sean Carroll.]

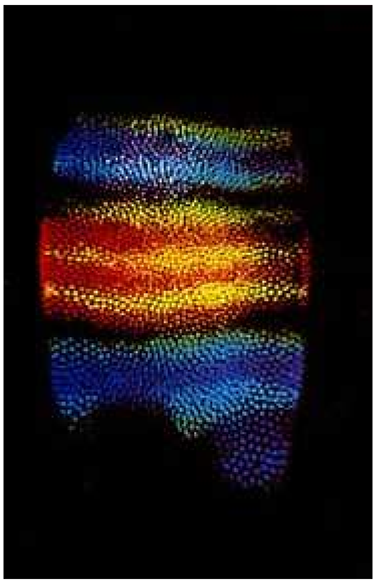

Figure 12.30
Combined patterns of expression of *hairy* (green), *Krüppel* (red), and *giant* (blue) in a *Drosophila* embryo approximately 3 hours after fertilization. Already considerable linear differentiation is apparent in the patterns of gene expression.
[Courtesy of James Langeland, Stephen Paddock, and Sean Carroll.]

Figure 12.31
Expression of the segment-polarity gene engrailed partitions the early Drosophila embryo into 14 regions. These eventually differentiate into three head segments, three thoracic segments, and eight abdominal segments.
[Courtesy of James Langland, Stephen Paddock, and Sean Carroll.]

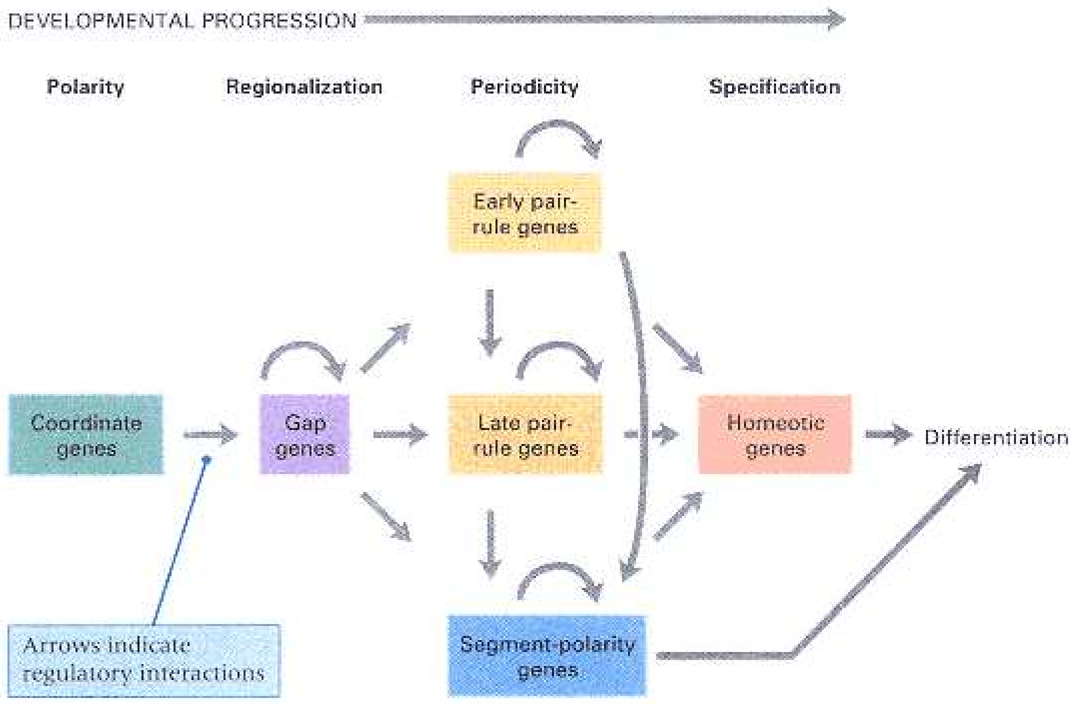

Figure 12.32
Hierachy of regulatory interactions among genes controlling early development in *Drosophila*. Each gene is controlled by a unique combination of other genes. The terms *polarity, regionalization, periodicity*, and *specification* refer to the major developmental determinations that are made in each time interval.

Homeotic Genes

As with most other insects, the larvae and adults of *Drosophila* have a segmented body plan consisting of a head, three thoracic segments, and eight abdominal segments (Figure 12.33). Metamorphosis makes use of about 20 structures called **imaginal disks** present inside the larvae (Figure

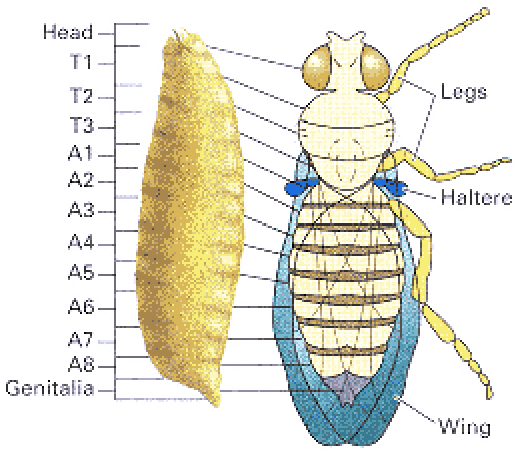

Figure 12.33
Relationship between larval and adult segmentation in *Drosophila*. Each of the three thoracic segments in the adult carries a pair of legs. The wings develop on the second thoracic segment (T2) and the halteres (flight balancers) on the third thoracic segment (T3).

12.34). Formed early in development, the imaginal disks ultimately give rise to the principal structures and tissues in the adult organism. Examples of imaginal disks are the pair of wing disks (one on each side of the body), which give rise to the wings and related structures; the pair of eye-antenna disks, which give rise to the eyes, antennae, and related structures; and the genital disk, which gives rise to the reproductive apparatus. During the pupal stage, when many larval tissues and organs break down, the imaginal disks progressively unfold and differentiate into adult strctures. The morphogenic events that take place in the pupa are initiated by the hormone ecdysone, secreted by the larval brain.

Cell determination in *Drosophila* also takes place within bounded units called **compartments.** Cells in the body segments and imaginal discs do not migrate across the boundaries between compartments. For example, the *Drosophila* wing disk includes five compartment boundaries, and most body segments include one

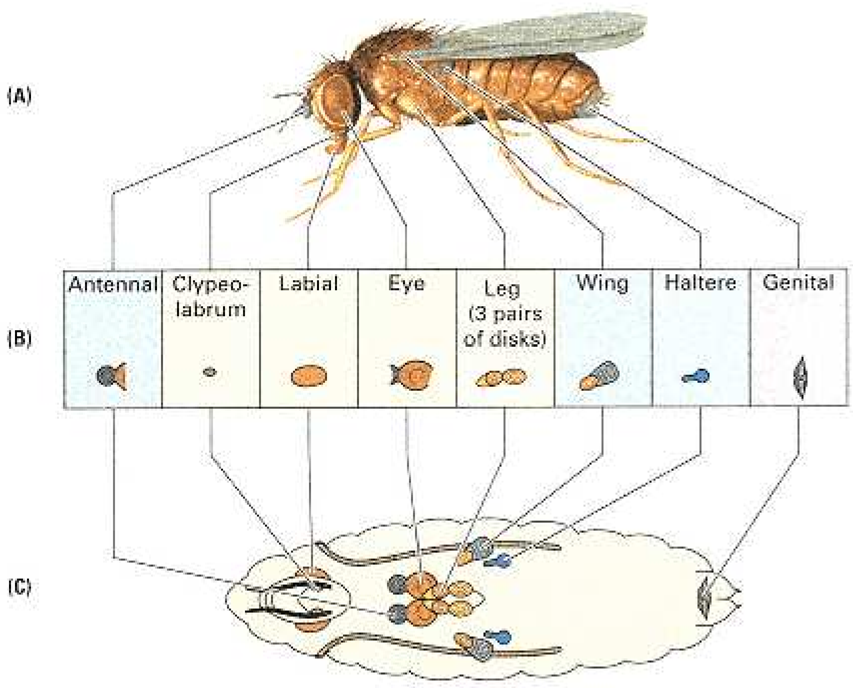

Figure 12.34
(A) Structures of the adult *Drosophila* larva and the adult structures derived from them. (B) Larval locations of the nine pairs of imaginal disks and one genital disk. (C) General morphology of the disks late in larval development.

boundary that divides the segment into anterior and posterior halves. The evidence for compartments comes from genetic marking of individual cells by means of mitotic recombination and observation of the positions of their descendants. Within each compartment, neighboring groups of cells not necessarily related by ancestry undergo developmental determination together.

As in the early embryo, overlapping patterns of gene expression and combinatorial control guide later events in *Drosophila* development. The expression patterns of two genes in the wing imaginal disk are shown in Figure 12.35. The expression of *apterous* is indicated in green, that of *vestigial* in red. Regions of overlapping expression are various shades of orange and yellow. Patterns of gene expression in imaginal disks are highly varied. Some genes are expressed in patterns with radial symmetry—for example, in alternating sectors or in concentric rings. The varied and overlapping patterns of expression ultimately yield the exquisitely fine level of cellular and morphological differentiation observed in the adult animal.

Among the genes that transform the periodicity of the *Drosophila* embryo into a body plan with linear differentiation are two small sets of **homeotic genes** (Figure 12.32). Mutations in homeotic genes result in the transformation of one body segment into another, which is recognized by the misplaced development of structures that are normally present elsewhere in the embryo. One class of homeotic mutation is

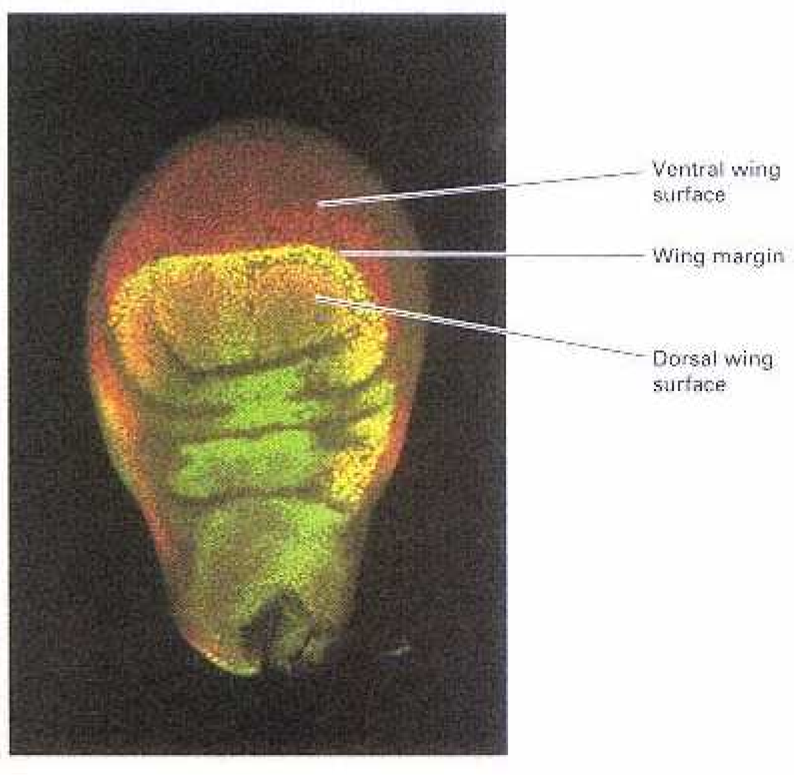

Figure 12.35
Expression of two genes that affect wing development in the wing imaginal disk. Expression of *apterous* is in green; that of *vestigial* is in red. Regions of overlapping expression are orange and yellow.
[Courtesy of James Langeland, Stephen Paddock, and Sean Carroll.]

illustrated by *bithorax*, which causes transformation of the anterior part of the third thoracic segment into the anterior part of the second thoracic segment, with the result that the halteres (flight balancers) are transformed into an extra pair of wings (Figure 12.36). The other class of homeotic mutation is illustrated by *Antennapedia*, which results in transformation of the antennae into legs. The normal *Antennapedia* gene specifies the second thoracic segment. *Antennapedia* mutations are dominant gain-of-function mutations in which the gene is overexpressed in the dorsal part of the head, transforming it into the second thoracic segment, with the antennae becoming transformed into a pair of misplaced legs.

Homeotic genes act within developmental compartments to control other genes concerned with such characteristics as rates of cell division, orientation of mitotic spindles, and the capacity to differentiate bristles, legs, and other features. Homeotic genes are also important in restricting the activities of groups of structural genes to definite spatial patterns. The homeotic genes represented by *bithorax* and *Antennapedia* are in fact gene clusters. The cluster containing *bithorax* is designated BX-C (stands for *bithorax*-complex), and that containing *Antennapedia* is called ANT-C (stands for *Antennapedia*-complex). Both gene clusters were initially discovered through their homeotic effects in adults. Later they were shown to affect the identity of larval segments. The BX-C is primarily concerned with the development of larval segments T3 through A8 (Figure 12.37) and has its principal effects in T3 and A1. The ANT-C is primarily concerned with the development of the head (H) and of thoracic segments T1 and T2.

Deletion (loss of function) and duplication (gain of function) of genes in the homeotic complexes help to define their functions. For example, deletion of the entire BX-C complex (Figure 12.37) results in a larva with a normal head (H), first thoracic segment (T1), and second thoracic segment (T2), but the remaining segments (T3 and A1-A8) differentiate in the manner of T2. Therefore, the function of the BX-C genes is to shift development progressively to more posterior types of segments. The BX-C region extends across approximately 300 kb of DNA yet contains only three essential coding regions. The rest of the region appears to consist of a complex series of enhancers and other regulatory elements that function to specify segment identity by activating the different coding regions to different degrees in particular parasegments.

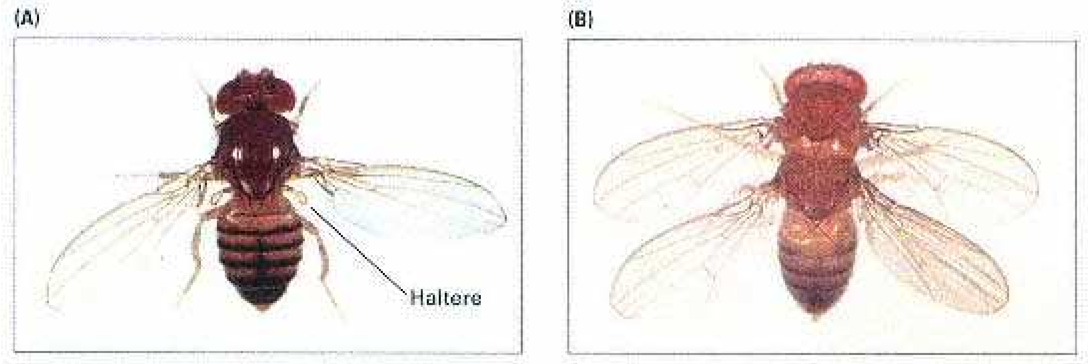

Figure 12.36
A) Wildtype *Drosophila* showing wings and halteres (the pair of knob-like structures protruding posterior to the wings). (B) A fly with four wings produced by mutations in the *bithorax* complex. The mutations convert the third thoracic segment into the second thoracic segment, and the halteres that are normally present on the third thoracic segment become converted into the posterior pair of wings.
[Courtesy of E. B. Lewis.]

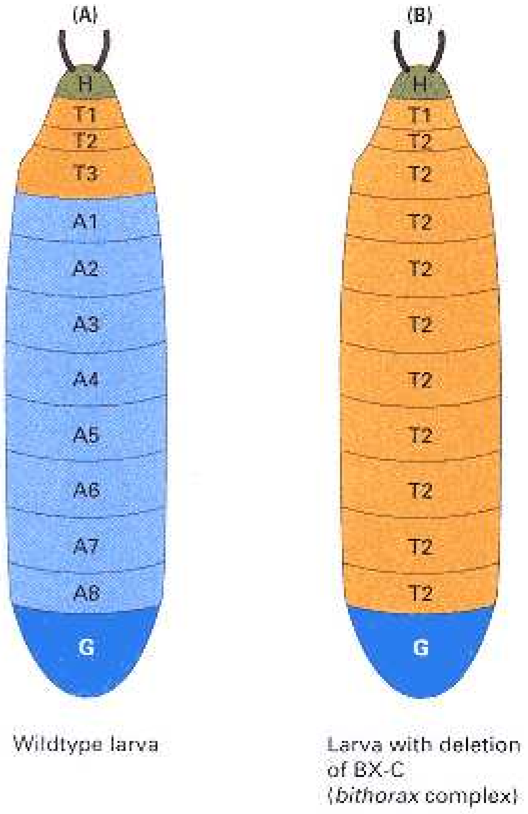

Figure 12.37
Segmentation patterns in *Drosophila* larvae. (A) A wildtype larva. (B) A mutant. The wildtype larva has three thoracic segments (T1-T3) and eight abdominal segments (A1-A8), in addition to head (H) and genital (G) segments. The mutant larva has a genetic deletion of most of the *bithorax* (BX-C) complex. The segments H, T1, and G are normal. All other segments develop as T2 segments.

The homeotic genes are transcriptional activators of other genes. Most homeotic genes contain one or more copies of a characteristic sequence of about 180 nucleotides called a **homeobox,** which is also found in key genes concerned with the development of embryonic segmentation in organisms as diverse as segmented worms, frogs, chickens, mice, and human beings. Homeobox sequences are present in exons; they code for a protein-folding domain that includes a helix-turn-helix DNA-binding motif (Chapter 11), as well as other transcriptional activation components that are not so well understood.

12.5— Genetic Control of Development in Higher Plants

Reproductive and developmental processes in plants differ significantly from those in other eukaryotes. For example, plants have an alternation of generations between the diploid sporophyte and the haploid gametophyte, and the plant germ line is not established in a discrete location during embryogenesis but rather at many locations in the adult organism. In a corn plant, for example, each ear contains germ-line cells that undergo meiosis to form the pollen and ovules.

In animals, as we have seen, most of the major developmental decisions are made early in life, in embryogenesis. In higher plants, however, differentiation takes place almost continuously throughout life in regions of actively dividing cells called **meristems** in both the vegetative organs (root, stem, and leaves) and the floral organs (sepal, petal, pistil, and stamen). The shoot and root meristems are formed during embryogenesis and consist of cells that divide in distinctive geometric planes and at different rates to produce the basic morphological pattern of each organ system. The floral meristems are established by a reorganization of the shoot meristem after embryogenesis and eventually differentiate

Figure 12.38
The ability of plant development to adjust to perturbations is illustrated by this tree. Encroached on by a fence, it eventually incorporates the fence into the trunk.
[Courtesy of Robert Pruitt.]

into floral structures characteristic of each particular species. One important difference between animal and plant development is that

> In higher plants, as groups of cells leave the proliferating region of the meristem and undergo further differentiation into vegetative or floral tissue, their developmental fate is determined almost entirely by their position relative to neighboring cells.

The critical role of positional information in development of higher plants stands in contrast to animal development, in which cell lineage often plays a key role in determinating cell fate.

The plastic or "indeterminate" growth patterns of higher plants are the result of continuous production of both vegetative and floral organ systems. These patterns are conditioned largely by day length and the quality and intensity of light. The plasticity of plant development gives plants a remarkable ability to adjust to environmental insults. Figure 12.38 shows a tree that, over time, adjusts to the presence of a nearby fence by engulfing it into the trunk Higher plants can also adjust remarkably well to a variety of genetic aberrations. For example, transgenic plants of *Arabidopsis thaliana* (a member of the mustard family) have been created that either overexpress or underexpress cyclin B. Overexpression of cyclin B results in an accelerated rate of cell division; underexpression of cyclin B results in a decelerated rate. Plants with the faster rate of cell division contain more cells and are somewhat larger than their wildtype counterparts, but otherwise they look completely normal. Likewise, plants with the decreased rate of cell division have less than half the normal number of cells, but they grow at almost the same rate and reach almost the same size as wildtype plants, because as the number of cells decreases, each individual cell gets larger. One of the important consequences of plants being able to adjust to abnormal growth conditions is that plant cells rarely undergo a transformation into proliferative cancer cells, as often happens in animals. The common plant tumors are produced only as a result of complex interactions with pathogens such a *Agrobacterium* (Chapter 9).

Flower Development in Arabidopsis

Genetic analysis of *Arabidopsis* has revealed some important principles in the genetic determination of floral structures. As is typical of flowering plants, the flowers of *Arabidopsis* are composed of four types of organs arranged in concentric rings or whorls. Figure 12.39 illustrates the geometry, looking down at a flower from the top. From outermost to innermost, the whorls are designated 1, 2, 3, and 4 (Figure 12.39A). In the development of the flower, each whorl gives rise to a different floral organ (Figure 12.39B). Whorl 1 yields the sepals (the green, outermost floral leaves), whorl 2 the petals (the white, inner floral leaves), whorl 3 the stamens (the male organs, which form pollen), and whorl 4 the carpels (which fuse to form the ovary).

Mutations that affect floral development fall into three major classes, each with a characteristic phenotype (Figure 12.40). Compared with the wildtype flower (panel A), one class lacks sepals and petals (panel B), another class lacks petals and stamens (panel C), and the third class lacks stamens and carpels (panel D). On the basis of crosses between homozygous mutant organisms, these classes of mutants can be assigned to four complementation groups, each of which defines a different gene (Chapter 2). Each gene and the phenotype of a plant homozygous for a recessive mutation in the gene are shown in Table 12.1. The phenotype lacking sepals and petals is caused by mutations in the gene *ap2 (apetala-2).* The phenotype lacking stamens and petals is caused by a mutation in either of two genes, *ap3 (apetala-3)* and *pi (pistillata).* The phenotype lacking stamens and carpels is caused by mutations in the gene *ag (agamous).* Each of these genes has been cloned and sequenced. They are all transcription factors. The transcription factors encoded by *ap3, pi,* and *ag* are members of what is called the *MADS box* family of transcription factors; each member of this family contains a sequence of 58 amino acids in which common features can be identified. MADS box transcription factors are very common in plants but are also found, less frequently, in animals.

Combinatorial Determination of the Floral Organs

The role of the *ap2, ap3, pi,* and *ag* transcription factors in the determination of floral organs can be inferred from the phenotypes of the mutations. The logic of the inference is based on the observation (see Table 12.1) that mutation in any of the genes eliminates two floral organs that arise from adjacent whorls. This pattern suggests that *ap2* is necessary for sepals and petals, *ap3* and *pi* are both necessary for petals and stamens, and *ag* is necessary for stamens and carpels. Because the mutant phenotypes are caused by loss-of-function alleles of the genes, it may be inferred that *ap2* is expressed in whorls 1 and 2, that *ap3* and *pi* are expressed in whorls 2 and 3, and that *ag* is expressed in whorls 3 and 4. The overlapping patterns of expression are shown in Table 12.2.

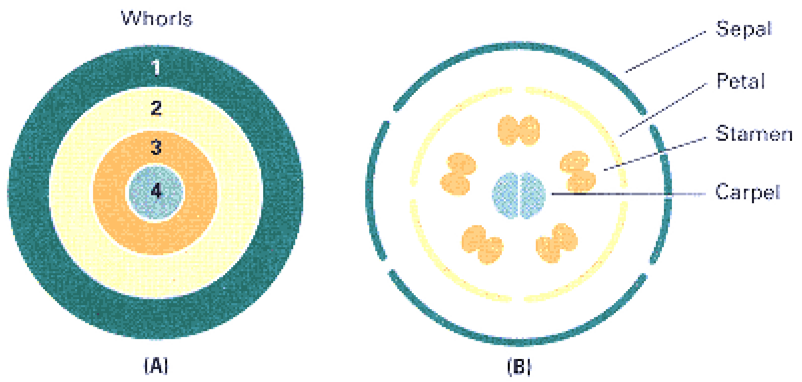

Figure 12.39
(A) The organs of a flower are arranged in four concentric rings, or whorls. (B) Whorls 1, 2, 3, and 4 give rise to sepals, petals, stamens, and carpels, respectively.

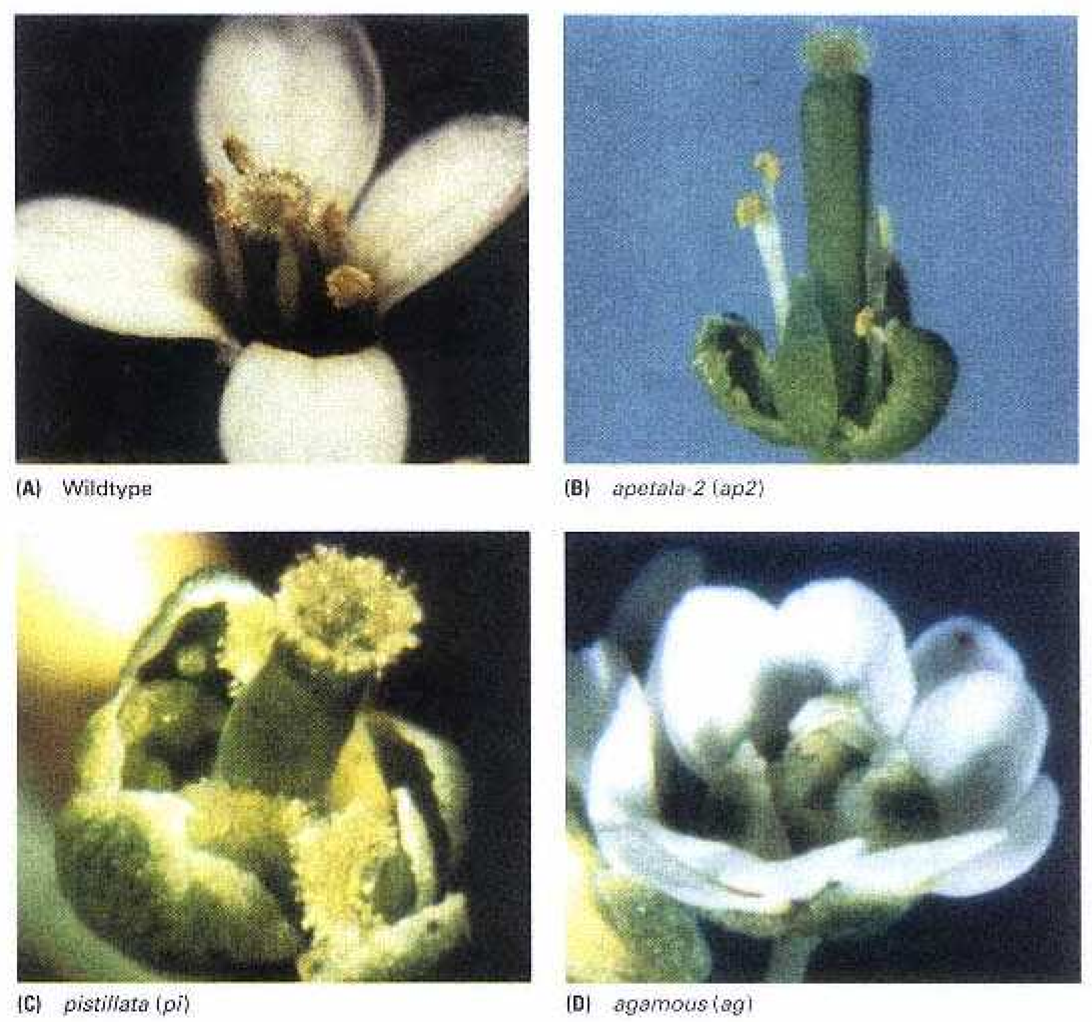

Figure 12.40
Phenotypes of the major classes of floral mutations in *Arabidopsis*. (A) The wildtype floral pattern consists of concentric whorls of sepals, petals, stamens, and carpels. (B) The homozygous mutation *ap2 (apetala-2)* results in flowers missing sepals and petals. (C) Genotypes that are homozygous for either *ap3 (apetala-3*) or *pi (pistillala*) yield flowers that have sepals and carpels but lack petals and stamens. (D) The homozygous mutation *ag (agamous*) yields flowers that have sepals and petals but lack stamens and carpels.
[Courtesy of Elliot M. Meyerowitz and John Bowman.
Part B from Elliot M. Meyerowitz. 1994. The genetics of flower development. *Scientific American,* 271: 56 (November 1994).]

Table 12.1 Floral development in mutants of *Arabidapsis*

	Whorl			
Genotype	**1**	**2**	**3**	**4**
wildtype	sepals	petals	stamens	carpels
ap2/ap2	carpels	stamens	stamens	carpels
ap3/ap3	sepals	sepals	carpels	carpels
pi/pi	sepals	sepals	carpels	carpels
ag/ag	sepals	petals	petals	sepals

Table 12.2 Domains of expression of genes determining floral development

Whorl	Genes expressed	Determination
1	*ap2*	sepal
2	*ap2* + *ap3* and *pi*	petal
3	*ap3* and *pi* + *ag*	stamen
4	*ag*	carpel

The model vcof gene expression in Table 12.2 suggests that floral development is controlled in combinatorial fashion by the four genes. Sepals develop from tissue in which only *ap2* is active; petals are evoked by a combination of *ap2*, *ap3*, and *pi;* stamens are determined by a combination of *ap3*, *pi* and *ag;* and carpels derive from tissue in which only *ag* is expressed. This model is illustrated graphically in Figure 12.41.

You may have noted already that the model in Table 12.2 does not account for all the phenotypic features of the *ap2* and *ag* mutations in Table 12.1. In particular, according to the commbinatorial model in Table 12.2, the development of carpels and stamens from whorls 1 and 2 in homozygous *ap2* plants would require expression of *ag* in whorls 1 and 2. Similarly, the development of petals and sepals from whorls 3 and 4 in homozygous *ag* plants

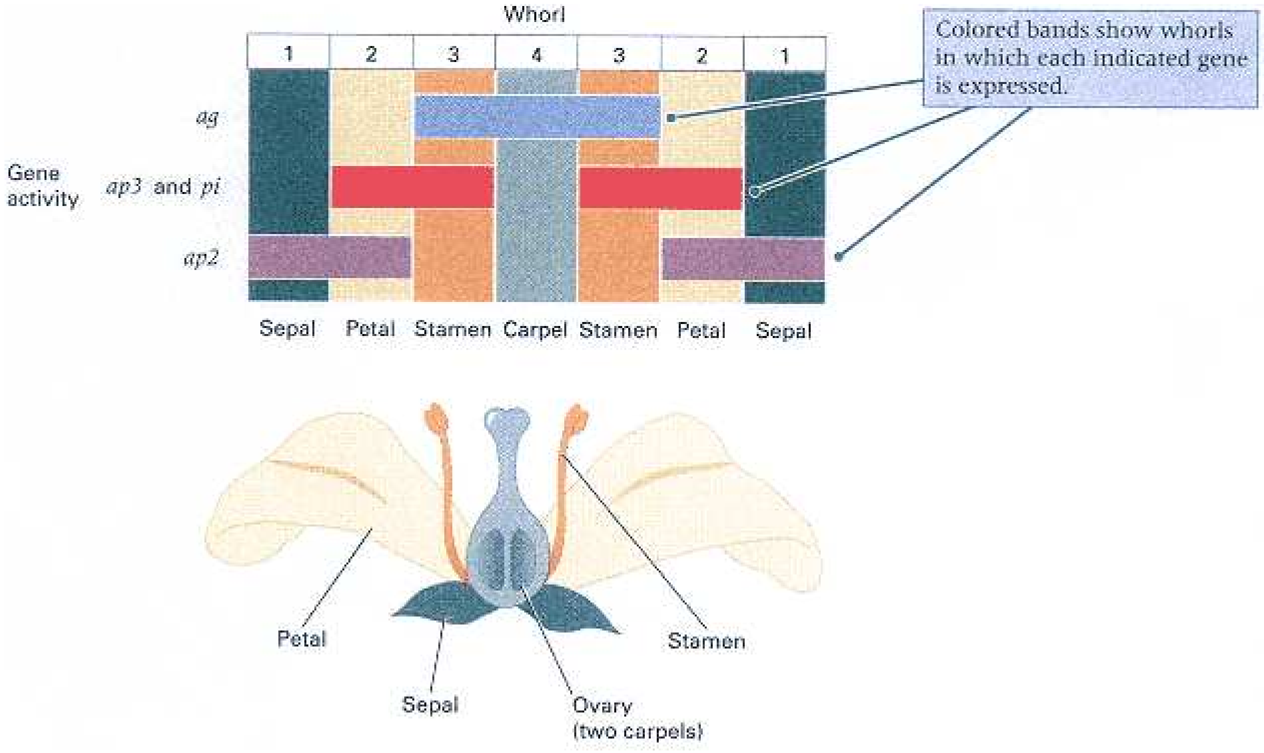

Figure 12.41
Control of floral development in *Arabidopsis* by the overlapping expression of four genes. The sepals, petals, stamens, and carpels are floral organ systems that form in concentric rings or whorls. The developmental identity of each concentric ring is determined by the genes *ap2*, *ap3* and *pi*, and *ag*, each of which is expressed in two adjacent rings. Gene *ap2* is expressed in the outermost two rings, *ap3* and *pi* in the middle two, and ag in the inner two. Therefore, each whorl has a unique combination of active genes.

would require expression of *ap2* in whorls 3 and 4. This discrepancy can be explained if it is assumed that *ap2* expression and *ag* expression are mutually exclusive: If the presence of the *ap2* transcription factor, *ag* is repressed, and in the presence of the *ag* transcription factor, *ap2* is repressed. If this were the case, then, in *ap2* mutants, *ag* expression would spread into whorls 1 and 2, and, in *ag* mutants, *ap2* expression would spread into whorls 3 and 4. This additional assumption, enables us to explain the phenotypes of the single and even double mutants.

With the additional assumption we have made about *ap2* and *ag* interaction, the model in Table 12.2 fits the data, but is the model true? For these genes, the patterns of gene expression, assayed by *in situ* hybridization of RNA in floral cells with labeled probes for each of the genes, fits the patterns in Table 12.2. In particular, *ap2* is expressed in whorls 1 and 2, *ap3* and *pi* in whorls 2 and 3, and *ag* in whorls 3 and 4. (The *ap2* gene is also expressed in nonfloral tissue, but its role in these tissues is unknown.) Furthermore, the seemingly *ad hoc* assumption about *ap2* and *ag* expression being mutually exclusive turns out to be true. In *ap2* mutants, *ag* is expressed in whorls 1 and 2; reciprocally, in *ag* mutants, *ap2* is expressed in whorls 3 and 4. It is also known how *ap3* and *pi* work together. The active transcription factor that corresponds to these genes is a dimeric protein composed of Ap3 and Pi polypeptides. Each component polypeptide, in the absence of the other, remains inactive in the cytoplasm. Together, they form an active dimeric transcription factor that migrates into the nucleus.

Figure 12.42
The homozygous triple mutant *ap2 pi ag* lacks all of the transcription factors needed for floral development. Hence the "flowers" lack all of the wildtype floral structures. They have no sepals, petals, stamens, or carpels. Without the floral genetic determinants, the flowers consist entirely of leaves arranged in concentric whorls [Courtesy of Elliot M. Meyerowitz and John Bowman.]

Given the critical role of the Ap2, Ap3/Pi, and Ag transcription factors in floral determination, it might be speculated that triple mutants lacking all three types of transcription factors would have very strange flowers. The phenotype of the *ap2 pi ag* triple mutant is shown in the photograph in Figure 12.42. The flowers have none of the normal floral organs. They consist merely of leaves arranged in concentric whorls.

Chapter Summary

The genotype determines the developmental potential of the embryo. By means of a developmental program that results in different sets of genes being expressed in different types of cells, the genotype controls the developmental events that take place and their temporal order. Mutations that interrupt developmental processes identify genetic factors that control development.

The early development of the animal embryo establishes the basic developmental plan for the whole organism. The earliest events in embryonic development depend on the correct spatial organization of numerous constituents present in the oocyte. Developmental genes that are needed in the mother for proper oocyte formation and zygotic development are maternal-effect genes. Genes that are required in the zygote nucleus are zygotic genes. Fertilization of the oocyte initiates a series of mitotic cleavage divisions that form the "hollow ball" blastula, which rapidly undergoes a restructuring into the gastrula. Accompanying these early morphological events are a series of molecular events that determine the developmental fates that cells undergo. Execution of a developmental state may be autonomous (genetically programmed), or it may require positional information supplied by neighboring cells or the local concentration of one or more morphogens.

The soil nematode *Caenorhabditis elegans* is used widely in studies of cell lineages because many lineages in the organism undergo virtually autonomous development and the developmental program is identical from one organism to the next. Most lineages are affected by many genes, including genes that control the sublineages into which the lineage can differentiate. Mutations that affect cell lineages define several types of developmental mutants: (1) execution mutants, which prevent a developmental program from being carried out; (2) transformation mutants, in which a wrong developmental program is executed; (3) segregation mutants, in which developmental determinants fail to segregate normally between sister cells or mother and daughter cells; (4) apoptosis (programmed cell death) mutants, in which cells fail to undergo a normal developmental program that normally leads to their death; and (5) heterochronic mutants, in which execution of a normal developmental program is either precocious or retarded in time.

Genes that control key points in development can often be identified by the unusual feature that recessive alleles (ideally, loss-of-function mutations) and dominant alleles (ideally, gain-of-function mutations) have opposite effects on phenotype. For example, if loss of function results in failure to execute a developmental program in a particular anatomical position, then gain of function should result in execution of the program in an abnormal location. The *lin-12* gene in *C. elegans* is an important developmental control gene identified by these criteria. The *lin-12* gene controls the developmental "decision" of a pair of cells whether to become anchor cells or ventral uterine precursor cells. The gene codes for a transmembrane receptor protein that shares some domains with the mammalian peptide hormone epidermal growth factor, shares other domains with the *Notch* protein in *Drosophila* (which controls epidermal or neural commitment), and shares still other domains with proteins that control the cell cycle. In mutations that inactivate *lin-12,* both precursors develop into anchor cells; in mutations in which *lin-12* is overexpressed, both become ventral uterine precursor cells.

Early development in *Drosophila* includes the formation of a syncytial blastoderm by early cleavage divisions without cytoplasmic division, the setting apart of pole cells that form the germ line at the posterior of the embryo, the migration of most nuclei to the periphery of the syncytial blastoderm, cellularization to form the cellular blastoderm, and determination of the blastoderm fate map at or before the cellular blastoderm stage. Metamorphosis into the adult fly makes use of about 20 imaginal disks present in the larva that contain developmentally committed cells that divide and develop into the adult structures. Most imaginal disks include several discrete groups of cells or compartments separated by boundaries that progeny cells do not cross.

Early development in *Drosophila* to the level of segments and parasegments requires four classes of segmentation genes: (1) coordinate genes that establish the basic anterior-posterior and dorsal-ventral aspect of the embryo, (2) gap genes for longitudinal separation of the embryo into regions, (3) pair-rule genes that establish an alternating on/off striped pattern of gene expression along the embryo, and (4) segment-polarity genes that refine the patterns of gene expression within the stripes and determine the basic layout of segments and parasegments. For example, the *bicoid* gene is a maternal-effect gene in the coordinate class that controls anterior-posterior differentiation of the embryo and determines position on the blastoderm fate map. The *bicoid* gene product is a transcriptional activator protein that is present in a concentration gradient from anterior to posterior along the embryo. Genes that are regulated by *bicoid* have different sensitivities to its concentration, depending on the number and types of high-affinity and low-affinity binding sites that they contain. The segmentation genes can regulate themselves, other members of the same class, and genes of other classes farther along the hierarchy. Together, the segmentation genes control the homeotic genes that initiate the final stages of developmental specification.

Mutations in homeotic genes result in the transformation of one body segment into another. For example, *bithorax* causes transformation of the anterior part of the third thoracic segment into the anterior part of the second thoracic segment, and *Antennapedia* results in transformation of the dorsal part of the head into the second thoracic segment. Homeotic genes act within developmental compartments to control other genes concerned with such characteristics as rates of cell division, orientation of mitotic spindles, and the capacity to differentiate bristles, legs, and other features. Both the *bithorax* complex (BX-C) and the *Antennapedia* complex (ANT-C) are clusters of genes, each containing homeoboxes that are characteristic of developmental-control genes in many organisms and that code for

DNA-binding and other protein domains.

Developmental processes in higher plants differ significantly from those in animals in that developmental decisions continue throughout life in the meristem regions of the vegetative organs (root, stem, and leaves) and the floral organs (sepal, petal, pistil, and stamen). However, genetic control of plant development is mediated by transcription factors analogous to those in animals. For example, control of floral development in *Arabidopsis* is through the combinatorial expression of four genes in each of a series of four concentric rings, or whorls, of cells that eventually form the sepals, petals, stamens, and carpels.

Key Terms

apoptosis	homeobox	segmentation gene
autonomous determination	homeotic gene	segment-polarity gene
blastoderm	imaginal disk	segregation mutation
blastula	ligand	syncytial blastoderm
cascade	lineage	transformation mutation
cell fate	lineage diagram	transmembrane receptor
cleavage division	loss-of-function mutation	zygotic gene
compartment	maternal-effect genes	
coordinate gene	meristem	
embryonic induction	morphogen	
execution mutation	pair-rule gene	
fate map	parasegment	
gain-of-function mutation	pattern formation	
gap gene	pole cell	
gastrula	positional information	
heterochronic mutation	programmed cell death	

Review the Basics

• What is meant by *polarity* in the mature oocyte?

• What is a loss-of-function mutation? What is a gain-of-function mutation? In developmental genetics, what is the significance of the observation that loss-of-function alleles and gain-of-function alleles of the same gene have opposite phenotypes?

• How does knowledge of the complete cell lineage of nematode development demonstrate the importance of programmed cell death (apoptosis) in development?

• Among genes that control embryonic development in *Drosophila*, distinguish between coordinate genes, gap genes, pair-rule genes, and segment-polarity genes. Generally speaking, what is the temporal order of expression of these classes of genes?

• What is a homeotic mutation? Give an example from *Drosophila*. Do homeotic mutations occur in organisms other than *Drosophila?*

• Do plants have a germ line in the same sense as animals? Explain.

• What does it mean to say that a cell in an organism is *totipotent?* Are all cells in a mature higher plant totipotent?

• What is the genetic basis of the developmental determination of sepals, petals, stamens, and carpels in floral development in *Arabidopsis?*

Guide to Problem Solving

Problem 1: What is the logic behind the following principle of developmental genetics: "For a gene affecting cell determination, if loss-of-function alleles eliminate a particular cell fate, and gain-of-function alleles induce the fate, then the product of the gene must be both necessary and sufficient for expression of the fate."

Answer: A gene product is necessary for cell fate if its absence prevents normal expression of the fate. Hence the effect of loss-of-function alleles means that the gene product is necessary for the fate. On the other hand, a gene product is sufficient for cell determination if its presence at the wrong time, in the wrong tissues, or in the wrong amount results in expression of the cell fate. The fact that gain-of-function mutations induce a particular cell fate means that the gene product is sufficient to trigger the fate.

Problem 2: A mutation *m* is called a recessive maternal-effect lethal if eggs from homozygous *mm* females are unable to support normal embryonic development, irrespective of the genotype of the zygote. What kinds of crosses are necessary to produce *mm* females?

Answer: Although the eggs from *mm* females are abnormal, those from m^+m females allow normal embryonic development because the wildtype m^+ allele supplies the cytoplasm of the oocyte with sufficient products for embryogenesis. Therefore, homozygous *mm* females can be obtained by crossing heterozygous m^+ females with either m^+m or *mm* males. The first cross produces 1/4 *mm* females, the second 1/2 *mm.*

Problem 3: The *bicoid* mutation in *Drosophila* results in absence of head structures in the early embryo. A similar phenotype results when cytoplasm is removed from the anterior end of a *Drosophila* embryo. What developmental effect would you expect in each of the following circumstances?

(a) A *bicoid* embryo was injected in the anterior end with some cytoplasm taken from the anterior end of a wildtype embryo.

(b) A wildtype *Drosophila* embryo was injected in the middle with some cytoplasm taken from the anterior end of another wildtype embryo.

Answer: The experimental results imply that the anterior cytoplasm, including the *bicoid* gene product, induces development of head structures.

(a) Anterior cytoplasm from a wildtype embryo should be able to rescue *bicoid,* so head structures should be formed.

(b) Anterior cytoplasm injected into the middle of the embryo should induce head structures in that location, so the embryo should have head structures at the anterior end and also in the middle.

Analysis and Applications

12.1 Distinguish between a developmental fate determined by autonomous development and one determined by positional information. What two types of surgical manipulation are used to distinguish between the processes experimentally?

12.2 What are lineage mutations and how are they detected?

12.3 What is the *Drosophila* blastoderm, and what is the blastoderm fate map? Does the existence of a fate map imply that *Drosophila* developmental decisions are autonomous?

12.4 What is the result of a maternal-effect lethal allele in *Drosophila?* If an allele is a maternal-effect lethal, how can a fly be homozygous for it?

12.5 What is the principal consequence of a failure in programmed cell death?

12.6 With regard to its effects on cell lineages, what kind of mutation is a homeotic mutation in *Drosophila?*

12.7 A cell, A, in *Caenorhabditis elegans* normally divides and the daughter cells differentiate into cell types B and C. What developmental pattern would result from a mutation in cell A that prevents sister-cell differentiation? From a mutation in cell A that prevents parent-offspring segregation?

12.8 A mutation is found in which the developmental pattern is normal but slow. Does this qualify as a heterochronic mutation? Explain.

12.9 Why is transcription of the zygote nucleus dispensable in *Drosophila* early development but not in early development of the mouse?

12.10 Distinguish between a loss-of-function mutation and a gain-of-function mutation. Can the same gene undergo both types of mutations? Can the same allele have both types of effects?

12.11 A particular gene is necessary, but not sufficient, for a certain developmental fate. What is the expected phenotype of a loss-of-function mutation in the gene? Is the allele expected to be dominant or recessive?

12.12 A particular gene is sufficient for a certain developmental fate. What is the expected phenotype of a gain-of-function mutation in the gene? Is the allele expected to be dominant or recessive?

12.13 What is the phenotype of a *Drosophila* mutation in the *gap* class? Of a mutation in the *pair-rule* class?

12.14 The drug actinomycin D prevents RNA transcription but has little direct effect on protein synthesis. When fertilized sea urchin eggs are immersed in a solution of the drug, development proceeds to the blastula stage, but gastrulation does not take place. How would you interpret this finding?

12.15 A mutation in the axolotl designated *o* is a maternal-effect lethal because embryos from *oo* females die at gastrulation, irrespective of their own genotype. However, the embryos can be rescued by injecting oocytes from *oo* females with an extract of nuclei from either o^+o^+ or o^+o eggs. Injection of cytoplasm is not as effective. Suggest an explanation for these results.

12.16 The nuclei of brain cells in the adult frog normally do not synthesize DNA or undergo mitosis. However, when transplanted into developing oocytes, the brain cell nuclei behave as follows: (a) In rapidly growing premeiosis oocytes, they synthesize RNA. (b) In more mature oocytes, they do not synthesize DNA or RNA, but their chromosomes condense and they begin meiosis. How would you explain these results?

Chapter 12 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. Browse the *C. elegans* database, ACeDB, to learn more about *lin-12* and other developmental-control genes in this organism. Choose *Locus* and then the alphabetical sublist containing *lin-12.* Read the entry for *lin-12.* If assigned to do so, pick any other of the *lin* mutations, and write a short description of its phenotype and genetic map position.

2. Great scanning electron microscopic images of *Drosophila* **embryogenesis** can be found at this site. Examine the process of gastrulation from the dorsal and lateral views. Note the prominence of the pole cells. If assigned to do so, pick one of the mutants (*bcd* or *ftz*), and sketch a mutant and wildtype embryo at approximately the same stage of development, pointing out the differences.

3. Arabidopsis is an important model organism for genetic studies of plant development. Use this site to find the genetic map positions of *ap2, ap3, pi,* and *ag.* If assigned to do so, show the rela-

(text box continued on next page)

Challenge Problems

12.17 The autosomal gene *rosy* (*ry*) in *Drosophila* is the structural gene for the enzyme xanthine dehydrogenase (XDH), which is necessary for wildtype eye pigmentation. Files of genotype *ry/ry* lack XDH activity and have rosy eyes. The X-linked gene *maroonlike* (*mal*) is also necessary for XDH activity, and *mal/mal;* ry^+/ry^+ females and *mal*/Y; ry^+/ry^+ males also lack XDH activity; they have maroonlike eyes. The cross mal^+/mal; *ry/ry/* × *mal*/Y; ry^+/ry^+ produces *mal/mal;* ry^+/ry females and *mal*/Y; ry^+/ry males that have wildtype eye color even though they lack active XDH enzyme. Suggest an explanation.

12.18 You wish to demonstrate that during segmentation of the *Drosophila* embryo, normal pair-rule patterns of expression require normal expression of the gap-genes, whereas gap gene expression does not require pair-rule expression. You have the following four mutations available:

(a) A mutation in the zygotic-effect gap gene *knirps* (*kni*).

(b) A mutation in the zygotic-effect pair-rule gene *fushi tarazu* (*ftz*).

(c) A transgene consisting of a reporter gene (*lacZ*) fused to the enhancer elements of *kni.*

(d) A transgene consisting of a reporter gene (*lacZ*) fused to the enhancer elements of *ftz.*

Describe the strains you would need and how you would use them to show that *kni* is epistatic to *ftz.* You do not need to give details of the crosses.

Further Reading

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(text box continued from previous page)

tive positions of these genes on the genetic map of the entire genome.

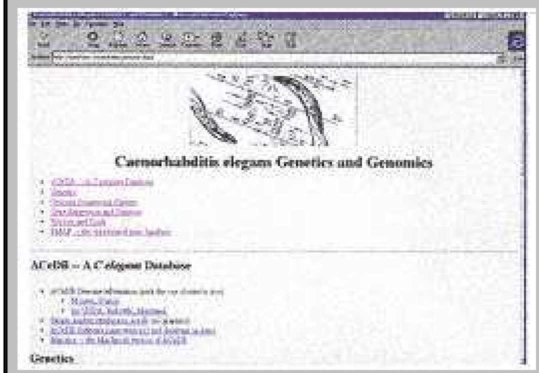
4. Check out this site for further illustrations of **floral development,** including sites of expression of some of the major genes. If assigned to do so, find which of the genes involved in floral development is expressed in all four whorls and describe the hypotheses put forward to explain this unexpected observation.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 12, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 12.



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This hummingbird is a rare, mutant form of the ruby-throated humming-bird found in the eastern United States. It has no pigment in its piumage. The round tail is a sign that this bird is a female. [Courtesy of Steve and Dave Maslowski.]

Chapter 13— Mutation, DNA Repair, and Recombination

CHAPTER OUTLINE

GeNETics on the web

PRINCIPLES

• Substitution of one base for another is an important mechanism of spontaneous mutation. A single base substitution in a coding region may result in an amino acid replacement; a single-base deletion or insertion results in a shifted reading frame.

• Transposable element insertion is also an important mechanism of spontaneous mutation.

• Mutations can be induced by various agents, including highly reactive chemical and x rays.

• Most mutagens are also carcinogens.

• Cells contain enzymatic pathways for the repair of different types of damage to DNA. Among the most important repair systems is mismatch repair of duplex DNA, in which a nucleotide that contains a mismatched base is excised and replaced with the correct nucleotide.

• Recombination between homologous DNA molecules includes the invasion of a duplex by one or both strands from another, forming a heteroduplex whose junction can migrate until it is finally resolved by breakage and reunion. Mismatch repair in the heteroduplex region can lead to 3 : 1 or 1 : 3 segregation of alleles; these types of aberrant segregation can be observed directly in organisms in which all four products of meiosis remain together, such as in fungal asci.

CONNECTIONS

CONNECTION: X-Ray Daze

Hermann J. Muller 1927
Artificial transmutation of the gene

CONNECTION: Replication Slippage in Unstable Repeats

Micheline Strand, Tomas A. Prolla, R. Michael Liskay, and Thomas D. Petes 1993
Destabilization of tracts of simple repetitive DNA in yeast by mutations affecting DNA mismatch repair

In preceding chapters, numerous examples were presented in which the information contained in the genetic material had been altered by mutation. A **mutation** is any heritable change in the genetic material. In this chapter, we examine the nature of mutations at the molecular level. You will learn how mutations are created, how they are detected phenotypically, and the means by which many mutations are corrected by special DNA repair enzymes almost immediately after they occur. You will see that mutations can be induced by radiation and a variety of chemical agents that produce strand breakage and other types of damage to DNA. The breakage and repair of DNA serve as an introduction to the process of recombination at the DNA level.

13.1— General Properties of Mutations

Mutations can happen at any time and in any cell. The phenotypic effects can range from minor alterations that are detectable only by biochemical methods to drastic changes in essential processes that cause, at one extreme, unrestrained cell proliferation (cancer) or, at the other extreme, the death of the cell or organism. Mutations that produce clearly defined effects are regularly used, when genetic phenomena are studied in the laboratory, but most mutations are not of this type. The effect of a mutation is determined by the type of cell containing the mutant allele, by the stage in the life cycle or development of the organism that the mutation affects, and, in diploid organisms, by the dominance or recessiveness of the mutant allele. A recessive mutation is usually not detected until a later generation when two heterozygous genotypes mate. Dominance does not complicate the expression of mutations in bacteria and haploid eukaryotes.

Mutations can be classified in a variety of ways. In multicellular organisms, one distinction is based on the type of cell in which the mutation first occurs: those that arise in cells that ultimately form gametes are **germ-line mutations,** all others are **somatic mutations.** A somatic mutation yields an organism that is genotypically, and for many dominant mutations phenotypically, a mixture of normal and mutant tissue. Because reproductive cells are not affected, such a mutant allele will not be transmitted to the progeny and may not be detected or be recoverable for genetic analysis. In higher plants, however, somatic mutations can often be propagated by vegetative means (without going through seed production), such as grafting or the rooting of stem cuttings. This process has been the source of valuable new varieties such as the 'Delicious' apple and the 'Washington' navel orange.

Among the mutations that are most useful for genetic analysis are those whose effects can be turned on or off at will. These are called **conditional mutations** because they produce changes in phenotype in one set of environmental conditions (called the **restrictive conditions**) but not in another (called the **permissive conditions**). A **temperature-sensitive mutation,** for example, is a conditional mutation whose expression depends on temperature. Usually, the restrictive temperature is high, and the organism exhibits a mutant phenotype above this critical temperature. The permissive temperature is lower, and under permissive conditions the phenotype is wildtype or nearly wildtype. Temperature-sensitive mutations are frequently used to block particular steps in biochemical pathways in order to test the role of the pathways in various cellular processes, such as DNA replication.

An example of temperature sensitivity is found in the ordinary Siamese cat, with its black-tipped paws, ears, and tail. In this breed of cat, the biochemical pathway leading to black pigmentation is temperature-sensitive and inactivated at normal body temperature. Consequently, the pigment is not present in the hair over most of the body. The tips of the legs, ears, and tail are cooler than the rest of the body, so the pigment is deposited in the hair in these areas.

Mutations can also be classified by other criteria, such as the kinds of alterations in the DNA, the kinds of phenotypic effects produced, and whether the mutational events are spontaneous in origin or were induced by exposure to a known **mutagen** (a mutation-causing agent). Such classifications are often useful in discussing aspects of the mutational process. *Spontaneous* usually means that the event that caused a mutation is unknown, and **spontaneous mutations** are those that take place in the absence of any known mutagenic agent. The properties of spontaneous mutations and of **induced mutations** will be described in later sections.

13.2— The Molecular Basis of Mutation

All mutations result from changes in the nucleotide sequence of DNA or from deletions, insertions, or rearrangement of DNA sequences in the genome. Some types of major rearrangements in chromosomes were discussed in Chapter 7. In this section, we discuss mutations whose molecular basis can be specified.

Base Substitutions

The simplest type of mutation is a **base substitution,** in which a nucleotide pair in a DNA duplex is replaced with a different nucleotide pair. For example, in an A $\rightarrow$ G substitution, an A is replaced with a G in one of the DNA strands. This substitution temporarily creates a mismatched G-T base pair; at the very next replication, the mismatch is resolved as a proper G-C base pair in one daughter molecule and as a proper A-T base pair in the other daughter molecule. In this case, the G-C base pair is the mutant and the A-T base pair is non-mutant. Similarly, in an A $\rightarrow$ T substitution, an A is replaced with a T in one strand, creating a temporary T-T mismatch, which is resolved by replication as T-A in one daughter molecule and A-T in the other. In this example, the T-A base pair is mutant and the A-T base pair is nonmutant. The T-A and the A-T are not equivalent, as can be seen by considering the nucleotide context. If the original unmutated DNA strand has the sequence 5'-GAC-3', for example, then the mutant strand has the sequence 5'-GTC-3' (which we have written as T-A), and the nonmutant strand has the sequence 5'-GAC-3' (which we have written as A-T).

Some base substitutions replace one pyrimidine base with the other or one purine base with the other. These are called **transition mutations.** The possible transition mutations are

$$\begin{array}{c} T \rightarrow C \quad \text{or} \quad C \rightarrow T \\ (\text{pyrimidine} \rightarrow \text{pyrimidine}) \\[6pt] A \rightarrow G \quad \text{or} \quad G \rightarrow A \\ (\text{purine} \rightarrow \text{purine}) \end{array}$$

Other base substitutions replace a pyrimidine with a purine or the other way around. These are called **transversion mutations.** The possible transversion mutations are

$$\begin{array}{c} T \rightarrow A,\ T \rightarrow G,\ C \rightarrow A, \quad \text{or} \quad C \rightarrow G \\ (\text{pyrimidine} \rightarrow \text{purine}) \\[6pt] A \rightarrow T,\ A \rightarrow C,\ G \rightarrow T, \quad \text{or} \quad G \rightarrow C \\ (\text{purine} \rightarrow \text{pyrimidine}) \end{array}$$

Altogether, there are four possible transitions and eight possible transversions. Therefore, if base substitutions were strictly random, then one would expect a 1 : 2 ratio of transitions to transversions. However,

> Spontaneous base substitutions are biased in favor of transitions. Among spontaneous base substitutions, the ratio of transitions to transversions is approximately 2 : 1.

Examination of the genetic code (Table 10.2) shows that the bias toward transitions has an important consequence for base substitutions in the third position of codons. In all codons with a pyrimidine in the third position, it does not matter which pyrimidine is present; likewise, in most codons that end in a purine, either purine will do. This means that most transition mutations in the third codon position do not change the amino acid that is encoded. Such mutations change the nucleotide sequence without changing the amino acid sequence; these are called **silent mutations** or **silent substitutions** because they are not detectable as changes in phenotype.

Mutational changes in nucleotides that are outside of coding regions can also be silent. In noncoding regions, which include introns and the DNA between genes, the precise nucleotide sequence is often not critical. These sequences can undergo base substitutions, small deletions or additions, insertions of transposable elements, and other rearrangements, and yet the mutations may have no detectable effect on phenotype. On the other hand, some noncoding sequences do have essential functions —for example, promoters, enhancers, transcription termination signals, and intron splice junctions. Mutations in these sequences often do have phenotypic effects.

Most base substitutions in coding regions do result in changed amino acids; these are called **missense mutations.** A change in the amino acid sequence of a protein may alter the biological properties of the protein. The classic example of a phenotypic effect of a single amino acid change is that responsible for the human hereditary disease **sickle-cell anemia,** which we discussed in Chapter 1. This condition, characterized by a change in shape of the red blood cells into an elongate form that blocks capillaries, results from the replacement of a glutamic acid by a valine at position 6 in the β-hemoglobin chain.

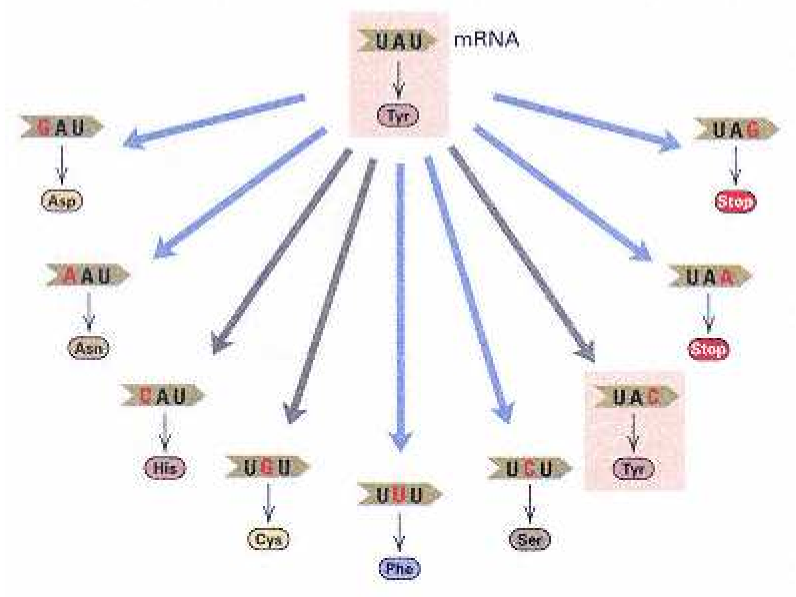


Figure 13.1
The nine codons that can result from a single base change in the tyrosine codon UAU. Blue arrows indicate transversions, gray arrows, transitions. Tyrosine codons are in boxes. Two possible stop ("nonsense") codons are shown in red. Altogether, the codon UAU allows for six possible missense mutations, two possible nonsense mutations, and one silent mutation.

On the other hand, an amino acid replacement does not always create a mutant phenotype. For instance, replacement of one amino acid for another with the same charge (say, lysine for arginine) may in some cases have no effect on either protein structure or phenotype. Whether the substitution of a similar amino acid for another produces an effect depends on the precise role of that particular amino acid in the structure and function of the protein. Any change in the active site of an enzyme usually decreases enzymatic activity.

Figure 13.1 illustrates the nine possible codons that can result from a single base substitution in the UAU codon for tyrosine. One mutation is silent (box), and six are missense mutations that change the amino acid inserted in the polypeptide at this position. The other two mutations create a stop codon resulting in premature termination of translation and production of a truncated polypeptide. A base substitution that creates a new stop codon is called a **nonsense mutation.** Because nonsense mutations cause premature chain termination, the remaining polypeptide fragment is almost always nonfunctional.

Insertions and Deletions

The genomes of most higher eukaryotes contain, at a very large number of locations, tandem repeats of any of a number of short nucleotide sequences; a particularly prevalent repeat in the human genome is that of the dinucleotide CA. A repeat of this type is called a *simple tandem repeat polymorphism*, or *STRP* (Chapter 4). In most organisms in which STRPs are found, the number of copies of the repeat often differs from one chromosome to the next. Hence, populations are usually highly polymorphic for the number of repeating units. The high level of polymorphism makes these repeats very useful in such applications as linkage mapping (Chapter 4), DNA typing (Chapter 15), and family studies to localize genes that influence multifactorial traits (Chapter 16).

STRPs are usually polymorphic because they are susceptible to errors in replication or recombination that change the number of repeats or the length of the run. For example, a run of consecutive CA dinucleotides may have extra copies added or a few deleted. Any short sequence repeated in tandem a number of times may gain or lose a few copies because of these types of errors, although the rate of change in the number of repeats depends, in some unknown manner, on the sequence in question as well as on its location in the genome. The umbrella term generally used to describe the processes leading to a change in the number of copies of a short repeating unit is **replication slippage.** One specific process that can result in such additions or deletions is unequal crossing-over, discussed in Chapter 7 (see Figure 7.20).

An increase in the number of repeating units in an STRP is genetically equivalent to an insertion, and a decrease in the number of copies is equivalent to a deletion. Nonrepeating DNA sequences are also subject to insertion or deletion. The phenotypic consequences of insertion or deletion mutations depend on their location. In nonessential regions, no effects may be seen. STRPs are usually present in noncoding regions. When insertions or deletions happen in regulatory or coding regions, however, their effects may be significant.

When they take place in coding regions, small insertions or deletions add or delete amino acids from the polypeptide, provided that the number of nucleotides added or deleted is an exact multiple of three (the length of a codon). Otherwise, the insertion or deletion shifts the phase in which the ribosome reads the triplet codons and, consequently, alters all of the amino acids downstream from the site of the mutation. As noted in Chapter 10, such mutations are called *frameshift* mutations because they shift the reading frame of the codons in the mRNA. A common type of frameshift mutation is a single-base addition or deletion. The consequences of a frameshift can be illustrated by the insertion of an adenine in this simple mRNA sequence:

Leu	Leu		Leu	Leu	
. . . CUG	CUG		CUG	CUG	. . .
. . . CUG	C**A**U	↓	GCU	GCU	G . . .
Leu	**His**		**Ala**	**Ala**	

Because of the frameshift, all the amino acids downstream from the insertion are different from the original. Any addition or deletion that is not a multiple of three nucleotides will produce a frameshift. Unless it is very near the carboxyl terminus of a protein, a frameshift mutation results in the synthesis of a nonfunctional protein.

Transposable-Element Mutagenesis

All organisms contain multiple copies of several different kinds of transposable elements, which are DNA sequences capable of readily changing their positions in the genome. The structure and function of transposable elements were examined in Chapter 6 (eukaryotic transposable elements), Chapter 8 (prokaryotic transposable elements), and Chapter 9 (in connection with genetic engineering). Transposable elements are also important agents of mutation. For example, in some genes in *Drosophila,* approximately half of all spontaneous mutations that have visible phenotypic effects result from insertions of transposable elements.

Among the ways in which transposable elements can cause mutations are the mechanisms illustrated in Figure 13.2. Most transposable elements are present in nonessential regions of the genome and usually do little or no harm. When an element transposes, however, it can insert into an essential region and disrupt that region's function. Figure 13.2A shows the result of transposition into a coding region of DNA (an exon). The insert interrupts the coding region. Because most transposable elements contain coding regions of their own, either transcription of the transposable element interferes with transcription of the gene into which it is inserted or transcription of the gene terminates within the transposable element. Even if transcription proceeds through the element, the phenotype will be mutant because the coding region then contains incorrect sequences.

Another mechanism of transposable-element mutagenesis results from recombination (Figure 13.2B and C). Transposable elements can be present in multiple copies, often with two or more in the same chromosome. If the copies are near enough

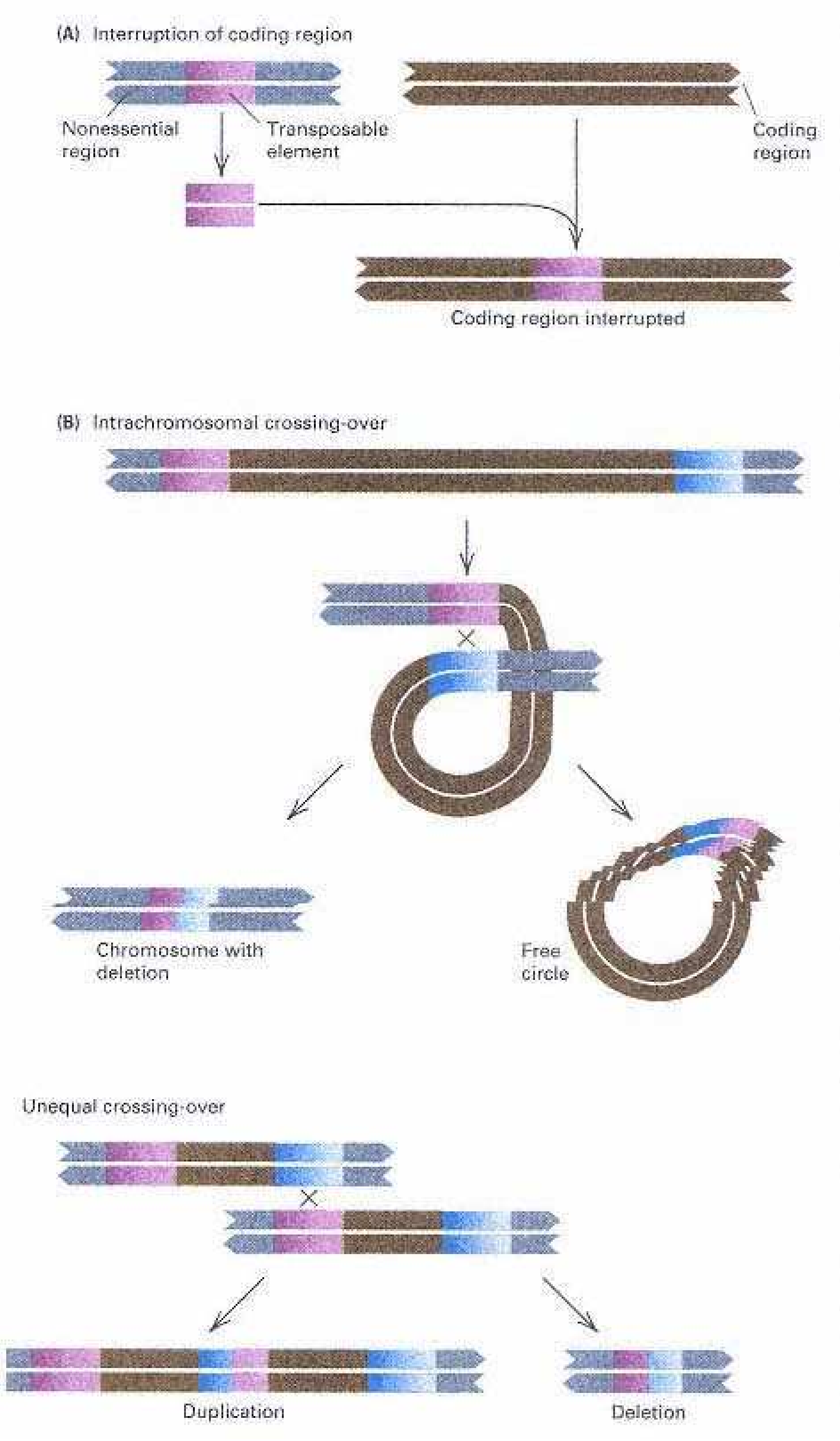


Figure 13.2
Some types of mutations resulting from transposable elements. (A) Insertion into the coding region of a gene eliminates gene function. (B) Crossing-over between two copies of a transposable element present in the same orientation in a single chromatid results in the formation of a single "hybrid" copy and the deletion of the DNA sequence between the copies. (C) Unequal crossing-over between homologous elements present in the same orientation in different chromatids results in products that contain either a duplication or a deletion of the DNA sequences between the elements. Crossing-over takes place at the four-strand stage of meiosis (Chapter 4), but in parts B and C, only the two strands participating in the crossover are shown.

together, they can pair during synapsis and undergo crossing-over, with the result that the region of DNA between the elements is deleted. Figure 13.2B shows what happens when elements located within the same chromosome undergo pairing and crossing-over. Alternatively, the downstream element in one chromosome can pair with the upstream element in the homologous chromosome, diagrammed in Figure 13.2C. An exchange within the paired elements results in one chromatid that is missing the region between the elements (right) and one chromatid in which the region is duplicated (left).

13.3— Spontaneous Mutations

Mutations are statistically random events. There is no way of predicting when, or in which cell, a mutation will take place. However, every gene mutates spontaneously at a characteristic rate, so it is possible to assign probabilities to particular mutational events. Hence there is a definite probability that a specified gene will mutate in a particular cell and, likewise, a definite probability that a mutant allele of a specified gene will appear in a population of a designated size. The various kinds of mutational alterations in DNA differ substantially in complexity, so their probabilities of occurrence are quite different. The mutational process is also random in the sense that whether a particular mutation happens is unrelated to any adaptive advantage it may confer on the organism in its environment. A potentially favorable mutation does not arise *because* the organism has a need for it. The experimental basis for this conclusion will be presented in the next section.

The Nonadaptive Nature of Mutation

The concept that mutations are spontaneous, statistically random events unrelated to adaptation was not widely accepted until the late 1940s. Before that time, it was believed that mutations occurred in bacterial populations *in response to* particular selective conditions. The basis for this belief was the observation that when antibiotic-sensitive bacteria are spread on a solid growth medium containing the antibiotic, some colonies form that consist of cells having an inherited resistance to the drug. The initial interpretation of this observation (and similar ones) was that these adaptive variations were *induced* by the selective agent itself.

Several types of experiments showed that adaptive mutations take place spontaneously and hence were present at low frequency in the bacterial population even *before* it was exposed to the antibiotic. One experiment utilized a technique developed by Joshua and Esther Lederberg called **replica plating** (Figure 13.3). In this procedure, a suspension of bacterial cells is spread on a solid medium. After colonies have formed, a piece of sterile velvet mounted on a solid support is pressed onto the surface of the plate. Some bacteria from each colony stick to the fibers, as shown in Figure 13.3A. Then the velvet is pressed onto the surface of fresh medium, transferring some of the cells from each colony, which give rise to new colonies that have positions identical with those on the first plate. Figure 13.3B shows how this method was used to demonstrate the spontaneous origin of phage T1-r mutants. A master plate containing about 10^7 cells growing on nonselective medium (lacking phage) was replica-plated onto a series of plates that had been spread with about 10^9 T1 phages. After incubation for a time sufficient for colony formation, a few colonies of phages-resistant bacteria appeared in the same positions on each of the selective replica plates. This meant that the T1-r cells that formed the colonies must have been transferred from corresponding positions on the master plate. Because the colonies on the master plate had never been exposed to the phage, the mutations to resistance must have been present, by chance, in a few original cells not exposed to the phage.

The replica-plating experiment illustrates the principle that

> Selective techniques merely select mutants that preexist in a population.

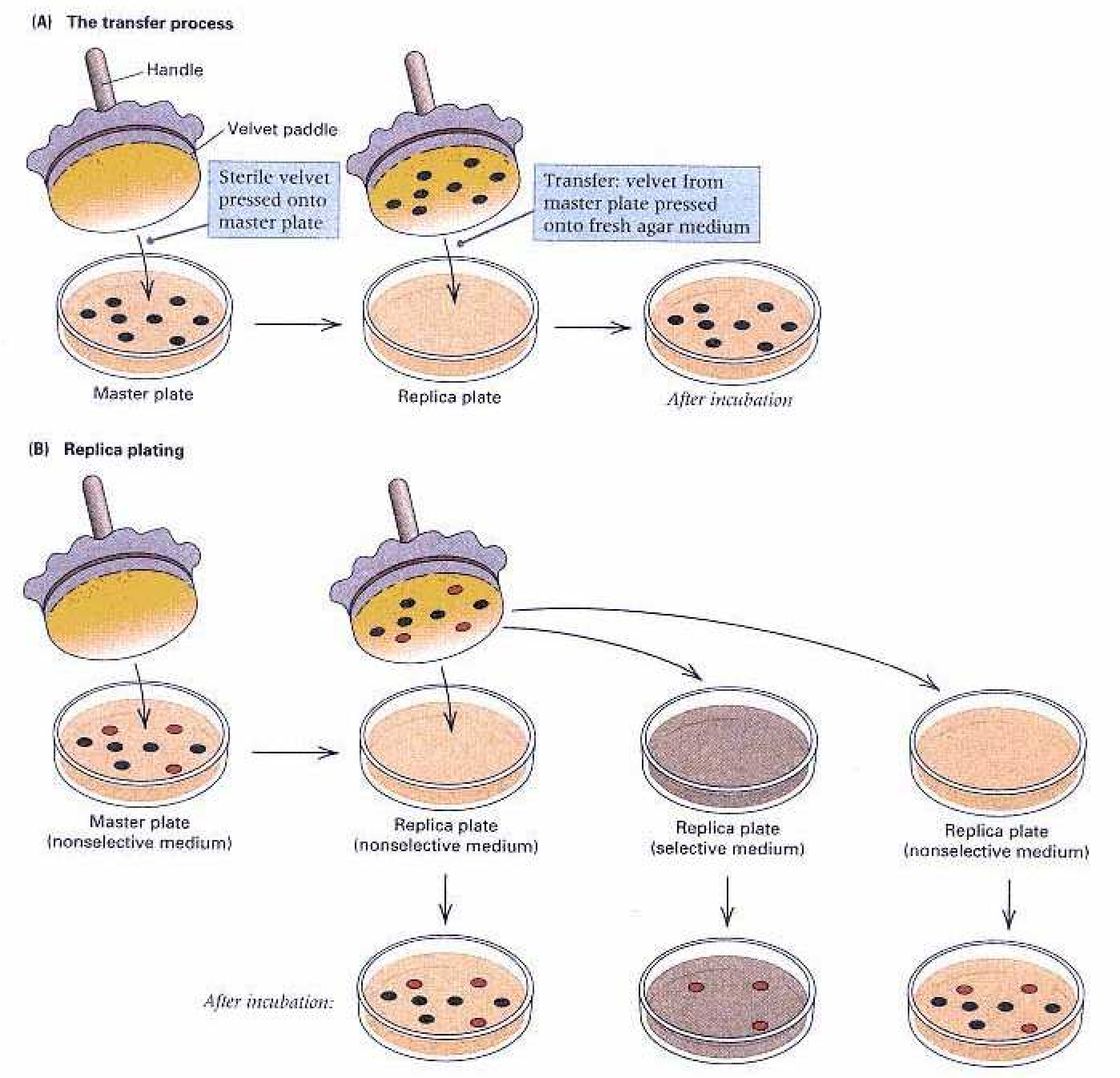


Figure 13.3

Replica plating. (A) In the transfer process, a velvet-covered disk is pressed onto the surface of a master plate in order to transfer cells from colonies on that plate to a second medium. (B) For the detection of mutants, cells are transferred onto two types of plates containing either a nonselective medium (on which all form colonies) or a selective medium (for example,one spread with T1 phages). Colonies form on the nonselective plate in the same pattern as on the master plate. Only mutant cells (for example, T1-r) can grow on the selective plate; the colonies that form correspond to certain positions on the master plate. Colonies consisting of mutant cells are shown in red.

This principle is the basis for understanding how natural populations of rodents, insects, and disease-causing bacteria become resistant to the chemical substances used to control them. A familiar example is the high level of resistance to insecticides, such as DDT, that now exists in many insect populations. It is the result of selection for spontaneous mutations affecting behavioral, anatomical, and enzymatic traits that enable the insect to avoid or resist the chemical. Similar problems are encountered in controlling plant pathogens. For example, the introduction of a new variety of a crop plant that is resistant to a particular strain of disease-causing fungus results in only temporary protection against the disease. The resistance inevitably breaks down because of the occurrence of spontaneous mutations in the fungus that enable it to attack the new plant genotype. Such mutations have a clear selective advantage, and the mutant alleles rapidly become widespread in the fungal population.

Measurement of Mutation Rates

Spontaneous mutations tend to be rare, and the methods used to estimate the frequency with which they arise require large populations and often special techniques. The **mutation rate** is the probability that a gene undergoes mutation in a single generation or in forming a single gamete. Measurement of mutation rates is important in population genetics, in studies of evolution, and in analysis of the effect of environmental mutagens.

One of the earliest techniques to be developed for measuring mutation rates is the ***CIB*** **method.** *CIB* refers to a special X chromosome of *Drosophila melanogaster;* it has a large inversion *(C)* that prevents the recovery of crossover chromosomes in the progeny from a female heterozygous for the chromosome (as described in Section 7.6); a recessive lethal *(l);* and the dominant marker Bar *(B),* which reduces the normal round eye to a bar shape. The presence of a recessive lethal in the X chromosome means that males with that chromosome and females homozygous for it cannot survive. The technique is designed to detect mutations arising in a normal X chromosome.

In the *CIB* procedure, females heterozygous for the *CIB* chromosome are mated with males that carry a normal X chromosome (Figure 13.4). From the progeny produced, females with the Bar phenotype are selected and then individually mated with normal males. (The presence of the Bar phenotype indicates that the females are heterozygous for the *CIB* chromosome and the normal X chromosome from the male parent.) A ratio of two females to one male is expected among the progeny from such a cross, as shown in the lower part of the illustration. The critical observation in determining the mutation rate is the fraction of males produced in the second cross. Because the *CIB* males die, all of the males in this generation must contain an X chromosome derived from the X chromosome of the initial normal male (top row of illustration). Furthermore, it must be a nonrecombinant X chromosome because of the inversion *(C)* in the *CIB* chromosome. Occasionally, the progeny include no males, and this means that the X chromosome present in the original sperm underwent a mutation somewhere along its length to yield a new recessive lethal. The method provides a quantitative estimate of the rate at which mutation to a recessive lethal allele occurs *at any of the large number of genes* in the X chromosome. About 0.15 percent of the X chromosomes acquire new recessive lethals in spermatogenesis; that is, the mutation rate is 1.5×10^{-3} recessive lethals per X chromosome per generation. Note that the *CIB* method tells us nothing about the mutation rate for a particular gene, because the method does not reveal how many genes on the X chromosome would cause lethality if they were mutant. Since the time that the *CIB* method was devised, a variety of other methods have been developed for determining mutation rates in *Drosophila* and other organisms. Of significance is the fact that mutation rates vary widely from one gene to another. For example, the yellow-body mutation in *Drosophila* occurs at a frequency of 10^{-4} per gamete per generation, whereas mutations to streptomycin resistance in *E. coli.* occur at a frequency of 10^{-9} per cell per generation. Furthermore, within a single organism, the frequency can vary enormously, ranging in *E. coli,*

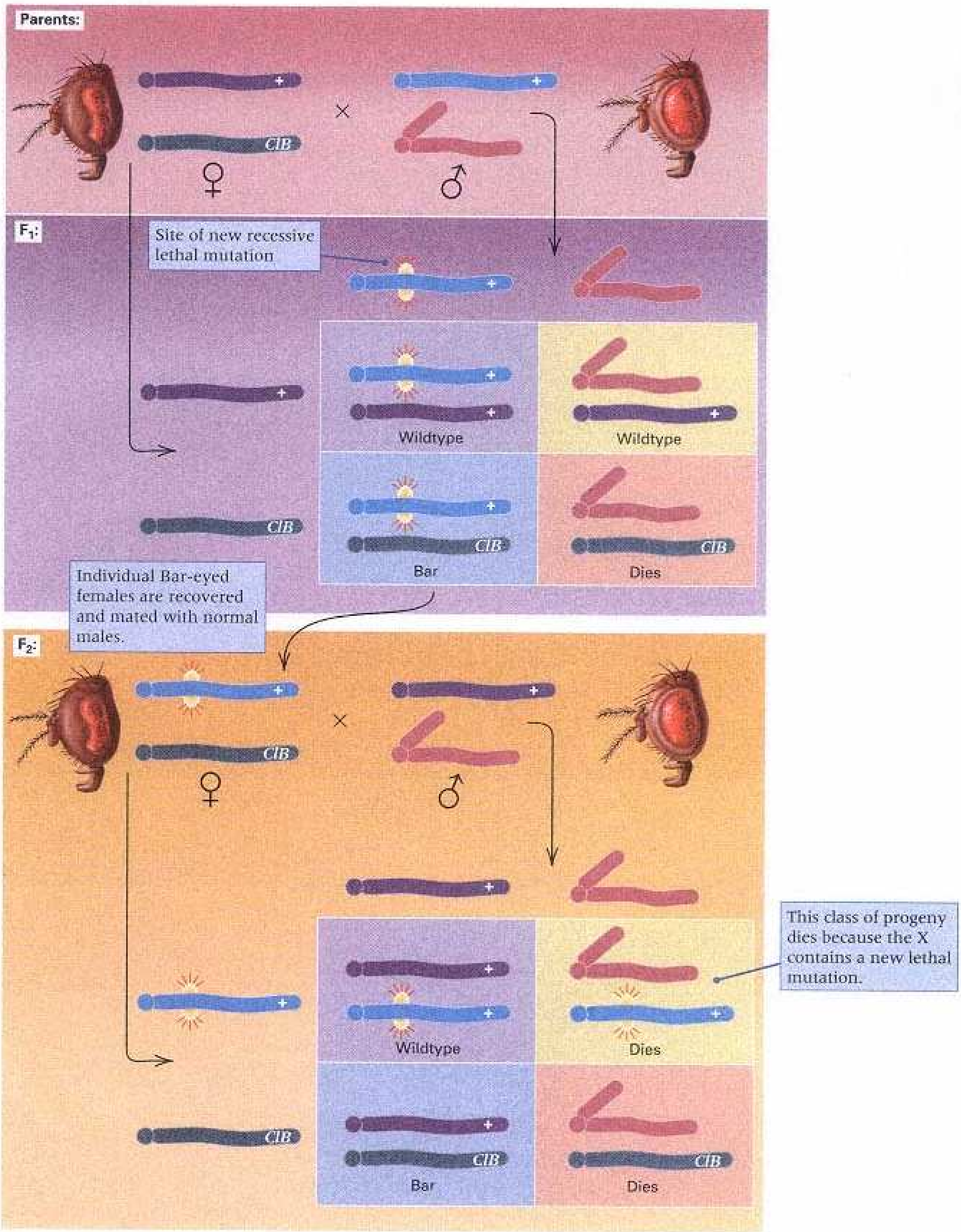


Figure 13.4
The *CIB* method for estimating the rate at which spontaneous recessive lethal mutations arise on the *Drosophila* X chromosome.

from 10^{-5} for some genes to 10^{-9} for others.

Hot Spots of Mutation

Certain DNA sequences are called mutational **hot spots** because they are more likely to undergo mutation than others. Mutational hot spots include monotonous runs of a single nucleotide and tandem repeats of short sequences (STRPs), which may gain or lose copies by any of a variety of mechanisms. Hot spots are found at many sites throughout the genome and within genes. For genetic studies of mutation, the existence of hot spots means that a relatively small number of sites accounts for a disproportionately large fraction of all mutations. For example, in the analysis of the *rII* gene in bacteriophage T4 discussed in Section 8.6, one extreme hot spot accounted for approximately 20 percent of all the mutations obtained.

Sites of cytosine methylation are usually highly mutable, and the mutations are usually GC $\rightarrow$ AT transitions. In many organisms, including bacteria, maize, and mammals (but not *Drosophila*), about 1 percent of the cytosine bases are methylated at the carbon-5 position, yielding 5-methylcytosine instead of ordinary cytosine (Figure 13.5). Special enzymes add the methyl groups to the cytosines in certain target sequences of DNA. In DNA replication, the 5-methylcytosine pairs with guanine and replicates normally. The genetic function of cytosine methylation is not entirely clear, but regions of DNA high in cytosine methylation tend to have reduced gene activity. Examples include the inactive X chromosome in female mammals (Chapter 7), genes that are imprinted during mammalian gametogenesis (Chapter 11), and inactive copies of certain transposable elements in maize.

Cytosine methylation is an important contributor to mutational hot spots, as illustrated in Figure 13.5. Both cytosine and 5-methylcytosine are subject to occasional loss of an amino group. For cytosine, this loss yields uracil. Because uracil pairs with adenine instead of guanine, replication of a molecule containing a GU base pair would ultimately lead to substitution

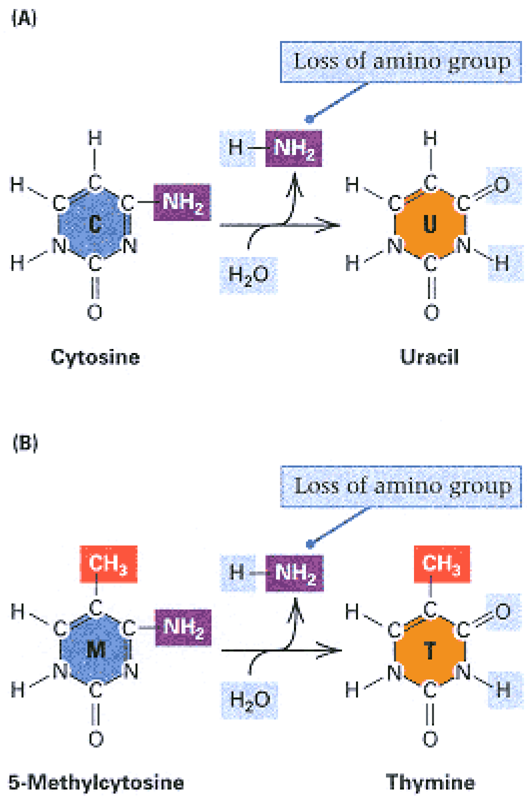


Figure 13.5
(A) Spontaneous loss of the amino group of cytosine to yield uracil. (B) Spontaneous loss of the amino group of 5-methylcytosine to yield thymine.

of an AT pair for the original GC pair (by the process GU $\rightarrow$ AU $\rightarrow$ AT in successive rounds of replication). However, cells possess a special repair enzyme that specifically removes uracil from DNA. The repair process is shown in Figure 13.6. Figure 13.6A shows deamination of cytosine leading to the presence of a uracil-containing base. Repair is initiated by an enzyme called **DNA uracil glycosylase,** which recognizes the incorrect GU base pair and cleaves the offending uracil from the deoxyribose sugar to which it is attached (Figure 13.6B). It is known how this enzyme works. It scans along duplex DNA until a uracil nucleotide is encountered, at which point a specific arginine residue in the enzyme intrudes into the DNA through the minor groove. The intrusion compresses the DNA backbone flanking the uracil and results in the flipping out of the

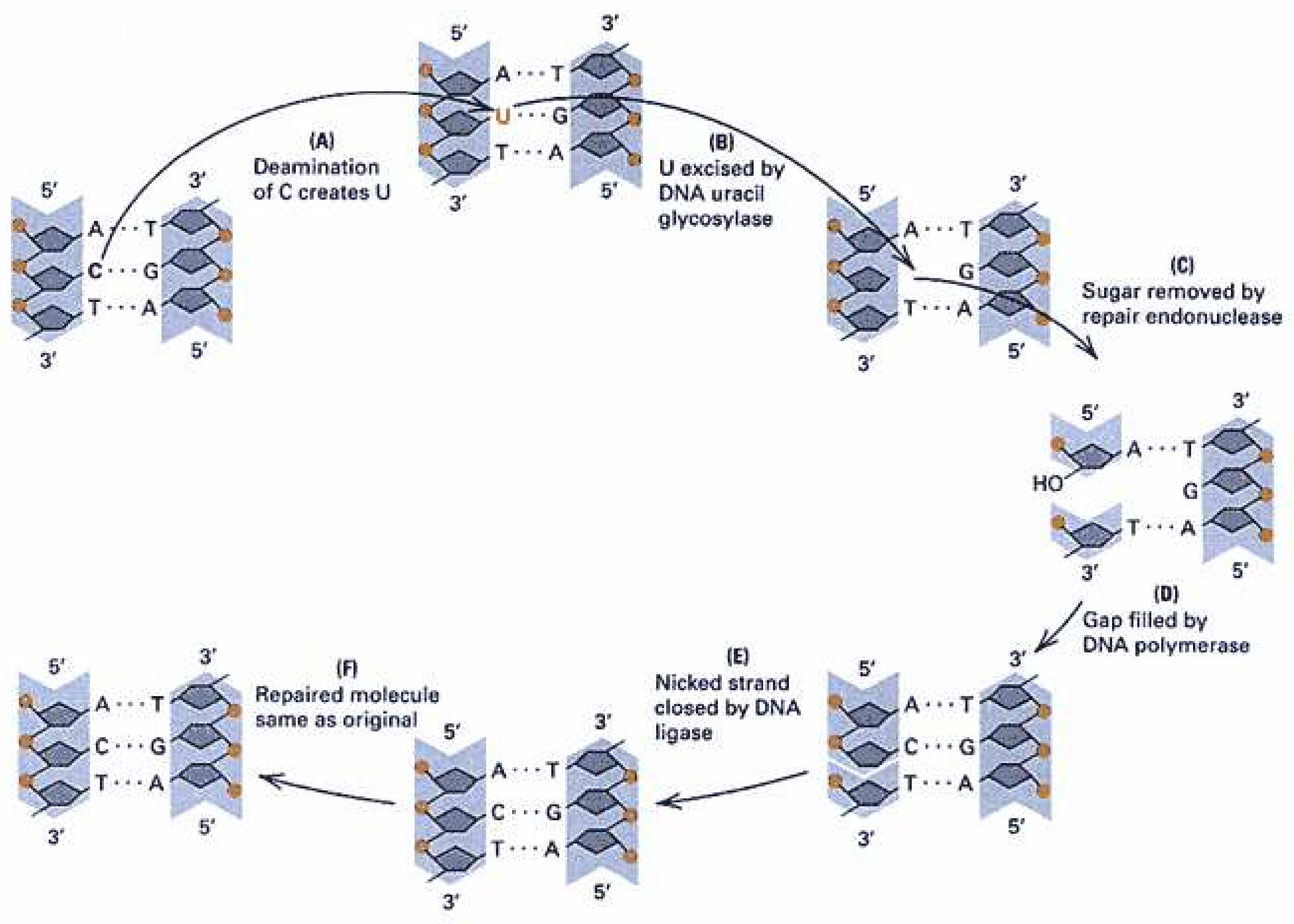


Figure 13.6
Mechanism by which uracil-containing nucleotides are formed in DNA and removed. (A) Deamination of cytosine produces uracil. (B) Uracil is cleaved from the deoxyribose sugar by DNA uracil glycosylase. (C) The deoxyribose sugar that does not contain a base is removed from the DNA backbone by another enzyme. (D) The gap in the DNA strand is filled by DNA polymerase, using the unagapped strand as template. (E) The nick remaining at the end of the gap-filling process is closed by DNA ligase. (F) The repair process restores the original sequence.

uracil base so that it sticks out of the helix. The flipping out makes it accessible to cleavage by the uracil glycosylase. After the uracil is removed, another enzyme, called the **AP endonuclease,** cleaves the baseless sugar from the DNA (Figure 13.6C), leaving a single-stranded gap that is repaired by one of the DNA polymerases (Figure 13.6D). The gap filling leaves one nick remaining in the repair strand, which is closed by DNA ligase (Figure 13.6E). The net result of the repair is that the original C → U conversion rarely leads to mutation because the U is removed and replaced with C (Figure 13.6F). Unfortunately, loss of the amino group of 5-methylcytosine yields thymine (Figure 13.5B), which is a normal DNA base and hence is not removed by the DNA uracil glycosylase. Thus the original GC pair becomes a GT pair and, in the next round of replication, gives GC (wildtype) and AT (mutant) DNA molecules.

13.4— Induced Mutations

The first evidence that external agents could increase the mutation rate was presented in 1927 by Hermann Muller, who showed that x rays are mutagenic in

Drosophila. (He later was awarded the Nobel Prize for this work.) Since then, a large number of physical agents and chemicals have been shown to increase the mutation rate. The use of these mutagens, several of which will be discussed in this section, is a means of greatly increasing the number of mutants that can be isolated. Because some environmental contaminants are mutagenic, as are numerous chemicals found in tobacco products, mutagens are also of great importance in public health.

Base-Analog Mutagens

A **base analog** is a molecule sufficiently similar to one of the four DNA bases that it can be incorporated into a DNA duplex in the course of normal replication. Such a substance must be able to pair with a base in the template strand. Some base analogs are mutagenic because they are more prone to mispairing than are the normal nucleotides. The molecular basis of the mutagenesis can be illustrated with 5-bromouracil (Bu), a commonly used base analog that is efficiently incorporated into the DNA of bacteria and viruses.

The base 5-bromouracil is an analog of thymine, and the bromine atom is about the same size as the methyl group of thymine (Figure 13.7A). If cells are grown in a medium containing 5-bromouracil, a thymine is sometimes replaced by a 5-bromouracil in the replication of the DNA, resulting in the formation of an A-Bu pair at a normal AT site (Figure 13.7B). The subsequent mutagenic activity of the incorporated 5-bromouracil stems from a rare shift in the configuration of the base. This shift is influenced by the bromine atom and happens in 5-bromouracil more frequently than it does in thymine. In the mutagenic configuration, 5-bromouracil pairs preferentially with guanine (Figure

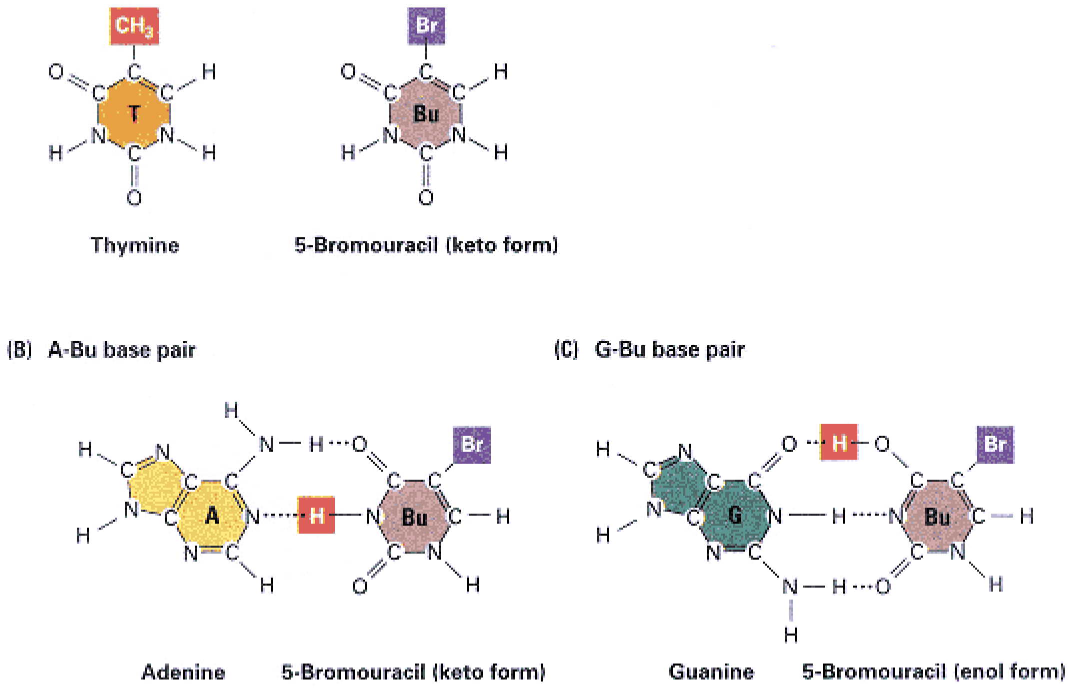


Figure 13.7
Mispairing mutagenesis by 5-bromouracil. (A) Structures of thymine and 5-bromouracil. (B) A base pair between adenine and the keto form of 5-bromouracil. (C) A base pair between guanine and the rare enol form of 5-bromouracil. One of 5-bromouracil's hydrogen atoms changes position to create the keto form.

13.7C), so in the next round of replication, one daughter molecule has a GC pair at the altered site, yielding an AT → GC transition (Figure 13.8).

Other experiments suggest that 5-bromouracil is also mutagenic in another way. The concentration of the nucleoside triphosphates in most cells is regulated by the concentration of thymidine triphosphate (dTTP). This regulation results in appropriate relative amounts of the four triphosphates for DNA synthesis. One part of this complex regulatory process is inhibition of the synthesis of deoxycytidine triphosphate (dCTP) by excess dTTP. The 5-bromouracil nucleoside triphosphate also inhibits production of dCTP. When 5-bromouracil is added to the growth medium, dTTP continues to be synthesized by cells at the normal rate, whereas the synthesis of dCTP is significantly reduced. The ratio of dTTP to dCTP then becomes quite high, and the frequency of misincorporation of T opposite G increases. Although repair systems can remove some incorrectly incorporated thymine, in the presence of 5-bromouracil the rate of misincorporation can exceed the rate of correction. An incorrectly incorporated T pairs with A in the next round of DNA replication, yielding a GC → AT transition in one of the daughter molecules. Thus 5-bromouracil usually induces transitions in both directions: AT → GC by the route in Figure 13.7, and GC → AT by the misincorporation route.

Just as a base analog is incorporated into DNA in place of a normal base, a **nucleotide analog** can be incorporated in place of a normal nucleotide. Modern DNA-sequencing methods are based on the use of dideoxy nucleotide analogs to terminate strand elongation at predetermined sites (Chapter 5). Dideoxy nucleotide analogs are also used in the clinical treatment of various diseases, including cancer and viral infections (Section 5.9). One of the most important drugs in the treatment of AIDS is the dideoxy analog AZT (Figure 13.9), which slows down replication of the HIV virus because the polymerase of the

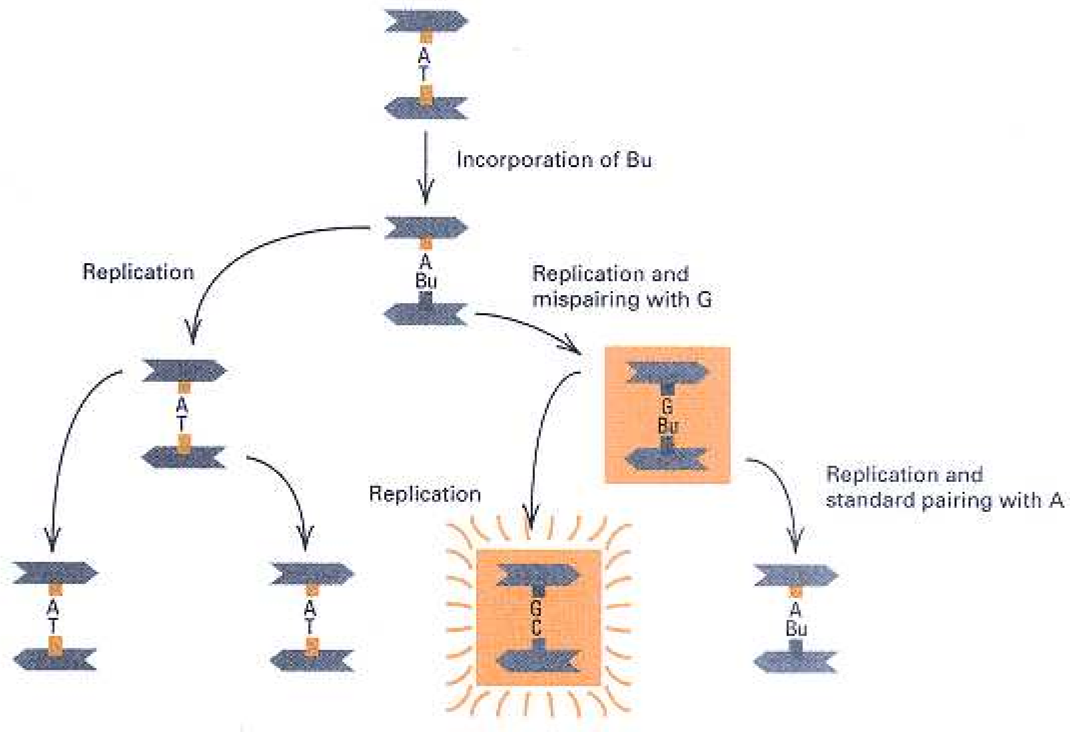


Figure 13.8

One mechanism for mutagenesis by 5-bromouracil (Bu). An AT → GC transition is produced by the incorporation of 5-bromouracil in DNA replication, forming an A – Bu pair. In the mutagenic round of replication, the Bu (in its rare enol form) pairs with G. In the next round of replication, the G pairs with C, completing the transition mutation.

virus incorporates AZT in preference to the normal deoxythymidine.

Chemical Agents That Modify DNA

Many mutagens are chemicals that react with DNA and change the hydrogenbonding properties of the bases. These mutagens are active on both replicating and nonreplicating DNA, in contrast to the base analogs, which are mutagenic only when DNA replicates. Several of these chemical mutagens, of which nitrous acid is a well-understood example, are highly specific in the changes they produce. Other—for example, the alkylating agents—react with DNA in a variety of ways and produce a broad spectrum of effects.

Nitrous acid (HNO_2) acts as a mutagen by converting amino (NH_2) groups of the bases adenine, cytosine, and guanine into keto (=O) groups. This process of *deamination* alters the hydrogen-bonding specificity of each base. Deamination of adenine

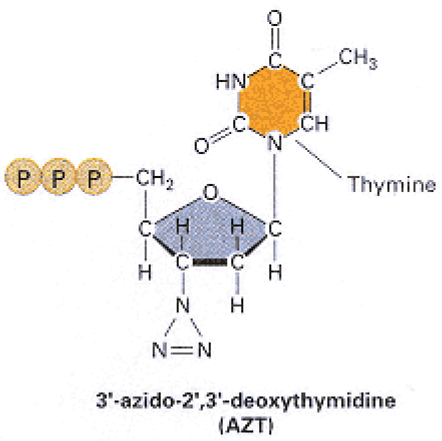


Figure 13.9
Chemical structure of AZT, a nucleotide analog used in the clinical treatment of AIDS.

yields a base, hypoxanthine, that pairs with cytosine rather than thymine, resulting in an AT → GC transition (Figure 13.10). As discussed earlier, deamination of cytosine yields uracil (Figure 13.5A), which pairs with adenine instead of guanine, producing a GC → AT transition.

In genetic research, chemical mutagens such as nitrous acid (and many others) are

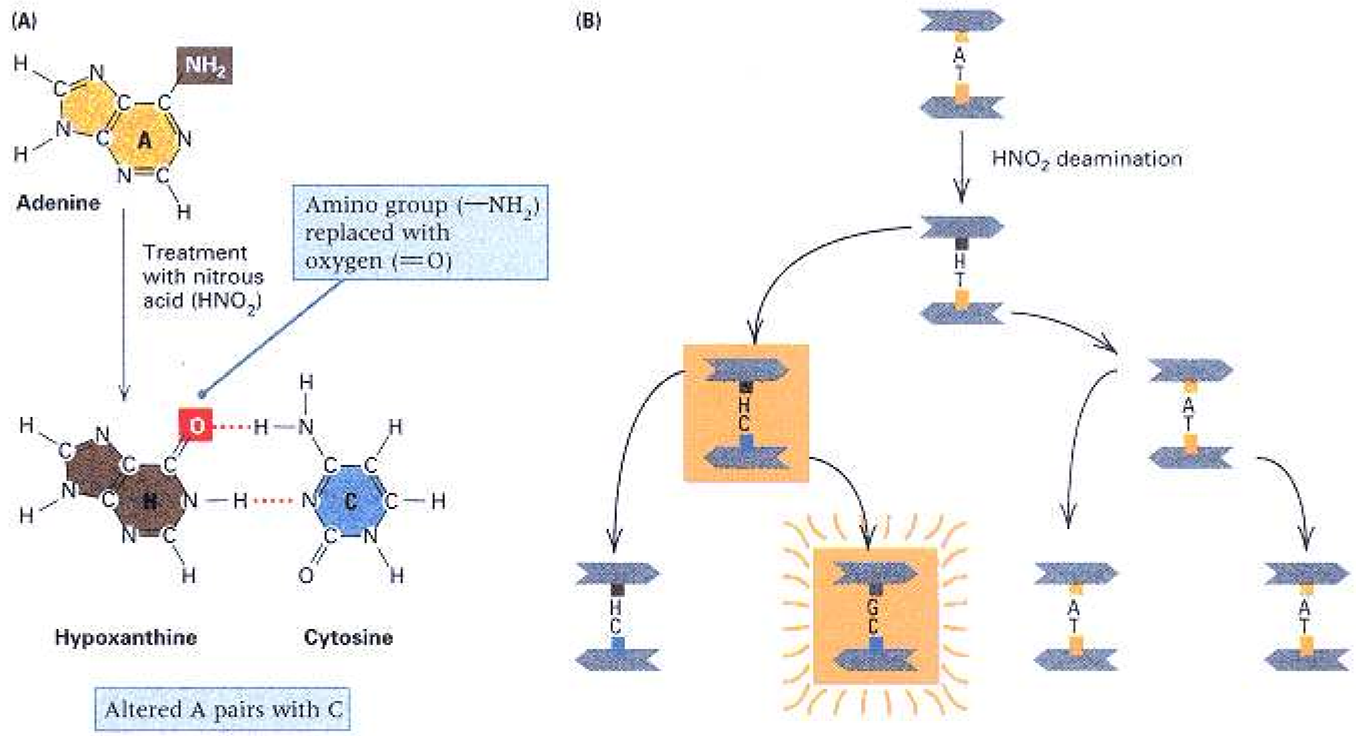
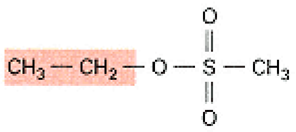
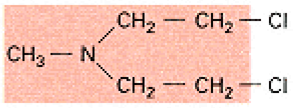


Figure 13.10

Nitrous acid mutagenesis. (A) Conversion of adenine into hypoxanthine, which pairs with cytosine. (The cytosine $\rightarrow$ uracil conversion is shown in Figure 13.5.) (B) Production of an AT $\rightarrow$ GC transition. In the mutagenic round of replication, the hypoxanthine (H) pairs with C. In the next round of replication, the C pairs with G, completing the transition mutation.



Ethyl methane sulfonate



Nitrogen mustard

Figure 13.11
The chemical structures of two highly mutagenic alkylating agents; the alkyl groups are shown in red.

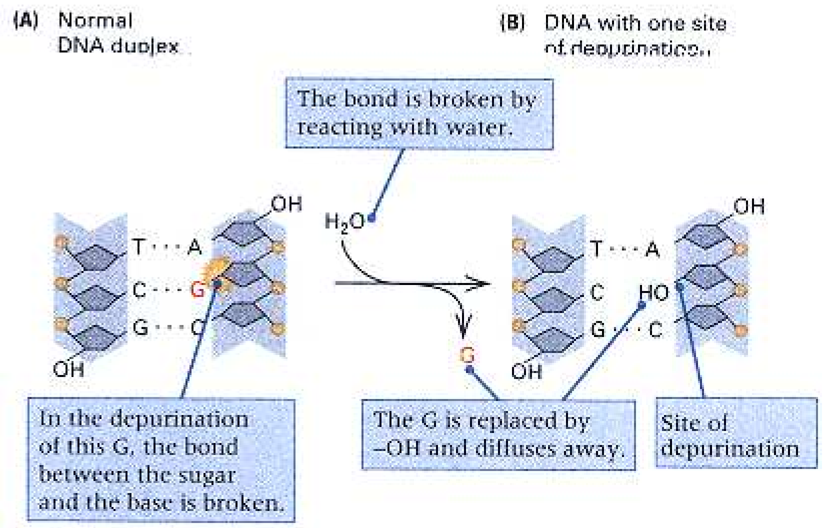


Figure 13.12
Deoxyribose-purine bonds are somewhat unstable and prone to undergo spontaneous reaction with water (hydrolysis), which results in loss of a purine base from the DNA (depurination). (A) Part of a DNA molecule prior to depurination. The bond between the labeled G and the deoxyribose to which it is attached is about to be hydrolyzed. (B) Hydrolysis of the bond releases the G purine, which diffuses away from the molecule and leaves a hydroxyl (– OH) in its place in the depurinated DNA.

exceedingly useful in prokaryotic systems but are not particularly useful as mutagens in higher eukaryotes because the chemical conditions necessary for reaction are not easily obtained. However, alkylating agents are highly effective in eukaryotes, as well as prokaryotes. **Alkylating agents,** such as ethyl methane sulfonate (EMS) and nitrogen mustard (the structures of which are shown in Figure 13.11), are potent mutagens that have been used extensively in genetic research. These agents add to the DNA bases various chemical groups that either alter their base-pairing properties or cause structural distortion of the DNA molecule. Alkylation of either guanine or thymine causes mispairing, leading to the transitions AT $\rightarrow$ GC and GC $\rightarrow$ AT. EMS reacts less readily with adenine and cytosine.

Another phenomenon that results from the alkylation of guanine is **depurination,** or loss of the alkylated base from the DNA molecule by breakage of the bond joining the purine nitrogen with the deoxyribose. The process is illustrated in Figure 13.12. Depurination is not always mutagenic, because the loss of the purine can be repaired by excision of the deoxyribose (a function of the AP endonuclease discussed in Section 13.3) and gap repair by DNA polymerase. If, however, the replication fork reaches the apurinic site before repair has taken place, then replication almost always inserts an adenine nucleotide in the daughter strand opposite the apurinic site. After another round of replication, the original GC pair becomes a TA pair, an example of a transversion mutation.

DNA is susceptible to spontaneous depurination also. In air, the rate of spontaneous depurination is approximately 3 × 10^{-9} depurinations per purine nucleotide per minute. This rate is at least tenfold greater than any other single source of spontaneous degradation. At this rate, the half-life of a purine nucleotide exposed to air is about 300 years. It is for this reason that reports of PCR amplification of ancient DNA from any material exposed to the elements for more than a few thousand years should be treated with some skepticism. There are, however, a few

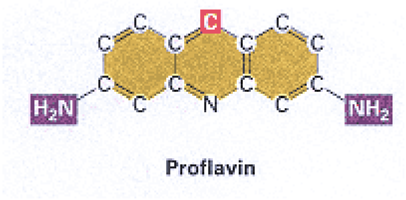


Figure 13.13
The structure of proflavine, an acridine derivative. Other mutagenic acridines have additional atoms on the NH_2 group and on the C of the central ring. Hydrogen atoms are not shown.

substances in which DNA that may be present degrades at a slower rate than normal. For example, the DNA in organisms preserved in amber may be more resistant to degradation.

Misalignment Mutagenesis

The **acridine** molecules, of which proflavin is an example (Figure 13.13), are planar, three-ringed molecules whose dimensions are roughly the same as those of a purine-pyrimidine pair. These substances insert between adjacent base pairs of DNA, a process called **intercalation.** The effect of the intercalation of an acridine molecule is to cause the adjacent base pairs to move apart by a distance roughly equal to the thickness of one pair (Figure 13.14). When DNA that contains intercalated acridines is replicated, the template and daughter DNA strands can misalign, particularly in regions where nucleotides are repeated. The result is that a nucleotide can be either added or deleted in the daughter strand. In a coding region, the result of a single-base addition or deletion is a frameshift mutation (Section 13.2).

Ultraviolet Irradiation

Ultraviolet (UV) light produces both mutagenic and lethal effects in all viruses and cells. The effects are caused by chemical changes in the bases resulting from absorption of the energy of the light. The major products formed in DNA after UV irradiation are covalently joined pyrimidines (**pyrimidine dimers**), primarily thymine (Figure 13.15A), that are adjacent in the same polynucleotide strand. This chemical linkage brings the bases closer together, causing a distortion of the helix (Figure 13.15B), which blocks transcription and transiently blocks DNA replication.

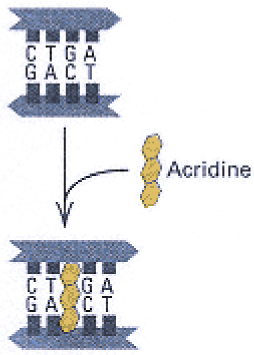


Figure 13.14
Separation of two base pairs caused by intercalation of an acridine molecule.

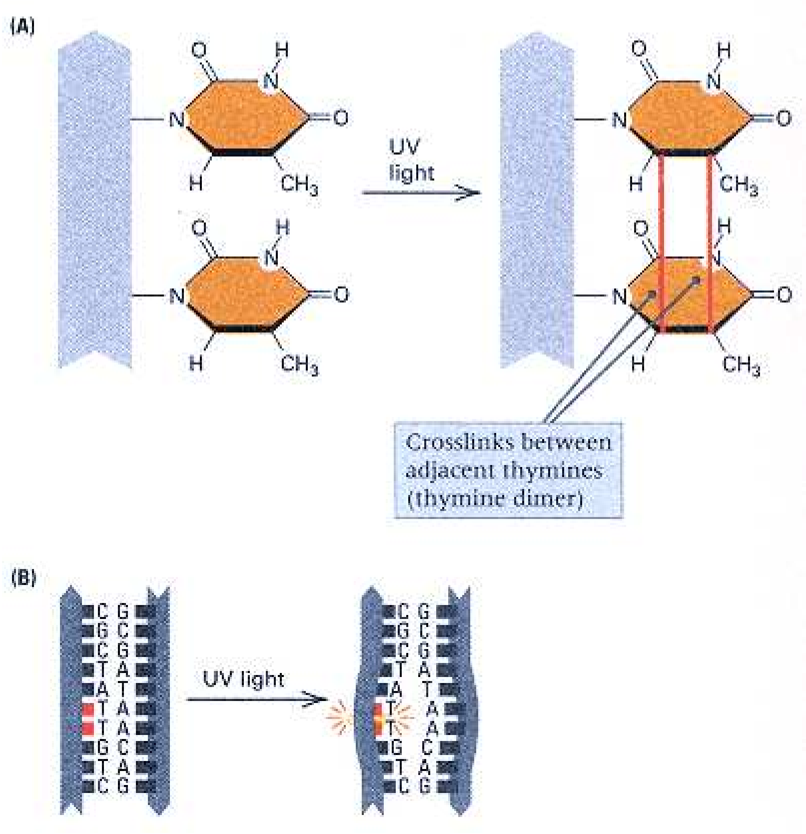


Figure 13.15
(A) Structural view of the formation of a thymine dimer. Adjacent thymines in a DNA strand that have been subjected to ultraviolet (UV) irradiation are joined by formation of the bonds shown in red. Other types of bonds between the thymine rings also are possible. Although not drawn to scale, these bonds are considerably shorter than the spacing between the planes of adjacent thymines, so the double-stranded structure becomes distorted. The shape of each thymine ring also changes. (B) The distortion of the DNA helix caused by two thymines moving closer together when joined in a dimer.

Pyrimidine dimers can be repaired in ways discussed later in this chapter.

In human beings, the inherited autosomal recessive disease **xeroderma pigments** is the result of a defect in a system that repairs ultraviolet-damaged DNA. Persons with this disease are extremely sensitive to sunlight, and this sensitivity results in excessive skin pigmentation and the development of numerous skin lesions that frequently become cancerous. Removal of pyrimidine dimers does not occur in the DNA of cells cultured from patients with the disease, and the cells are sensitive to much lower doses of ultraviolet light than are cells from normal persons.

Even in normal people, excessive exposure to the UV rays in sunlight increases the risk of skin cancer. Ozone, a form of oxygen (O_3) present high in the atmosphere (at a height of 15 to 30 kilometers), absorbs UV rays and ameliorates, to some extent, the damaging effects of sunlight. The thinning of the ozone layer (the "ozone hole"), which is especially pronounced at certain latitudes, greatly increases the effective exposure to UV and correspondingly increases the rate of skin cancer. The thinning of the ozone layer can be attributed to the destructive effects of chlorofluorocarbons (CFCs) used as pressurizers in aerosol spray cans and as refrigerants, when these agents ascend to the upper atmosphere.

Ionizing Radiation

Ionizing radiation includes x rays and the particles and radiation released by radioactive elements (α and β particles and γ rays). When x rays were first discovered late in the nineteenth century, their power to pass through solid materials was regarded as a harmless entertainment and a source of great amusement:

> By 1898, personal x rays had become a popular status symbol in New York. The *New York Times* reported that "there is quite as much difference in the appearance of the hand of a washerwoman and the hand of a fine lady in an x-ray picture as in reality." The hit of the exhibition season was Dr. W. J. Morton's full-length portrait of "*the x-ray lady*," a "fashionable woman who had evidently a scientific desire to see her bones." The portrait was said to be a "fascinating and coquettish" picture, the lady having agreed to be photographed without her stays and corset, the better to satisfy the "longing to have a portrait of well-developed ribs." Dr. Morton said women were not afraid of x rays: "After being assured that there is *no danger* they take the rays without fear."
>
> The titillating possibility of using x rays to see through clothing or to invade the privacy of locked rooms was a familiar theme in popular discussions of x rays and in cartoons and jokes. Newspapers carried advertisements for "*x-ray proof underclothing*" for those seeking to protect themselves from x-ray inspection.
>
> The luminous properties of radium soon produced a full-fledged radium craze. A famous woman dancer performed *radium dances* using veils dipped in fluorescent salts containing radium. *Radium roulette* was popular at New York casinos, featuring a "roulette wheel washed with a radium solution, such that it glowed brightly in the darkness; an unseen hand cast the ball on the turning wheel and sparks marked its course as it bounded from pocket to glimmery pocket." A patent was issued for a process for making women's gowns luminous with radium, and Broadway producer Florenz Ziegfeld snapped up the rights for his stage extravaganzas.
>
> Even while the unrestrained use of x rays and radium was growing, evidence was accumulating that the new forces might not be so benign after all. Hailed as tools for fighting cancer, they could also cause cancer. Doctors using x rays were the first to learn this bitter lesson. (Quoted from S. Hilgartner, R. C. Bell, and R. O'Connor. 1982. *Nukespeak.* Sierra Club Books, San Francisco.)

Doctors were indeed the first to learn the lesson. Many suffered severe x-ray burns or required amputation of overexposed hands or arms. Many others died from radiation poisoning or from radiation-induced cancer. By the mid-1930s, the number of x-ray deaths had grown so large that a monument to the "x-ray martyrs" was erected in a hospital courtyard in Germany. Yet the full hazards of x-ray exposure were not widely appreciated until the 1950s.

When ionizing radiation interacts with water or with living tissue, highly reactive ions called **free radicals** are formed. The free radicals react with other molecules, including DNA, which results in the carcinogenic and mutagenic effects. The intensity of a beam of ionizing radiation can be described quantitatively in several ways. There are, in fact, a bewildering variety of units in common use (Table 13.1). Some of the units (becquerel, curie) deal with the number of disintegrations emanating from a material, others (roentgen) with the number of ionizations the radiation produces in air, still others (gray, rad) with the amount of energy imparted to material exposed to the radiation, and some (rem, sievert) with the effects of radiation on living tissue. The types of units have proliferated through the years in attempts to encompass different types of radiation, including nonionizing radiation, in a common frame of reference. The units in Table 13.1 are presented only for ease of interpreting the multitude of units found in the literature on radiation genetics. They need not be memorized.

Genetic studies of ionizing radiation support the following general principle:

> Over a wide range of x-ray doses, the frequency of mutations induced by x rays is proportional to the radiation dose.

One type of evidence supporting this principle is the frequency with which X-chromosome recessive lethals are induced in *Drosophila* (Figure 13.16). The mutation

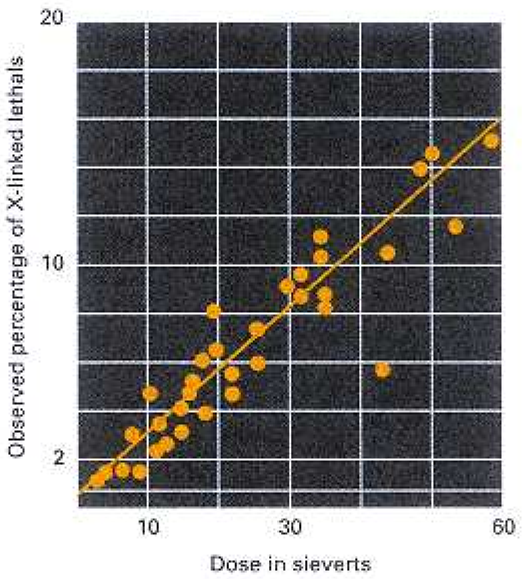


Figure 13.16
The relationship between the percentage of X-linked recessive lethals in D.*melanogaster* and x-ray dose, obtained from several experiments. The frequency of spontaneous X-linked lethal mutations is 0.15 percent per X chromosome per generation.

rate increases linearly with increasing x-ray does. For example, an exposure of 10 sieverts increases the frequency from the spontaneous value of 0.15 percent to about 3 percent. The mutagenic and lethal effects of ionizing radiation at low to moderate doses

Table 13.1 Units of radiation

Unit (abbreviation)	**Magnitude**
Becquerel (Bq)*	1 disintegration/second = 2.7×10^{-11} Ci
Curie (Ci)	3.7×10^{10} disintegrations/second 3.7×10^{10} Bq
Gray (Gy)*	1 joule/kilogram = 1000 rad
Rad (rad)	100 ergs/gram = 0.01Gy
Rem (rem)	damage to living tissue done by 1 rad = 0.01 Sv
Roentgen (R)	produces 1 electrostatic unit of charge per cubic centimeter of dry air under normal conditions of pressure and temperature. (By definition, 1 electrostatic unit repels with a force of 1 dyne at a distance of 1 centimeter.)
Sievert (Sv)	100 rem

* Units officially recognized by the International System of Units as defined by the General Conference on Weights and Measures.

Connection X-Ray Daze

Hermann J. Muller 1927

University of Texas, Austin, Texas
Artificial Transmutation of the Gene

Mutagenesis, which means the deliberate induction of mutations, plays an important role in genetics because it makes possible the identification of genes that control biological processes, such as development and behavior. For the demonstration that x rays could be used to induce new mutations, Muller was awarded a Nobel Prize in 1946. His discovery also had important practical implications in regard to x rays (and other mutagenic agents) present in the environment. After World War II ended with the nuclear bombing of Hiroshima and Nagasaki, Muller became a leader in the fight against the above-ground testing of nuclear bombs because of the release of radioactivity into the atmosphere. He also condemned the casual use of x rays, such as for the fitting of children's shoes, and he campaigned for the engineering of safer x-ray machines for medical use. Muller's prose could be turgid at time, thanks to his use of subordinate clauses. This excerpt captures the style.

Most modern geneticists will agree that gene mutations form the chief basis of organic evolution, and therefore of most of the complexities of living things. Unfortunately for the geneticists, however, the study of these mutations, and, through them, of the genes themselves, has heretofore been very seriously hampered by the extreme infrequency of their occurrence under ordinary conditions, and by the general unsuccessfulness of attempts to modify decidedly, and in a sure and detectable way, this sluggish "natural" mutation rate. . . . On theoretical grounds, it has

Treatment of the sperm with relatively heavy doses of x-rays induces the occurrence of true "gene mutations" in a high proportion the treated germ cells

appeared to the present writer that radiations of short wave length should be especially promising for the production of mutational changes, and for this and other reasons a series of experiments concerned with this problem has been undertaken during the past year on the fruit fly, *Drosophila melanogaster.* . . . It has been found quite conclusively that treatment of the sperm with relatively heavy doses of x-rays induces the occurrence of true "gene mutations" in a high proportion of the treated germ cells. Several hundred mutants have been obtained in this way in a short time and considerably more than a hundred have been followed through three, four or more generations. They are (nearly all of them, at any rate) stable in their inheritance. . . . Regarding the types of mutations produced, it was found that the recessive lethals greatly outnumbered the nonlethals producing a visible morphological abnormality. . . . In addition to gene mutations, it was found that there is also caused by x-ray treatment a high proportion of rearrangements in the linear order of the genes. . . . The transmuting action of x-rays on the genes is not confined to the sperm cells, for treatment of the unfertilized females causes mutations about as readily as treatment of the males. . . . In conclusion, the attention of those working along classical genetic lines may be drawn to the opportunity, afforded them by the use of x-rays, of creating in their chosen organisms a series of artificial races for use in the study of genetics. If, as seems likely on genetic considerations, the effect is common to most organisms, it should be possible to produce, "to order," enough mutations to furnish respectable genetic maps.

Source: Science 66: 84–87

result primarily from damage to DNA. Three types of damage in DNA are produced by ionizing radiation: single-strand breakage (in the sugar-phosphate backbone), double-strand breakage, and alterations in nucleotide bases. The single-stand breaks are usually efficiently repaired, but the other damage is responsible for mutation and lethality. In eukaryotes, ionizing radiation also results in chromosome breaks. Although systems exist for repairing the breaks, the repair often leads to translocations, inversions, duplications and deletions. In human cells in culture, a dose of 0.2 sievert results in an average of one visible chromosome break per cell.

Although the effects of extremely low levels of radiation are extremely difficult to measure because of the background of spontaneous mutation, most experiments support the following principle:

> There appears to be no threshold exposure below which mutations are not induced. Even very low doses of radiation induce some mutations.

Ionizing radiation is widely used in tumor therapy. The basis for the treatment is the increased frequency of chromosomal breakage (and the consequent lethality) in cells undergoing mitosis compared with cells in interphase. Tumors usually contain many more mitotic cells than most normal tissues, so more tumor cells than normal cells are destroyed. Because not all tumor cells are in mitosis at the same time, irradiation is carried out at intervals of several days to allow interphase tumor cells to enter mitosis. Over a period of time, most tumor cells are destroyed.

Table 13.2 gives representative values of doses of ionizing radiation received by human beings in the United States in the course of a year. The unit of measure is the millisievert, which equals 0.1 rem. The exposures in Table 13.2 are on a yearly basis, so over the course of a generation, the total exposure is approximately 100 millisieverts. Note that, with the exception of diagnostic x rays, which yield important compensating benefits, most of the total radiation exposure comes from natural sources, particularly radon gas. Less than 20 percent of the average radiation exposure comes from artificial sources. Nevertheless, there are dangers inherent in any exposure to ionizing radiation, particularly in an increased risk of leukemia and certain other cancers in the exposed persons. In regard to increased genetic diseases in future generations the risk that a small amount of additional radiation will have mutagenic effects is low enough that most geneticists are currently more concerned about the effects of the many mutagenic (as well as carcinogenic) chemicals that are introduced into the environment from a variety of sources.

The National Academy of Sciences of the United States regularly updates the estimated risks of radiation exposure. The latest estimates are summarized in Table 13.3. The message is that an additional 10 millisieverts of radiation per generation (about a 10 percent increase in the annual exposure) is expected to cause a relatively modest increase in diseases that are wholly or in part due to genetic factors. The most common conditions in the table are heart disease and cancer. No estimate for the radiation-induced increase is given for either of these traits because the genetic contribution to the total is still very uncertain.

Genetic Effects of the Chernobyl Nuclear Accident

The city of Chernobyl in the Ukraine has become a symbol for nuclear disaster. On April 26, 1986, a nuclear power plant near the city exploded, heavily contaminating the immediate area and sending clouds of radioactive debris over long distances. It was the largest publicly acknowledged nuclear accident in history. The meltdown is estimated to have released between 50 and 200 million curies of radiation, along with a wide variety of heavy-metal and chemical pollutants. Iodine-131 and cesium-137 were the principal radioactive contaminants. People living in the area were evacuated almost immediately, but many were heavily exposed to radiation.

Table 13.2 Annual exposure of human beings in the United States to various forms of ioniszing radiation

Source	**Dose (in millisieverts*)**
Natural radiation	
Radon gas	2.06
Cosmic rays	0.27
Natural radioisotopes in the body	0.39
Natural radioisotopes in the soil	0.28
Total natural radiation	3.00
Other radiation sources	
Diagnostic x rays	0.39
Radiopharmaceuticals	0.14
Consumer products (x rays from TV, radioisotopes in clock dials) and building materials	0.10
Fallout from weapon tests	<0.01
Nuclear power plants	<0.01
	0.63

Total from non-natural sources	
Total from all sources	3.63

*One millisievert (mSV) equals 1 1000 sievert, or 0.1 rem.

Source: From National Research Council. Committee on the Biological Ellects of Ionizing Radiations. Health Effects of Exposure to Low Levels of Inoizing Radiation (*BEIR V*) National Academy Press, 1990.

Within a short time there was a notable increase in the frequency of thyroid cancer in children, and other health effects of radiation exposure were also detected.

At first, little attention was given to possible genetic effects of the Chernobyl disaster, because relatively few people were acutely exposed and because the radiation dose to people outside the immediate area was considered too small to worry about. In the district of Belarus, some 200 kilometers north of Chernobyl, the average exposure to iodine-131 was estimated at approximately 0.185 sievert per person. This is a fairly high dose, but the exposure was brief because the half-life of iodine-131 is only 8 days. Radiation from the much longer-lived (30-year half-life) cesium-137 was estimated at less than 5 millisieverts per year. On the basis of data from laboratory animals and studies of the survivors of the Hiroshima and Nagasaki atomic bombs, little detectable genetic damage was expected from these exposures.

Nevertheless, 10 years after the meltdown, studies of people living in Belarus did indicate a remarkable increase in the mutation rate. The observations focused on STRP polymorphisms because of their intrinsically high mutation rate due to replication slippage and other factors. Each STRP was examined in parents and their offspring to detect any DNA fragments that increased or decreased in size, indicative of an increase or decrease in the number of repeating units contained in the DNA fragment. Five STRP polymorphisms were studied, with the results summarized in Figure 13.17. Two of the loci (blue dots) showed no evidence for an increase in the mutation rate. This is the expected result. The unexpected finding was that three of the loci (red dots) did show a significant increase, and the level of increase was consistent with an approximate doubling of the mutation rate. At the present time, it is still not known whether the increase was detected because replication slippage is

Table 13.3 Estimated genetic effects of an additional 10 millisieverts per generation

		Additina cases per million liveborn per 10 mSv per generation	
Type of disorder	**Current incidence per million liveborn**	**First generation**	**At equilibrium**
Autosomal dominant			
Clinically severe	2500	5–20	25
Clinically mild	7500	1–15	75
X-linked	400	<1	<5
Autosomal recessive	2500	<1	Very slow increase
Chromosomal			
Unbalanced translocations	600	<5	Very little increase
Trisomy	3800	<1	<1
Congenital abnormalities (multifactorial)	20,000–30,000	10	10–100
Other multifactorial disorders			
Heart disease	600,000	Unknown	Unknown
Cancer	300,000	Unknown	Unknown
Others	300,000	Unknown	Unknown

Source: Health Effects of Exposure to Low Levels of Ionizing Radiation (BEIR V), National Research Council, Washinton, D. C.

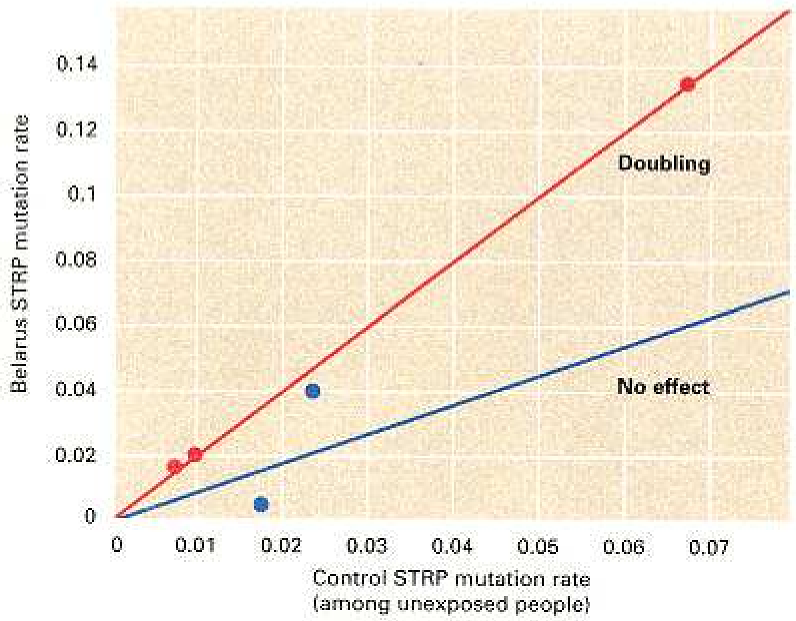


Figure 13.17
Mutation rates of five STRP polymorphisms among people of Belarus exposed to radiation from Chernobyland among unexposed British people. Three of the loci (red dots) show evidence of an approximately two fold increase in the mutation rate.
[Data from Y. E. Dubrova, V. N. Nesterov, N. G. Krouchinsky, V. A. Ostapenko, R. Neumann, D. L. Neil, and A. J. Jeffreys, *Nature* 1996. 380: 183.]

more sensitive to radiation than other types of mutations or because the effective radiation dose sustained by the Belarus population was much higher than originally thought.

The finding of increased mutation among the people of Belarus is paralleled by observations carried out in two species of voles, small rodents related to hamsters and gerbils, that live in the immediate vicinity of Chernobyl. These animals live in areas still highly contaminated by radiation, heavy metals, and chemical pollutants. The animals eat food that is so heavily contaminated that they themselves become extremely radioactive. Yet they survive and reproduce in this harsh environment.

Among the voles of Chernobyl, the rate of mutation in the cytochrome *b* gene present in mitochondrial DNA was estimated at 2×10^{-4} mutations per nucleotide per generation. This rate is at least 200 times higher than that found among animals living in a relatively uncontaminated area just 30 kilometers to the southwest of Chernobyl. It is so high that, on average, each mitochondrial DNA molecule of approximately 17 kb undergoes three new mutations in each generation. It is as yet unknown whether this high rate of mutation is due to radiation exposure, to nonradioactive pollutants, or to the synergistic effects of both.

13.5—
Mechanisms of DNA Repair

Spontaneous damage to DNA in human cells takes place at a rate of approximately 1 event per billion nucleotide pairs per minute (or, per nucleotide pair, at a rate of 1×10^{-9} per minute). This may seem like quite a small rate, but it implies that every 24 hours, in every human cell, the DNA is damaged at approximately 10,000 different sites.

Fortunately for us, and for all living organisms, much of the damage done to DNA by spontaneous chemical reactions in the nucleus, chemical mutagens, and radiation can, under ordinary circumstances, be repaired. We have already seen (Section 13.3) one example of repair in the manner in which DNA uracil glycosylase and the AP endonuclease act in tandem first to remove and then to replace uracil nucleotides that get into DNA, such as by the deamination of cytosine. In this section, we introduce a number of other processes that are important in the repair of aberrant or damaged DNA.

Mismatch Repair

Base-substitution mutations usually start as a single mismatched (mispaired) base in a double-stranded DNA molecule—for example, a GT base pair resulting from spontaneous loss of an amino group from 5-methylcytosine (Figure 13.5). However, many of these mismatched bases need not persist through DNA replication to be resolved. The mismatched bases can be detected and corrected by enzymes that carry out *mismatch repair,* a process of excision and resynthesis illustrated in Figure 13.18. When a mismatched base is detected. one of the strands is cut in two places and a region around the mismatch is removed. The excised region is variable in size. In *E. coli*, an enzyme cleaves the DNA at the 5' end of a GATC sequence on either side of the mismatch, provided the site is within about 1 kb of the mismatch. Then an exonuclease degrades the cleaved DNA strand to a position on the other side of the mismatch (the excision step), creating a single-stranded gap. After excision, DNA polymerase fills in the gap by using the remaining strand as template.

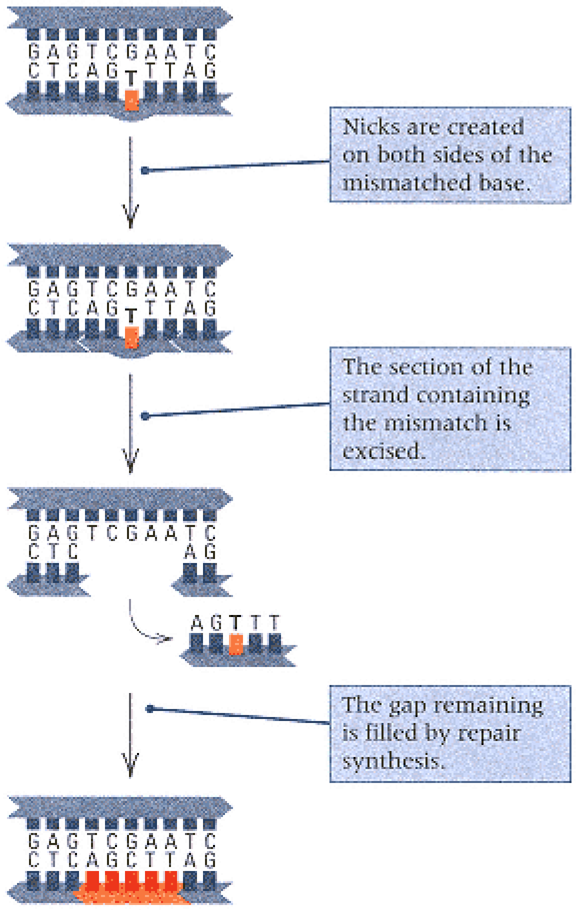


Figure 13.18
Mismatch repair consists of the excision of a segment of a DNA strand that contains a base mismatch, followed by repair synthesis. The length of the excised stand is typically 13 nucleotides in prokaryotes and 29 nucleotides in eukaryotes. Either strand can be excised and corrected, but in newly synthesized DNA, methylated bases in the template strand often direct the environmentalism the newly synthesized strand that contains the incorrect nucleotide.

Connection Replication Slippage in Unstable Repeats

Micheline Strand,[1], Tomas A. Prolla,[2] R. Michael Liskay,[2], and Thomas D. Petes[1] 1993

[1]University of North Carolina, Chapel Hill, North Carolina
[2]Yale University, New Haven, Connecticut
Destabilization of Tracts of Simple Repetitive DNA in Yeast by Mutations Affecting DNA Mismatch Repair

One of the most important principles of genetics is that many fundamental processes are carried out in much the same way in many different organisms. Basic cellular processes are similar in all eukaryotic cells, and the basic mechanisms of DNA replication and repair are fundamentally the same in eukaryotes and prokaryotes. The similarity of mismatch repair among organisms motivated this paper. The authors report that yeast cells deficient in mismatch repair are very deficient in their ability to replicate faithfully tracts of short repeating sequences. Because this type of instability also characterizes certain human hereditary colon cancers, the authors correctly predicted that the mutant genes causing the colon cancer would be genes involved in mismatch repair.

The genomes of all eukaryotes contain tracts of DNA in which a single base or a small number of bases is repeated. Expansions of such tracts have been associated with several human disorders including the fragile-X syndrome. In addition, simple repeats are unstable in certain forms of colorectal cancer, suggesting a

The instability of poly(GT) tracts in yeast is increased by mutations in the mismatch repair genes

defect in DNA replication of repair. . . . One mechanism [for changes in repeat number] is DNA polymerase slippage: during replication of the tract, the primer and template strand transiently dissociate and then reassociate in a misaligned configuration. If the unpaired bases are in the primer strand, continued synthesis results in an elongation of the tract whereas unpaired bases in the template strand result in a deletion. . . . The most common simple repeat in most eukaryotes is poly$(GT)_{10-30}$. We examined the genetic control of the stability of poly(GT) tracts. . . . Three mutation affecting mismatch repair in yeast were examined, *pms1* and *mlh1* (homologs of *MutL*) and *msh2* (a homolog of *MutS*). . . . In two different genetic backgrounds using two different instability assay systems, all three mutations lead to 100- to 700-fold elevated levels of tract instability. . . . Yeast contains two DNA polymerases that have 3'–5' exonuclease ("proofreading") activities. In strains with mutations in the nuclease portion of these genes, neither affects tract stability as dramatically as mutations that reduce mismatch repair. . . . The results show that the instability of poly(GT) tracts in yeast is increased by mutations in the mismatch repair genes but is relatively unaffected by mutations affecting the proofreading functions of DNA polymerase. These results indicate that DNA polymerase *in vivo* has a very high rate of slippage on templates containing simple repeats, but most of the errors are corrected by cellular mismatch repair systems. Thus, the instability of simple repeats observed for some human diseases may be a consequence of either an increased rate of DNA polymerase slippage or a decreased efficiency of mismatch repair.

Source: Nature 365: 274–276

thereby eliminating the mismatch. The mismatch-repair system also corrects most single-base insertions or deletions.

If the DNA strand that is removed in mismatch repair is chosen at random, then the repair process sometimes creates a mutant molecule by cutting the strand that contains the correct base and using the mutant strand as template. However, in newly synthesized DNA, there is a way to prevent this from happening because the daughter strand is less methylated than the parental strand. The mismatch-repair system recognizes the degree of methylation of a strand and *preferentially excises nucleotides from the undermethylated strand.* This helps ensure that incorrect nucleotides incorporated into the daughter strand in replication will be removed and repaired. The daughter strand is always the under-methylated strand because its methylation lags somewhat behind the moving replication fork, whereas the parental strand was fully methylated in the preceding round of replication.

The mismatch-repair system, like the proofreading function of DNA polymerase (Chapter 5) and all other biochemical systems, is not perfect. Some spontaneous base substitutions still arise as a result of the incorporation of errors that escape correction. Figure 13.19 summarizes the

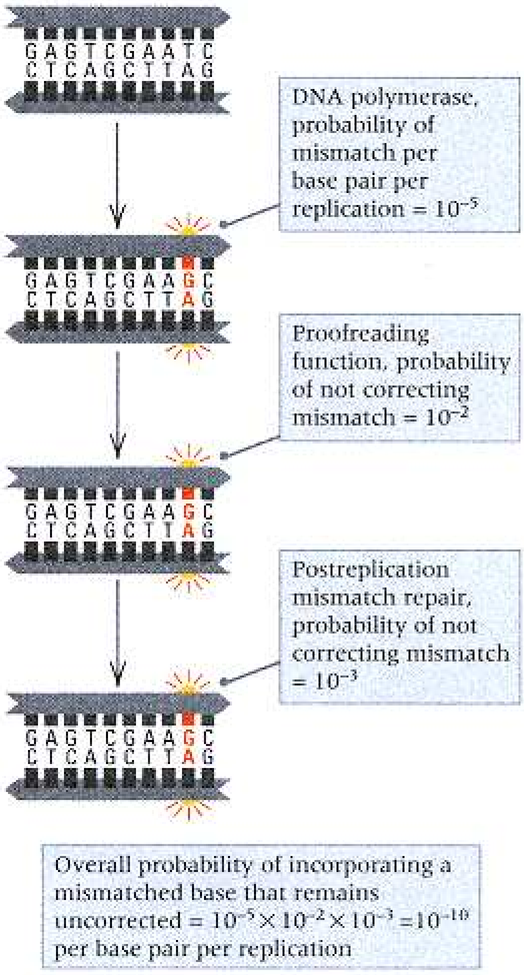


Figure 13.19
Summary of rates of error in DNA polymerization, proofreading, and postreplication mismatch repair. The initial rate of nucleotide misincorporation is 10^{-5} per base pair per replication. The proofreading function of DNA polymerase corrects 99 percent of these errors and of those that remain, postreplication mismatch} repair corrects 99.9 percent. The overall rate of misincorporated nucleotides that are not repaired is 10^{-10} per base pair per replication.

relevant error rates in *E. coli.* The rate of initial misincorporation is about 10^{-5} per nucleotide per replication. This consists of two components: chemical mispairing of the bases, which takes place at a rate of about 10^{-2} per nucleotide, and selection of the incorrect nucleotide by the DNA polymerase, which occurs at a rate of about 10^{-3} per nucleotide. Once there is an initial mismatch (probability 10^{-5}), the chance that the mismatch escapes the proofreading function is approximately 10^{-2}, and the chance that it escapes the postreplication mismatch-repair system is approximately 10^{-3}. Overall, the probability of incorporating a mismatched base that remains uncorrected by either mechanism is $10^{-5} \times 10^{-2} \times 10^{-3} = 10^{-10}$ per base pair per replication.

The mechanism of mismatch repair has been studied extensively in bacteria. Mutants defective in the process were identified as having high rates of spontaneous mutation. The products of two genes, *mutL* and *mutS*, recognize and bind to a mismatched base pair. This triggers the excision of a tract of nucleotides from the newly synthesized strand. Experiments carried out in yeast by Thomas Petes and colleagues revealed that simple sequence repeats (tandem repeats of short nucleotide sequences) are a thousandfold less stable in mutants deficient in mismatch repair than in wildtype yeast. By that time it was already known that some forms of human hereditary colorectal cancer result in decreased stability of simple repeats, and the yeast reseachers suggested that these high-risk families might be segregating for an allele causing a mismatch-repair deficiency. Within less than two years, scientists in several laboratories had identified four human genes homologous to *mutL* or *mutS*, any one of which, when mutated, results in hereditary non-polyposis colorectal cancer (HNPCC). Most cases of this type of cancer may be caused by mutations in one of these four mismatch-repair genes.

Photoreactivation

Various enzymes can recognize and catalyze the direct reversal of specific DNA damage. A classic example is the reversal of UV-induced pyrimidine dimers by **photo-reactivation,** in which an enzyme breaks the bonds that join the pyrimidines in the dimer and thereby restores the original

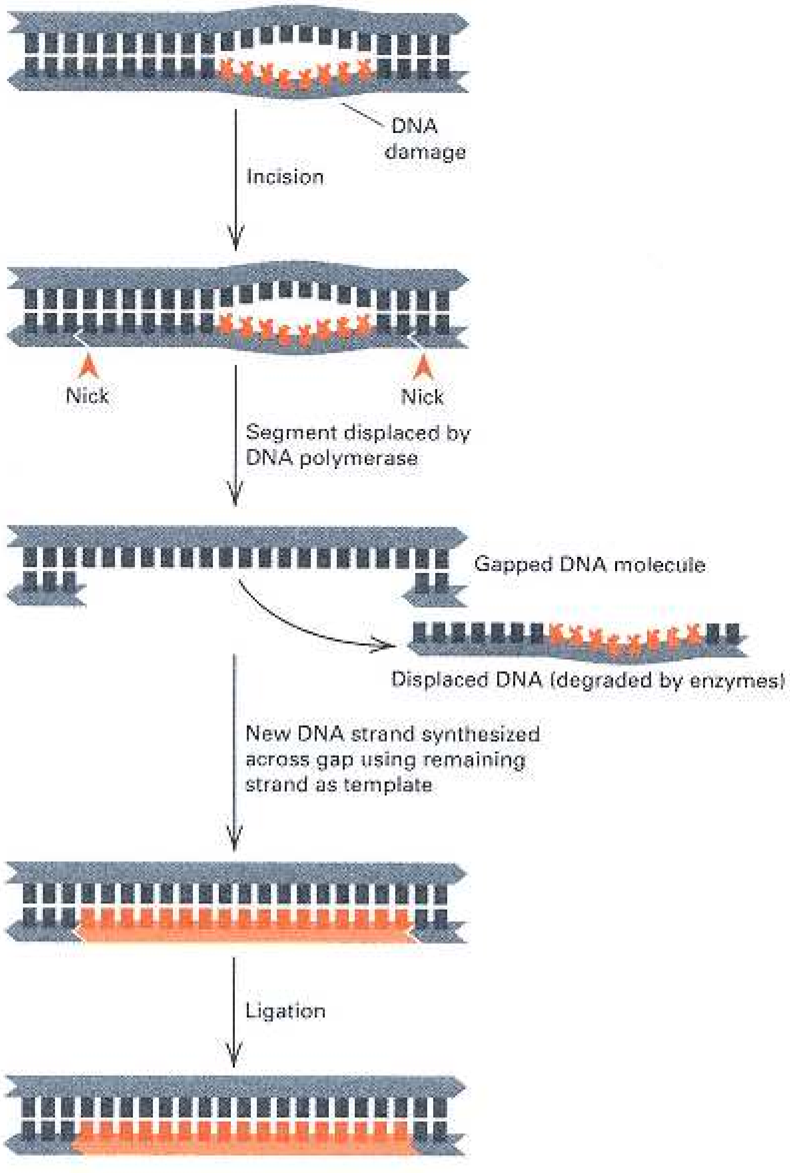


Figure 13.20
Mechanism of excision repair of DNA damage.

bases. The enzyme binds to the dimers in the dark but then utilizes the energy of blue light to cleave the bonds. Another important example is the reversal of guanine methylation in O^6-methyl guanine by a methyl transferase enzyme, which otherwise would pair like adenine in replication.

Excision Repair

Excision repair is a ubiquitous multistep enzymatic process by which a stretch of a damaged DNA strand is removed from a duplex molecule and replaced by resynthesis using the undamaged strand as a template. Mismatch repair of single mispaired bases is one important type of excision repair. The overall process of excision repair is illustrated in Figure 13.20. The DNA damage can be due to anything that produces a distortion in the duplex molecule; one example is a pyrimidine dimer. In excision repair, a repair endonuclease recognizes the distortion produced by the DNA damage and makes one or two cuts in the sugar-phosphate backbone, several nucleotides away from the damage on either side. A 3'-OH group is produced at the 5' cut,

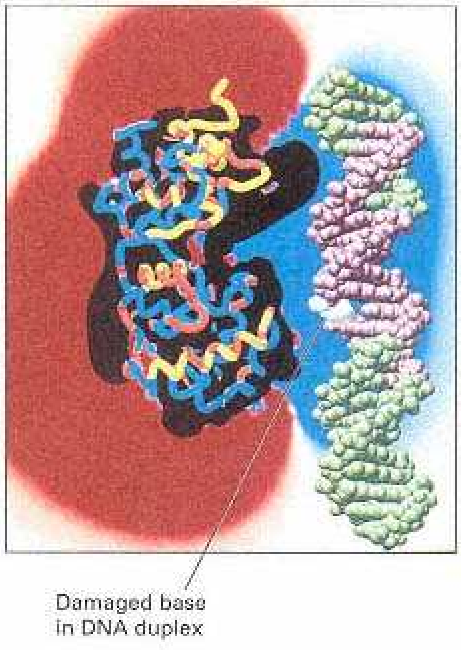


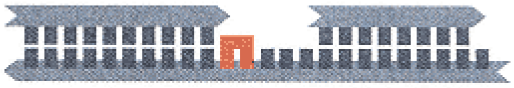
Figure 13.21
Model of endonuclease III from *E. coli* (left) in close proximity with a region of 15 base pairs (purple) of DNA (right) that it contacts in carrying out its function. The region of binding is asymmetrical around a damaged base (white), which the enzyme removes while producing a nick at the same site in the DNA backbone. The blue cloud indicates a region of strong electrostatic attraction; the red cloud indicates a region of electrostatic repulsion. [Courtesy of John A. Tainer. Computer graphic model by Michael Pique, Cindy Fisher, and John A. Tainer. From C.-F. Kuo, D. E. McRee, C. L. Fisher, S. F. O'Handley, R. P. Cunningham and J. A. Tainer 1992. *Science* 258:434.]

which DNA polymerase uses as a primer to synthesize a new strand while displacing the DNA segment that contains the damage. The final step of the repair process is the joining of the newly synthesized segment to the contiguous strand by DNA ligase.

Figure 13.21 shows a computer model based on the crystal structure of the repair enzyme endonuclease III from *E. coli*. The purple region in the DNA (right) is a region of 15 base pairs contacted by the enzyme in carrying out the repair of the damaged base (white). The enzyme removes the damaged base and introduces a single-strand nick at the site from which the damaged base was removed. The endonuclease is unusual in containing a cluster of four iron and four sulfur atoms, which are represented by the cluster of yellow and red spheres near the top of the enzyme.

Postreplication Repair

Sometimes DNA damage persists rather than being reversed or removed, but its harmful effects may be minimized. This often requires replication across damaged areas, so the process is called **post-replication repair.** For example, when DNA polymerase reaches a damaged site (such as a pyrimidine dimer), it stops synthesis of the strand. However, after a brief time, synthesis is reinitiated beyond the damage and chain growth continues, producing a gapped strand with the damaged spot in the gap.



The gap can be filled by strand exchange with the parental strand that has the same polarity, and the secondary gap produced in that strand can be filled by repair synthesis (Figure 13.22). The products of this exchange and resynthesis are two intact single strands, each of which can then serve in the next round of replication as a template for the synthesis of an undamaged DNA molecule.

The SOS Repair System

SOS repair, which is found in *E. coli.* and related bacteria, is a complex set of processes that includes a bypass system that allows DNA replication to take place across pyrimidine dimers or other DNA distortions, but at the cost of the fidelity of replication. Even though intact DNA strands are formed, the strands are often defective. SOS repair is said to be *error prone.* A significant feature of the SOS repair system is that it is not always active but is induced by DNA damage. Once activated, the SOS system allows the growing fork to advance across the damaged region, adding nucleotides that are often incorrect because the damaged template strand cannot be replicated properly. The proofreading system of DNA polymerase is relaxed to allow polymerization to proceed across the damage, despite the distortion of the helix.

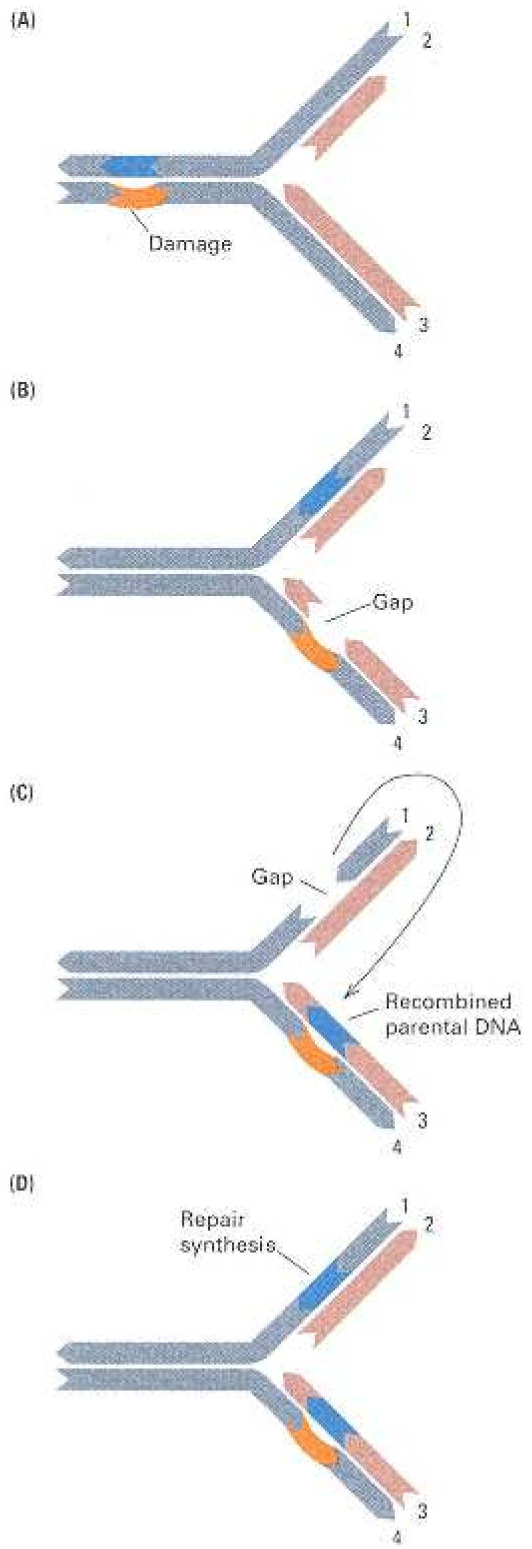


Figure 13.22
Postreplication repair. (A) A molecule with DNA damage in strand 4 is being replicated. (B) By reinitiation of synthesis beyond the damage a gap is formed in strand 3. (C) A segment of parental strand 1 is excised and inserted in strand 3. (D) The gap in strand 1 is next filled in by repair synthesis.

13.6— Reverse Mutations and Suppressor Mutations

In most of the mutations we have considered so far, a wildtype (normal) gene is changed into a form that results in a mutant phenotype, an event called a **forward mutation.** Mutations are frequently reversible, and an event that restores the wildtype phenotype is called a **reversion.** A reversion may result from a **reverse mutation,** an exact reversal of the alteration in base sequence that occurred in the original forward mutation, restoring the wildtype DNA sequence. A reversion may also result from the occurrence, at some other site in the genome, of a second mutation that in any of several ways compensates for the effect of the original mutation. Reversion by the exact reversal mechanism is infrequent. The second-site mechanism is much more common, and a mutation of this kind is called a **suppressor mutation.** A suppressor mutation can occur at a different site in the gene containing the mutation that it suppresses (*intra*genic suppression) or in a different gene in either the same chromosome or a different chromosome (*inter*genic suppression). Most suppressor mutations do not fully restore the wildtype phenotype, for reasons that will be apparent in the following discussion of the two kinds of suppression.

Intragenic Suppression

Reversion of frameshift mutations usually results from **intragenic suppression;** the mutational effect of the addition (or deletion) of a nucleotide pair in changing the reading frame of the mRNA is rectified by a compensating deletion (or addition) of a second nucleotide pair at a nearby site in the gene (Section 10.6). A second type of intragenic suppression is observed when loss of activity of a protein, caused by one amino acid change, is at least partly restored by a change in a second amino acid. An example of this type of suppression in the protein product of the *trpA* gene in *E. coli* is shown in Figure 13.23. The A polypeptide, composed of 268 amino acids, is one of two polypeptides that make up the enzyme tryptophan synthase in *E. coli.*

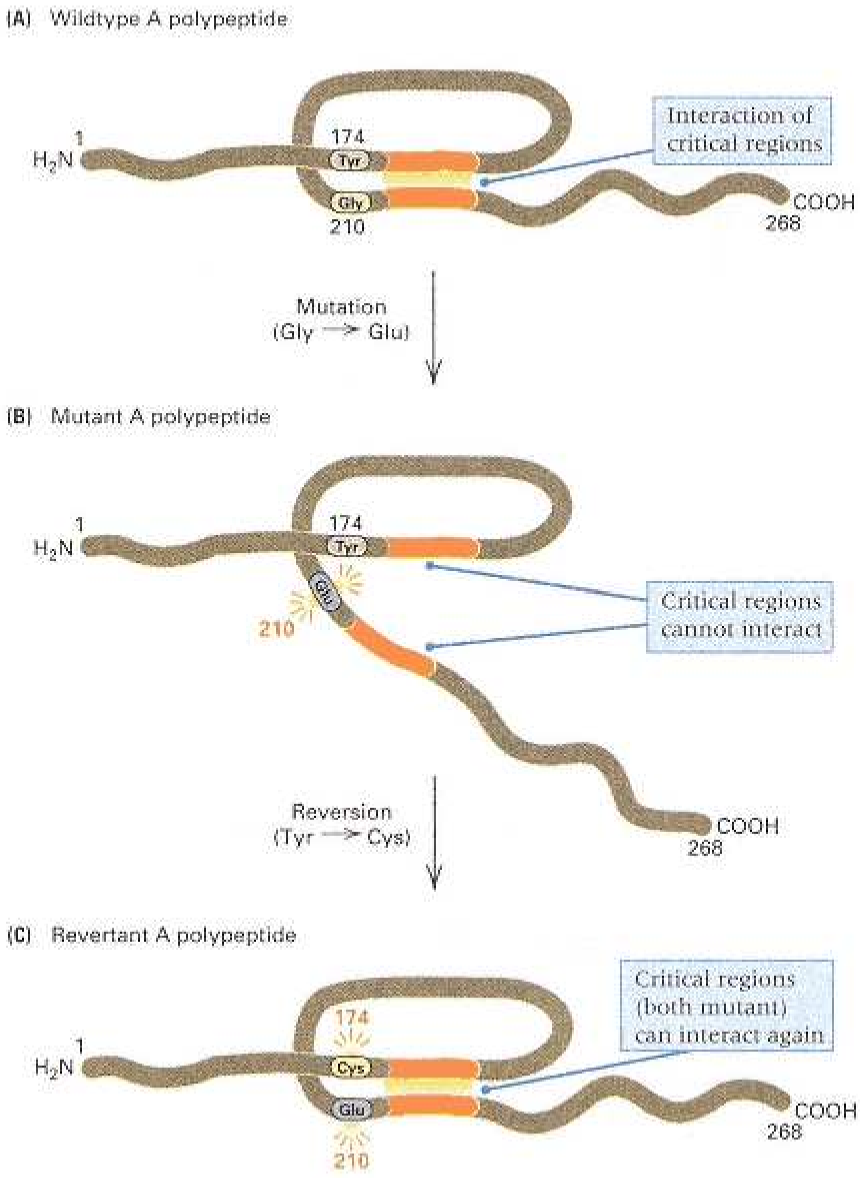


Figure 13.23
Simplified model for the effect of mutation and intragenic suppression on the folding and activity of the A protein of tryptophan synthase in *E. coli*. (A) The wildtype protein in which two critical regions (orange segments) of the polypeptide chain interact. (B) Disruption of proper folding of the polypeptide chain by a substitution of amino acid 210 prevents the critical regions from interacting. (C) Suppression of the effect of the original forward mutation by a subsequent change in amino acid 174. The structure of the region that contains this amino acid is altered bringing the critical regions together again.

The mutation shown, one of many that inactivate the enzyme, is a change of amino acid number 210 from glycine in the wildtype protein to glutamic acid. This glycine is not in the active site; rather, the inactivation is caused by a change in the folding of the protein, which indirectly affects the active site. The activity of the mutant protein is partly restored by a second mutation, in which amino acid number 174 changes from tyrosine to cysteine. A protein in which only the tyrosine has been replaced by cysteine is inactive, again because of a change in the shape of the protein. This change at the second site, found repeatedly in different reversions of the original mutation, restores activity because the two regions containing amino acids 174 and 210 interact fortuitously to produce a protein with the correct folding. This type of reversion can usually be taken to mean that the two regions of the protein interact, and studies of the amino acid sequences of mutants and revertants are often informative in elucidating the structure of proteins.

Intergenic Suppression

Intergenic suppression refers to a mutational change *in a different gene* that eliminates or suppresses the mutant phenotype. Many intergenic suppressors are very specific in being able to suppress the effects of mutations in only one or a few other genes. For example, phenotypes resulting from the accumulation of intermediates in a metabolic pathway as a consequence of mutational inactivation of one step in the pathway can sometimes be suppressed by mutations in other genes that function earlier in the pathway, which reduces the concentration of the intermediate. These suppressors act only to suppress mutations that affect steps further along the pathway, and they suppress all alleles of these genes.

Other intergenic suppressors are more general in being able to suppress mutations in many functionally unrelated genes—and usually in only a subset of alleles of these genes. For example, certain intergenic suppressors in *Drosophila* affect transcription factors that control the transcription of particular families of transposable elements. Alleles that are mutant because they contain a transposable element whose transcription interferes with that of the gene can be suppressed by an intergenic suppressor that prevents transcription of the inserted transposable element.

In prokaryotes, the best-understood type of intergenic suppression is found in some tRNA genes, mutant forms of which can suppress the effects of particular mutant alleles of many other genes. These suppressor mutations change the anticodon sequence in the tRNA and thus the specificity of mRNA codon recognition by the tRNA molecule. Mutations of this type were first detected in certain strains of *E. coli*, which were able to suppress particular phage - T4 mutants that failed to form plaques on standard bacterial strains. These strains were also able to suppress particular alleles of numerous other genes in the bacterial genome. The suppressed mutations were, in each case, nonsensemutations—thosein which a stop codon (UAA, UAG, or UGA) had been introduced within the coding sequence of a gene, with the result that polypeptide synthesis was prematurely terminated and only an amino-terminal fragment of the polypeptide was synthesized. Because they suppressed nonsense mutations, the suppressors were called nonsense suppressors.

Nonsense suppression takes place when the anticodon of one of the normal tRNA molecules is mutated so that it pairs with a nonsense codon and inserts an amino acid, allowing translation to continue (Chapter 10). Such mutations are not lethal because most tRNA genes are present in several copies, so nonmutated copies of the tRNA are also present to allow correct translation of the codon.

Reversion as a Means of Detecting Mutagens and Carcinogens

In view of the increased number of chemicals present as environmental contaminants, tests for the mutagenicity of these substances have become important. Furthermore, most carcinogens are also mutagens, so mutagenicity provides an intial screening for potential hazardous agents. One simple method of screening large numbers of substances for mutagenicity is a reversion test that uses nutritional mutants of bacteria. In the simplest type of reversion test, a compound that is a potential mutagen is added to solid growth medium, a known number of a mutant bacterium is plated, and the number of revertant colonies is counted. An increase in the reversion frequency significantly greater than that obtained in the absence of the test compound identifies the substance as a mutagen. However, simple tests of this type fail to demonstrate the mutagenicity of a large number of potent carcinogens. The explanation for this failure is that many substances are not directly mutagenic (or carcinogenic); rather, they require a conversion into mutagens by enzymatic reactions that take place in the livers of animals and that have no counterpart in bacteria. The normal function of these enzymes is to protect the organism from various naturally occurring harmful substances by converting them into soluble nontoxic substances that can be disposed of in the urine. However, when the enzymes encounter certain human-made and natural compounds, they convert these substances, which may not be harmful in themselves, into mutagens or carcinogens. The enzymes of liver cells, when added to the bacterial growth medium, activate the compounds and allow their mutagenicity to be recognized. The addition of liver extract is one step in the Ames test for carcinogens and mutagens.

In the **Ames test** histidine-requiring (His^-) mutants of the bacterium *Salmonella typhimurium,* containing either a base substitution or a frameshift mutation, are tested for reversion to His^+. In addition, the bacterial strains have been made more sensitive to mutagenesis by the incorporation of several mutant alleles that inactivate the excision-repair system and that make the cells more permeable to foreign molecules. Because some mutagens act only on replicating DNA, the solid medium used contains enough histidine to support a few rounds of replication but not enough to permit the formation of a visible colony. The medium also contains the potential mutagen to be tested and an extract of rat liver. If the test substance is a mutagen or is converted into a mutagen, then some colonies form. A quantitative analysis of reversion frequency can also be carried out by incorporating various amounts of the potential mutagen in the medium. The reversion frequency generally depends on the concentration of the substance being tested and, for a known carcinogen or mutagen, correlates roughly with its carcinogenic potency in animals.

The Ames test has been used with thousands of substances and mixtures (such as industrial chemicals, food additives, pesticides, hair dyes, and cosmetics), and numerous unsuspected substances have been found to stimulate reversion in this test. A high frequency of reversion does not necessarily indicate that the substance is definitely a carcinogen but only that it has a high probability of being so. As a result of these tests, many industries have reformulated their products. For example, the cosmetics industry has changed the formulation of many hair dyes and cosmetics to render them nonmutagenic. Ultimate proof of carcinogenicity is determined by testing for tumor formation in laboratory animals. However, only a few percent of the substances known from animal experiments to be carcinogens fail to increase the reversion frequency in the Ames test.

13.7— Recombination

Genetic recombination may be regarded as a process of breakage and repair between two DNA molecules. In eukaryotes, the process takes place early in meiosis after each molecule has replicated, and with respect to genetic markers, it results in two molecules of the parental type and two recombinants (Chapter 4). For genetic studies of recombination, fungi such as yeast or *Neurospora* are particularly useful because all four products of meiosis are contained in a four-spore or eight-spore ascus (Section 4.5). Most asci from heterozygous *Aa* individuals contain a ratio of

2 *A* : 2 *a* in four-spored asci,

or

4 *A* : 4 *a* in eight-spored asci

because of normal Mendelian segregation. Occasionally, however, aberrant ratios are also found, such as

3 *A* : 1 *a* or 1 *A* : 3 *a* in four-spored asci

and

5 *A* : 3 *a* or 3 *A* : 5 *a* in eight-spored asci

Different types of aberrant ratios can also occur. The aberrant asci are said to result from **gene conversion** because it appears as though one allele has "converted" the other allele into a form like itself. Gene conversion is frequently accompanied by recombination between genetic markers on either side of the conversion event, even when the flanking markers are tightly linked. This implies that gene conversion can be one consequence of the recombination process. The accepted explanation of gene conversion is that it results from mismatch repair in a small segment of duplex DNA containing one strand from each parental molecule (Figure 13.24). The region of hybrid DNA is called a **heteroduplex region.**

Molecular models of recombination are quite complex because heteroduplexes may or may not lead to recombination between flanking genetic markers. The models also strive to account for all the types of aberrant asci that are observed and for their relative frequencies. We will consider three models that differ in the type of DNA breakage that initiates the process.

The Holliday Model

In the **Holliday model** of recombination, each of the participating DNA duplexes (Figure 13.25A) undergoes a symmetrically positioned single-stranded nick in strands of the same polarity (Figure 13.25B). The nicked strands unwind in the region of the nicks, switch pairing partners, and form a heteroduplex region in each molecule (Figure 13.25C). Ligation anchors the ends of the exchanged strands in place, and further unwinding increases the length of the heteroduplex region (termed *branch migration*, Figure 13.25D). Two more nicks and rejoinings are necessary to resolve the structure in Figure 13.25D. As the

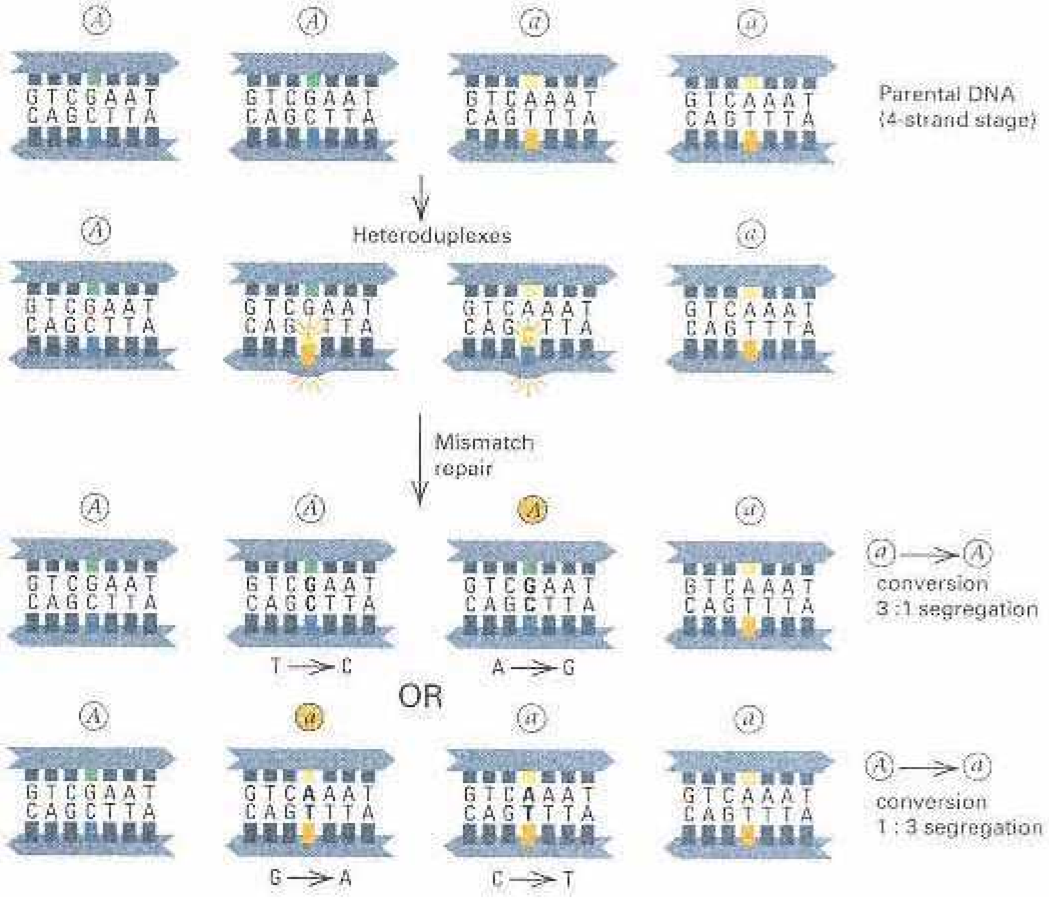


Figure 13.24
Mismatch repair resulting in gene conversion. Only a small part of the heteroduplex region is shown. Each mismatched heteroduplex can be repaired to give an *A allele* or an *a allele*. The patterns of repair leading to 3 : 1 and 1 : 3 segregation are shown.

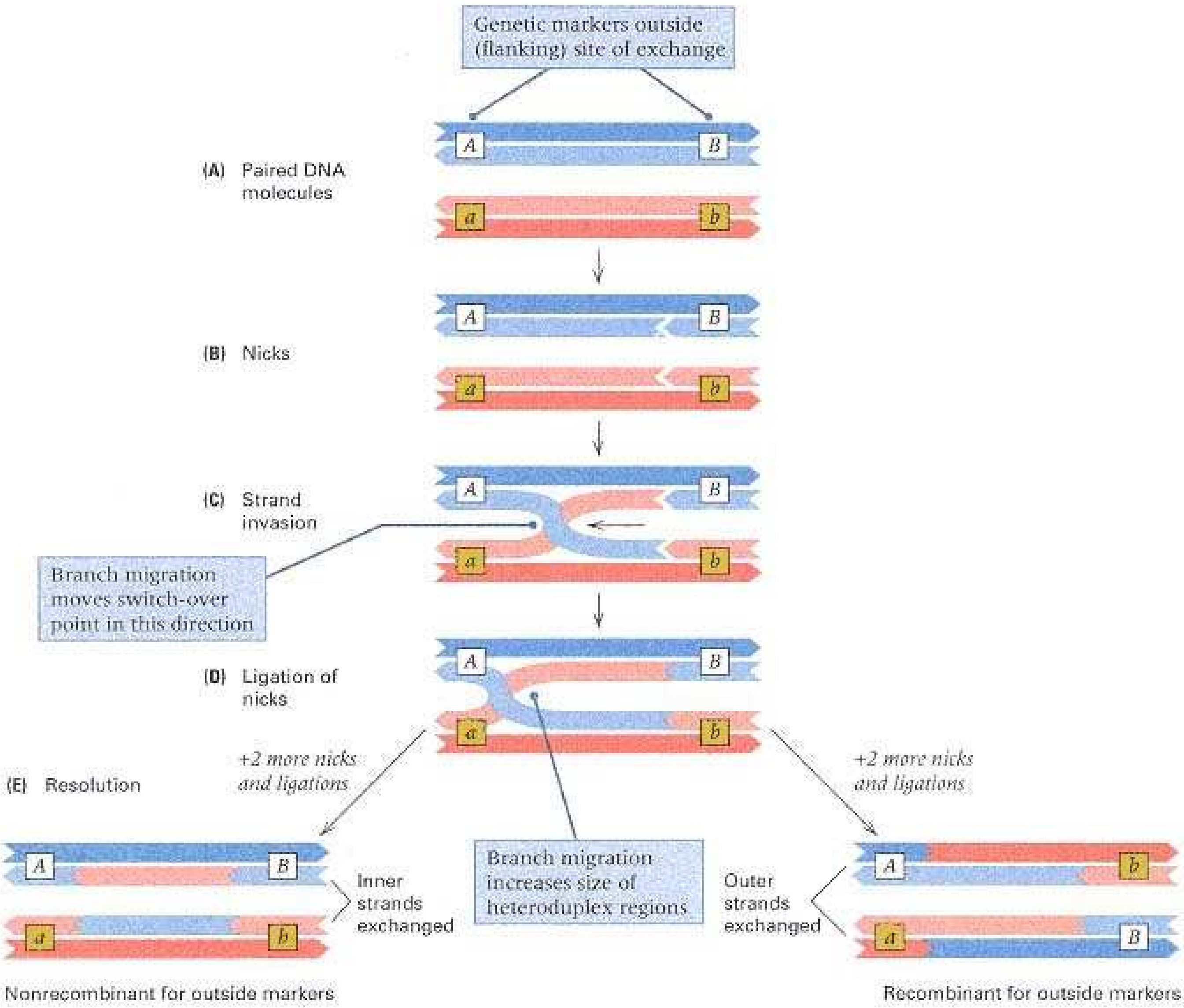


Figure 13.25
Holliday model of recombination. (A) Paired DNA molecules. (B) The exchange process is initiated by single-stranded breaks in strands of the same polarity. (C) Strands exchange pairing partners (referred to as "invasion"). (D) Ligation and branch migration. (E) Exchange is resolved either by the breaking and rejoining of the inner strands (resulting in molecules that are nonrecombinant for outside markers) or by the breaking and rejoining of the outer strands (resulting in recombinant molecules). Either type of resolution gives a heteroduplex region of the same length in each participating molecule.

configuration is drawn, these breaks can be in the inner strands, resulting in molecules that are *nonrecombinant* for markers flanking the heteroduplex region, or they can be in the outer strands, resulting in molecules that are *recombinant* for outside markers (Figure 13.25E).

Of the two possibilities for resolution of the structure in Figure 13.25D, one results in recombination of the outside markers and the other does not. These outcomes are equally likely because there is no topological difference between the inner strands and the outer strands. To understand why, note that an end-for-end rotation of the lower duplex in Figure 13.26 gives the configuration in Figure 13.26. This type of configuration is known as a **Holliday**

structure. Drawing the structure in this manner makes it clear that breakage and rejoining of the east-west strands or the north-south strands are topologically equivalent. However, north-south resolution results in recombination of the outside markers (*A b* and *a B* gametes), whereas east-west resolution gives nonrecombinants (*A B* and *a b* gametes).

An electron micrograph of a Holliday structure formed between two DNA duplexes is shown in Figure 13.27A. This is the physical structure of DNA corresponding to the diagram in Figure 13.26. An interpretation of the structure that preserves the double-helical structure of duplex DNA is shown in Figure 13.27B. A pair of genetic markers—*A, a* and *B, b*—flanking the exchange are shown. One of the duplexes that participates in the exchange carries the alleles *A B*, and the other duplex carries *a b.* Near the site of exchange, each of the double-stranded molecules contains a short heteroduplex region. The key feature of the structure is that there are four short, single-stranded regions where the duplexes come together. The Holliday structure can be resolved either by breakage and reunion in the single-stranded regions on the top and bottom (called north and south in Figure 13.26) or by breakage and reunion in the single-stranded regions on the left and right (called east and west in Figure 13.26). The left-right breakage and rejoining result in two duplexes that are nonrecombinant for the outside markers (*A b* and *a b*,) whereas the top-bottom breakage and rejoining result in two duplexes that are recombinant for the outside markers (*A b* and *a B*).

Several types of enzymes are required for recombination. In *E. coli,* the exchange of strands is mediated by the recA protein. Another enzyme, called the recBCD nuclease because it contains three protein subunits (B, C, and D), acts as an endonuclease, an exonuclease, and a helicase; recBCD has been implicated in strand exchange and branch migration. The PolI DNA polymerase also promotes strand displacement in branch migration, and DNA ligase is necessary to connect the strands with a covalent bond. Finally, the resolution of the Holliday structure is an enzymatic function, carried out by the

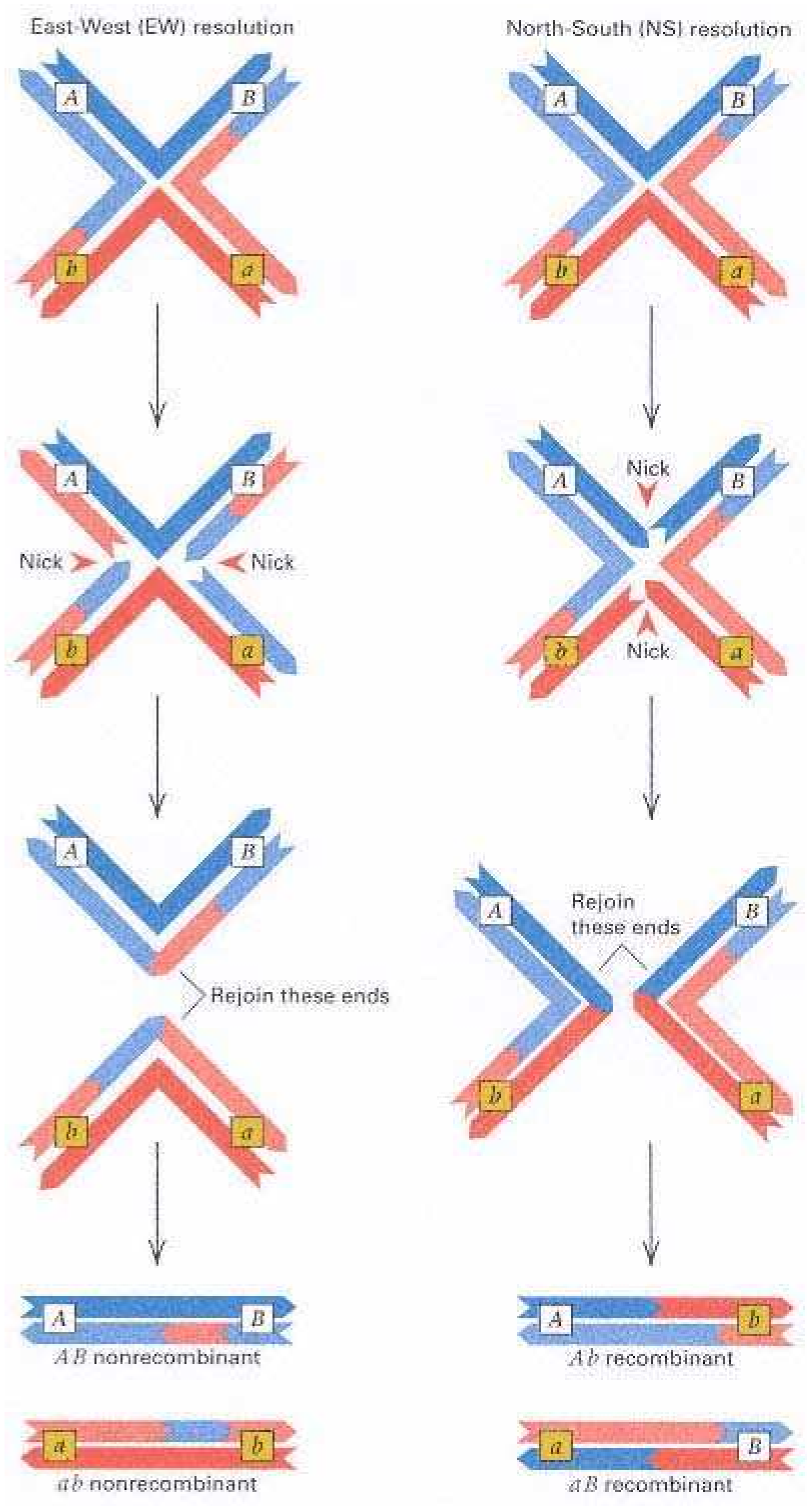


Figure 13.26
The cross-shaped configuration shows the topological equivalence of outer-strand and inner-strand resolution of the Holliday structure in Figure 13.25D. East-west resolution corresponds to inner-strand breakage and rejoining; north-south resolution corresponds to outer-strand breakage and rejoining.

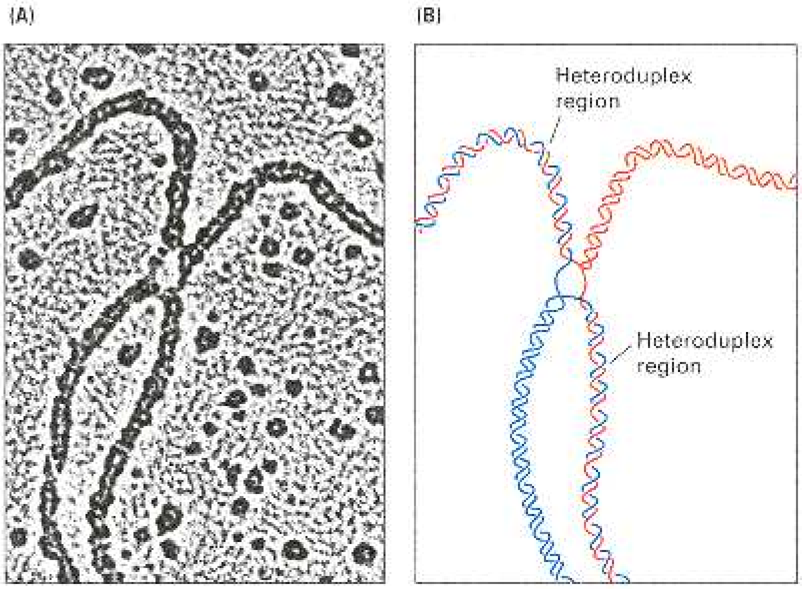


Figure 13.27
(A) Electron micrograph of a Holliday structure. Note the short, single-stranded region of DNA on each side of the oval-shaped structure where the duplexes come together. (B) Interpretative drawing of a Holliday structure in terms of double-helical molecules. Note the single-stranded regions where the duplexes come together, as well as the short heteroduplex region in two of the duplexes.

Holliday junction-resolving enzyme, which is denoted RuvC.

An important feature of the Holliday model is that heteroduplex regions are found in the region of the exchange. Mismatch repair in this region can result in gene conversion, and the equal likelihood of north-south and east-west resolution (the RuvC function) implies that gene conversion will frequently be accompanied by recombination of flanking markers.

Asymmetrical Single-Strand Break Model

Observed patterns of gene conversion indicate that the heteroduplex region is not always the same length in the DNA duplexes participating in an exchange. This is a difficulty for the Holliday model, because in its original form, the heteroduplex regions are symmetric. However, in the single-strand break and repair model (Figure 13.28), the heteroduplex regions can be asymmetrical. Starting with a single-strand nick (Figure 13.28A), progressive unwinding and DNA synthesis (red strand) liberate one DNA strand from a duplex. This strand can "invade" the other duplex (Figure 13.28B), resulting in displacement of the original strand and formation of a **D loop** (**displacement loop**). Nucleases degrade the D loop, and the invading strand is ligated in place (Figure 13.28C). Branch migration (Figure 13.28D) forms a second heteroduplex region, but the heteroduplex regions in the participating duplexes are of different lengths. As with the Holliday model, the configuration in Figure 13.28D can be resolved by RuvC in either the east-west or the north-south direction, yielding nonrecombinant or recombinant products, respectively.

Double-Strand Break Model

In yeast, duplex DNA molecules with double-stranded breaks are very efficient in initiating recombination with a homologous unbroken duplex. A model for this process is shown in Figure 13.29. The initiating molecule will typically contain a gap with missing sequences and strands of unequal length at the ends because of exonuclease activity. The ends are free to invade a homologous duplex, forming a D loop (Figure 13.29B), which increases in extent until it bridges the gap (Figure 13.29C). Repair synthesis and ligation in both duplexes gives the configuration in Figure 13.29D, which can be resolved in four ways because it contains two Holliday-type junctions, each of which can be resolved in east-west or north-south orientation. Two resolutions yield nonrecombinant duplexes and two yield recombinant duplexes, so genetic recombination is a frequent outcome.

Although each of the three models of recombination accounts for the principal features of gene conversion and recombination, each falls short of completely explaining all of the details and the aberrations that are occasionally found. For example, the Holliday model predicts an excess of two-strand double recombination, which is not observed. All three models may represent alternative pathways of recombination whose frequencies depend on whether the initiating event consists of a pair of single-stranded nicks or a double-stranded gap.

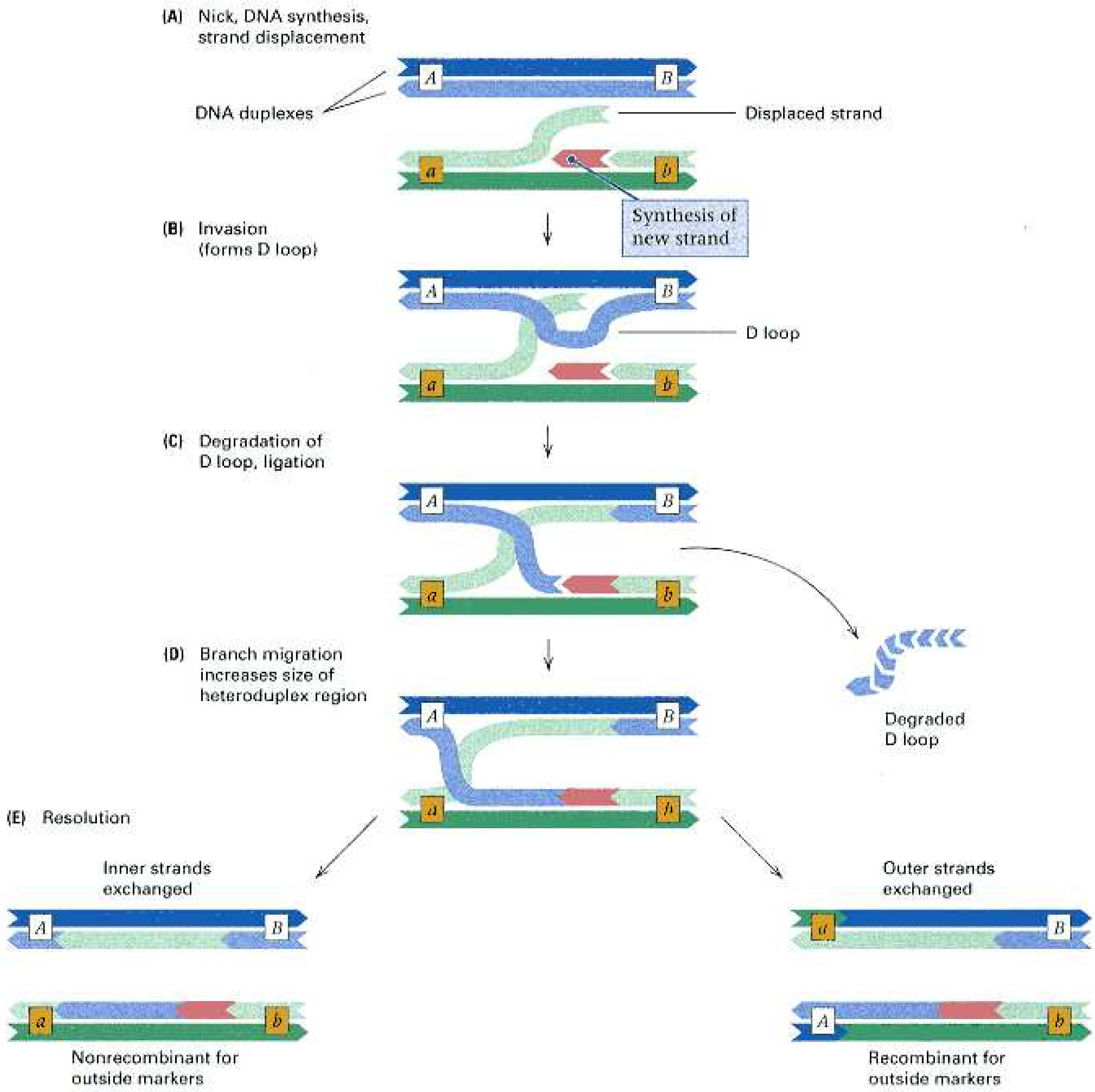


Figure 13.28
Model of recombination initiated by a break in one strand of a pair of DNA molecules (the layout is like that in Figure 13.25). (A) A freed DNA strand is replaced by new synthesis and invades the partner molecule. (B) Displacement of a strand in the partner molecule forms a D (displacement) loop. (C) The D loop is degraded, and free DNA ends are ligated. (D) Branch migration increases the length of the heteroduplex region. (E) Resolution occurs in either of two topologically equivalent ways. In this model, the heteroduplex regions are not of equal length.

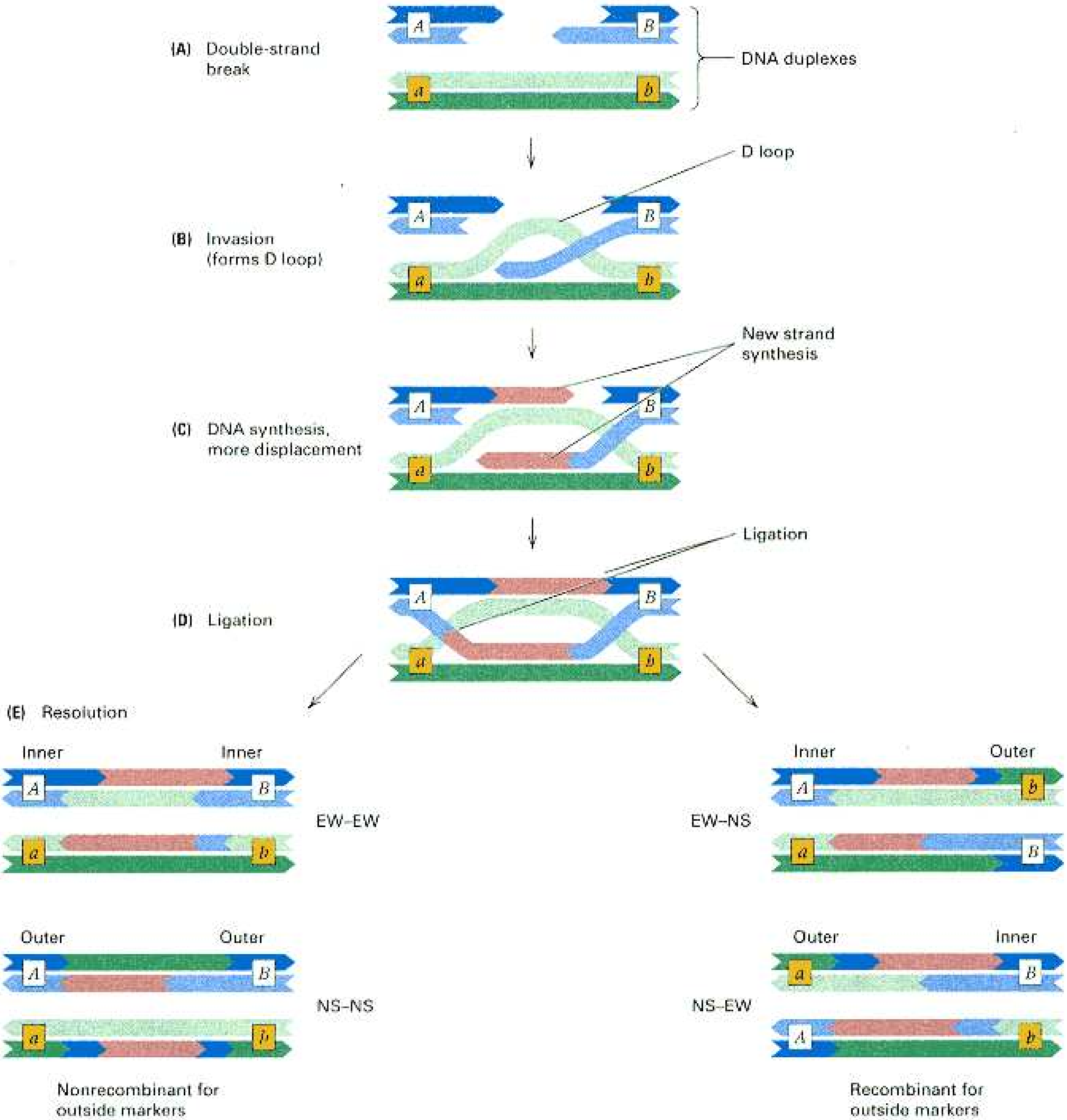


Figure 13.29

Model of recombination initiated by a double-stranded break (the layout is like that in Figure 13.25). (A) The double-stranded break is trimmed back asymmetrically by nucleases. (B) One of the strands invades the partner molecule, forming a displacement loop. (C) The invading strand is elongated by new synthesis, and the displacement loop serves as a template for new synthesis in the other duplex. (D) Free DNA ends are ligated. (E) Two Holliday junctions are formed which can be resolved in any one of four topologically equivalent ways, two of which result in recombination of outside makers. In this model, the heteroduplex regions are unequal in length, and all of them contain some newly synthesized DNA

Chapter Summary

Mutations can be classified in a variety of ways in terms of (1) how they come about, (2) the nature of the chemical change, or (3) how they are expressed. Conditional mutations cause a change in phenotype under restrictive but not permissive conditions. For example, temperature-sensitive mutations are expressed only above a particular temperature. Spontaneous mutations are of unknown origin, whereas induced mutations result from exposure to chemical reagents or radiation. All mutations ultimately result from changes in the sequence of nucleotides in DNA, including base substitutions, insertions, deletions, and rearrangements. Base-substitution mutations are single-base changes that may be silent (for example, in a noncoding region or a synonymous codon position), may change an amino acid in a polypeptide chain (a missense mutation), or may cause chain termination by production of a stop codon (a nonsense mutation). A transition is a base-substitution mutation in which a pyrimidine is substituted for another pyrimidine or a purine for another purine. In a transversion, a pyrimidine is substituted for a purine, or the other way around. A mutation may consist of the insertion or deletion of one or more bases; in a coding region, if the number of bases is not a multiple of three, the mutation is a frameshift.

Spontaneous mutations are random. Favorable mutations do not take place in response to harsh environmental conditions. Rather, the environment selects for particular mutations that arise by chance and happen to confer a growth advantage. Spontaneous mutations often arise by errors in DNA replication that fail to be corrected either by the proofreading system or by the mismatch-repair system. The proofreading system removes an incorrectly incorporated base immediately after it is added to the growing end of a DNA strand. The mismatch-repair system removes incorrect bases at a later time. Methylation of parental DNA strands and delayed methylation of daughter strands allows the mismatch-repaid system to operate selectively on the daughter DNA strand.

Mutations can be induced chemically by direct alteration of DNA—for example, by nitrous acid, which deaminates bases. Base analogs are incorporated into DNA in replication. They undergo mispairing more often than do the normal bases, which gives rise to transition mutations. The base analog 5-bromouracil is an example of such a mutagen. Alkylating agents often cause depurination; this type of mutation is usually repaired, but often an adenine is inserted opposite a depurination site, which results in a transversion. Acridine molecules intercalate (interleaf) between base pairs of DNA and cause misalignment of parental and daughter strands in DNA replication, giving rise to frameshift mutations, usually of one or two bases. Ionizing radiation results in oxidative free radicals that cause a variety of alterations in DNA, including double-stranded breaks. Although the amount of genetic damage from normal background and other sources of radiation is believed to be low, studies of persons exposed to fallout from the Chernobyl nuclear meltdown indicate an increased mutation rate, at least for simple tandem repeat polymorphisms (STRPs).

A variety of systems exist for repairing damage to DNA. In excision repair, a damaged stretch of a DNA strand is excised and replaced with a newly synthesized copy of the undamaged strand. In photoreactivation, there is direct cleavage of the pyrimidine dimers produced by ultraviolet radiation. Damage to DNA that results in the formation of O^6-methyl guanine is repaired by a special methyl transferase. In all of these systems, the correct template is restored. Postreplication repair is an exchange process in which gaps in one daughter strand produced by aberrant replication across damaged sites are filled by nondefective segments from the parental strand of the other branch of the newly replicated DNA. Thus a new template is produced by this system. The SOS repair system in bacteria is error prone (mutagenic) because it facilitates replication across stretches in the template DNA that are too damaged to serve as a normal template.

Mutant organisms sometimes revert to the wildtype phenotype. Reversion is normally due to an additional mutation at another site. Reversion by a second mutation is called suppression, and the secondary mutations are called suppressor mutations. These can be intragenic or intergenic. In intragenic suppression, a mutation in one region of a protein alters the folding of the protein, and a change in another amino acid causes correct folding again. Intergenic suppression takes place through a variety of mechanisms. For example, in nonsense suppression, a chain-termination mutation (a stop codon) is read by a mutant tRNA molecule with an anticodon that can hydrogen-bond with the stop codon, and an amino acid is inserted at the site of the stop codon.

The Ames test measures reversion as an indicator of mutagens and carcinogens; it uses an extract of rat liver, which in mammals occasionally converts intrinsically harmless molecules into mutagens and carcinogens.

Genetic recombination is intimately connected with DNA repai because the process always includes breakage and rejoining of DNA strands. Three current models of recombination differ in the types of DNA breaks that initiate the process and in predictions about the types and frequencies of gene-conversion events. In gene conversion, one allele becomes converted into a homologous allele, which is detected by aberrant segregation in fungal asci, such as 3 : 1

or 1 : 3. Gene conversion is the outcome of mismatch repair in heteroduplexes. All recombination models include the creation of heteroduplexes in the region of the exchange and predict that 50 percent of gene conversions will be accompanied by recombination of genetic markers flanking the conversion event. The Holliday model assumes that single-strand nicks occur at homologus positions in the participating duplexes. Other models assume initiation with a single- strand nick or a double-stand break. All three processes may be operative.

Key Terms

acridine	Holliday model	photoreactivation
alkylating agent	Holliday structure	postreplication repair
Ames test	hot spot	pyrimidine dimer
base analog	induced mutation	replica plating
base-substitution mutation	intercalation	replication slippage
ClB method	intergenic suppression	restrictive condition
conditional mutation	itragenic suppression	reverse mutation
depurination	ionizing radiation	reversion
displacement loop	mismatch repair	sickle-cell anemia
D loop	missense mutation	silent mutation
DNA uracil glycosylase	mutagen	somatic mutation
excision repair	mutation	SOS repair
forward mutation	mutation rate	spontaneous mutation
free radical	nitrous acid	suppressor mutation
gene conversion	nonsense mutation	temperature-sensitive mutation
germ-line mutation	nonsense suppression	transition mutation
heteroduplex region	nucleotode analog	transversion mutation
Holliday junction-resolving enzyme	permissive condition	xeroderma pigmentosum

Review the Basics

• What is a spontaneous mutation? An induced mutation?

• What is meant by the term *mutation rate?*

• How was the technique of replica plating used to demonstrate that antibiotic-resistance mutations are present at low frequency in populations of bacteria even in the absence of the antibiotic?

• What is mismatch repair and how does it happen? Name one human disease associated with a defective mismatch-repair system.

• What features of DNA polymerase could qualify it as a repair enzyme?

• What is a hot spot of mutation? Why might a mutational hot spot coincide with the site of a methylated cytosine in the DNA?

• Explain the following statement: "There is no threshold dose for the mutagenic effect of radiation."

• What unusual types of tetrads provide evidence that a region of heteroduplex DNA is created in the process of recombination?

Guide to Problem Solving

Problem 1: The molecule 2-aminopurine (Ap) is an analog of adenine that pairs with thymine. It also occasionally pairs with cytosine. What types of mutations will be induced by 2-aminopurine?

Answer: In problems of this sort, first note the base that is replaced, and then follow what can happen in subsequent founds of DNA replication when the base analog pairs normally or abnormally. In this example, the 2-aminopurine Ap) is incorporated in place of adenine opposite thymine, forming an Ap-T base pair. In the following rounds or replication, the Ap will usually pair with T, but when it occasionally pairs with C, it forms an Ap-C pair. In the next round of replication, the Ap again pairs with T (forming an Ap-T pair), but the C pairs with G (forming a mutant G-C pair). Hence the type of mutation induced by 2-aminopurine is a change from an A-T pair to a G-C pair, which is a transition mutation.

Problem 2: Twenty spontaneous white-eye (*w*) mutants of *Drosophila* are examined individually to determine their reversion frequencies. Of these, 12 revert spontaneously at a frequency of 10^{-3} to 10^{-5}. Another 6 do not revert spontaneously at detectable frequency but can be induced to revert at frequencies ranging from 10^{-5} to 10^{-6} by treatment with alkylating agents. The remaining 2 cannot be made to revert

Chapter 13 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the gen
information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of th
information available at the site. This report, or an alternative, may be assigned by your instructor.

1. Search this database for **HNPCC** (hereditary nonpolyposis colorectal cancer) to learn more about the homologs of
coli mutS and *mutL* involved in human colon cancer. Check out each of the entries to learn whether the genetic basis
mutation in a *mutS* homolog, a *mutL* homolog, or neither. If assigned to do so, choose either HNPCC Type 1 or HNP
Type 2, and write a 150-word report on how the genetic basis of the disease was discovered.

2. Some inherited defects in **DNA repair** are included in this database of pediatric disorders. Find the list of disorder
and follow the links to ataxia-talangiectasia, Bloom syndrome, Fanconi anemia, and xeroderma pigmentosum. If assi
to do so, pick any one of these conditions, and write a 250-word report on its incidence, genetic basis, clinical featur
and treatment.

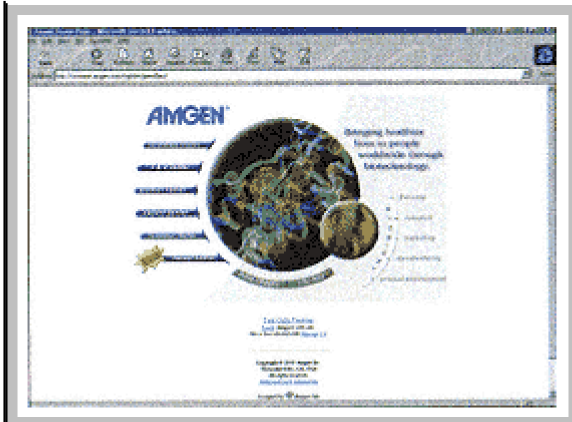
3. You may view an animated **Holliday structure** in the process of formation and resolution at this keyword site. If
assigned to do so, draw key stages of the process using the colors in the animation; draw similar pictures showing ho
the Holliday structure is resolved without an exchange of genetic markers flanking the heteroduplex region.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of gene
resources available on the World Wide Web. Select the **Mutable Site** for Chapter 13, and you will be linked to the
current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 13.



under any conditions. Answer the following questions, and explain your answers.

(a) Which class of mutants would you suspect to result from the insertion of transposable elements?

(b) Which class of mutants are probably missense mutations?

(c) Which class of mutants are probably deletions?

Answer:

(a) Insertions of transposable elements in eukaryotes are often characterized by genetic instability and high reversion Therefore, the twelve mutants with the highest reversion frequencies are probably transposable-element insertions.

(b) The fact that alkylating agents can induce the second class of mutations to revert suggests that they are missense mutations. Although they do not revert spontaneously, this is probably because the exact reversal of a base substitutio an exceedingly rare spontaneous event.

(c) Deletion mutations cannot revert under any conditions because there is too much missing genetic information to b restored. Therefore, the class of mutations that does not revert probably consists of deletions.

Analysis and Applications

13.1 Occasionally, a person is found who has one blue eye and one brown eye or who has a sector of one eye a different color from the rest. Can these phenotypes be explained by new mutations? If so, in what types of cells must the mutations occur?

13.2 A mutation is isolated that cannot be induced to revert. What types of molecular changes might be responsible?

13.3 There are 12 possible substitutions of one nucleotide pair for another (for example, A $\rightarrow$ G). Which changes are transitions and which transversions?

13.4 Do all nucleotide substitutions in the second position of a codon necessarily produce an amino acid replacement?

13.5 Do all nucleotide substitutions in the first position of a codon necessarily produce an amino acid replacement? Do all of them necessarily produce a nonfunctional protein?

13.6 In a coding sequence made up of equal proportions of A, U, G, and C, what is the probability that a random nucleotide substitution will result in a chain-termination codon?

13.7 If a mutagen produces frameshift mutations, what kinds of mutations will it induce to revert?

13.8 If a deletion eliminates a single amino acid in a protein, how many base pairs are missing?

13.9 Does the nucleotide sequence of a mutation tell you anything about its dominance or recessiveness?

13.10 What are two ways in which pyrimidine dimers are removed from DNA in *E. coli?*

13.11 This problem illustrates how conditional mutations can be used to determine the order of genetically controlled steps in a developmental pathway. A certain organ undergoes development in the sequence of stages A $\rightarrow$ B $\rightarrow$ C, and both gene *X* and gene *Y* are necessary for the sequence to proceed. A conditional mutation *X'* is sensitive to heat (the gene product is inactivated at high temperatures), and a conditional mutation *Y'* is sensitive to cold (the gene productis nactivated at cold temperatures). The double mutant *X'/X'Y'/Y'* is created and reared at either high or low temperatures. How far would development proceed in each of the following cases at the high temperature and at the low temperature?

(a) Both *X* and *Y* are necessary for the A $\rightarrow$ B step.

(b) Both *X* and *Y* are necessary for the B $\rightarrow$ C step.

(c) *X* is necessary for the A $\rightarrow$ B step, and *Y* for the B $\rightarrow$ C step.

(d) *Y* is necessary for the A $\rightarrow$ B step, and *X* for the B $\rightarrow$ C step.

13.12 A Lac^+ culture of *E. coli* gives rise to a Lac^- strain that results from a UGA nonsense mutation in the *lacZ* gene. The Lac^- strain gives rise to Lac^+ colonies at the rate of 10^{-8} per cell per generation, and 90 percent of the latter are caused by suppressor tRNA molecules. What is the rate of production of suppressor mutations in the original Lac^+ culture?

13.13 If a gene in a particular chromosome has a probability of mutation of 5×10^{-5} per generation, and if the allele in a particular chromosome is followed through successive generations.

(a) What is the probability that the allele does not undergo a mutation in 10,000 consecutive generations?

(b) What is the average number of generations before the allele undergoes a mutation?

13.14 In the mouse, a dose of approximately 1 gray (Gy) of x rays produces a rate of induced mutation equal to the rate of spontaneous mutation. Taking the spontaneous mutation rate into account, what is the total mutation rate at 1 Gy? What dose of x rays will increase the mutation rate by 50 percent? What dose will increase the mutation rate by

10 percent?

13.15 Among several hundred missense mutations in the gene for the A protein of tryptophan synthase in *E. coli*, fewer than 30 of the 268 amino acid positions are affected by one or more mutations. Explain why the number of positions affected by amino acid replacements is so low.

13.16 Human hemoglobin C is a variant in which a lysine in the β-hemoglobin chain is substituted for a particular glutamic acid. What single-base substitution can account for the hemoglobin-C mutation?

13.17 How many amino acids can substitute for tyrosine by a single-base substitution? (Do not assume that you know which tyrosine codon is being used.)

13.18 Which of the following amino acid replacements would be expected with the highest frequency among mutations induced by 5-bromouracil? (1) Met $\rightarrow$ Leu, (2) Met $\rightarrow$ Lys, (3) Leu $\rightarrow$ Pro, (4) Pro $\rightarrow$ Thr, (5) Thr $\rightarrow$ Arg.

13.19 Dyes are often incorporated into a solid medium to determine whether bacterial cells can utilize a particular sugar as a carbon source. For instance, on eosin-methylene blue (EMB) medium containing lactose, a Lac^+ cell yields a purple colony and a Lac^- cells yields a pink colony. If a population of Lac^+ cells is treated with a mutagen that produces Lac^- mutants and the population is allowed to grow for many generations before the cells are placed on EMB-lactose medium, a few pink colonies are found among a large number of purple ones. However, if the mutagenized cells are plated on the medium immediately after exposure to the mutagen, some colonies appear that are *sectored* (half purple and half pink). Explain how the sectored colonies arise.

13.20 Mutant alleles of two genes, *a* and *b,* in a haploid organism interact as follows: Most ab^+ and a^+b combinations are mutant, but a few *ab* combinations are wildtype. Each *a* allele that is suppressed requires a particular *b* allele, and each *b* allele that is suppressed requires a particular *a* allele. However, most *a* and *b* alleles are not suppressible by any alleles of the other gene. What kind of interaction between the *a* and *b* gene products can explain these results?

13.21 The following technique has been used to obtain mutations in bacterial genes that are otherwise difficult to isolate. Suppose a mutation is desired in a gene, *a,* and suppose a mutation, b^-, in a closely linked gene is available. The method is to grow phage P1 on a b^+ strain to obtain a^+ b^+ transducing particles. The P1 phages containing the transducing particles are exposed to a mutagen, such as hydroxylamine, and the b^- strain is infected with the phages. What step or steps would you carry out next to isolate and a^- mutation? On what genetic phenomenon does this procedure depend?

Challenge Problems

13.22 A *Neurospora* strain that is unable to synthesize arginine (and that therefore requires this amino acid in the growth medium) produces a revertant colony able to grow in the absence of arginine. A cross is made between the revertant and a wildtype strain. What proportion of the progeny from this cross would be arginine-independent if the reversion occurred by each of the following mechanisms?

(a) a precise reversal of the nucleotide change that produced the original arg^- mutant allele

(b) a mutation to a suppressor of the arg^- allele occurring in another gene located in a different chromosome

(c) a suppressor mutation occurring in another gene located 10 map units away from the arg^- allele in the same chromosome

13.23 Which of the following 8-spore asci might arise from gene conversion at the *m* locus? Describe how these asci might arise.

Ascus type	**A**	**B**	**C**	**D**	**E**	**F**
Spore 1	+	+	+	+	+	+
Spore 2	+	+	+		+	+
Spore 3	+	*m*	+	+	*m*	*m*
Spore 4	+	*m*	*m*	*m*	*m*	*m*
Spore 5	*m*	*m*	*m*	+	+	*m*
Spore 6	*m*	*m*	*m*	*m*	+	*m*
Spore 7	*m*	+	*m*	*m*	*m*	*m*
Spore 8	*m*	+	*m*	*m*	*m*	*m*

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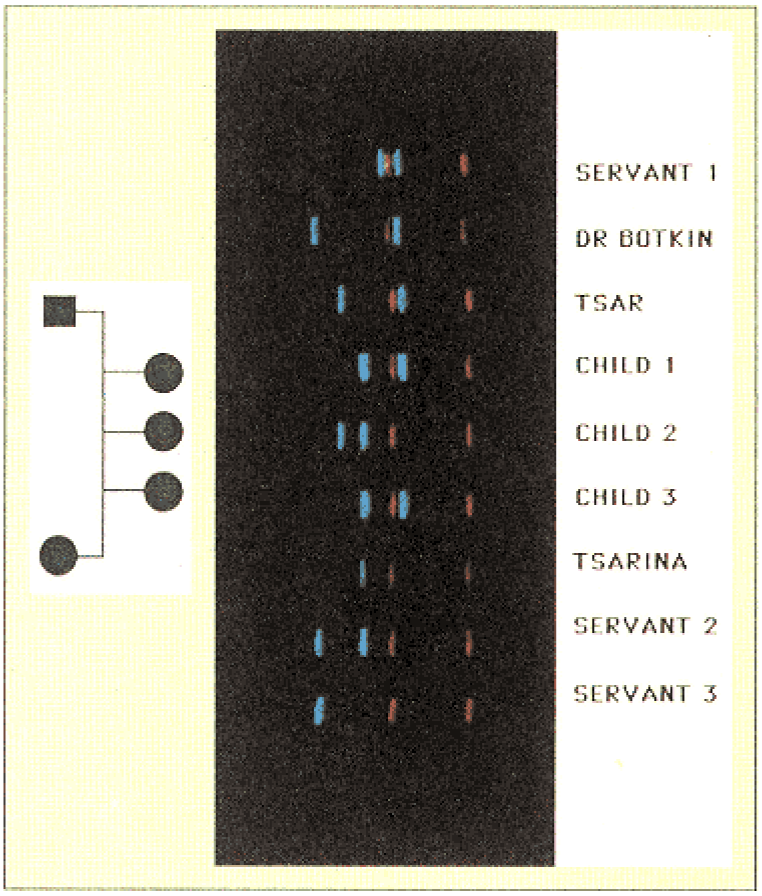


DNA typing confirmed the suspected identities of the skeletons of six adults and three children found in a common, shallow, unmarked grave near Ekaterinburg, Russia, in July 1991. The red bands across the gel are size markers that are the same in all lanes. The blue bands are from DNA markers that are polymorphic in the human population. Sex was identified separately by PCR amplification of genes in the X and Y chromosomes. The identities of the Tsar, Tsarina, and Dr. Botkin were inferred from DNA typing of living relatives. Because the three children, all girls, share one band from the Tsar and one from the Tsarina for this DNA marker, as well as four other markers, they are undoubtedly three of the four daughters. The remains had been buried since 1918. The skeletons of the fourth daughter and of the son affected with hemophilia have not been found. [Courtesy of Peter D. Gill.]

Chapter 14— Extranuclear Inheritance

CHAPTER OUTLINE

PRINCIPLES

• Some inherited traits are determined by DNA molecules present in organelles, principally mitochondria and chloroplasts. A hallmark of such traits is uniparental inheritance, either maternal or paternal.

• Progeny resembling the mother is characteristic of both maternal inheritance and maternal effects. Although maternal inheritance results from the transmission of non-nuclear genetic factors, maternal effects result from nuclear genes expressed in the mother that determine the phenotype of the progeny.

• Mitochondria, photosynthetic plastids, and certain other cellular organelles are widely believed to have originated through independent episodes of symbiosis in which a prokaryotic cell, the progenitor of the present organelle, became an intracellular symbiont of the proto-eukaryotic cell.

• Evidence from DNA sequence variation in human mitochondrial DNA suggests that the mitochondrial DNA present in all modern human beings originated from a single female who lived approximately 150,000 to 300,000 years ago, probably in Africa.

CONNECTIONS

CONNECTION: Chlamydomonas Moment

Ruth Sager and Zenta Ramanis 1965
Recombination of nonchromosomal genes in Chlamydomonas

CONNECTION: A Coming Together

Lynn Margulis (formerly Lynn Sagan) 1967
The origin of mitosing cells

Most traits in eukaryotes show Mendelian inheritance because they are determined by genes in the nucleus that segregate in meiosis. Less common are traits determined by factors located outside the nucleus. These include traits determined by the DNA in mitochondria and in chloroplasts—the self-replicating cytoplasmic **organelles** specialized for respiration and for photosynthesis, respectively. Each of these organelles contains its own DNA that codes for numerous proteins and RNA molecules. The DNA is replicated within the organelles and transmitted to daughter organelles as they form. The organelle genetic system is separate from that of the nucleus, and traits determined by organelle genes show patterns of inheritance quite different from the familiar Mendelian ratios.

Many organisms also contain intracellular parasites or symbionts, including cytoplasmic bacteria, viruses, and other elements. In some cases, inherited traits of the infected organism are determined by such cytoplasmic entities. Organelle heredity and other examples of the diverse phenomena that make up **extranuclear,** or **cytoplasmic, inheritance** are presented in this chapter.

14.1— Recognition of Extranuclear Inheritance

Other than a non-Mendelian pattern of inheritance, no criterion is universally applicable to distinguish extranuclear from nuclear inheritance. In higher organisms, the transmission of a trait through only one parent, **uniparental inheritance,** usually indicates extranuclear inheritance. Genetic transmission of cytoplasmic factors, such as mitochondria or chloroplasts, is determined by maternal and paternal contributions at the time of fertilization, by mechanisms of elimination from the zygote, and by the irregular sorting of cyto-

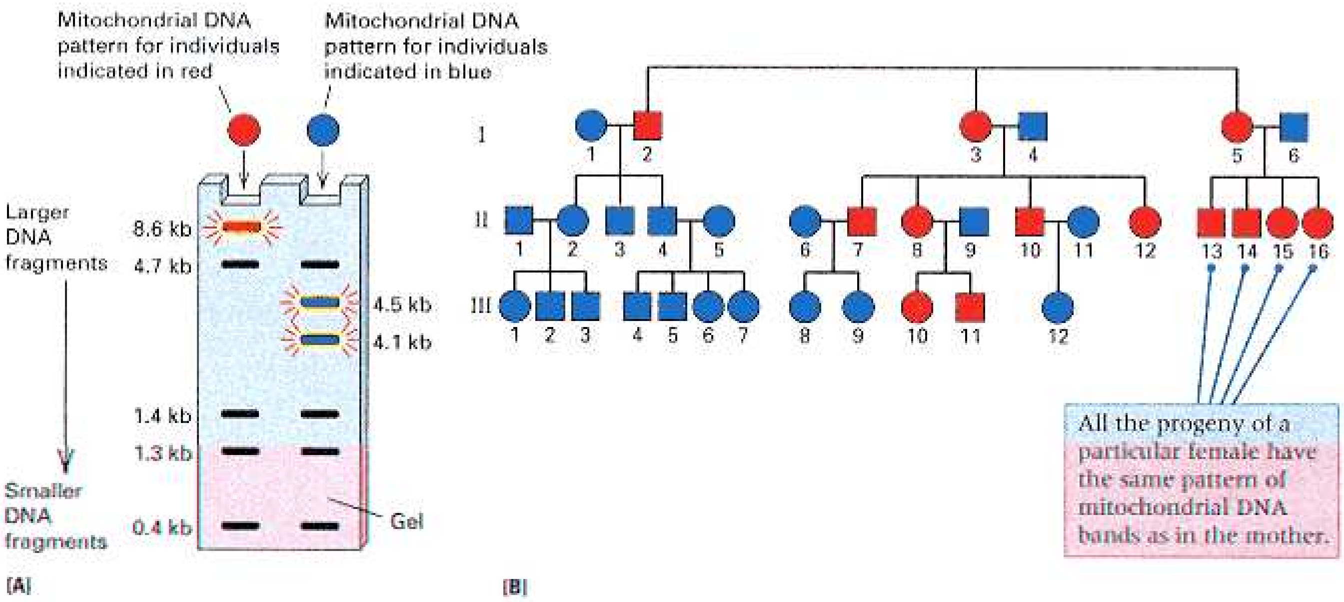


Figure 14.1
Maternal inheritance of human mitochondrial DNA. (A) Pattern of DNA fragments obtained when mitochondrial DNA is digested with the restriction enzyme *Hae*II. The DNA type at the left includes a fragment of 8.6 kb (red). The DNA type at the right contains a cleavage site for *Hae*II within the 8.6-kb fragment, which results in smaller fragments of 4.5 kb and 4.1 kb (blue). (B) Pedigree showing maternal inheritance of the DNA pattern with the 8.6-kb fragment (red symbols). The mitochondrial DNA type is transmitted only through the mother.
[After D. C. Wallace. 1989. *Trends in Genetics*, 5: 9.]

plasmic elements during cell division. Uniparental transmission through the mother constitutes **maternal inheritance;** uniparental transmission through the father is **paternal inheritance.**

Mitochondrial Genetic Diseases

In higher animals, mitochondria are typically inherited through the mother because the egg is the major contributor of cytoplasm to the zygote. Therefore, mitochondria usually show maternal inheritance in which the progeny of a mutant mother and a normal father are mutant, whereas the progeny of a normal mother and a mutant father are normal. A typical pattern of maternal inheritance of mitochondria is shown in the human pedigree in Figure 14.1. When human mitochondrial DNA is cleaved with the restriction enzyme *Hae*II, the cleavage products include either one fragment of 8.6 kb or two fragments of 4.5 kb and 4.1 kb (Figure 14.1A). The pattern with two smaller fragments is typical, and it results from the presence of a *Hae*II cleavage site within the 8.6-kb fragment. Maternal inheritance of the 8.6-kb mitochondrial DNA fragment is indicated in Figure 14.1B, because males (I-2, II-7, and II-10) do not transmit the pattern to their progeny, whereas females (I-3, I-5, and II-8) transmit the pattern to all of their progeny. Although the mutation in the *Hae*II site yielding the 8.6-kb fragment is not associated with any disease, a number of other mutations in mitochondrial DNA do cause diseases and have similar patterns of mitochondrial inheritance. Most of these conditions decrease the ATP-generating capacity of the mitochondria and affect the function of muscle and nerve cells, particularly in the central nervous system, leading to blindness, deafness, or stroke. Table 14.1 summarizes some of the human diseases caused by mitochondrial mutations. Many of these conditions are lethal in the absence of some normal mitochondria, and there is variable expressivity because of differences in the proportions of normal and mutant mitochondria among affected persons.

An example of a human pedigree for inheritance of a mitochondrially inherited disease is shown in Figure 14.2A. The dis-

Table 14.1 Phenotypes associated with some mitochondrial mutations

Nucleotide changed	Mitochondrial component affected	Phenotype[a]
3460	ND1 of Complex I[b]	LHON
11778	ND4 of Complex I	LHON
14484	ND6 of Complex I	LHON
8993	ATP6 of Complex V[b]	NARP
3243	$tRNA^{Leu(UUR)}$[c]	MELAS, PEO
3271	$tRNA^{Leu(UUR)}$	MELAS
3291	$tRNA^{Leu(UUR)}$	MELAS
3251	$tRNA^{Leu(UUR)}$	PEO
3256	$tRNA^{Leu(UUR)}$	PEO
5692	$tRNA^{Asn}$	PEO
5703	$tRNA^{Asn}$	PEO, myopathy
5814	$tRNA^{Cys}$	Encephalopathy
8344	$tRNA^{Lys}$	MERRF
8356	$tRNA^{Lys}$	MERRF
9997	$tRNA^{Gly}$	Cardiomyopathy
10006	$tRNA^{Gly}$	PEO
12246	$tRNA^{Ser(AGY)}$[c]	PEO
14709	$tRNA^{Glu}$	Myopathy
15923	$tRNA^{Thr}$	Fatal infantile multisystem disorder
15990	$tRNA^{Pro}$	Myopathy

[a]LHON Leber's hereditary optic neuropathy

NARP Neurogenic muscle weakness, astaxia, retinitis pigmentosa

MERRF Myoclonic epilepsy and ragged-red fiber syndrome

MELAS Mitochondrial myopathy, encephalopathy, lactic acidosis, stroke-like episodes

PEO Progressive external ophthalmoplegia

[b]Complex I is NADH dehydrogenase. Complex V is ASTP synthase.

[c]In $tRNA^{Leu(UUR)}$, the R stands for either A or G; in $tRNA^{Ser(AGY)}$, the Y stands for either T or C.

ease is myoclonic epilepsy associated with ragged-red muscle fibers (MERRF), a disease of the central nervous system and skeletal muscle. The cause of the disease is a mutation in a $tRNA^{Lys}$ gene encoded in mitochondrial DNA (mtDNA). Figure 14.3 shows the organization of genes in human mtDNA. The $tRNA^{Lys}$ mutation is very pleiotropic because it affects synthesis of all proteins encoded in mitochondrial DNA in all cells. Like many mitochondrial mutations, MERRF is lethal in the absence of normal mitochondria. All of the individuals

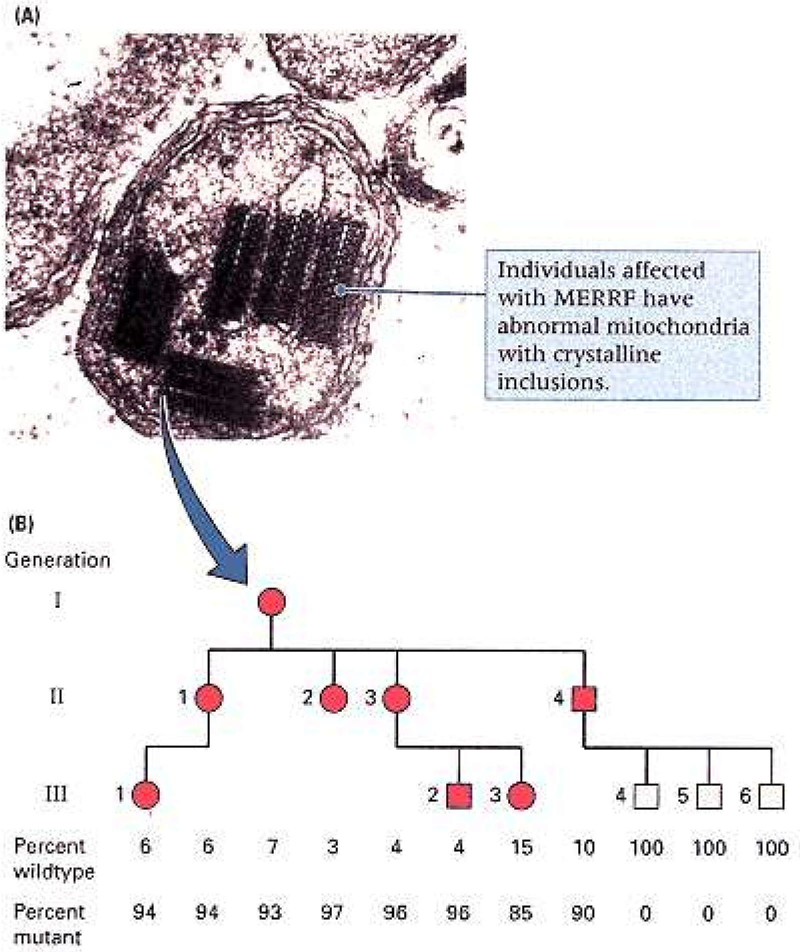


Figure 14.2
Inheritance of myoclonic epilepsy with ragged-red fiber disease (MERFF) in humans. (A) Electron micrograph of an abnormal MERRF mitochondrion containing paracrystalline inclusions. (B) The pedigree shows inhertance of MERFF in one family and the percentage of the mitochondria in ach person found to be wildtype or mutant.
[Micrograph courtesy of D. C. Wallace, from J. M. Shoffner, M. T. Lott, A. M. S. Lezza, P. Seibel, S. W. Ballinger, and D. C. Wallace. 1990. *Cell* 61:931.]

indicated by red symbols in the pedigree manifest some symptoms of the disease, which include hearing loss, seizures, fatigue, dementia, tremors, and jerkiness of movement. The affected individuals in the pedigree vary considerably in the symptoms they express. This variable expressivity is due to differences in the proportions of normal and mutant mitochondria in the cells of affected persons. Those most severely affected have the highest proportion of mutant mitochondria; those least affected have the lowest, as indicated in Figure 14.2B. An electron micrograph of an abnormal mitochondrion from one of the affected patients is shown in Figure 14.2A. It contains numerous paracrystalline inclusion bodies formed from abnormal protein aggregates.

Consideration of the phenotypes caused by the mutations listed in Table 14.1, and the variable expressivity seen in pedigrees of mitochondrial diseases, indicate that mutations in different mitochondrial genes can result in similar phenotypes and that mutations in the same mitochondrial gene can result in different phenotypes. Some of the variation appears to arise from differences among affected persons in the ratio of mutant to wildtype mitochondria, as seen in Figure 14.2B. Another source of variation results from differences in the structural and metabolic consequences of each individual mutation.

Heteroplasmy

The condition in which two or more genetically different types of mitochondria (or other organelle) are present in the same cell is known as **heteroplasmy.** Heteroplasmy of mitochondria is unusual among animals. For example, a typical human cell contains from 1000 to 10,000 mitochondria, and all of them are normally genetically identical.

The rarity of heteroplasmy made it possible to identify the remains of the last Russian Tsar, Nicholas II, who, together with his family and servants, was secretly executed by a Bolshevik firing squad on the night of July 16, 1918. Their bodies were hidden in a shallow grave and remained undisturbed until they were discovered in 1992.

DNA analysis of the exhumed bone samples thought to be from members of the murdered family resulted in identification of nine bodies, five of whom were related. Three were female siblings and another was the mother, the putative Tsarina; all had mitochondrial DNA sequences similar to that of the present Duke of Edinburgh, the maternal grandnephew of the Tsarina. The male relative of the three female siblings was thought to be the father, the Tsar Nicholas. He proved to be heteroplasmic for mitochondrial DNA. At the time of the

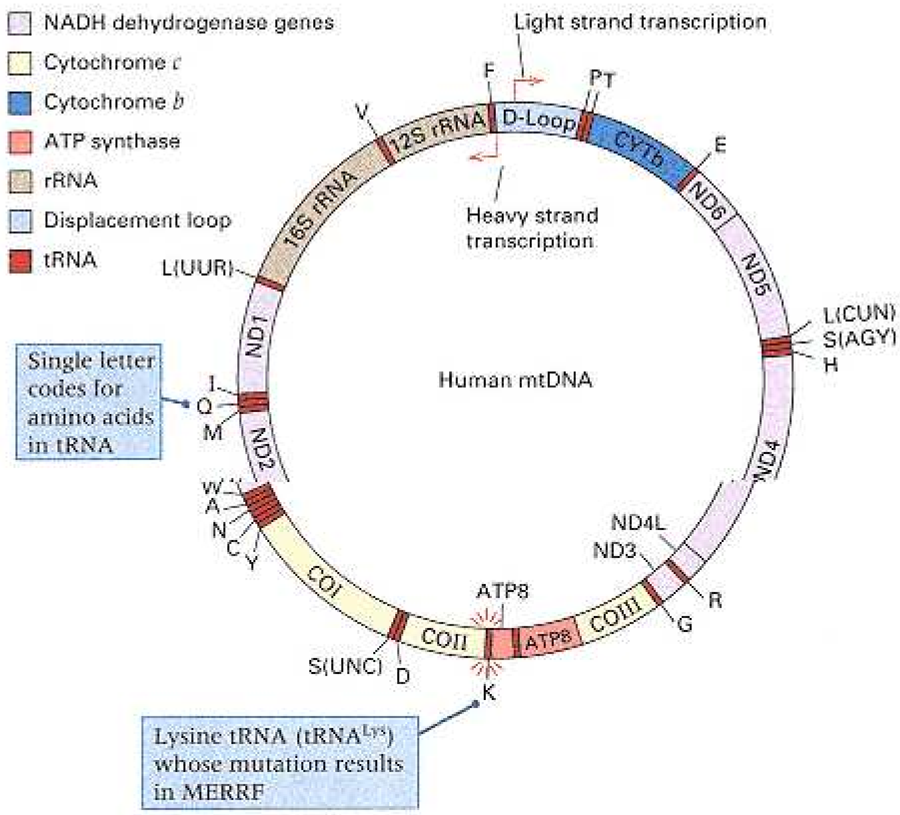


Figure 14.3
Genes in human mitochondrial DNA. The tRNA genes are indicated by the one-letter amino acid symbols; hence $tRNA^{Lys}$ is denoted K. The positions of these and other genes in the mitochondrial DNA are indicated by color according to the key at the upper left. The arrows indicate the promoters for transcription of the heavy (HSP) and light (LSP) strands. [From N-G. Larsson and D. A. Clayton. 1995. *Ann. Rev. Genet*. 29: 151.]

1992 report, no reference DNA sample was available to confirm identification of the Tsar. In 1994, however, the body of the Tsar's brother, Georgij, who had died of tuberculosis in 1899, was exhumed. The mitochondrial DNA sequence of Georgij Romanov not only matched that of the putative Tsar but was also heteroplasmic at the identical base pair, providing powerful evidence supporting identification of the remains as those of the Tsar. The two brothers differed in the ratios of the two mitochondrial types: Georgij Romanov had about 62 percent T and 38 percent C at the heteroplasmic base pair, whereas Nicholas had 28 percent T and 72 percent C. The heteroplasmy is not present in two living members of this matrilineage, who are separated by two or three additional generations from the brothers. This suggests that random segregation of mitochondria within the lineage has finally resulted in homoplasmy.

Maternal Inheritance and Maternal Effects

A maternal pattern of inheritance usually indicates extranuclear inheritance, but not always. The difficulty is in distinguishing between *maternal inheritance* and *maternal effects.* The influence of the mother's nuclear genotype on the phenotype of the progeny is referred to as a **maternal effect.** Such effects may be mediated either through substances present in the egg that affect early development or through the effects of nurture. Examples of

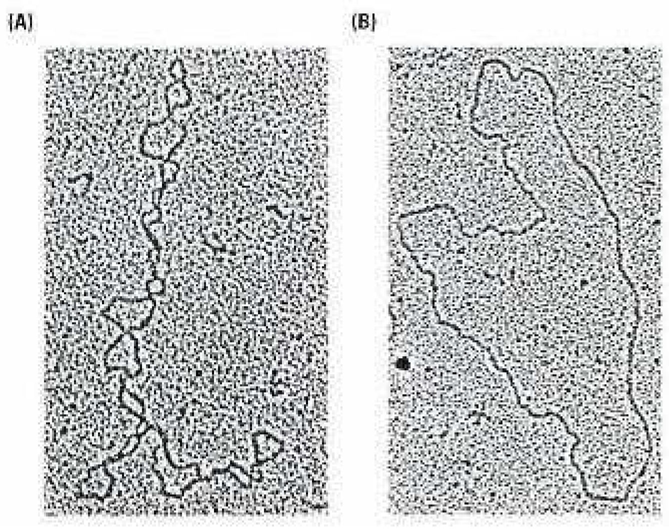


Figure 14.4
DNA from rat liver mitochondria. (A) Native supercoiled configuration. (B) Relaxed configuration produced by a nick in the DNA strand in the process of isolation. The size of each molecule is 16.298 nucleotide pairs. [Courtesy of David Wolstenholme.]

developmental and nurturing effects are, respectively, the intrauterine environment of female mammals and their ability to produce milk. The distinction between maternal inheritance and maternal effects is as follows:

1. In *maternal inheritance*, the hereditary determinants of a trait are extranuclear, and genetic transmission is only through the maternal cytoplasm; Mendelian segregation is not observed.

2. In a *maternal effect*, the nuclear genotype of the mother determines the phenotype of the progeny. The hereditary determinants are nuclear genes transmitted by both sexes, and in suitable crosses the trait undergoes Mendelian segregation.

In higher plants, organelle inheritance is often uniparental, but the particular type of uniparental inheritance depends on the organelle and the organism. For example, among angiosperms, mitochondria typically show maternal inheritance, but chloroplasts may be inherited maternally (the predominant mode), paternally, or from both parents, depending on the species. In conifers, chloroplast DNA is usually inherited paternally, but a few offspring result from maternal transmission. The red-wood *Sequoia sempervirens* shows paternal transmission of both mitochondrial and chloroplast DNA. Whatever the pattern of cytoplasmic transmission, most species have a few exceptional progeny, which indicates that the predominant mode of transmission is usually not absolute.

14.2— Organelle Heredity

As noted earlier, mitochondria are respiratory organelles, and chloroplasts are photosynthetic organelles. The DNA of these organelles is usually in the form of super-coiled circles of double-stranded molecules (Figure 14.4). The chloroplast DNA in most plants ranges in size from 120 to 160 kb. Mitochondrial genomes are usually smaller and have a greater size range. For example, the mitochondrial genome in mammals is about 16.5 kb, that in *D. melanogaster* is about 18.5 kb, and that in some higher plants is more than 100 kb. The mitochondrial genomes of higher plants are exceptional in consisting of two or more circular DNA molecules of different sizes. Several copies of the genome are present in each organelle. There are typically from 2 to 10 copies in mitochondria and from 20 to 100 copies in chloroplasts. In addition, most cells contain multiple copies of each organelle.

Organelle genes code for DNA polymerases that replicate the organelle DNA. They also code for other components essential to their function or replication, but many organelle components are determined by nuclear genes. These components are synthesized in the cytoplasm and transported into the organelle. Mitochondrial DNA contains a relatively small number of genes. For example, both the human mitochondrial genome and that of yeast contain approximately 40 genes. Chloroplast genomes are generally larger than those of mitochondria and contain more genes. Many genes in mitochondria and chloroplasts code for the ribosomal

RNA and transfer RNA components used in protein synthesis (Chapter 10). Figure 14.5 shows the organization of genes in the 120-kb chloroplast genome of the liverwort *Marchantia polymorpha.*

RNA Editing

The mRNAs of mitochondria and chloroplasts often are modified post-transcriptionally in processes called **RNA editing.** The mechanisms of modification and the predominant patterns of modification differ between plants and animals. In most land plants, except algae and mosses, mRNAs may be extensively modified by conversion of cytosine residues to uracils. Because of RNA modification, an organelle gene may carry a CGG codon for arginine, but the mRNA will carry a UGG codon for tryptophan. The modification appears to be accomplished by deamination of selected cytosine residues in the mRNA, leaving uracil residues in their places. In the mitochondria of the trypanosome protozoans that cause sleeping sickness, uridine residues are inserted and/or deleted as templated by a guide RNA in a process that resembles mRNA splicing (Chapter 10). Whatever the mechanism, the outcome is an mRNA whose sequence may differ substantially from that of the DNA from which it was transcribed.

The Genetic Codes of Organelles

Translation in organelles follow the standard decoding process in which the organelle tRNA molecules translate codon by codon along the mRNA. However, the genetic code in mitochondria may differ somewhat from that used to encode proteins in prokaryotes, in archaea, and in nuclear genes of eukaryotes. Human mitochondria depart from the standard code in three principal ways: (1) The UGA codon is not a stop codon but encodes tryptophan, (2) the AGA and AGG codons are stop codons rather than arginine codons, and (3) AUA encodes methionine rather than isoleucine. The genetic code in yeast mitochondria resembles that in humans in that UGA is a tryptophan codon; however, in yeast, AUA is an isoleucine codon, AGA

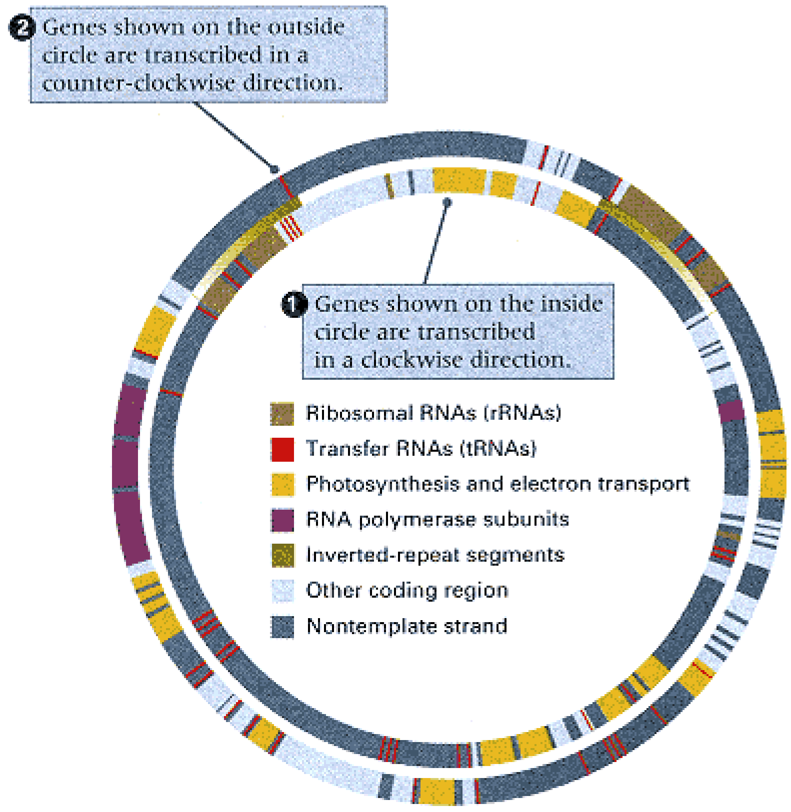


Figure 14.5
Organization of genes in the chloroplast genome of the liverwort *Marchantia polymorpha*. The large inverted-repeat segments (green) that include the genes coding for ribosomal RNA (light brown) are a feature of many other chloropla st genomes. The red lines show the locations of the genes for transfer RNA. Yellow bars are the genes for components of the photosynthetic and electron transport systems; purple bars are the genes for subunits of RNA polymerase. Genes shown inside the circle are transcribed in a clockwise direction; those outside are transcribed counter-clockwise. The length of the entire molecule is 121,024 nucleotide pairs. [Data from K. Ohyama, H. Fukuzawa, T. Kohchi, H. Shirai, T. Sano, S. Sano, K. Umesono, Y. Shiki, M. Takeuchi, Z. Chang, S. Aota, H. Inokuchi, and H. Ozeki. 1986. *Nature* 322: 752.]

and AGG are arginine codons (as in the standard code), and the four codons of the form CUN code for threonine rather than leucine. Although the mitochondrial genetic code may differ from the standard genetic code from one animal species to the next, some are identical to the standard code. On the other hand, plant species employ the standard genetic code in both mitochondria and chloroplasts.

Because the structure and function of mitochondria and chloroplasts depend on both nuclear and organelle genes, it is sometimes difficult to distinguish these contributions. In addition, numerous mitochondria or chloroplasts are present in a zygote, but only one of them may contain a mutant allele. The result is that traits determined by organelle genes exhibit a pattern of determination and transmission very different from simple Mendelian traits determined entirely by nuclear genes.

In the following sections, we briefly consider some examples of traits determined by chloroplast and mitochondrial genes.

Leaf Variegation in Four-O'clock Plants

Chloroplast transmission accounts for the unusual pattern of inheritance observed with leaf variegation in a strain of the four-o'clock plant, *Mirabilis jalapa*. Variegation refers to the appearance, in the leaves and stems of plants, of white regions that result from the lack of green chlorophyll (Figure 14.6). On the variegated plants, some branches are completely green, some completely white, and others variegated. Branches of all three types produce flowers, and these flowers can be used to perform nine possible crosses (Table 14.2). From each cross, the seeds are collected and planted, and the phenotypes of the progeny are examined. The results of the crosses are summarized in Table 14.2. Two significant observations are that

1. Reciprocal crosses yield different results. For example, the cross

green ♀ × white ♂

yields green plants, whereas the cross

white ♀ × green ♂

yields white plants (which usually die shortly after germination because they lack chlorophyll and hence photosynthetic activity).

2. The phenotype of the branch bearing the female parent in each case determines the phenotype of the progeny (compare columns 1 and 3 in the table). The male makes no contribution to the phenotype of the progeny.

Furthermore, when flowers present on either the green or the variegated progeny plants are used in subsequent crosses, the

Figure 14.6
Leaf variegation in Mirabilis jalapa. The sectors result from cytoplasmic segregation of normal (green) and defective (white) chloroplasts.

patterns of transmission are identical with those of the original crosses. These observations suggest direct transmission of the trait through the mother, or maternal inheritance.

The genetic explanation for the variegation is as follows:

1. Green color depends on the presence of chloroplasts, and pollen contains no chloroplasts.

2. Segregation of chloroplasts into daughter cells is determined by cytoplasmic division and is somewhat irregular.

3. The cells of white tissue contain mutant chloroplasts that lack chlorophyll.

These points enable us to draw the model shown in Figure 14.7. Variegated plants germinate from embryos that are

Table 14.2 Crosses and progeny phenotypes in variegated four-o'clock plants

Phenotype of branch bearing egg parent	Phenotype of branch bearing pollen parent	Phenotype of progeny
white	white	white
white	green	white
white	variegated	white
green	white	green
green	green	green
green	variegated	green
variegated	white	variegated, green, or white
variegated	green	variegated, green, or white
variegated	variegated	variegated, green, or white

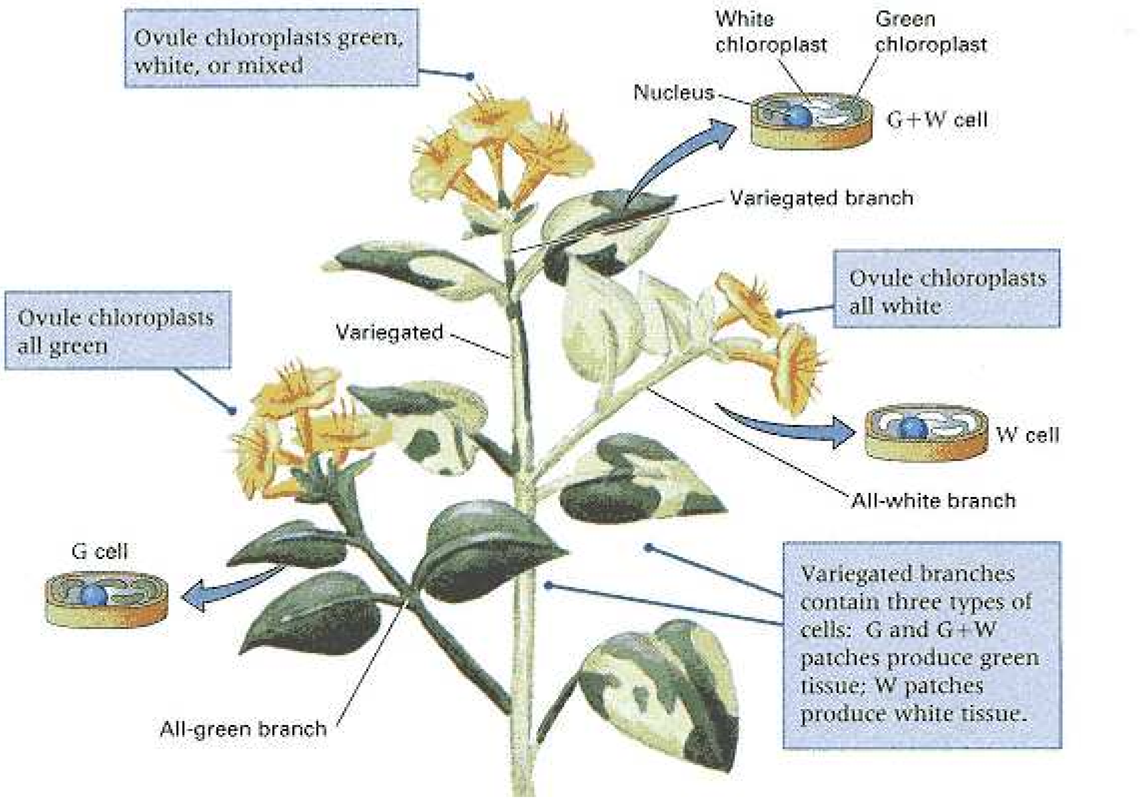


Figure 14.7
Genetic model for leaf variegation. Branches that are all green, all white, and variegated form on the same plant. Flowers form on all three kinds of branches. The insets show the chloroplast composition of cells in each type of branch. Cells in all-green or all-white branches contain only green

or white chloroplasts, respectively, whereas cells in variegated branches are heteroplasmic.

Connection Chlamydomonas Moment

Ruth Sager and Zenta Ramanis 1965

Columbia University,
New York, New York
Recombination of Nonchromosomal Genes in Chlamydomonas

Before the widespread use of polymorphic DNA markers in mitochondria and chloroplasts, the study of extranuclear inheritance was hindered in most organisms not only by uniparental inheritance of cellular organelles but also by the lack of mutant phenotypes showing extranuclear transmission. The alga Chlamydomonas proved to be suitable for studies of extranuclear inheritance. Mutants showing nonchromosomal inheritance could be obtained, and strains were discovered in which the inheritance of nonchromosomal genes was biparental, making it possible to examine segregation of nonchromosomal alleles. The compelling evidence for the existence of genetic systems apart from genes in the nucleus was the discovery of intragenic recombination in nonchromosomal genes. In the matings described here, the parental genotypes for the nonchromosomal genes were ac_1 sd *(slow growth on acetate, streptomycin requiring) and* ac_2 sr *(acetate requiring, streptomycin resistant). The intragenic recombinant products were either wildtype* ac^+ *(acetate independent) versus the double mutant* ac_1 ac_2 *or they were wildtype* ss *(streptomycin sensitive) versus the double mutant sd sr. The authors leave no doubt about their opinion (which turned out to be correct) that the nonchromosomal genetic systems are due to DNA in the cellular organelles.*

The analysis of nonchromosomal heredity has made slow progress, often under severe attack, [in part because of] the difficulty in obtaining mutations of nonchromosomal genes. . . . Within the past few years, a concerted attack has been made on this problem with the alga *Chlamydomonas.* . . . Two findings have been of key importance in our investigation. First, streptomycin was developed as a mutagen for nonchromosomal (NC) genes. . . . Second, the existence of an exceptional class of zygotes was dis-

We view our results as providing evidence of the nucleic acid nature of a nonchromosomal genetic system.

covered that transmits NC genes from both parents to the progeny, in contrast to the standard pattern of maternal inheritance. With this material it was established that the NC genes are particulate in nature, genetically autonomous, and stable. . . . In this paper we describe what is to our knowledge the first evidence of recombination of nonchromosomal genes. . . . The occurrence of linked recombination demonstrates that NC genes, like the chromosomal ones, are capable of close pairing and precise reciprocal exchange. . . . The salient features of our results are the following: (1) Segregation of NC genes did not occur during meiosis. . . . (2) NC gene segregation began at the first or second mitotic division after meiosis, with pure clones arising so long as any heterozygous cells remained. (3) The average segregation ratio is 1 : 1. although the progeny of individual zygotes gave ratios significantly different from 1 : 1. (4) The acetate alleles segregated independently of the streptomycin alleles. (5) Novel classes of progeny were recovered that were phenotypically indistinguishable from wildtype ac^+ and from wildtype *ss.* New mutant phenotypes were also observed with new levels of acetate requirement and new levels of streptomycin resistance and dependence. [These were the reciprocal, double-mutant types.] . . . From existing knowledge, we infer that intragenic recombination, in which the wildtype gene is reconstituted, requires a close intermolecular pairing and exchange of the sort only known to occur between nucleic acids. Consequently, we view our results as providing evidence of the nucleic acid nature of a nonchromosomal genetic system. These genetic findings have been paralleled by the discovery of DNA of high molecular weight in chloroplasts and mitochondria. It is a reasonable surmise that these organelle DNA's carry primary genetic information.

Source: Proceedings of the National Academy of Sciences of the USA 53: 1053–1061 heteroplasmic for green and white (mutant) chloroplasts. Random segregation of the chloroplasts in cell division results in some branches derived from cells that contain only green chloroplasts and some branches derived from cells that contain only mutant chloroplasts. The flowers on green branches produce ovules with normal chloroplasts, so all progeny are green. Flowers on white branches produce ovules with only mutant chloroplasts, so only white progeny are formed. The ovules formed by flowers on variegated branches are of three types: those with only chlorophyll-containing chloroplasts, those with only mutant chloroplasts, and those

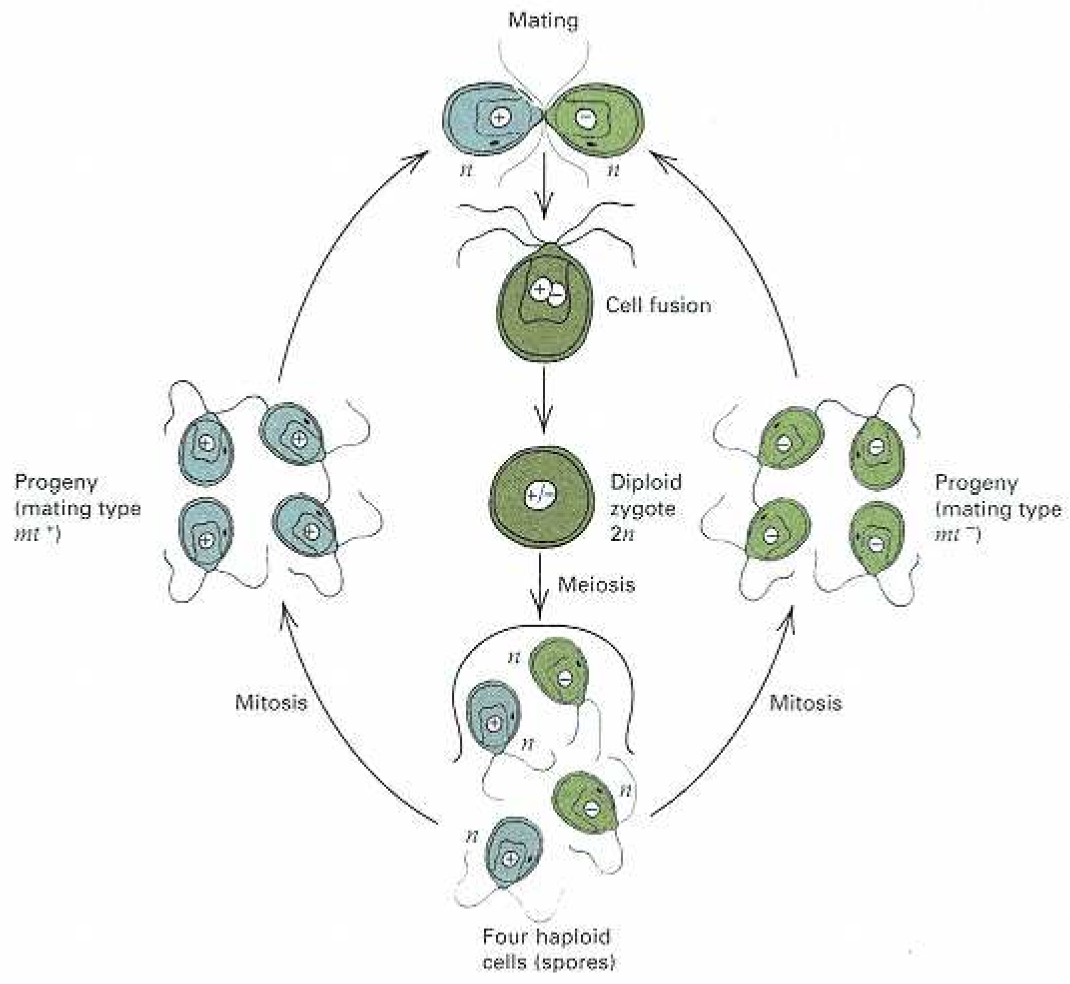


Figure 14.8
Life cycle of Chlamydomonas reinhardtii.

that are heteroplasmic. The heteroplasmic class again yields variegated plants.

Drug Resistance in Chlamydomonas

Cytoplasmic inheritance has been studied extensively in the unicellular green alga *Chlamydomonas.* Cells of this organism have a single large chloroplast that contains from 75 to 80 copies of a double-stranded DNA molecule approximately 195 kb in size. The life cycle of *Chlamydomonas* is shown in Figure 14.8. There are two mating types, *mt*$^{+}$ and *mt*$^{-}$. These cells are of equal size and appear to contribute the same amount of cytoplasm to the zygote. Cells of opposite mating type fuse to form a diploid zygote, which undergoes meiosis to form a tetrad of four haploid cells. The cells can be grown on solid growth medium to form visible clusters of cells (colonies) that can indicate the genotype. Reciprocal crosses between certain antibiotic-resistant strains (mutant) and antibiotic-sensitive strains (wildtype) yield different results. For example, for streptomycin resistance (*str-r*),

str-r mt^{+} × *str-s mt*$^{-}$ yields only
str-r progeny

str-s mt^{+} × *str-r mt*$^{-}$ yields only
str-s progeny

This is a clear example of uniparental inheritance of the streptomycin resistance or sensitivity, because the resistance or sensitivity comes from the *mt*$^+$ parent. In contrast, the *mt* alleles behave in strictly Mendelian fashion, yielding 1 : 1 progeny ratios, as expected of nuclear genes.

A large number of antibiotic-resistance markers with uniparental inheritance have been examined in *Chlamydomonas.* In each case, the antibiotic resistance has been traced to the chloroplast. Haploid *Chlamydomonas* contains only one chloroplast. If the chloroplast could be derived from the *mt*$^+$ or *mt*$^-$ cell with equal probability, then uniparental inheritance would not be observed because the chloroplast could come from either parent. However, studies in which DNA markers are used to distinguish the chloroplasts of the two mating types have shown that the chloroplast of the *mt*$^-$ parent is preferentially lost after mating, which accounts for the fact that the antibiotic-resistance phenotype of the *mt*$^+$ parent is the one transmitted to the progeny.

About 5 percent of the progeny from *Chlamydomonas* crosses are heteroplasmic. Both parental chloroplasts are retained, although they ultimately segregate in later cell divisions. The heteroplasmic cells have been valuable for genetic analysis, because recombination can take place between the DNA in the two chloroplasts. Analysis of the recombination frequencies for the various phenotypes has made possible the construction of fairly detailed genetic maps of the chloroplast genome.

An interesting inference can be made from the study of *Chlamydomonas* mutants that are resistant to myxothiazol and mucidin, which are known to be inhibitors of the cytochrome *b* of mitochondria. The mutations that confer resistance are therefore in the mitochondrial DNA. As before, reciprocal crosses can be made to demonstrate the pattern of inheritance of the resistance trait. When crosses are made between antibiotic-resistant (*MUD2*) and antibiotic-sensitive (*mud2*) strains (the *mud2* strains are wildtype), the results are as follows:

MUD2 mt$^+$ × *mud2 mt*$^-$ yields only
mud2 (sensitive) progeny

mud2 mt$^+$ × *MUD2 mt*$^-$ yields only
MUD2 (resistant) progeny

In each cross, the tetrads yield 2 *mt*$^+$: *2 mt*$^-$ spores. However, all spore progeny have the resistance phenotype of the *mt*$^-$ parent. Inheritance of resistance to mucidin or myxothiazol therefore shows uniparental inheritance, because the phenotype is the same as that of the *mt*$^-$ parent. Resistance to myxothiazol or mucidin results from a change in the mitochondrial DNA, so we can also infer that the meiotic progeny must receive their mitochondrial DNA from the *mt*$^-$ parent, because the resistance comes from this parent. In contrast, as indicated by the streptomycin example above, they receive their chloroplast DNA from the *mt*$^+$ parent. It is presently unknown whether there is any biological significance in the fact that in *Chlamydomonas,* meiotic progeny inherit mitochondria from one parent and chloroplasts from the other parent, or whether the patterns are merely a matter of chance evolutionary accident.

Respiration-Defective Mitochondrial Mutants

Very small colonies of the yeast *Saccharomyces cerevisiae* are occasionally observed when the cells are grown on solid medium that contains glucose. These are called **petite mutants.** Microscopic examination indicates that the cells are of normal size. Physiological studies show that the cells grow normally during early stages of colony formation when they are fermenting the glucose to ethanol but cease to grow thereafter because of a defect in oxygen-requiring respiration that is needed for further metabolism of the ethanol. The colonies are quite small, because the yield of ATP is much higher during respiration than during fermentation. There are several types of petite mutants. We will discuss two of them.

Genetic analysis of crosses between petites and wildtype distinguishes the two major types of petites illustrated in Figure 14.9. One type, called **segregational petites,** exhibits the typical segregation of a simple Mendelian recessive. In a cross between a segregational petite and

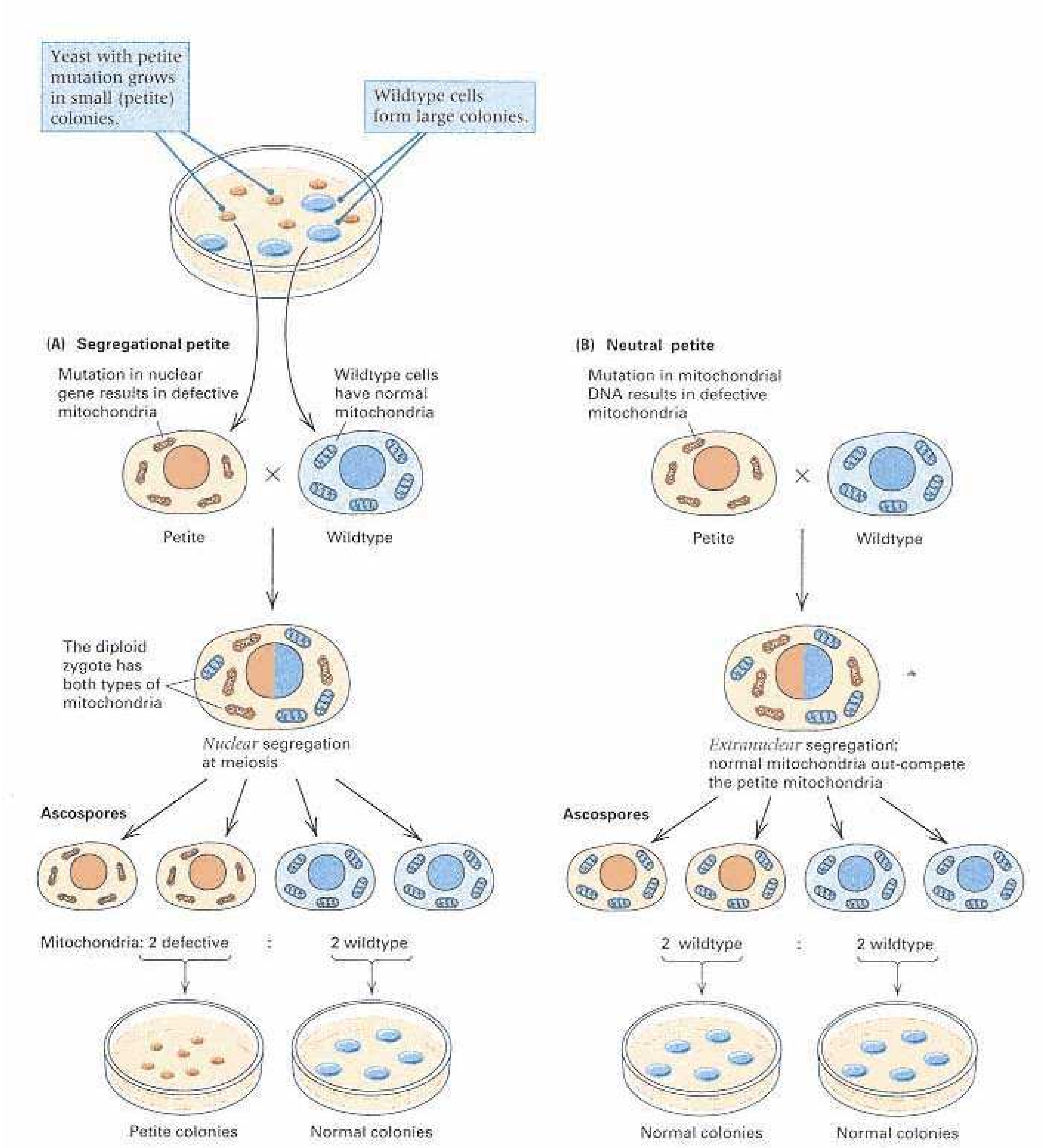


Figure 14.9
Behavior of petite mutations in genetic crosses. The brown circles represent petite cells. The brown nucleus in part A contains an allele resulting in petite colonies. The blue cells have normal mitochondria.

wildtype, the diploid zygotes are normal, and when the diploids undergo meiosis, half of the ascospores in an ascus produce petite colonies and half form wildtype colonies. The 1 : 1 ratio (actually 2 : 2) indicates that these petites are the result of a nuclear mutation and that the allele for petite is recessive.

The second type, **neutral petites,** is quite different: In a cross with wildtype, all diploid zygotes produce wildtype ascospores (4 : 0 segregation). The same pattern is found when the progeny from such a cross are backcrossed with neutral petites. In both cases, the phenotype that is inherited is that of the normal parent, so the inheritance is uniparental. The explanation for these results is that the majority of neutral petites lack most or all mitochondrial DNA, in which many of the genes determining oxidative respiration are encoded. When a neutral-petite cell mates with a wildtype cell, the cytoplasm of the latter is the source of the normal mitochondrial DNA in the resulting progeny spores.

Petites arise at a frequency of roughly 10^{-5} per generation. The neutral petite phenotype is the result of large deletions in mitochondrial DNA. For unknown reasons, yeast mitochondria often fuse and fragment, which may cause occasional deletions of DNA. The production of neutral petites is apparently a result of the segregation of aberrant mitochondrial DNA from normal DNA and the further sorting out of mutant mitochondria in the course of cell division.

Cytoplasmic Male Sterility in Plants

An example of extranuclear inheritance that is important in agriculture is **cytoplasmic male sterility** in plants, a condition in which the plant does not produce functional pollen, but the female reproductive organs and fertility are normal. This type of sterility is used extensively in the production of hybrid corn seed, because it circumvents the need for manual "detasseling" of the plants to be used as females to produce the hybrid. Because the tassel is the pollen-bearing organ, the detasseled female plants cannot produce pollen and so cannot fertilize themselves. The detasseled female plants must therefore use pollen from the still-tasseled male plants used to produce the hybrid.

The genetically transmitted male sterility is not controlled by nuclear genes but is transmitted through the egg cytoplasm from generation to generation. The pattern of inheritance of cytoplasmic male sterility in corn is summarized in Figure 14.10. The key observation is that repeated backcrossing of a male-sterile variety, using pollen from a male-fertile variety, does not restore male fertility. The resulting plants, in which nearly all nuclear genes from the male-sterile variety have been replaced with those from the male-fertile variety, remain male-sterile. Because a small amount of functional pollen is produced by the male-sterile variety, it is also possible to carry out the reciprocal cross:

male-fertile ♀ × male-sterile ♂

In this case, the progeny are fully male-fertile. The difference between the reciprocal crosses means that male sterility or fertility in these varieties is maternally inherited.

Cytoplasmic male sterility in maize results from rearrangements in mitochondrial DNA. For example, one DNA rearrangement fuses two mitochondrial genes and creates a novel protein that causes male sterility. Although cytoplasmic male sterility is maternally inherited, certain nuclear genes called **fertility restorers** can suppress the male-sterilizing effect of the cytoplasm. Different types of cytoplasmic male sterility can be classified on the basis of their response to restorer genes. For example, in one case, restoration of fertility requires the presence of dominant alleles of two different nuclear genes, and plants with male-sterile cytoplasm and both restorer alleles produce normal pollen. In another case, suppression requires the dominant allele of a different restorer gene, but this gene acts only in the gametophyte; hence plants with male-sterile cytoplasm that are heterozygous for the restorer allele produce a 1 : 1 ratio of normal to aborted pollen grains.

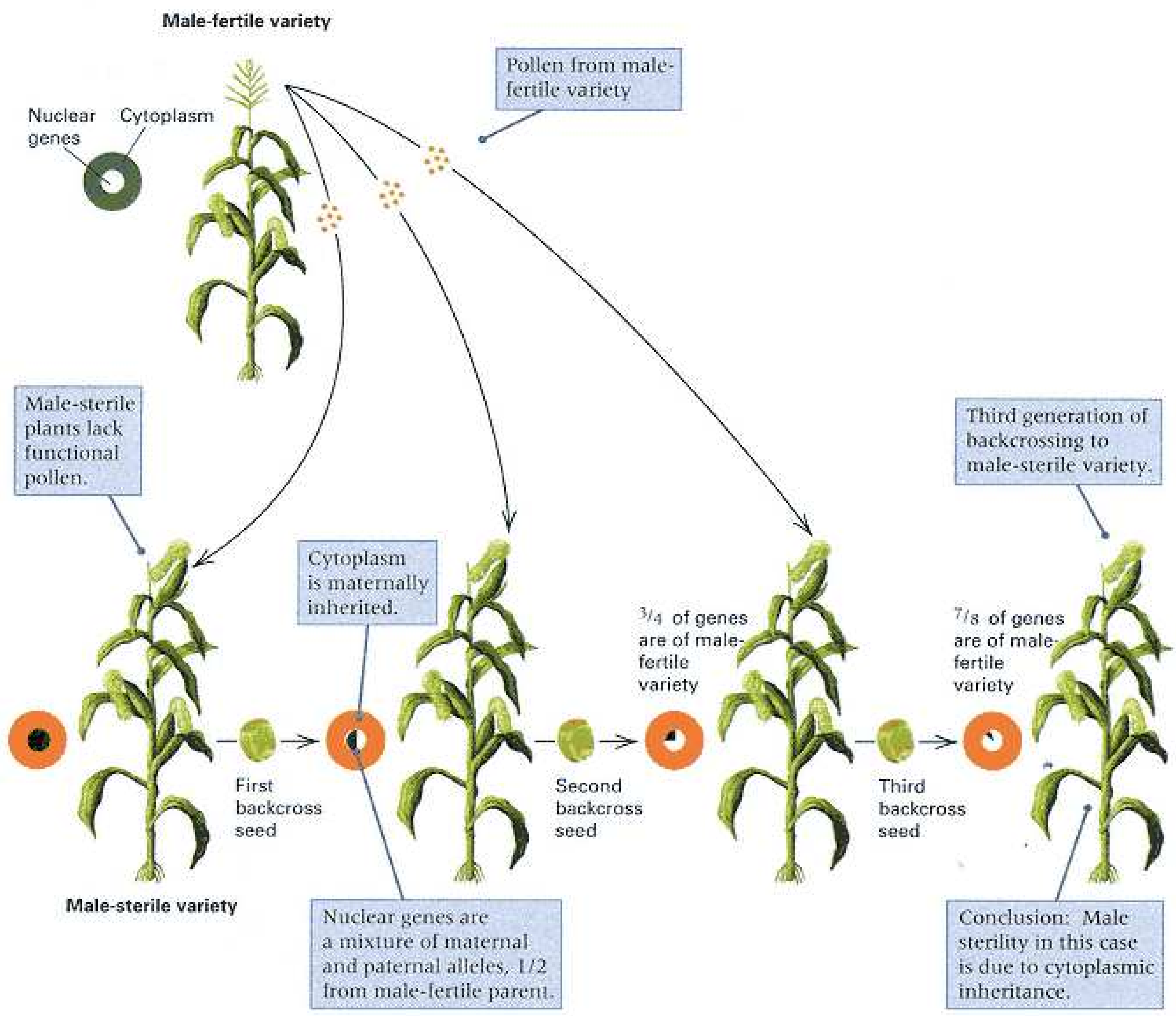


Figure 14.10
Diagram of an experiment demonstrating maternal inheritance of male sterility (orange cytoplasm) in maize.

14.3—
The Evolutionary Origin of Organelles

A widely accepted theory holds that organelles originally evolved from prokaryotic cells that lived inside primitive eukaryotic cells in symbiosis, a mutually beneficial interaction. This is the **endosymbiont theory.** Mitochondria and chloroplasts are thought to have originated from bacteria and cyanobacteria (formerly called blue-green algae), respectively. The archaea, which comprise the third major type of life form (the other two are eubacteria and eukaryotes), are good candidates for the primitive hosts of endosymbionts that ultimately evolved into eukaryotes, including animals, plants, and fungi. They share some with eubacteria, including their prokaryotic cytological organization. However, in their macromolecular syntheses—replication, transcription, and translation—the archaea resemble the eukaryotes.

One line of evidence for the endosymbiont theory is that mitochondrial and chloroplast organelles of eukaryotes share numerous features with prokaryotes that distinguish them from eukaryotic cells. Among these common features:

1. The genomes are composed of circular DNA that is not extensively complexed with histone-like proteins.

2. The genomes are organized with functionally related genes close together and often expressed as a single unit.

3. The ribosome particles on which protein synthesis takes place have major subunits whose size is similar in organelles and in prokaryotes but differs from that in cytoplasmic ribosomes in eukaryotic cells.

4. The nucleotide sequences of the key RNA constituents of ribosomes are similar in chloroplasts, cyanobacteria such as *Anacystis nidulans,* and even bacteria such as *E. coli.*

The genomes of today's organelles are small compared with those of bacteria and cyanobacteria. For example, chloroplast genomes are only from 3 to 5 percent as large as the genomes of cyanobacteria. Organelle evolution was probably accompanied by major genome rearrangements, some genes being transferred into the nuclear genome of the host and others being eliminated. The transfer of organelle genes into the nucleus differed from one lineage to the next. For example, the gene for one subunit of the mitochondrial ATP synthase is located in the mitochondrial genome in yeast but in the nuclear genome in *Neurospora.*

14.4— The Cytoplasmic Transmission of Symbionts

In eukaryotes, a variety of cytoplasmically transmitted traits result from the presence of bacteria and viruses living in the cytoplasm of certain cells. One of the best known is the killer phenomenon found in certain strains of the protozoan *Paramecium aurelia.* Killer strains of *Paramecium* release to the surrounding medium a substance that is lethal to many other strains of the protozoan. The killer phenotype requires the presence of cytoplasmic bacteria referred to as **kappa particles,** whose maintenance is dependent on a dominant nuclear gene, *K.* The kappa particles are cells of *Caedobacter taeniospiralis,* which produce the killer substance. Why killer strains are immune to the substance has not yet been determined.

Paramecium is a diploid protozoan that undergoes sexual exchange through a mating process called **conjugation** (Figure 14.11A). Initially, each cell has two diploid micronuclei. Many protozoans contain micronuclei (small nuclei) and a macronucleus (a large nucleus with specialized function), but only the micronuclei are relevant to the genetic processes described here. When two cells come into contact for conjugation, the two micronuclei in each cell undergo meiosis, forming eight micronuclei in each cell. Seven of the micronuclei and the macronucleus disintegrate, leaving each cell with one micronucleus, which then undergoes a mitotic division. The cell membrane between the two cells breaks down slightly, and the cells exchange a single micronucleus; then the nuclei fuse to form a diploid nucleus. After nuclear exchange, the two new cells (the exconjugants) are genotypically identical. In this sequence of events, only the micronuclei are exchanged, without any mixing of cytoplasm.

Single cells of *Paramecium* also occasionally undergo an unusual nuclear phenomenon called **autogamy** (Figure 14.11B). Meiosis takes place and again seven of the micronuclei disintegrate. The surviving nucleus then undergoes mitosis and nuclear fusion to recreate the diploid state.

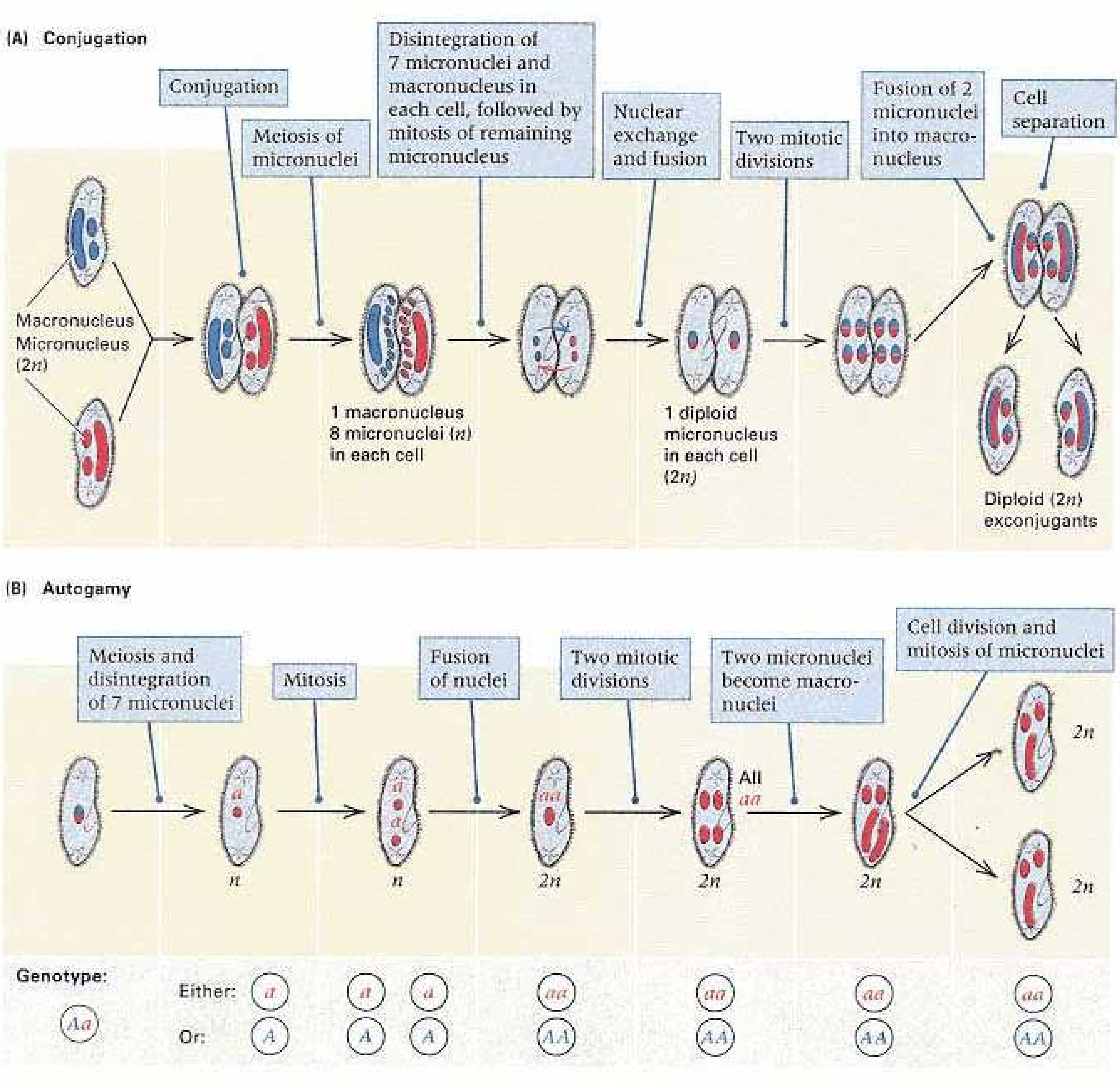


Figure 14.11
(A) Conjugation in *Paramecium* results in reciprocal fertilization and the formation of two genetically identical exconjugant cells. (B) Autogamy in *Paramecium*. The alleles in the heterozygous *Aa* cell segregate in meiosis, with the result that the progeny cells become homozygous for either *A* or *a*. The production of *aa* progeny cells is shown in the diagram. The initial steps resemble those in conjugation in part A: The micronuclei undergo meiosis, after which seven of the products and the macronucleus degenerate.

The genetic importance of autogamy is that even if the initial cell is heterozygous, the newly formed diploid nucleus becomes homozygous because it is derived from a single haploid meiotic product. In a population of cells undergoing autogamy, the surviving micronucleus is selected randomly, so for any pair of alleles, half of the new cells contain one allele and half contain the other.

Conjugation usually allows an exchange *only* of micronuclei at the surface of

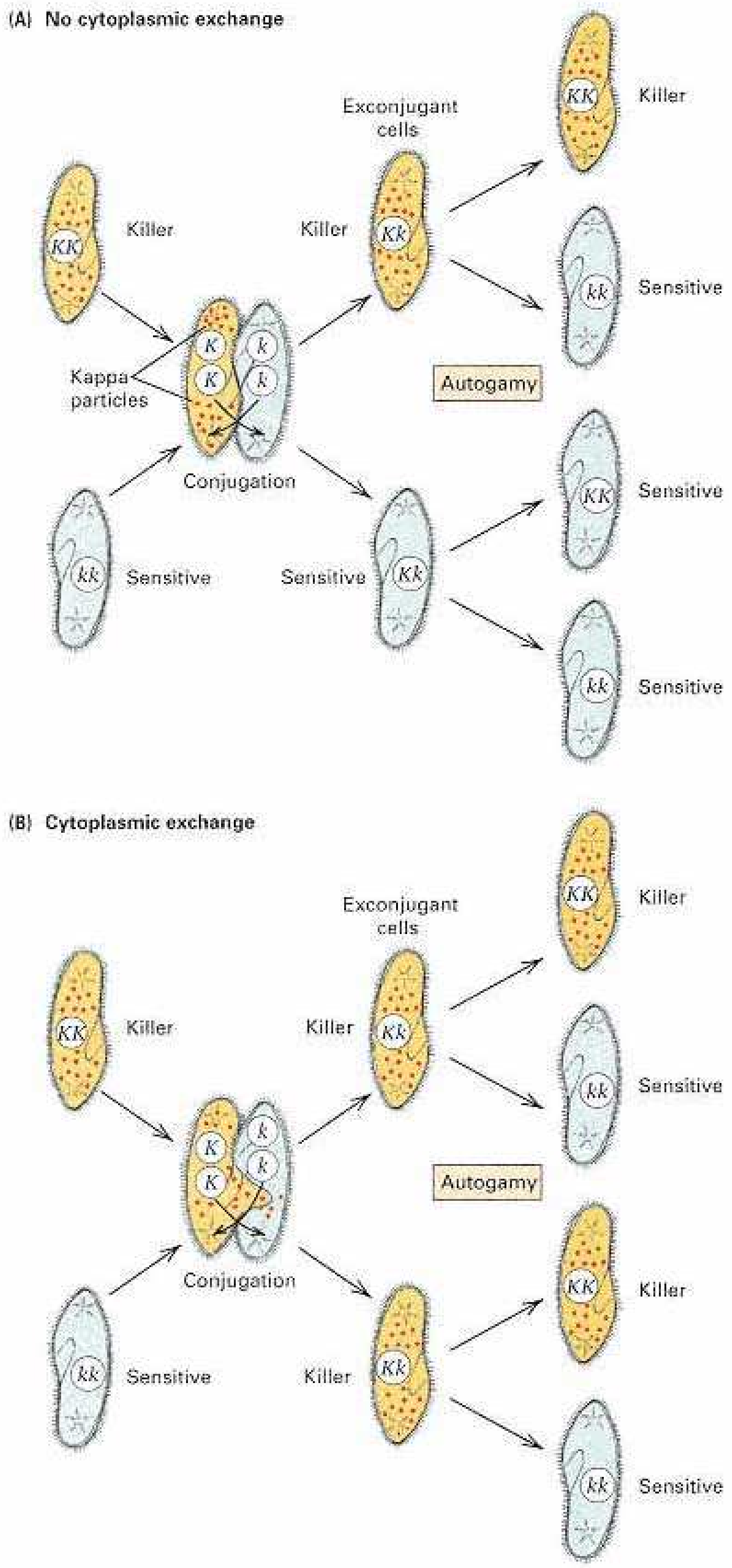


Figure 14.12
Crosses between killer (*KK*) and sensitive (*kk*) *Paramecium*. (A) The result when no cytoplasmic exchange takes place in conjugation. (B) The result when cytoplasmic exchange occurs. Cells containing kappa are shown in yellow. The genotype and phenotype of each cell are indicated.

contact between the two cells. Occasionally, small amounts of cytoplasm also are exchanged. A comparison of the phenotypes following conjugation of killer (*KK*) cells and sensitive (*kk*) cells, with and without cytoplasmic exchange, yields evidence for cytoplasmic inheritance of the killer phenotype (Figure 14.12). Without cytoplasmic mixing (Figure 14.12A), the expected 1 : 1 ratio of killer to sensitive exconjugant cells is observed; with cytoplasmic mixing (Figure 14.12B), both exconjugants are killer cells because each cell has kappa particles derived from the cytoplasm of the killer parent. Conjugation is eventually followed by autogamy of each of the exconjugants (Figure 14.12). Autogamy of *Kk* killer cells results in an equal number of *KK* and *kk* cells, but the *kk* cells cannot maintain the kappa particle and so become sensitive.

Another example of a symbiont-related cytoplasmic effect is a condition in *Drosophila* called **maternal sex ratio,** which is characterized by the production of almost no male progeny. The daughters of sex-ratio females pass on the trait, whereas the occasional sons that are produced do not. Cytoplasm taken from the eggs of sex-ratio females transmits the condition when injected into the female embryos from unaffected cultures of the same or other *Drosophila* species. A species of bacteria has been isolated from the cytoplasm of sex-ratio females; when these bacteria are allowed to infect other *Drosophila* females, these acquire the sex-ratio trait. The causative agent of the sex-ratio condition is not the bacteria themselves but rather a virus that multiplies in them. When released by the bacterial cells, this virus kills most male *Drosophila* embryos. Why female embryos are not killed by the virus is still a mystery.

14.5— Maternal Effect in Snail-Shell Coiling

There are also some examples in which maternal transmission takes place, but the maternal transmission is not mediated by cytoplasmic organelles or symbionts but rather through the cytoplasmic transmission of the products of nuclear genes. An example is the determination of the direction of coiling of the shell of the snail *Limnaea peregra*. The direction of coiling, as viewed by looking into the opening of the shell, may be either to the right (dextral coiling) or to the left (sinistral coiling). Reciprocal crosses between homozygous strains give the following results:

dextral ♀ × sinistral ♂ → all F_1 dextral
sinistral ♀ × dextral ♂ → all F_1 sinistral

In these crosses, the F_1 snails have the same genotype, but the reciprocal crosses give different results in that the direction of coiling is the same as that of the mother. This result is typical of traits with maternal inheritance. However, in this case, all F_2 progeny from both crosses exhibit dextral coiling, a result that is inconsistent with cytoplasmic inheritance. The F_3 generation provides the explanation.

The F_3 generation, obtained by self-fertilization of the F_2 snails (the snail is hermaphroditic), indicates that the inheritance of coiling direction depends on nuclear genes rather than extranuclear factors. Three-fourths of the F_3 progeny have dextrally coiled shells and one-fourth have sinistrally coiled shells (Figure 14.13). This is a typical 3 : 1 ratio, which indicates Mendelian segregation in the F_2. Dextral coiling (+/+ and *s*/+) is dominant to sinistral coiling (*s*/*s*). The reason why reciprocal crosses give different results, and why the production of the 3 : 1 Mendelian ratio is delayed for a generation, is as follows:

> The coiling phenotype of an individual snail is determined by the genotype of its mother.

This means that the direction of shell coiling is not a case of maternal inheritance but

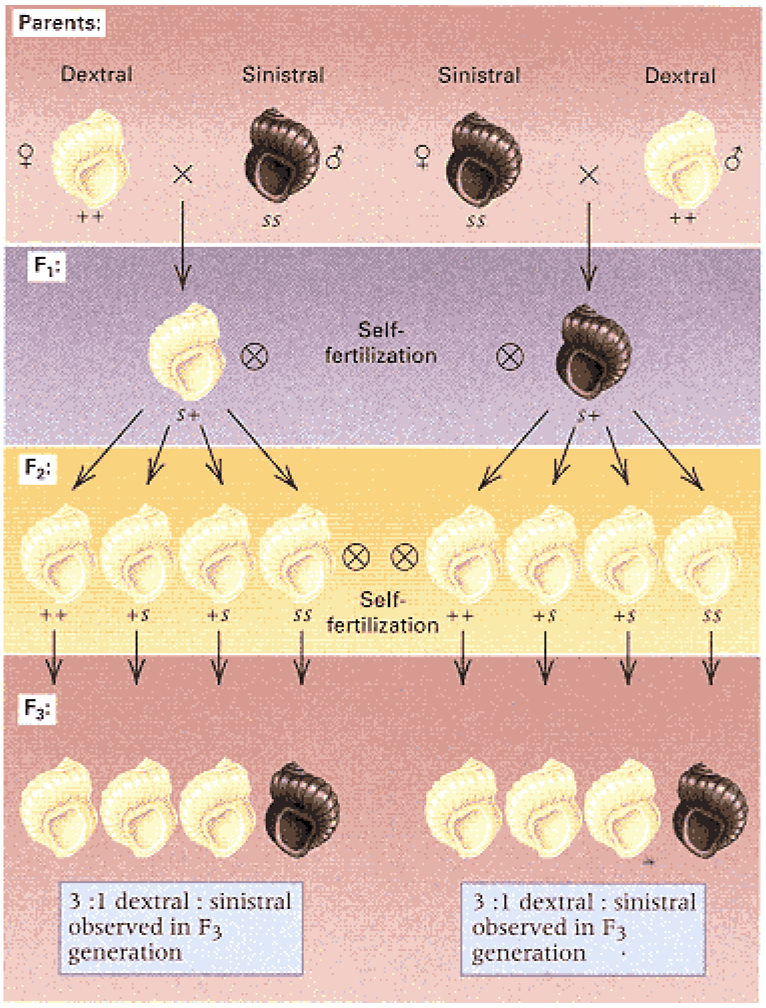


Figure 14.13
Inheritance of the direction of shell coiling in the snail *Limnaea*. Sinistral coiling is determined by the recessive allele *s*, dextral coiling by the wild-type + allele. However, the direction of coiling is determined by the nuclear genotype of the mother, not by the genotype of the snail itself or by extranuclear inheritance. The F_2 and F_3 generations can be obtained by self-fertilization because the snail is hermaphroditic and can undergo either self-fertilization or cross-fertilization.

rather a case of maternal effect (Section 14.1). Cytological analysis of developing eggs has provided the explanation: The genotype of the mother determines the orientation of the spindle in the initial mitotic division after fertilization, and this in turn controls the direction of shell coiling of the offspring (Figure 14.14). This classic example of maternal effect indicates that more than a single generation of crosses is needed to provide conclusive evidence of extranuclear inheritance of a trait.

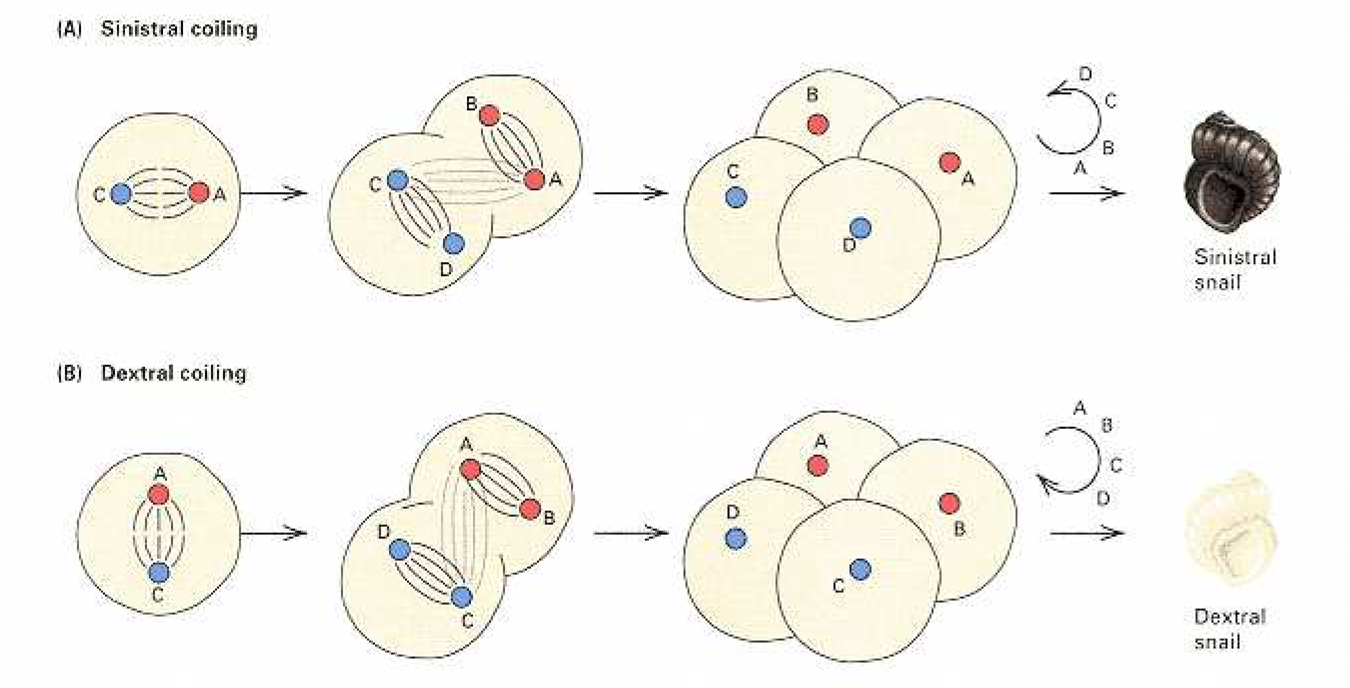


Figure 14.14
The direction of shell coiling in *Limnaea* is determined by the orientation of the mitotic spindles during the second cleavage division in the zygote. The orientation is predetermined in the egg cytoplasm as a result of the mother's genotype. The spiral patterns established by the cleavage result in (A) sinistral (leftward) or (B) dextral (rightward) coiling.

14.6— In Search of Mitochondrial "Eve"

Mitochondrial DNA has a number of features that make it useful for studying the genetic relationships among organisms in natural populations, including human populations. First, it is maternally inherited and does not undergo genetic recombination, and hence the DNA molecule in any mitochondrion derives from a single ancestral molecule. Second, mammalian mitochondrial DNA evolves considerably faster than that of nuclear genes; for example, differences in mitochondrial DNA sequences accumulate among human lineages at a rate of approximately 1 change per mitochondrial lineage every 1500 to 3000 years.

To put this rate of evolution into per-spective, consider the people of Australia and New Guinea, who have been relatively isolated genetically from each other since an aboriginal people colonized these areas approximately 30,000 years ago (Australia) and 40,000 years ago (New Guinea). The total time of separation between the populations is therefore 70,000 years (30,000 years in Australia and 40,000 years in New Guinea). If one nucleotide change accumulates every 1500 to 3000 years, the total number of differences in the mitochondrial DNA of Australian aborigines and New Guineans is expected to range from 23 nucleotides (calculated as 70,000/3000) to 47 nucleotides (calculated as 70,000/1500). This example shows how the rate of mitochondrial DNA evolution can be used to predict the number of differences between populations. In practice, the calculation is done the other way around, and the observed number of differences between pairs of populations is used to estimate the rate of mitochondrial evolution.

Nucleotide differences in mitochondrial DNA have been used to reconstruct the probable historical relationships among human populations. Figure 14.15 is an example that includes people from five native populations: African, Asian, Australian, Caucasian (European), and

CONNECTION A Coming Together

Lynn Margulis (a.k.a. Lynn Sagan) 1967

Boston University, Boston,
Massachusetts
The Origin of Mitosing Cells

By the 1960s certain features of the genetic apparatus of cellular organelles were already known to resemble those in prokaryotes. A number of scientists had suggested that the similarities may reflect an ancient ancestral relationship. Lynn Margulis took these ideas a giant step further by proposing a comprehensive hypothesis of the origin of the eukaryotic cell as a series of evolved symbiotic relationships. The main players are the mitochondria, the kinetosomes (basal bodies) of the flagella and cilia, and the photosynthetic plastids. All are hypothesized to have originated as prokaryotic symbionts. When the paper was submitted for publication, the anonymous reviewers were so strongly negative that it was almost rejected. Through the years, specific predictions of the hypothesis have been experimentally verified. A few revisions have been necessary; for example, current evidence suggests that the earliest symbiont was kinetosome-bearing, not a protomitochondrion. However, no major inconsistencies have been uncovered. Today the serial endosymbiont theory is widely accepted.

This paper presents a theory of the origin of the discontinuity between eukaryotic and prokaryotic cells. Specifically, the mitochondria, the basal bodies [kinetosomes] of the flagella, and the photosynthetic plastids can be considered to have derived from free-living cells, and the eukaryotic cell is the result of the evolution of ancient symbioses. Although these ideas are not new, in this paper they have been synthesized in such a way as to be consistent with recent data on the biochemistry and cytology of subcellular organelles. . . . Many aspects of this theory are verifiable by modern techniques of molecular biology. . . . Prokaryotic cells containing DNA, synthesizing protein or ribosomes, and using messenger RNA as intermediate between DNA and protein, are ancestral to all extant cellular life. Such cells arose under reducing conditions of the primitive terrestrial atmosphere. . . . Eventually, a population of cells arose using photoproduced ATP with water as the source of hydrogen atoms in

The eukaryotic cell is the result of the evolution of ancient symbioses.

the reduction of CO_2 for the production of cell material. This led to the formation of gaseous oxygen as by-product of photosynthesis. . . . The continued production of free oxygen resulted in a crisis. . . . It is suggested that the first step in the origin of eukaryotes from prokaryotes was related to survival in the new oxygen-containing atmosphere: an aerobic prokaryotic microbe (the protomitochondrion) was ingested into the cytoplasm of a heterotrophic anaerobe. This symbiosis became obligate and resulted in the first aerobic amoeboid [without, at this stage, mitosis]. . . . Some of the amoeboids ingested certain motile prokaryotes. The genes of the parasite coded for its characteristic morphology, (9 + 2) fibrils in cross section. This parasite also became symbiotic, forming primitive amoebo-flagellates . . . that could actively pursue their own food. . . . The replicating nucleic acid of the endosymbiont genes (which determines its characteristic 9 + 2 structure of sets of microtubules) was eventually utilized to form the chromosomal centromeres and centrioles of eukaryotic mitosis and to distribute newly synthesized host nuclear chromatin to host daughters. . . . Eukaryotic plant cells acquired photosynthesis by symbiosis with photosynthetic prokaryotes (protoplastids), which themselves evolved from organisms related to cyanobacteria. . . . In order to document this theory, the following, at the very least, are required: (1), the theory must be consistent with geological and fossil records; (2), each of the three symbiotic organelles (the mitochondria, the (9 + 2) homologues, and the plastids) must demonstrate general features characteristic of cells originating in hosts as symbionts. None may have features that conflict with such an origin; (3), predictions based on this account of the origin of eukaryotes must be verified. The rest of this paper [24 pages] discusses the evidence for this theory in terms of assumptions based on molecular biology which can be made concerning evolutionary mechanisms.

Source: Journal of Theoretical Biology 14:225–274

New Guinean. The diagram is *not* a pedigree, although it superficially resembles one. Rather, the diagram is a hypothetical **phylogenetic tree,** which depicts the lines of descent that connect the mitochondrial DNA sequences. The "true" phylogenetic tree connecting the mitochondrial DNA sequences is unknown, so the tree illustrated should be regarded as an approximation (or an "estimate") of the true tree. It is approximate because it is inferred from analysis of the mitochondrial DNA, and this analysis requires a number of assumptions, including, for example, uniformity of rates of evolution in different branches. Nevertheless, the tree in Figure 14.15 has two interesting features: (1) Each population contains multiple mitochondrial types connected to the tree at widely dispersed positions, and (2) one of the primary branches issuing from the inferred common ancestor is composed entirely of

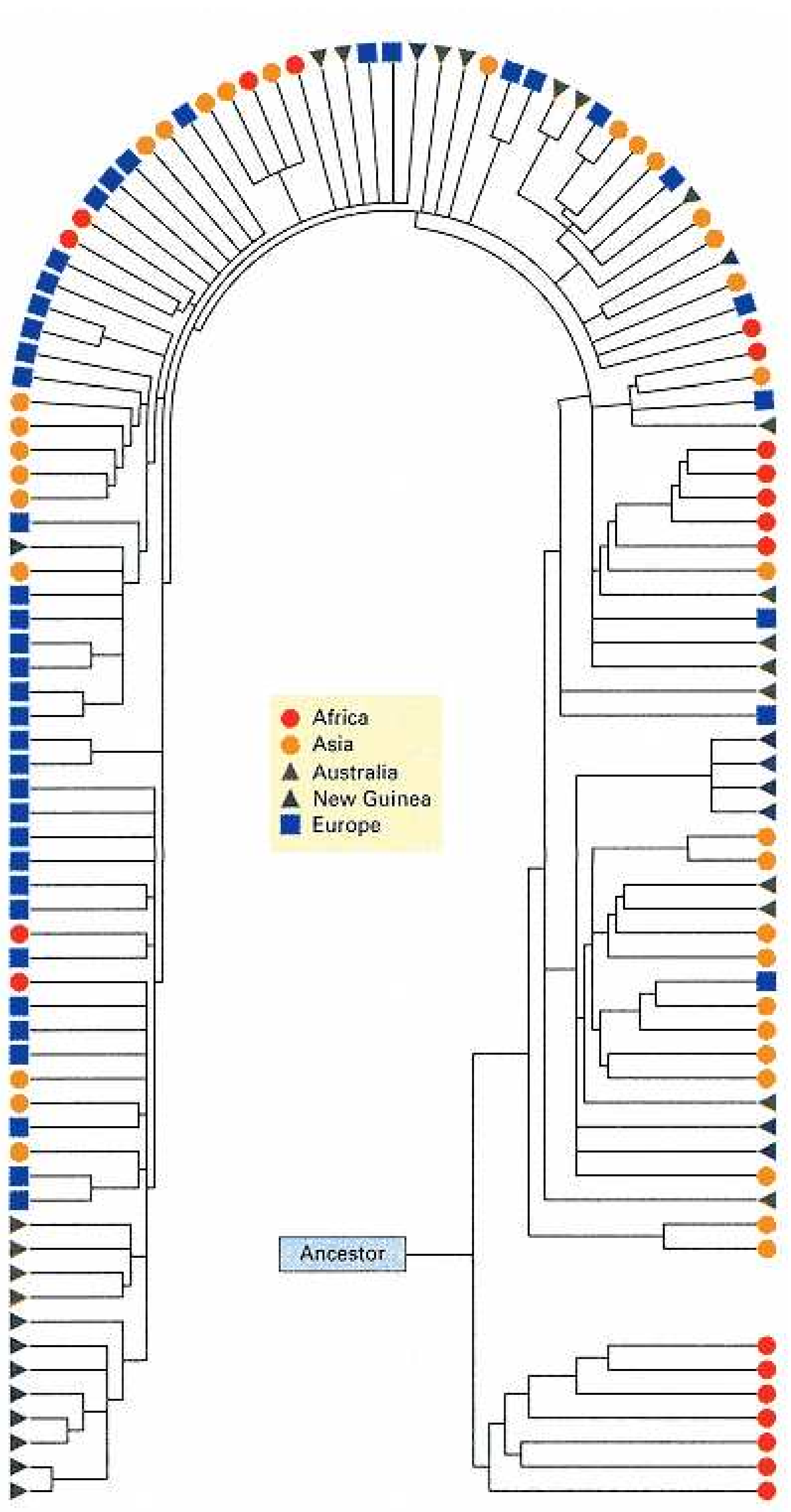


Figure 14.15
Possible phylogenetic tree of human mitochondria based on analysis of presence or absence of restriction sites for 12 different restriction enzymes in mitochondrial DNA from 134 persons.
[From R. L. Cann, M. Stoneking, and A. C. Wilson. 1987. *Nature* 325:31.]

Africans. These features can be explained most readily by postulating that the mitochondrial DNA type of the common ancestor was African. Furthermore, because the inferred common ancestor links mitochondrial DNA types that differ at an average of 95 nucleotides, the time since the existence of the common ancestor can be estimated as between 95 × 1500 = 142,500 years and 95 × 3000 = 285,000 years. Therefore, the common ancestor of all human mitochondrial DNA was a woman who lived in Africa between 142,500 and 285,000 years ago. This woman has been given the name "Eve" by the popular press, although she was not the only female alive at the time. Nevertheless, Eve's mitochondrial DNA became the common ancestor of all human mitochondrial DNA that exists today, because during the intervening years, all other mitochondrial lineages except hers became extinct (probably by chance).

The hypothesis of an African mitochondrial Eve has been very controversial. The difficulty is not with postulating a common ancestor of human mitochondrial DNA: It is a mathematical certainty that all human mitochondrial DNA must, eventually, trace back to a single ancestral woman. This principle is true because there is differential reproduction, and therefore, each mitochondrial lineage has a chance of becoming extinct in every generation. Hence, given enough time, all lineages except one must eventually become extinct. Was the common ancestor all mitochondria an African? When did she live? These are the questions in dispute. Later analysis of the same data suggested that other trees were almost as good as the one in Figure 14.15 and that some trees were even better. The main criterion for comparing hypothetical trees based on DNA sequences is the total number of mutational steps in all the branches together: Trees with fewer mutational steps are generally regarded as better. Therefore, the finding of better trees than the one in Figure 14.15 was a serious blow to the hypothesis of an African Eve, because none of the better trees had an earliest branch composed exclusively of Africans. However, much of the uncertainty is intrinsic in the data themselves. Another hypothetical phylogenetic tree, with fewer mutational steps than that in

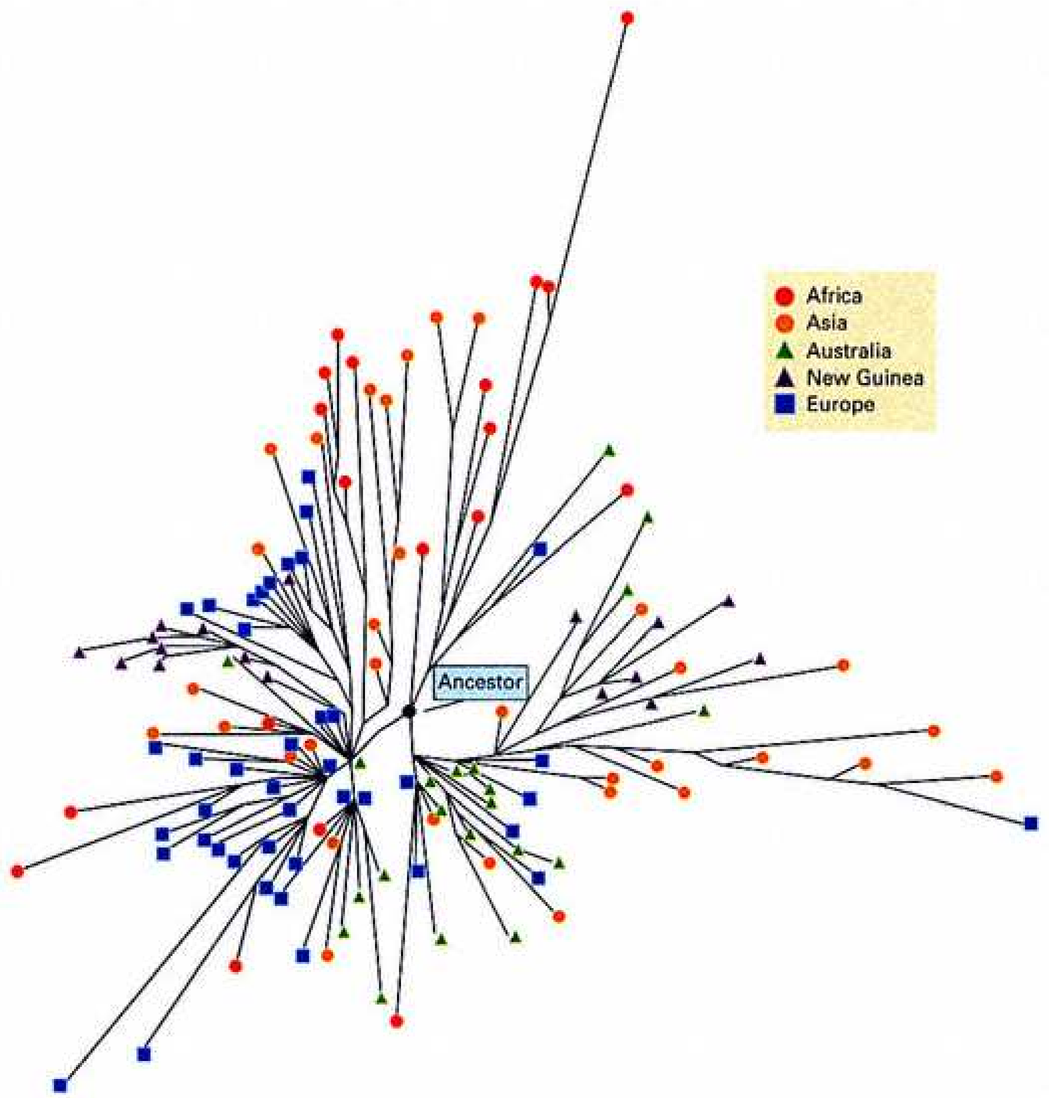


Figure 14.16
An improved (but still hypothetical) phylogenetic tree of human mitochondrial DNA represented as a starburst with the reader looking back in time. The symbols are the same as those in Figure 14.15. [From C. Wills. 1992. *Nature* 356:389.]

Figure 14.15, is shown in Figure 14.16. This tree is drawn with the lengths of the branches proportional to the number of mutational steps, and it resembles a starburst with the viewer looking back in time. The difficulty of identifying the earliest common ancestor is evident from this style of representation. Although there is an early branch (at a position near one o'clock in the starburst) composed primarily of Africans, this branch also includes two Asians, one Caucasian, and one Australian aborigine. On the other hand, this is no proof that the mitochondrial Eve was *not* an African, and considerable evidence of other kinds supports Africa as the cradle of human evolution and colonization. However, the evidence for postulating an African Eve solely on the basis of current mitochondrial DNA data is inconclusive.

Chapter Summary

The cytoplasm of eukaryotic cells contains complex organelles, the most prevalent of which are mitochondria, found in both plant and animal cells, and chloroplasts, found in plant cells. Mitochondria are responsible for respiratory metabolism. Chloroplasts are responsible for photosynthesis. Both mitochondria and chioroplasts contain DNA molecules that encode a variety of proteins. The genes contained in mitochondria and chloroplast DNA are not present in chromosomal DNA. Hence, certain traits determined by these genes are not inherited according to Mendelian principles but show extranuclear (cytoplasmic) inheritance instead. Extranuclear inheritance is also observed for traits that are determined by bacteria and viruses transmitted in the cytoplasm from one generation to the next. In eukaryotic microorganisms, cytoplasmic mixing usually takes place in zygote formation, and both parents contribute cytoplasmic particles to the zygote. However, in multicellular eukaryotes, the egg contains a large amount of cytoplasm and the sperm very little, so cytoplasmic transmission results in maternal inheritance.

Progeny resembling the mother is characteristic of both maternal inheritance and maternal effects. Maternal inheritance results from the transmission of extranuclear factors by the mother. Maternal effects occur when the nuclear genotype of the mother determines the phenotype of the offspring—for example, through the molecular organization of the egg or by nurturing ability. The direction of shell coiling in snails results from a maternal division in the zygote.

Variegation in the four-o'clock plant is caused by mutations in chloroplast DNA that eliminate chlorophyll. Variegated plants contain patches of green and while derived from wildtype cells and from cells with mutant chloroplasts, respectively. Inheritance of the variegated phenotype depends on the chloroplast content of the egg because pollen does not contain chloroplasts. The inheritance of resistance to certain antibiotics in *Chlamydomonas* is also cytoplasmic and depends on mutations in the chloroplast.

Mitochondrial defects account for the cytoplasmic inheritance of some petite mutations in yeast. Petites produce very small colonies because the defective mitochondria are unable to carry out aerobic metabolism. Petite dells can use only anaerobic metabolism, which is so inefficient that the cells grow very slowly, thereby producing small colonies. There are two major types of petites. Segragational petites result from a mutation in a nuclear gene and show strictly Mendelian inheritance. Neutral petites show non-Mendelian inheritance (4: 0 segregation in ascospores) and result from large deletions in mitochondrial DNA. Male sterility in maize also results from defective mitochondria that contain certain types of deletions or rearrangements in the DNA. Male sterility is very useful in plant breeding, because it makes it possible to cross-pollinate plants without removing immature anthers from the recipient plants.

The killer phenotype in *Paramecium* exemplifies cytoplasmic inheritance through a symbiotic bacterium that produces a product that is toxic to many other paramecia. Cells containing the bacterium are resistant to the killer substance. *Paramecium* metes by two cells coming in contact and exchanging haploid nuclei. After the nuclei exchange, the cells separate, Normally, the exchange takes place without any cytoplasmic mixing, in which case all segregating traits are inherited according to Mendelian principles. Sometimes cytoplasmic according to Mendelian principles. Sometimes cytoplasmic mixing occurs, and in this case one sees cytoplasmic inheritance. Maternal sex ration in *Drosophila,* expressed as a deficiency of males among progeny, is another bacterium. In this case, the bacterium contains a virus that inhibits the development of male embryos.

Human mitochondrial DNA types have been organized into a hypothetical phylogenetic tree in which one branch issuing from the inferred common ancestor is composed entirely of mitochondrial DNA from Africans. The woman whose mitochondrial DNA is thought to be the common ancestor of all human mitochondria has been named "Eve." The all-African branch in the phylogenetic tree and the number of nucleotide differences observed in mitochondrial DNA have been interpreted to mean that the mitochondrial Eve was present in a woman who lived in Africa between 142,500 and 285,000 years ago. On the other hand, additional analyses of the same data have yielded hypothetical phylogenetic trees that have fewer mutational steps overall and that do not contain exclusively African branches. Hence, the evidence that suggests a mitochondrial Eve who lived in Africa is inconclusive.

Key Terms

autogamy	heteroplasmy	Paternal inheritance
conjugation	kappa particle	petite mutant
cytoplasmic inheritance	maternal effect	phylogenetic tree
cytoplasmic male sterility	maternal inheritance	RNA editing
endosymbiont theory	maternal sex ratio	segregational petite

extranuclear inheritance	neutral petite	symbiosis
fertility restorer	organelle	uniparental inheritance

Review the Basics

• What are the normal functions of mitochondria and chloroplasts?

• What eukaryotic cells can survive without mitochondria? Can plant cells survive without chloroplasts?

• What is an inherited mitochondrial disease?

• Distinguish between maternal inheritance and maternal effect.

• What is the endosymbiont theory of the origin of certain cellular organelles?

• What is meant by variegation?

• How does leaf variegation in four-o'clock plants demonstrate heteroplasmy of chloroplasts in certain cell lineages and homoplasmy in others?

• Of what practical use is cytoplasmic male sterility in maize breeding?

• What does the occurrence of segregational petites in yeast imply about the relationship between the nucleus and mitochondria?

• What is meant by the term *Eve* with reference to human mitochondrial DNA? Do you think *Eve* is a suitable term for this concept? Why or why not?

Guide to Problem Solving

Problem 1: Two strains of maize exhibit male sterility. One is cytoplasmically inherited; the other is Mendelian. What crosses could you make to identify which is which?

Answer: Cytoplasmic male sterility is transmitted through the female, so outcrosses (crosses to unrelated strains) using the cytoplasmic male-sterile strain as the female parent will yield male-sterile progeny. Repeated outcrosses using the strain with Mendelian male sterility will result in either no male sterility (if the sterility gene is recessive) or 50 percent sterility (if it is dominant).

Problem 2: A snail of the species *Limnaea peregra* in which the shell coils to the left undergoes self-fertilization, and all of the progeny coil to the right. What is the genotype of the parent? If the progeny are individually self-fertilized, what coiling phenotypes are expected?

Answer: In this species, shell coiling is determined by the genotype of the mother. Because the coiling of the mother is leftward, its own mother had the genotype *ss,* so the individual must carry at least one *s* allele. Because the progeny coil to the right, the mother's genotype must be s^+. The progeny genotypes are 1/4 s^+s^+, 1/2 s^+s, and 1/4 *ss.* When selffertilized, the s^+s^+ and s^+s individuals yield rightward-coiled progeny, and the *ss* individuals yield leftward-coiled progeny.

Problem 3: Consider an inherited trait in *Drosophila* with the following characteristics: (1) It is transmitted from father to son to grandson. (2) It is not expressed in females. (3) It is not transmitted through females.

(a) Is the trait likely to result from extranuclear inheritance?

(b) Can male-limited expression explain the data?

(c) What hypothesis of Mendelian inheritance can explain the data?

(d) What observations could be made to confirm the Mendelian hypothesis? (*Hint:* Think about the consequences of nondisjunction.)

Answer:

(a) The trait is very unlikely to result from extranuclear inheritance. First, it would be very unusual for extranuclear inheritance in animals to be transmitted through the male. Second, if it were transmitted through sperm, both sexes should be affected.

(b) The trait is also unlikely to have male-limited expression. Although the observation that only males are affected is consistent with this hypothesis, a trait with male-limited expression could be transmitted through females.

(c) The father $\rightarrow$ son $\rightarrow$ grandson pattern of inheritance parallels that of the Y chromosome. Therefore, the data fit a Y-linked model of inheritance.

(d) Nondisjunction in an XY male results in XXY zygotes (which in *Drosophila* are female) and XO zygotes (which in *Drosophila* are male). Therefore, if the trait is Y-linked, then XXY females resulting from nondisjunction should be unaffected. Furthermore, the affected XXY females should give rise to affected sons (XY) and some affected daughters (XXY).

Problem 4: The rate of mitochondrial DNA evolution is given as 1 nucleotide change per mitochondrial lineage every 1500 to 3000 years. Given that human mitochondrial DNA is approximately 16,500 nucleotides, what is the rate of change per nucleotide per million years?

Answer: 16,500 nucleotides for 1500 years equals 2.5 x 10^7 nucleotide-years, and 16,500 nucleotides for 3000 years equals 5.0 x 10^7 nucleotide-years. The rate per nucleotide is therefore between 1/2.5 x 10^7 = 4 x 10^{-8} per year and 1/5.0 x 10^7 = 2 x 10^{-8} per year. That is, it is between 0.02 and 0.04 per million years, which is usually expressed as 2 percent to 4 percent nucleotide changes per million years.

Analysis and Applications

14.1 A mutant plant is found with yellow instead of green leaves. Microscopic and biochemical analyses show that the cells contain very few chloroplasts and that the plant manages to grow by making use of other pigments. The plant does not grow very well, but it does breed true. The inheritance of yellowness is studied in the following crosses:

green ♂ × yellow ♀ → all yellow

yellow ♂ × green ♀ → all green

What do these results suggest about the type of inheritance?

14.2 For the phenotype in the preceding problem, suppose that variegated progeny are found in the crosses at a frequency of about 1 per 1000 progeny. How might this be explained?

GeNETics on the web Chapter 14

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the Web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1 The complete, fully annotated sequence of **human mitochondrial DNA** can be found at this site, along with the sequences of many other mitochondrial genomes. Find the coordinates defining the complete nucleotide sequence of the lysine tRNA implicated in the MERRF type of myoclonic epilepsy, and determine whether the tRNA sequence matches the strand given or its complement. If assigned to do so, draw the self-paired structure of the tRNA in a "cloverleaf" configuration.

2 Genetic maps of the **Chalmydomonas** chloroplast and mitochondrion can be found by choosing the *Linkage group option*at this site. From the maps, deduce the approximate size of each molecule in kilobase pairs. If assigned to do, determine how many genes are in each organelle genome; use the search engine to find out what the mitochondrial *s* and L genes encode as well as the function of the chloroplast *tufA* gene.

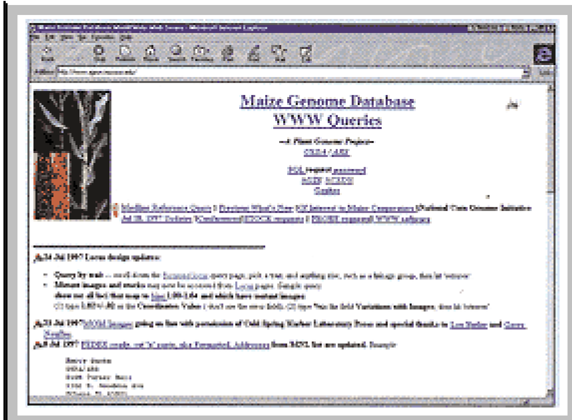
3. Search this maize site for **cytoplasmic male sterility,** and follow the links to the *cms* types of cytoplasmic male sterility. If assigned to do so, make a list of which *cms*types can be suppressed by fertility restorers and which types confer susceptibility to fungal infection.

MUTABLE SITE EXERCISES

The Mutable Site Exerise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 14, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 14.



14.3 What kinds of progeny result from the following crosses with *Mirabilis?*

(a) green ♀ × white ♂

(b) white ♀ × green ♂

(c) variegated ♀ × green ♂

(d) green ♀ × varigated ♂

14.4 In the maternally inherited leaf variegation in *Mirabilis*, how does the F^2 of the cross green ♀ × white ♂ differ from that which would be expected if white leaves were due to a converntinal X-linked recessive gene in animals?

14.5 Assuming that chloroplasts duplicate and segrate randomly in cell division, what is the probability that both replicas of a single mutant chloroplast will segregate to the same daughter cell during mitosis? After two divisions to same probability that at least one daughter cell will lack the mutant chloroplast?

14.6 If *a* and *b* are chloroplast markers in *Chalmydomonas*, what genotypes of progeny are expected from the cross a^+ $mt^+ \times a^- b\ ^+mt^-$?

14.8 An antibiotic-resistant haplod strain of yeast is isolated. Mating with a wildtype (antibiotic-sensitive) strain produces a diploid that, when grown for some generations and induced to undergo sporulation, produces tetrads containing either four antibiotic-resistant spores or four antibioticsensitive spores. What can you conclude about the inheritance of the antibiotic resistance?

14.9 What possible genotypes of the progeny of an *Aaparamecium* that undergoes autogamy?

14.10 Equal numbers of *AA* and *aa Paramecium* undergo conjugation in random pairs. What are the expected matings? What genotypes and freaquencies are expected after the conjugating pairs separate? What genotypes and frequencies are expected after the cells undergo autogamy?

14.11 Corn plants of genotype *AA bb* with cytoplasmic male sterility and normal *aa BB* plants are planted in alterate rows, and pollination is allowed to occur at random. What progeny genotypes would be expected from the two types of plants?

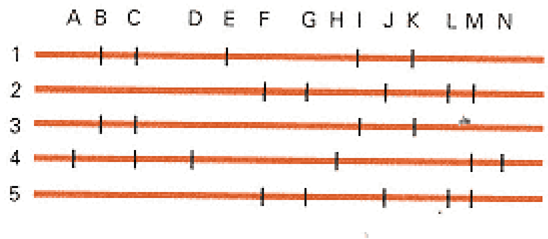
14.12 Females of a certain strain of *Drosophila* are mated with wildtype males. All progeny are female. There are three possible explanations: (1) The females arehomozygous for are autosomal allele that produces lethal male embryos (that is, a maternal effect). (2) The females are homozygous for an X-linked allele that is lethal in males. (3) The females carry a cytoplasmically inherited factor that kills male embryos. A cross is carried out between F_1 females and wildtype males, and only females progeny are produced. What mode of inheritance does this result imply?

14.13 A rightward-coiled snail, when self-fertilized, has all leftward-coiled progeny. What is the genotype of the rightward-coiled parent? What was the genotype of the parent's mother?

Challenge Problems

14.14 There are an average of 78 nucleotide difference in mitochondrial DNA between two Africans, an average of 58 differences between two Africans, and an average of 40 differences between two Caucasians. Assuming a rate of evolution of mitochondrial DNA of 1 nucleotide difference per 1500 to 3000 years, what is the estimated average age of the common ancestor of the mitochondrial DNA among Africans? Among Asians? Among Caucasians?

14.15 The illustration shows the locations of restriction sites A through N in the mitochondrial DNA molecules of five species of crabs. The presence of a restriction site is represented by a short vertical line. Assuming that the proportion of shared restriction sites between two molecules is an indication of their closeness of relationship, draw a diagram showing the inferred relationships among the species.



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This spectacular white baby emperor penguin is a mutant discovered by wildlife researchers in snow-covered sea ice in the western Ross Sea of Antarctica Other than its black eyes, the bird is completely white, including its bill and feet.
[Courtesy of Gerald Kooyman.]

Chapter 15— Population Genetics and Evolution

CHAPTER OUTLINE

PRINCIPLES

• Many genes in natural populations are polymorphic—they have two or more common alleles. Genetic polymorphisms can be used as genetic markers in pedigree studies and for individual identification (DNA typing).

• The Hardy-Weinberg principle asserts that, with random mating, the frequencies of the genotypes *AA*, *Aa*, and *aa* in a population are expected to be p^2, $2pq$, and q^2, respectively, where p and q are the allele frequencies of *A* and *a*. One implication of this principle is that the great majority of rare, harmful recessive alleles are present in heterozygous carriers.

• Inbreeding means mating between relatives. Through inbreeding, replicas of an allele present in a common ancestor may be passed down both sides of a pedigree and come together in an inbred organism. Such alleles are said to be identical by descent. Because inbreeding can result in alleles that are identical by descent, inbreeding results in an excess of homozygous genotypes relative to the frequencies of genotypes expected with random mating.

• Mutation and migration both introduce new alleles into populations. Because mutation rates are typically small, mutation is a weak force for changing allele frequencies.

• Natural selection and random genetic drift are the usual causes of change in allele frequency—the former in a systematic direction, the latter randomly. Selection can result in a stable genetic polymorphism if the heterozygous genotype has the highest fitness. Selection against recurrent harmful mutations leads to a mutation-selection balance in which the allele frequency of the harmful recessive remains constant generation after generation.

CONNECTIONS

CONNECTION: A Yule Message from Dr. Hardy

Godfrey H. Hardy 1908
Mendelian proportions in a mixed population

CONNECTION: Be Ye Son or Nephew?

Alec J. Jeffreys, John F. Y. Brookfield, and Robert Semeonoff 1985
Positive identification of an immigration test-case using human DNA fingerprints

In the genetic analyses we have examined so far, matings have been deliberately designed by geneticists. In most organisms, matings are not under the control of the investigator, and often the familial relationships among individuals are unknown. This situation is typical of studies of organisms in their natural habitat. Most organisms do not live in discrete family groups but exist as part of populations of individuals with unknown genealogical relationships. At first, it might seem that classical genetics with its simple Mendelian ratios could have little to say about such complex situations, but this is not the case. The principles of Mendelian genetics can be used both to interpret data collected in natural populations and to make predictions about the genetic composition of populations. Application of genetic principles to entire populations of organisms constitutes the subject of **population genetics.**

15.1—
Allele Frequencies and Genotype Frequencies

The term **population** refers to a group of organisms of the same species living within a prescribed geographical area. Sometimes the area is large, as when we refer, say, to the population of sparrows in North America; it may even include the entire earth, as in reference to the "human population." More commonly, the area is considered to be of a size within which individuals in the population are likely to find mates. Many geographically widespread species are subdivided into more or less distinct breeding groups, called **subpopulations,** that live within limited geographical areas. Each subpopulation is a **local population.** The complete set of genetic information contained within the individuals in a population is called the **gene pool.** The gene pool includes all alleles present in the population. This section begins with an analysis of a local population with respect to a phenotype determined by two alleles.

Allele Frequency Calculations

The genetic composition of a population can often be described in terms of the frequencies, or relative abundances, in which alternative alleles are found. The concepts will be illustrated by using the human MN blood groups because this genetic system is exceptionally simple. There are three possible phenotypes—M, MN, and N—corresponding to the combinations of M and N antigens that can be present on the surface of red blood cells. These antigens are unrelated to ABO and other red-cell antigens. The M, MN, and N phenotypes correspond to three genotypes of one gene: *MM*, *MN*, and *NN*, respectively. In a study of a British population, a sample of 1000 people yielded 298 M, 489 MN, and 213 N phenotypes. From the one-to-one correspondence between genotype and phenotype in this system, the genotypes can be directly inferred to be

298 *MM*   489 *MN*   213 *NN*

These numbers contain a surprising amount of information about the population—for example, whether there may be mating between relatives. One of the goals of population genetics is to be able to interpret this information. First, note that the sample contains two types of data: (1) the number of each of the three genotypes, and (2) the number of individual *M* and *N* alleles. Furthermore, the 1000 genotypes represent 2000 alleles of the gene because each genome is diploid. These alleles break down in the following way:

298 *MM* persons =	596 *M* alleles		
298 *MM* persons =	489 *M* alleles	+	489 *Nalleles*
298 *MM* persons =			426 *N alleles*
Totals =	1085 *M* alleles	+	915 *N* alleles

Usually, it is more convenient to analyze the data in terms of relative frequency than in terms of the observed numbers. For genotypes, the **genotype frequency** in a population is the proportion of organisms that have the particular genotype. For each allele, the **allele frequency** is the proportion of all alleles that are of the specified type. In the *MN* example, the genotype frequencies are obtained by dividing the observed numbers by the total sample size—in this case, 1000. Therefore, the genotype frequencies are

0.298 *MM*   0.489 *MN*   0.213 *NN*

Similarly, the allele frequencies are obtained by dividing the observed number of each allele by the total number of alleles (in this case, 2000), so

$$\begin{aligned}\text{Allele frequency of } M &= 1085/2000\\ &= 0.5425\end{aligned}$$

$$\begin{aligned}\text{Allele frequency of } N &= 915/2000\\ &= 0.4575\end{aligned}$$

Note that the genotype frequencies add up to 1.0, as do the allele frequencies; this is a consequence of their definition in terms of proportions, which must add up to 1.0 when all of the possibilities are taken into account. *Allele and genotype frequencies must always be between 0 and 1.* A population with an allele frequency of 1.0 for some allele is said to be **fixed** for that allele. If an allele frequency becomes 0, then the allele is said to be **lost.**

Allele frequencies can be used to make inferences about matings in a population and to predict the genetic composition of future generations. Allele frequencies are often more useful than genotype frequencies because genes, not genotypes, form the bridge between generations. Genes rarely undergo mutation in a single generation, so they are relatively stable in their transmission from one generation to the next. Genotypes are not permanent; they are disrupted in the processes of segregation and recombination that take place in each reproductive cycle. We know from simple Mendelian considerations what types of gametes must be produced from the *MM, MN,* and *NN* genotypes:

$$\begin{array}{lcl} MM \text{ genotypes} & \longrightarrow & \text{all } M \text{ gametes}\\ MN \text{ genotypes} & \longrightarrow & 1/2\ M \text{ gametes}\\ & \longrightarrow & 1/2\ N \text{ gametes}\\ NN \text{ genotypes} & \longrightarrow & \text{all } N \text{ gametes}\end{array}$$

Consequently, the *M*-bearing gametes in the population consist of all the gametes from *MM* individuals and half the gametes from *MN* individuals. Likewise, the *N*-bearing gametes consist of all the gametes from *NN* individuals and half the gametes from *MN* individuals. Therefore, in terms of gametes, the population produces the following frequencies:

$$\begin{array}{lcl} 0.298\ MM & \longrightarrow & \left.\begin{array}{l}0.298\ M\\ 0.2445\ M\end{array}\right\}\ 0.5425\ M \text{ gametes}\\ 0.489\ MM & \nearrow\!\!\searrow & \\ 0.213\ NN & \longrightarrow & \left.\begin{array}{l}0.2445\ N\\ 0.213\ N\end{array}\right\}\ 0.4575\ N \text{ gametes}\end{array}$$

Note that the allele frequencies among *gametes* equal the allele frequencies among *adults* calculated earlier, which must be true whenever each adult in the population produces the same number of functional gametes.

Enzyme Polymorphisms

An important method for studying genes in natural populations is **electrophoresis,** or the separation of charged molecules in an electric field (Section 5.7). This method is commonly used to study genetic variation in proteins coded by alternative alleles and to study the relative sizes of homologous DNA fragments produced by a restriction enzyme.

In protein electrophoresis, protein-containing tissue samples are placed near the edge of a gel and a voltage is applied for several hours. All charged molecules in the sample move in response to the voltage, and a variety of techniques can be used to locate particular substances. For example, an enzyme can be located by staining the gel with a reagent that is converted into a colored product by the enzyme; wherever the enzyme is located, a colored band appears.

Figure 15.1, a photograph of a gel stained to reveal the enzyme phosphoglucose isomerase in tissue samples from 16 mice, illustrates typical raw data from an electrophoretic study. The pattern of bands varies from one mouse to the next, but only three patterns are observed. Samples

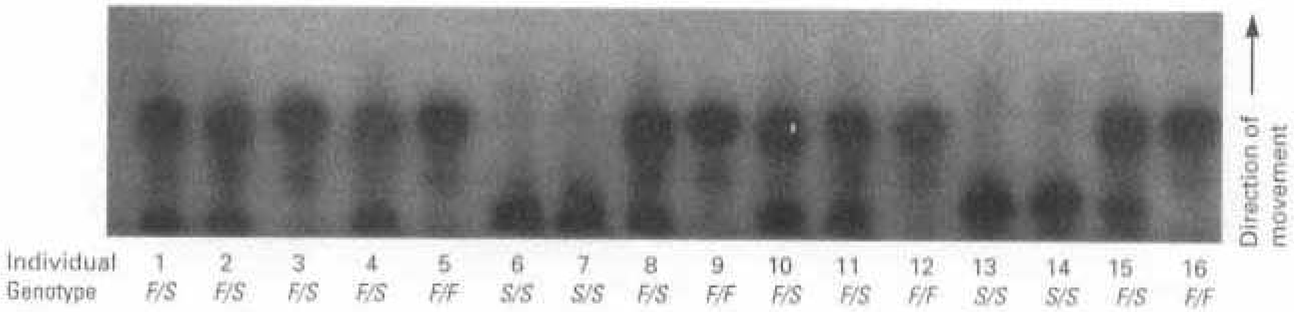


Figure 15.1
Electrophoretic mobility of glucose phosphate isomerase in a sample of 16 mice.
[Courtesy of S. E. Lewis and F. M. Johnson.]

from mice 1, 2, 4, 8, 10, 11, and 15 have two bands: one that moves fast (upper band) and one that moves slowly (lower band). A second pattern is seen for the samples from mice 3, 5, 9, 12, and 16, in which only the fast band appears. The samples from mice 6, 7, 13, and 14 show a third pattern, in which only the slow band appears. Electrophoretic analysis of many enzymes in a wide variety of organisms has shown that mobility variation almost always has a simple genetic basis: Each electrophoretic form of the enzyme contains one or more polypeptides with a genetically determined *amino acid replacement* that changes the electrophoretic mobility of the enzyme. Alternative forms of an enzyme coded by alleles of a single gene are known as **allozymes.** Alleles coding for allozymes are usually codominant, which means that heterozygotes express the allozyme corresponding to each allele. Therefore, in Figure 15.1, mice with only the fast allozyme are *FF* homozygotes, those with only the slow allozyme are *SS* homozygotes, and those with both fast and slow allozymes are *FS* heterozygotes. The 16 genotypes in Figure 15.1 are consequently

5 *FF*   7 *FS*   4 *SS*

and the allele frequency of F in this small sample is

$$\text{Allele frequency of } F = [(2 \times 5) + 7]/(2 \times 16)$$
$$= 0.53$$

Because only two alleles are represented in Figure 15.1, the allele frequency of *S* must equal 1 - 0.53 = 0.47, which can also be calculated directly as

$$\text{Allele frequency of } S = [7 + (2 \times 4)]/(2 \times 16)$$
$$= 0.47$$

Many genes in natural populations are **polymorphic,** which means that they have two or more relatively frequent alleles. Among plants and vertebrate animals, approximately 20 percent of enzyme genes are polymorphic (Table 15.1). The implication of this finding is that genetic variation among organisms is very common in natural populations of most organisms. Furthermore, among plants and vertebrate animals, the average proportion of genes that are heterozygous within an organism is about 0.05, which again emphasizes the prevalence of alternative alleles in populations. Invertebrate animals such as *Drosophila* have even more genetic variation.

DNA Polymorphisms

Polymorphisms can also be detected directly in DNA molecules. One method uses a *probe* DNA that is generally obtained from plasmid or phage vectors into which DNA fragments from the organism of interest have been cloned. The probe DNA is complementary in sequence to a segment of DNA in the organism of interest, but its presence in clones allows the probe to be

Table 15.1 Detection of genetic variation through use of protein electrophoresis

	Number of species examined	Proportion of protein-coding loci that are polymorphic	Proportion of heterozygous loci (average) ± standard deviation
Plants	15	0.26 ± 0.17	0.07 ± 0.07
Invertebrates	93	0.40 ± 0.20	0.11 ± 0.07
Vertebrates	135	0.17 ± 0.12	0.05 ± 0.04

Source: From Nevo, E. 1978. "Genetic variation in natural populations: Patterns and theory." *Theor. Pop. Biol.* 43: 121–177

obtained in large quantities and pure form. In the *Southern blot* procedure, in which genomic DNA is digested with a restriction enzyme and fragments of different sizes separated by electrophoresis, radioactive probe DNA hybridizes only with DNA fragments that contain complementary sequences and so identifies these restriction fragments (Section 5.7).

When this procedure is used to study natural populations, it is common to find restriction fragments complementary to a probe that differ in size among individuals. These differences result from differences in the locations of restriction sites along the DNA. Each region of the genome that hybridizes with the probe generates a restriction fragment. The restriction fragments that derive from corresponding positions in homologous chromosomes are analogous to alleles in that they segregate in meiosis: The offspring of a heterozygote may inherit a chromosome with either fragment size, but not both. Moreover, allelic restriction fragments of different sizes are codominant because both fragments are detected in heterozygotes; this is the best kind of situation for genetic studies, because genotypes can be inferred directly from phenotypes.

One type of variation in restriction fragment length is illustrated in Figure 15.2A. A single nucleotide change knocks out a restriction site, so the restriction fragment that is complementary to the probe stretches farther downstream to the next restriction site. Therefore, hybridization with probe DNA will reveal a shorter restriction fragment produced from allele 1 than from allele 2. As with any other type of allelic variation, an organism may be homozygous for allele 1, homozygous for allele 2, or heterozygous. Restriction fragments of different sizes produced from the alleles of a gene constitute a *restriction fragment length polymorphism,* or *RFLP,* of the type discussed in Section 4.4. RFLPs are extremely common in the human genome and in the genomes of most other organisms; hence they provide an extensive set of highly polymorphic, codominant genetic markers scattered throughout the genome, which can be studied with Southern blotting procedures identical except for the use of different restriction enzymes and different types of probe DNA. Figure 15.2B illustrates the segregation of alleles of a restriction fragment polymorphism within a human pedigree.

A second type of variation in DNA consists of a short sequence of nucleotides repeated in tandem a number of times, such as 5'-TGTGTGTGTGTG-3'. The number of repeats may differ from one chromosome to the next, so the number of repeats can be used to identify different chromosomes. A polymorphism of this type is called a *simple tandem repeat polymorphism,* or *STRP* (Section 4.4). Although there are many places in the genome where an STRP of TGs (or any other short repeat) may occur, each place is flanked by sequences that are unique in the genome. By means of oligonucleotide primers specific for the unique flanking sequences, any one of the STRPs in the genome can be amplified in the polymerase chain reaction, and the number of repeats in any chromosome can be determined by examining the length of the fragment that was amplified.

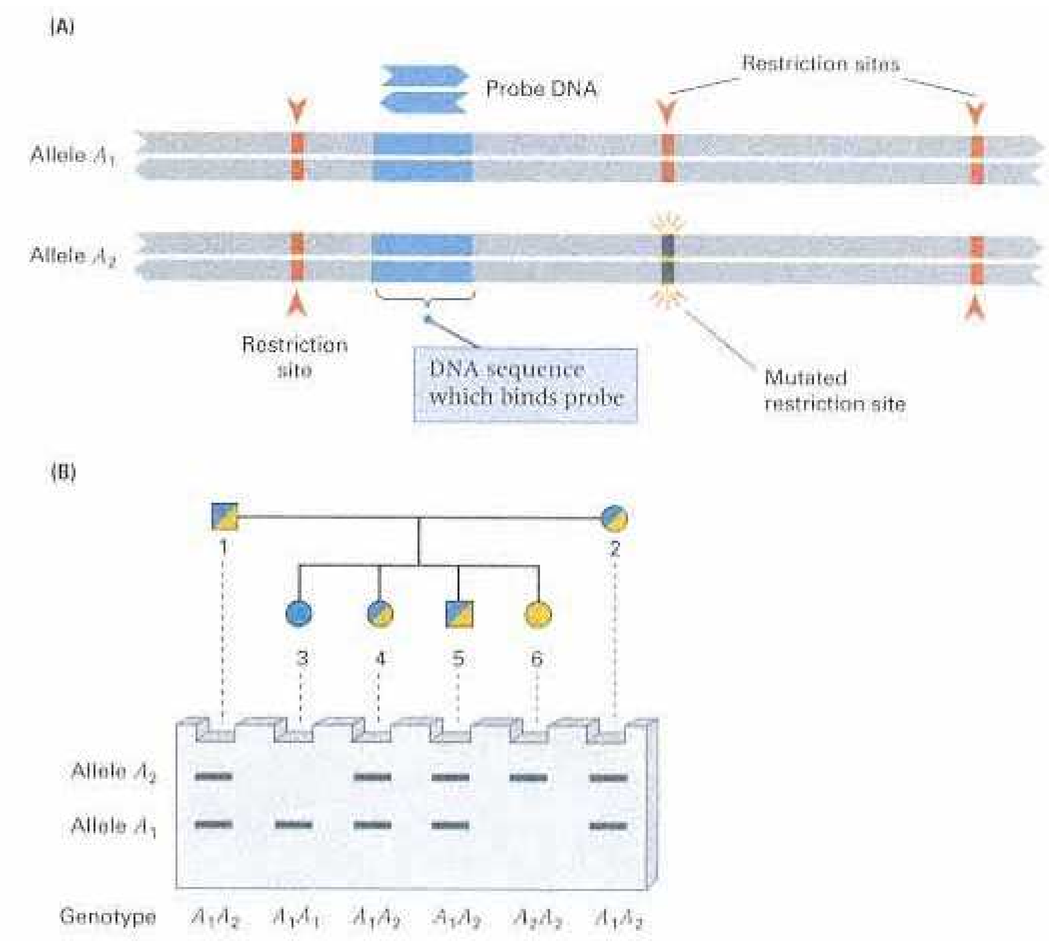


Figure 15.2
Restriction fragment size variation in DNA. (A) Nucleotide substitution in a DNA molecule (highlighted in A_2) eliminates a restriction site. The result is that in the region detected by the probe DNA, alleles 1 and 2 differ in the size of a restriction fragment. (B) Segregation of a restriction fragment length polymorphism like that in part A. In the pedigree, both parents (individuals 1 and 2) are heterozygous A_1A_2. Expected progeny resulting from segregationare $A_1A_1A_1A_2$, and A_2A_2 in the proportions 1:2:1. The diagram of the electrophoresis gel shows the expected patterns of bands in a Southern blot. Homozygous individuals numbered 3 (A_1A_1) and 6 (A_2A_2) yield a single restriction fragment that hybridizes with the probe DNA. The heterozygous individuals show both bands.

15.2— Systems of Mating

The transmission of genetic material from one generation to the next is analyzed in terms of alleles rather than genotypes because genotypes are broken up in each generation by the process of segregation and recombination. When gametes unite in the process of fertilization, the genotypes of the next generation are formed. The genotype frequencies in the zygotes of the progeny generation are determined by the frequencies with which the various types of parental gametes come together, and these frequencies are, in turn, determined by the **mating system** in which the genotypes pair as mates. Three types of mating systems predominate in natural populations: random mating, assortative mating, and inbreeding.

The mating system called **random mating** means that mating pairs are formed without regard to genotype. With random mating, a mating between any two genotypes takes place in proportion to the relative frequencies of the genotypes in the population. A population can undergo random mating with respect to some traits but, at the same time, nonrandom mating with respect to other traits. In a type of mating system called **assortative mating,** mating pairs come together on the basis of their degree of similarity in phenotype. In **positive assortative mating,** organisms tend to mate with others that are similar in phenotype. Positive assortative mating takes place for many traits in human populations. In the United States, these traits include skin color and height, traits for which mates tend to resemble each other more than would be expected by chance. In **negative assortative mating,** the mating pairs are more dissimilar in phenotype than would be expected by chance. Negative assortative mating is not so common as positive assortative mating, but a famous example is found in some species of primroses (*Primula)* and their relatives. In most populations of these species, two types of flowers are found in approximately equal proportions (Figure 15.3). One type, known as *pin,* has a long style and short stamens. The anthers are the organs that produce pollen and are located atop the stamens; the stigma is the organ that receives pollen and is located atop the style. Pin flowers produce pollen low in the flowers and receive pollen high in the flowers. In the other type of plant, known as *thrum,* the relative positions of stigma and anthers are reversed. Consequently, insect pollinators that work low in the flowers pick up mainly pin pollen and deposit it mainly on thrum stigmas, whereas pollinators that work high in the flowers pick mainly thrum pollen and deposit it mainly on pin stigmas. This behavior promotes negative assortative mating, which is further enhanced because pin pollen grows better in thrum stigmas, and vice versa. The genetic basis of the flower type resides in a single genetic region: pin is homozygous *ss,* and thrum is heterozygous *Ss.*

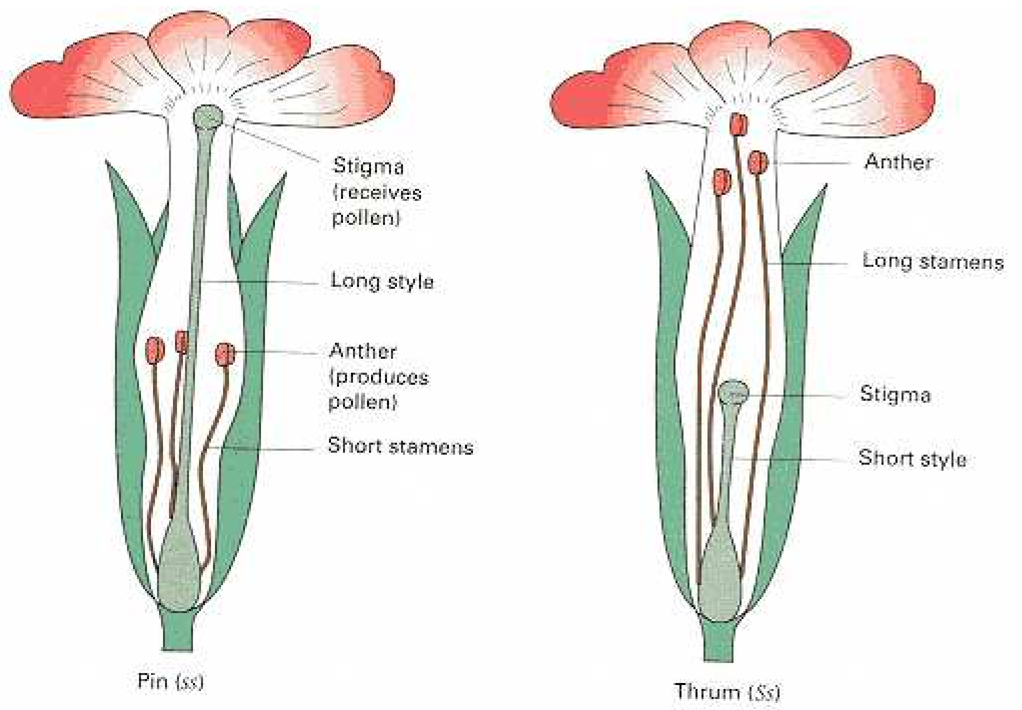


Figure 15.3
Pin and thrum forms of flowers of *Primula officalis*. Negative assortative mating is promoted by the anatomy of these flowers. Pollinators that work low in the flowers tend to pick up pin pollen and deposit it on thrum stigmas, and pollinators that work high in the flowers tend to do the reverse.

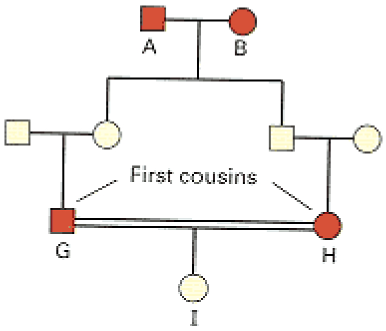


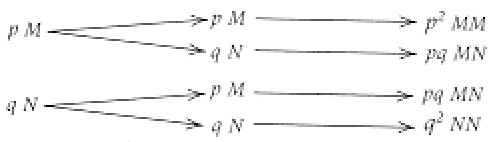
Figure 15.4
An inbreeding pedigree in which individual I is the result of a mating between first cousins.

A third important type of mating system is **inbreeding,** mating between relatives. In human pedigrees, a mating between relatives is often called a **consanguineous mating.** Figure 15.4 represents an example of inbreeding. In this case, the female I is the offspring of a first cousin mating (G with H). The closed loop in the pedigree is diagnostic of inbreeding, and the individuals designated A and B are called *common ancestors* of I, because they are ancestors of both of I's parents. Because A and B are common ancestors, a particular allele present in A (or in B) could, by chance, be transmitted in inheritance down both sides of the pedigree, to meet again in the formation of I. This possibility, which is the most important characteristic and consequence of inbreeding, will be discussed later in this chapter.

Random Mating and the Hardy-Weinberg Principle

In random mating, organisms form mating pairs independently of genotype; each type of mating pair is formed as often as would be expected by chance encounters. Random mating is by far the most prevalent mating system for most species of animals and plants, except for plants that regularly reproduce through self-fertilization.

With random mating, the relationship between allele frequency and genotype frequency is particularly simple because the *random mating of individuals is equivalent to the random union of gametes.* Conceptually, we may imagine all the gametes of a population as present in a large container. To form zygote genotypes, pairs of gametes are withdrawn from the container at random. To be specific, consider the alleles M and N in the MN blood groups, whose allele frequencies are p and q, respectively (remember that $p + q = 1$). The genotype frequencies expected with random mating can be deduced from the following diagram:



In this diagram, the gametes at the left represent the sperm and those in the middle the eggs. The genotypes that can be formed with two alleles are shown at the right, and with random mating, the frequency of each genotype is calculated by multiplying the allele frequencies of the corresponding gametes. However, the genotype *MN* can be formed in two ways—the M allele could have come from the father (top part of diagram) or from the mother (bottom part of diagram). In each case, the frequency of the *MN* genotype is *pq;* considering both possibilities, we find that the frequency of *MN* is $pq + pq = 2pq$. Consequently, the overall genotype frequencies expected with random mating are

$$MM{:}\ p^2 \qquad MN{:}\ 2pq \qquad NN{:}\ q^2 \qquad (1)$$

The frequencies p^2, $2pq$, and q^2 result from random mating for a gene with two alleles; they constitute what is called the **Hardy Weinberg principle** (after Godfrey Hardy and Wilhelm Weinberg, each of whom derived it independently of the other in 1908). Sometimes the Hardy-Weinberg principle is demonstrated by a Punnett square, as illustrated in Figure 15.5. Such a square is completely equivalent to the tree diagram used above. Although the Hardy-Weinberg principle is exceedingly simple, it has a number of important implications that are not obvious. These are described in the following sections.

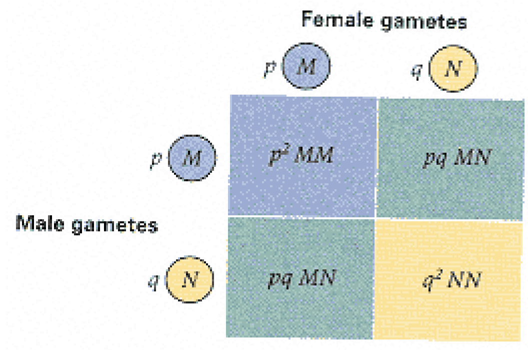


Figure 15.5
A cross-multiplication square (Punnett square) showing the result of the random union of male and female gametes, which is equivalent to the random mating of individuals.

Implications of the Hardy-Weinberg Principle

One important implication of the Hardy Weinberg principle is that *the allele frequencies remain constant from generation to generation.* Consider a gene with two alleles, *A* and *a,* that have frequencies *p* and *q* respectively ($p + q = 1$). With random mating, the frequencies of genotypes *AA*, *Aa*, and *aa* among zygotes are p^2, $2pq$, and q^2, respectively. Assuming equal *viability* (ability to survive) among the genotypes, these frequencies equal those among adults. If all of the genotypes are equally fertile, then the frequency, p', of allele *A* among gametes of the next generation will be

$$p' = p^2 + 2pq/2 = p(p + q) = p$$

(because $p + q = 1$). This argument shows that the frequency of allele *A* remains constant at the value of *p* through the passage of one (or any number of) complete generations. This principle depends on certain assumptions, of which the most important are

1. Mating is random; there are no subpopulations that differ in allele frequency.

2. Allele frequencies are the same in males and females.

3. All the genotypes are equal in viability and fertility; *selection* does not operate.

4. Mutation does not occur.

5. Migration into the population is absent.

6. The population is sufficiently large that the frequencies of alleles do not change from generation to generation because of chance.

A Test for Random Mating

As a practical application of the Hardy Weinberg principle, consider again the MN blood groups among British people, which we discussed in Section 15.1. The frequencies of alleles *M* and *N* among 1000 adults were 0.5425 and 0.4575, respectively; assuming random mating, the *expected* genotype frequencies can be calculated from Equation (1) as

$$\begin{aligned} MM&: \quad (0.5425)^2 = 0.2943 \\ MN&: \quad 2(0.5425)(0.4575) = 0.4964 \\ NN&: \quad (0.4575)^2 = 0.2093 \end{aligned}$$

Because the total number of people in the sample is 1000, the expected numbers of these genotypes are 294.3 MM, 496.4 MN, and 209.3 NN. The *observed*numbers are 298 MM, 489 MN, and 213 NN. Goodness of fit between observed numbers and expected numbers would normally be determined by means of the X^2 test described in Chapter 3. We will not apply the test here because the agreement between expected and observed numbers is quite good. The X^2 test yields a probability of about 0.67, which means that the hypothesis of random mating can account for the data. On the other hand, the X^2 test detects only deviations that are rather large, so a good fit to Hardy-Weinberg frequencies should not be overinterpreted, because:

> It is entirely possible for one or more assumptions of the Hardy-Weinberg principle to be violated, including the assumption of random mating, and still not produce deviations from the expected genotype frequencies that are large enough to be detected by the X^2 test.

The use of a X^2 test to determine departures from random mating may seem somewhat circular inasmuch as the data

Connection A Yule Message from Dr. Hardy

Godfrey H. Hardy 1908

Trinity College, Cambridge, England
Mendelian Proportions in a Mixed Population

An early argument against Mendelian inheritance was that dominant traits, even harmful ones, should eventually come to have a 3:1 ratio in any population. This is obviously not the case. There are many dominant traits in human beings that are rare now and have always been rare, such as brachydactyly (short fingers), which is the specific issue in this paper. Hardy, a mathematician, realized that with random mating, and assuming no differences in survival or fertility, the frequencies of dominant and recessive genotypes would have no innate tendency to change of their own accord. The form of the principle as written by Hardy is that if the genotype frequencies of AA, Aa, *and aa are* P, 2Q, *and* R *in any generation, then with random mating, the frequencies will be* $(P + Q)^2:2(P + Q)(Q + R): (Q + R)^2$ *in the next and subsequent generations. The allele frequencies* p *and* q *of* A *and a are not stated explicitly, so the familiar form of the principle as stated today*—$p^2:2pq:q^2$—*does not appear as such in Hardy's paper. But it is there implicitly, because* p = P + Q *and* q = Q + R.

I am reluctant to intrude in a discussion concerning matters of which I have no expert knowledge, and I should have expected the very simple point which I wish to make to have been familiar to biologists. However, some remarks of Mr. Udny Yule suggest that it may still be worth making . . ..Mr. Yule is reported to have suggested, as a criticism of the Mendelian position, that if brachydactyly is dominant "in the course of time one would expect, in the absence of counteracting forces, to get

There is not the slightest foundation for the idea that a dominant character should show a tendency to spread, or that a recessive should tend to die out.

three brachydactylous persons to one normal." . . . It is not difficult to prove, however, that such an expectation would be quite groundless. Suppose that *A, a* is a pair of Mendelian characters, *A* being dominant, and that in any given generation the numbers of pure dominants *(AA),* heterozygotes (*Aa),* and pure recessives (*aa)* are as *P:* 2*Q: R.* Finally, suppose that the numbers are fairly large, so that the mating may be regarded as random, that the sexes are evenly distributed among the three varieties, and that they are all equally fertile. A little mathematics of the multiplication table type is enough to show that in the next generation the numbers will be as

$$(P+Q)^2 :2(P+Q)(Q+R) :(Q+R)^2$$

or as $P_1:2Q_1:R_1$, say. The interesting question is—in what circumstances will this distribution be the same as that in the generation before? It is easy to see that the condition for this is $Q^2 = PR$. And since $Q_1^2 = P_1R_1$, whatever the values of *P, Q,* and *R* may be, the distribution will in any case continue unchanged after the second generation . . .. In a word, there is not the slightest foundation for the idea that a dominant character should show a tendency to spread, or that a recessive should tend to die out . . .. I have, of course, considered only the very simplest hypothesis possible. Hypotheses other than that of purely random mating will give different results, and, of course, if, as appears to be the case sometimes, the character is not independent of sex, or has an influence on fertility, the whole question may be greatly complicated. But such complications seem to be irrelevant to the simple issue raised by Mr. Yule's remarks.

Source: Science 28: 49–50

themselves are used in calculating the expected genotype frequencies. However, the use of the data is exactly compensated for by the simple expedient of deducting one degree of freedom for every parameter estimated from the data. In the MN case, there would ordinarily be two degrees of freedom, because there are three classes of data (the numbers of M, MN, and N phenotypes). The allele frequency *p* was estimated from the data, so we deduct one degree of freedom, which leaves one degree of freedom in the final X^2 test. (No deduction is made for q, because $q = 1 - p$.)

To verify that the X^2 test as employed is not circular, consider the following observed numbers of genotypes in the flower *Phlox cuspidata* for a gene coding for phosphoglucomutase: $Pgm^a\ Pgm^a$, 15; $Pgm^a\ Pgm^b$, 6; and $Pgm^b\ Pgm^b$, 14. You should verify for yourself that the allele frequencies of Pgm^a and Pgm^b are 0.51 and 0.49, respectively, and that the expected numbers of the genotypes with random mating are 9.1, 17.5, and 8.4, respectively. The X^2 value is 15.1, which has one degree of freedom because there are three classes of data (the observed number of each of the three genotypes) and one parameter (the allele frequency of either *Pgm*^{a} or *Pgm*^{b}) had to be estimated from the data. The corresponding probability is about 0.0002. The probability associated with the X^2 value is less than 0.05 (in this case, very much less), so these data must be judged significantly different from the Hardy-Weinberg expectations. The reason, as we will soon see, is inbreeding: In this plant, some of the ovules are fertilized by self-fertilization.

Frequency of Heterozygotes

Another important implication of the Hardy-Weinberg principle is that *for a rare allele, the frequency of heterozygotes far exceeds the frequency of the rare homozygote.* For example, when the frequency of the rarer allele is $q = 0.1$, the ratio of heterozygotes to homozygotes equals

$$2pq/q^2 = 2p/q = 2(0.9)/(0.1)$$

or approximately 20; when $q = 0.01$, this ratio is about 200; and when $q = 0.001$, it is about 2000. In other words,

> When an allele is rare, there are many more heterozygotes than there are homozygotes for the rare allele.

The reason for this perhaps unexpected relationship is shown in Figure 15.6, which plots the frequencies of homozygotes and heterozygotes with random mating. Note that at allele frequencies near 0 or 1, the frequency of the heterozygous genotype goes to 0 as a linear function, whereas that of the rarer, homozygous genotype goes to 0 more rapidly.

One practical implication of this principle is seen in the example of cystic fibrosis, an inherited secretory disorder of the pancreas and lungs. Cystic fibrosis is one of the

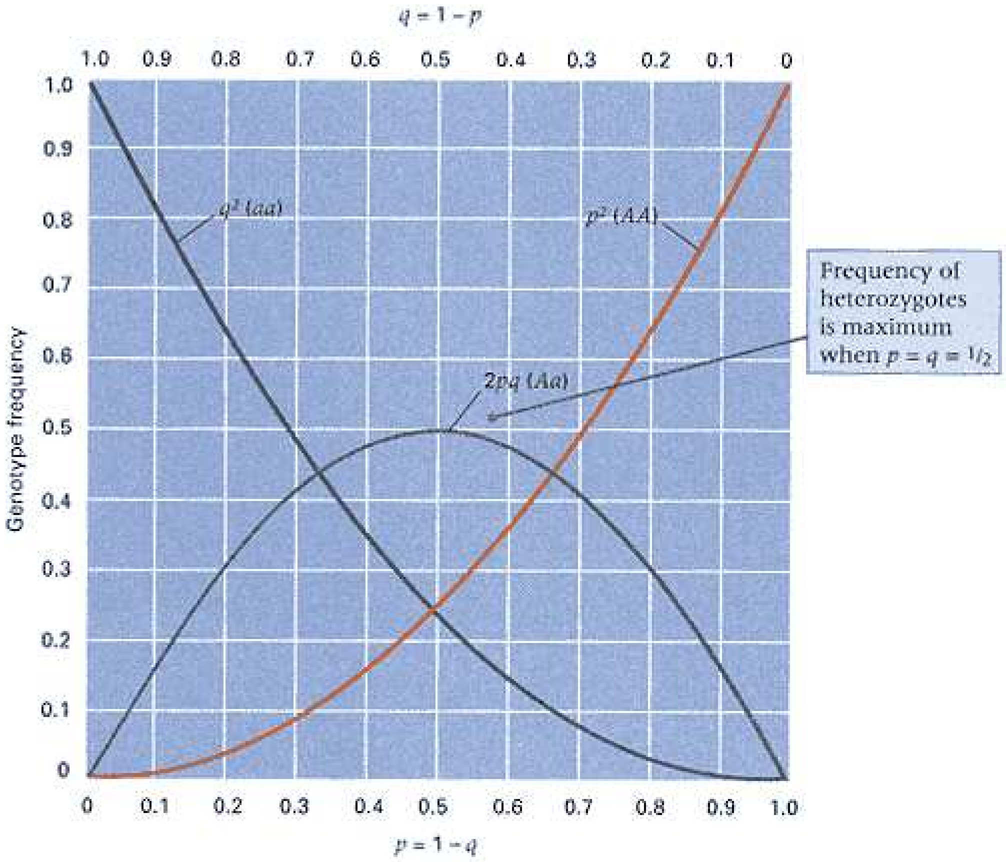


Figure 15.6
Graphs of p^2,2pq, and q^2. If the allele frequencies are between 1/3 and 2/3, then the hetero zygote is the genotype with the highest frequency.

most common recessively inherited severe disorders among Caucasians; it affects about 1 in 1700 newborns. In this case, the heterozygotes cannot readily be identified by phenotype, so a method of calculating allele frequencies that is different from the gene-counting method used earlier must be employed. This method is straightforward because, with random mating, the frequency of recessive homozygotes must correspond to q^2. In the case of cystic fibrosis,

$$q^2 = 1/1700 = 0.00059 \text{ or}$$
$$q = (0.00059)^{1/2} = 0.024$$

and consequently,

$$p = 1 - q = 1 - 0.024 = 0.976$$

The frequency of heterozygotes that carry the allele for cystic fibrosis is calculated as

$$2pq = 2(0.976)(0.024) = 0.047 = 1/21$$

This calculation implies that for cystic fibrosis, although only 1 person in 1700 is affected with the disease (homozygous), about 1 person in 21 is a carrier (heterozygous). The calculation should be regarded as approximate because it is based on the assumption of Hardy-Weinberg genotype frequencies. Nevertheless, considerations like these are important in predicting the outcome of population screening for the

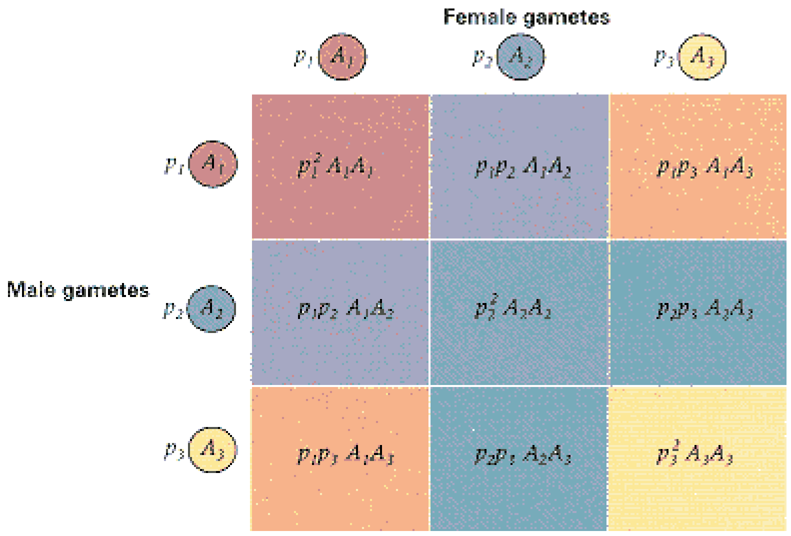


Figure 15.7
Punnett square showing the results of randommating with three alleles.

detection of carriers of harmful recessive alleles, which is essential in evaluating the potential benefits of such screening programs.

Multiple Alleles

Extension of the Hardy-Weinberg principle to multiple alleles of a single autosomal gene can be illustrated by a three-allele case. Figure 15.7 shows the results of random mating in which three alleles are considered. The alleles are designated A_1, A_2, and A_3, where the uppercase letter represents the gene and the subscript designates the particular allele. The allele frequencies are p_1, p_2, and p_3, respectively. With three alleles (as with any number of alleles), the allele frequencies of all alleles must sum to 1; in this case, $p_1 + p_2 + p_3$, = 1.0. As in Figure 15.5, the entry in each square is obtained by multiplying the frequencies of the alleles at the corresponding margins; any homozygote (such as A_1A_1) has a random-mating frequency equal to the square of the corresponding allele frequency (in this case, p_1^2). Any heterozygote (such as A_1A_2) has a random-mating frequency equal to twice the product of the corresponding allele frequencies (in this case, $2p_1p_2$). The extension to any number of alleles is straightforward:

Frequency of any homozygote =
square of allele frequency

Frequency of any heterozygote =
2 × product of allele frequencies (2)

Multiple alleles determine the human ABO blood groups (Chapter 2). The gene has three principal alleles, designated I^A, I^B, and I^o. In one study of 3977 Swiss people, the allele frequencies were found to be 0.27 I^A, 0.06 I^B, and 0.67 I^o. Applying the rules for multiple alleles, we can expect the genotype frequencies resulting from random mating to be

$$\left.\begin{aligned} I^AI^A &= (0.27)^2 = 0.0729 \\ I^AI^O &= 2(0.27)(0.67) = 0.3618 \end{aligned}\right\} \text{Type A} = 0.4347$$

$$\left.\begin{aligned} I^BI^B &= (0.06)^2 = 0.0036 \\ I^BI^O &= 2(0.06)(0.67) = 0.0804 \end{aligned}\right\} \text{Type B} = 0.0840$$

$$I^OI^O = (0.67)^2 = 0.4489 \quad \text{Type O} = 0.4489$$

$$I^AI^B = 2(0.27)(0.06) = 0.0324 \quad \text{Type AB} = 0.0324$$

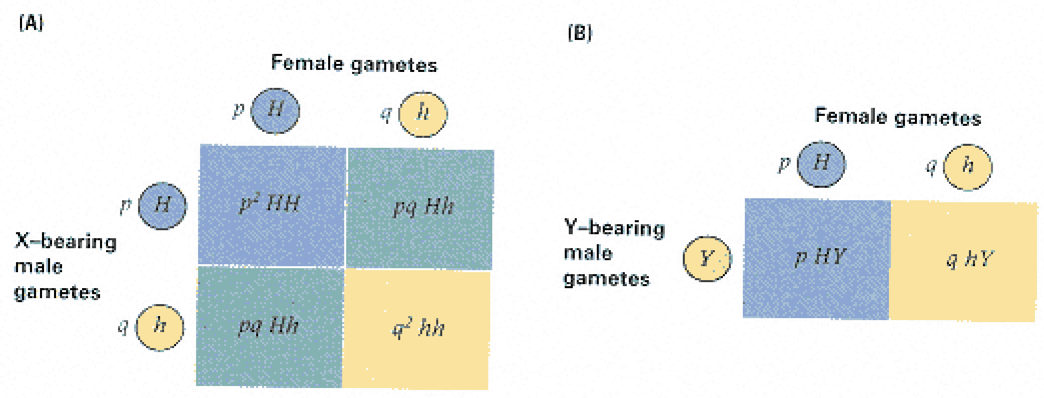


Figure 15.8
The results of random mating for an X-linked gene. (A) Genotype frequencies in females. (B) Genotype frequencies in males.

Because both I^A and I^B are dominant to I^o, the expected frequency of blood-group *phenotypes* is that shown at the right. Note that the majority of A and B phenotypes are heterozygous for the I^o allele.

X-Linked Genes

Random mating for two X-linked alleles (*H* and *h*) is illustrated in Figure 15.8. The principles are the same as those considered earlier, but male gametes carrying the X chromosome (Figure 15.8A) must be distinguished from those carrying the Y chromosome (Figure 15.8B). When the male gamete carries an X chromosome, the Punnett square is exactly the same as that for the two-allele autosomal gene in Figure 15.5. However, because the male gamete carries an X chromosome, all the offspring in question are female. Consequently, among females, the genotype frequencies are

$$HH: p^2 \quad Hh: 2pq \quad hh: q^2$$

When the male gamete carries a Y chromosome, the outcome is quite different (Figure 15.8B). All the offspring are male, and each has only one X chromosome, which is inherited from the mother. Therefore, each male receives only one copy of each X-linked gene, and the genotype frequencies among males are the same as the allele frequencies: *H* males with frequency *p*, and *h* males with frequency *q*.

An important implication of Figure 15.8 is that if *h* is a rare recessive allele, then many more males will exhibit the trait than females because the frequency of affected females (q^2) will be much smaller than the frequency of affected males (*q*). For example, the common form of X-linked color blindness in human beings affects about 1 in 20 males among Caucasians, so $q = 1/20 = 0.05$. The expected frequency of color-blind females is calculated as $q^2 = (0.05)^2 = 0.0025$, or about 1 in 400.

15.3— DNA Typing and Population Substructure

In principle, no two human beings are genetically identical. The only exceptions are identical twins, identical triplets, and so on. Each human genotype is unique because so many genes in human populations are polymorphic. One practical application of the theoretical principle of genetic uniqueness has come through the study of DNA polymorphisms. Small samples of human material from an unknown person (for example, material left at the scene of a crime) often contain enough DNA that the genotype can be determined for a number of polymorphisms and matched against those present among a group of suspects. Typical examples of crime-scene evidence include blood, semen, hair roots, and skin cells. Even a small number of cells is sufficient, because predetermined regions of DNA can be amplified by the polymerase chain reaction (Chapter 5).

If a suspect's DNA contains sequences that are clearly not present in the crime-scene sample, then the sample must have originated from a different person. On the other hand, if a suspect's DNA *does* match that of the crime-scene sample, then the suspect could be the source. The strength of the DNA evidence depends on the number of polymorphisms that are examined and on the number of alleles present in the population. The greater the number of polymorphisms that match, especially if they are highly polymorphic, the stronger the evidence linking the suspect to the sample taken from the scene of the crime. The use of polymorphisms in DNA to link suspects with samples of human material is called **DNA typing.** Although there is some controversy over certain technical aspects of the procedures, DNA typing is generally regarded as the most important innovation in criminal investigation since the development of fingerprinting more than a century ago. One of the controversies involves the application of population genetics to evaluate the significance of a matching DNA type.

Figure 15.9 illustrates the type of polymorphism commonly used in DNA typing. The restriction fragments that correspond to each allele differ in length because they contain different numbers of units repeated in tandem. For this reason, each polymorphism of this type is called a *VNTR,* which stands for *variable number of tandem repeats* (Section 4.4). VNTRs are of value in DNA typing because many alleles are possible, owing to the variable number of repeating units. Although many alleles may be present in the population as a whole, each person can have no more than two alleles for each VNTR polymorphism. An example

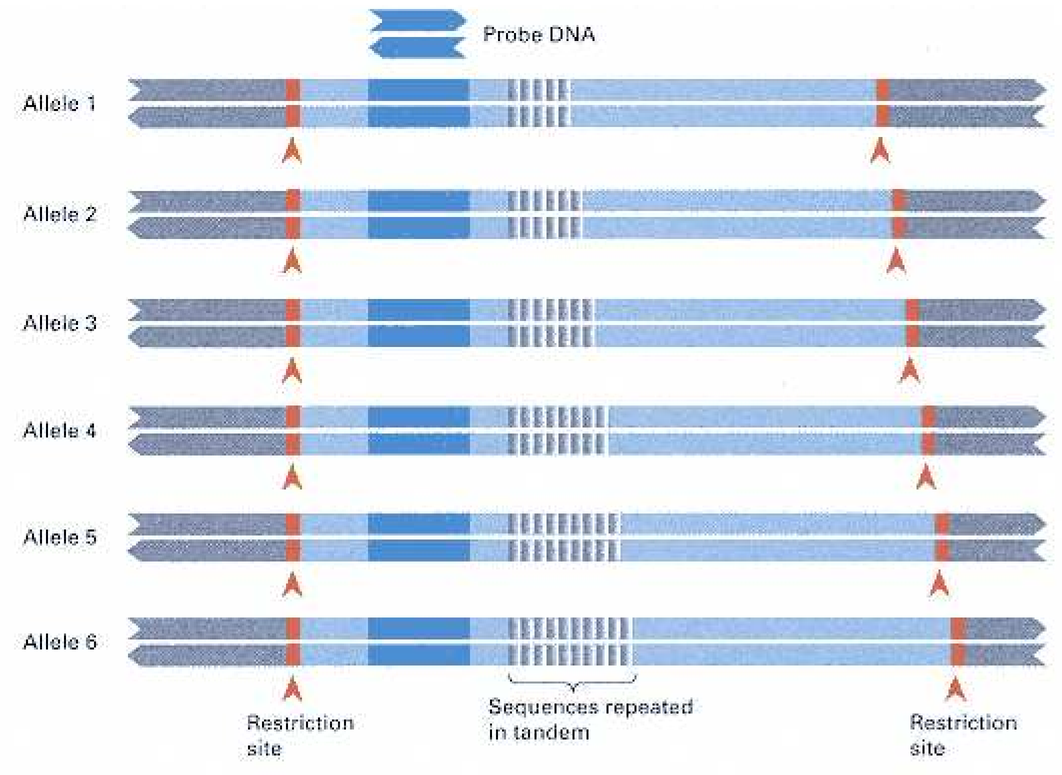


Figure 15.9
Allelic variation resulting from a variable number of units repeated in tandem in a nonessential region of a gene. The probe DNA detects a restriction fragment for each allele. The length of the fragment depends on the number of repeating units present.

Connection Be Ye Son or Nephew?

Alec J. Jeffreys, John F. Y. Brookfield, and Robert Semeonoff 1985

University of Leicester, Leicester, England
Positive Identification of an Immigration Test-Case Using Human DNA Fingerprints

One of the first practical applications of DNA typing was this immigration case. A boy from Ghana was refused permission to rejoin his mother in the United Kingdom because it was not clear whether he was the woman's son or nephew. One difficulty in the case was that the father was in Ghana and unavailable for DNA typing. As you will see, there was enough genetic information available from the mother's other children to infer the genotype of the father and to reach a conclusion about the parentage of the boy. The term "minisatellites" used in the excerpt refers to types of simple tandem repeat polymorphisms that are abundant in the human genome and highly variable ("hypervariable") from one person to the next. These features make minisatellites suitable for use in DNA typing.

The human genome contains a set of minisatellites, each of which consists of tandem repeats of a DNA segment . . .. [Analysis of] many hypervariable minisatellites simultaneously produces DNA fingerprints . . . suitable for individual identification in, for example, forensic science and paternity testing . . .. They can also be used to resolve immigration disputes arising from lack of proof of family relationships . . .. The case in question concerned a Ghanian boy born in the United Kingdom who emigrated to Ghana to join his father and subsequently returned to the United Kingdom to be reunited with his mother, brother and two sisters. However,

At the request of the family's solicitor, we therefore carried out a DNA fingerprint analysis to determine the maternity of the boy

there was evidence to suggest that a substitution might have occurred, either for an unrelated boy, or for a son of a sister of the mother; she has several sisters, all of whom live in Ghana. As a result the returning boy was not granted residence in the United Kingdom . . ..At the request of the family's solicitor, we therefore carried out a DNA fingerprint analysis to determine the maternity of the boy . . ..DNA fingerprints from blood DNA samples were taken from the mother M, brother B, sisters S1 and S2 and the boy X in dispute . . ..Although the father was unavailable, most of his DNA fingerprint could be reconstructed from paternal-specific DNA fragments present in at least one of B, S1 or S2 but absent from M. Of 39 paternal fragments so identified, approximately half were present in the DNA fingerprint of X. Since DNA fragments are seldom shared between unrelated individuals, this suggests very strongly that X has the same father as B, S1 and S2 . . .. There remained 40 DNA fragments in X, all of which were present in M. This in turn provides strong evidence that M is the mother of X, and therefore that X, B, S1 and S2 are true sibs . . .. How certain is this identification? . . . The odds that M is a sister of X's true mother but by chance contains all of X's maternal-specific bands are 6×10^{-6}. We therefore conclude that, beyond any reasonable doubt, M must be the true mother of X. This evidence . . . was provided to the immigration authorities, who dropped the case against X and granted him residence in the United Kingdom.

Source: Nature 317:818–819

of a VNTR used in DNA typing is shown in Figure 15.10. The lanes in the gel labeled M contain multiple DNA fragments of known size to serve as molecular-weight markers. Each numbered lane 1–9 contains DNA from a single person. Two typical features of VNTRs are to be noted:

1. Most people are heterozygous for VNTR alleles that produce restriction fragments of different sizes. Heterozygosity is indicated by the presence of two distinct bands. In Figure 15.10, only the person numbered 1 appears to be homozygous for a particular allele.

2. The restriction fragments from different people cover a wide range of sizes. The variability in size indicates that the population as a whole contains many VNTR alleles.

The reason why VNTRs are useful in DNA typing is also evident in Figure 15.10: Each of the nine people tested has a different pattern of bands and thus could be identified uniquely by means of this VNTR. On the other hand, the uniqueness of each person in Figure 15.10 is due in part to the high degree of polymorphism of the VNTR and in part to the small sample size. If more people were examined, then pairs that matched in their VNTR types by chance would certainly be found. Table 15.2 shows the frequencies of alleles for the VNTR designated *D1S80* in major population groups

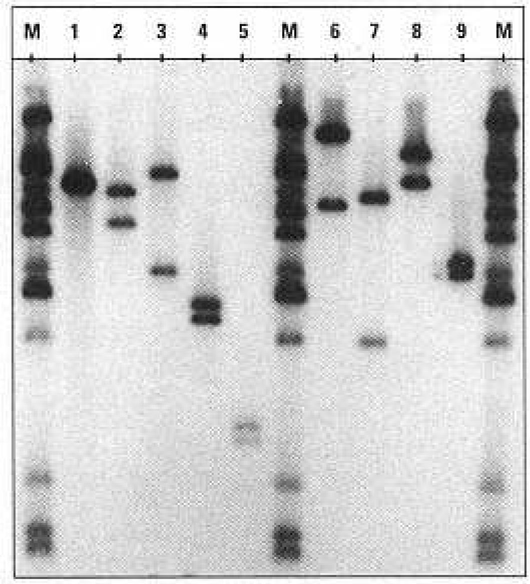


Figure 15.10
Genetic variation in a VNTR used in DNA typing. Each numbered lane contains DNA from a single person; the DNA has been cleaved with a restriction enzyme, separated by electrophoresis, and hybridized with radioactive probe DNA. The lanes labeled M contain molecular-weight markers.
[Courtesy of R. W. Allen.]

in the United States. The allele with 24 repeats has a frequency of 0.378 in Caucasians, so with random mating, more that 14 percent of the population would be homozygous for this allele. The same allele also occurs quite frequently in the other population groups. Because of the possibility of chance matches between VNTR types, applications of DNA typing are usually based on at least three—and preferably more—highly polymorphic loci.

Suppose a person is found whose DNA type for three VNTR loci matches that of a sample from the scene of a crime. How is the significance of this match to be evaluated? The significance of the match depends on the likelihood of its happening by chance, and hence matches of rare DNA types are more significant than matches of common DNA types. If random mating proportions can be assumed, and it is also assumed that each VNTR is independent of the others, then it is reasonable to use Equation (2) to calculate the expected frequency of the particular genotype for each VNTR and then to multiply the expected frequencies across each VNTR in order to calculate the expected frequency of the multilocus genotype. The factor p^2 for homozygous genotypes is usually modified, because when a single band is observed (as in person 1 in Figure 15.10), it is uncertain whether the genotype is really homozygous or is heterozygous with one DNA band not observed (perhaps because of running off the end of the electrophoresis gel). Thus the factor for single-band patterns is taken as $2p$ rather than the p^2 in Equation (2) because $2p$ is always greater than $p^2$for VNTR alleles.

With the assumptions of random mating and independence of VNTRs, if a DNA match included the allele A_i of VNTR *A* (a single-banded pattern); alleles B_j and B_k of VNTR *B;* and alleles C_l and C_m of VNTR *C*, then the expected frequency of the DNA type in the population as a whole would be calculated as

$$2 \times \text{Freq}(A_i) \times 2 \times \text{Freq}(B_j) \times \text{Freq}(B_k) \times 2 \times \text{Freq}(C_l) \times \text{Freq}(C_m) \quad (3)$$

where Freq(A_i) is the frequency of the allele A_i, Freq(B_j) is that of allele B_j, and so on. Because human populations can differ in their allele frequencies, the calculation would be carried out using allele frequencies among Caucasians for white suspects, using those among Blacks for black suspects, and using those among Hispanics for Hispanic suspects.

Differences among Populations

Equation (3) makes a number of assumptions about human populations: (1) that the Hardy-Weinberg principle holds for each locus, (2) that each locus is statistically independent of the others so that the multiplication across loci is justified, and (3) that the only level of population substructure that is important is that of race. The term **population substructure** refers to the extent to which a larger population is subdivided into smaller subpopulations that may differ in allele frequencies from one subpopulation to the next.

Critics of the method of calculation are worried about differences in the frequencies of VNTR alleles among different

Table 15.2 Allele frequencies for *D1S80* among U.S. population groups

Repeat number	Caucasian	Hispanic	African American	Asian
14	0	0	0	0
15	0	0.001	0	0
16	0.001	0.010	0.002	0.034
17	0.002	0.009	0.028	0.025
18	0.237	0.224	0.073	0.152
19	0.003	0.005	0.003	0.022
20	0.018	0.013	0.032	0.007
21	0.021	0.028	0.115	0.034
22	0.038	0.024	0.081	0.017
23	0.012	0.009	0.014	0.017
24	0.378	0.315	0.234	0.230
25	0.046	0.072	0.045	0.027
26	0.020	0.007	0.006	0
27	0.007	0.016	0.008	0.047
28	0.063	0.078	0.130	0.076
29	0.052	0.055	0.053	0.042
30	0.008	0.039	0.009	0.123
31	0.072	0.053	0.054	0.093
32	0.006	0.005	0.007	0.012
33	0.003	0.004	0.004	0.005
34	0.001	0.006	0.086	0.005
35	0.003	0	0.002	0.005
36	0.004	0.011	0.001	0.005
37	0.001	0.004	0	0.007
38	0	0	0	0
39	0.003	0.004	0.003	0.005
40	0	0	0	0
41	0	0.002	0.002	0.007
>41	0.001	0.006	0.007	0.002
Sample size	718	409	606	204

Source: Data from B. Budowle, Et al. 1995. *Journal of Forensic Science* 40:38

human subpopulations. For example, in Figure 15.11, each bar shows the range in frequency (minimum to maximum) of each allele of *D1S80* observed among approximately 30 subpopulations. For some alleles, the range is very great; for example, alleles with 21, 30, 31, or 34 repeats are very rare in some subpopulations but relatively common in others. Even the common alleles, with 18 and 24 repeats, can vary substantially in frequency from one subpopulation to the next. Critics

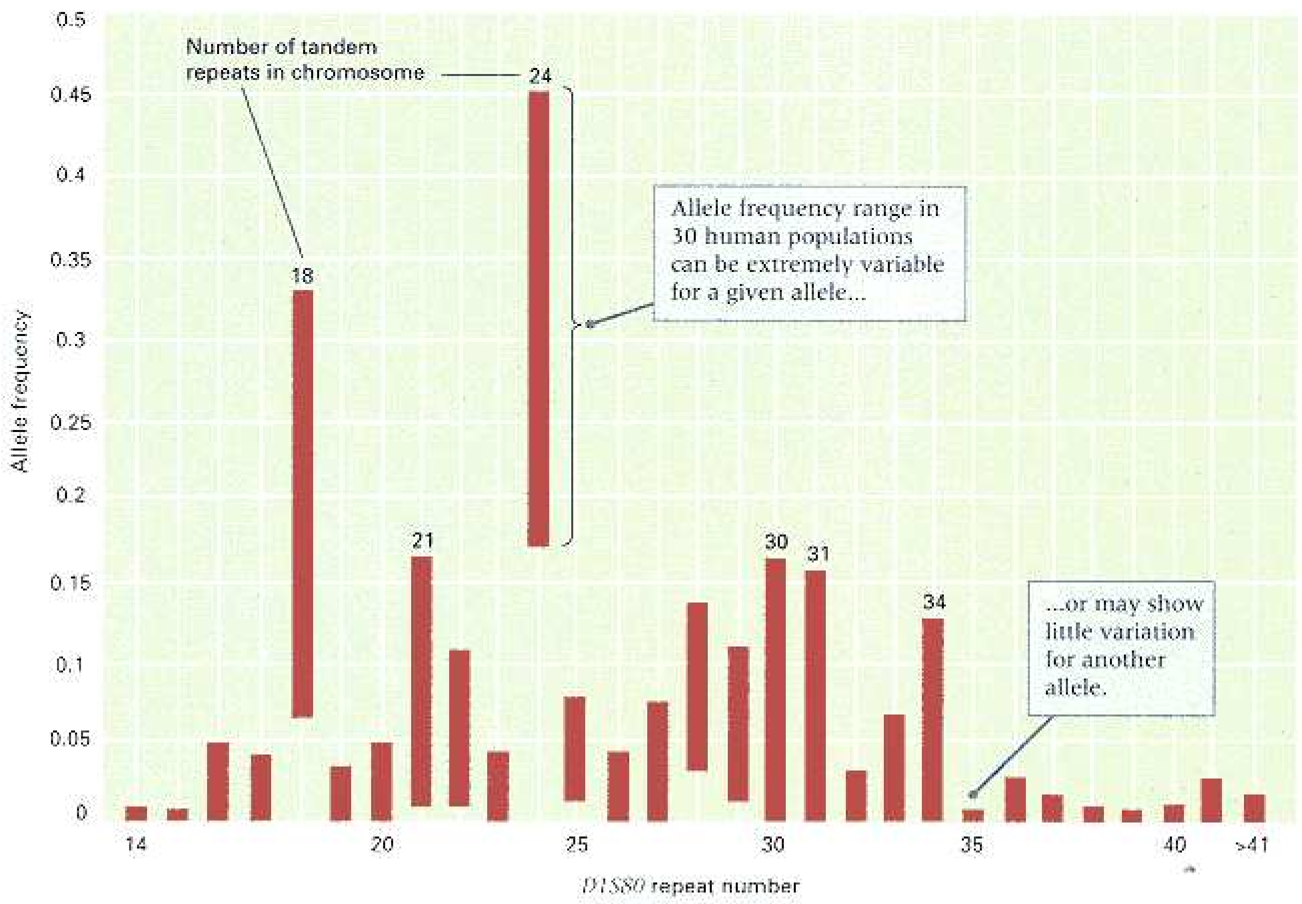


Figure 15.11
Range of allele frequencies found among human subpopulations for the VNTR D1S80.
[Data from B. Budowle et al. J. *Forensic Science* 1995. 40:38.]

argue that DNA matches across multiple VNTRs may be much more common among persons within a particular ethnic group than among persons drawn at random from the population as a whole, and so calculations of genotype frequency should be based on the ethnic group of the accused person, not on the race as a whole. Others argue that population substructure has a relatively minor effect on the final outcome of the calculation and that what matters most is not a high degree of accuracy but rather a general sense of whether a particular multilocus genotype is rare or common. The underlying issue is very serious: how to maintain a proper balance between the need to protect the rights of accused persons and the desire to get rapists and murderers off the streets.

DNA Exclusions

Issues of population genetics apply only when DNA types match. DNA types that do *not* match can exclude innocent suspects irrespective of any considerations about population substructure. For example, if the DNA type of a suspected rapist does not match the DNA type of semen taken from the victim, then the suspect could not be the perpetrator. Another example, which illustrates the use of DNA typing in paternity testing, is given in Figure 15.12. The numbers 1 and 2 designate different cases, in each of which a man was accused of fathering a child. In each case, DNA was obtained from the mother (M), the child (C), and the accused man (A). The pattern of bands obtained for one VNTR is shown.

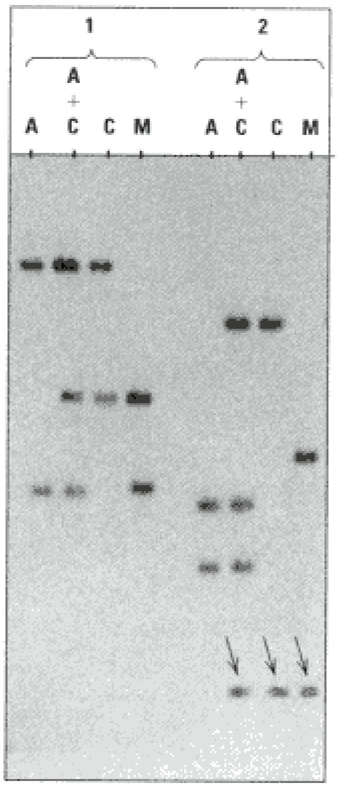


Figure 15.12
Use of DNA typing in paternity testing. The sets of lanes numbered 1 and 2 contain DNA samples from two different paternity cases. In each case, the lanes contain DNA fragments from the following sources: M, the mother; C, the child; A, the accused father. The lanes labeled A + C contain a mixture of DNA fragments from the accused father and the child. The arrows in case 2 point to bands of the same size present in lanes M, C, and A + C. Note that the accused male in case 2 could not be the father because neither of his bands is shared with the child. [Courtesy of R. W. Allen.]

The lanes labeled A + C contain a mixture of DNA from the accused man and the child. In case 1, the lower band in the child was inherited from the mother and the upper band from the father; because the upper band is the same size as one of those in the accused, the accused man could have contributed this allele to the child. This finding does not prove that the accused man is the father; it says only that he cannot be excluded on the basis of this particular gene. (On the other hand, if a large enough number of genes are studied, and the man cannot be excluded by any of the genes, then it makes it more likely that he really is the father.) Case 2 in Figure 15.12 is an exclusion. The band denoted by the arrows is the band inherited by the child from the mother. The other band in the child does not match either of the bands in the accused, so the accused man could not be the biological father. (In theory, mutation could be invoked to explain the result, but this is extremely unlikely.)

15.4— Inbreeding

Inbreeding means mating between relatives, such as first cousins. The principal consequence of inbreeding is that the frequency of heterozygous offspring is smaller than it is with random mating. This effect is seen most dramatically in repeated self-fertilization, which happens naturally in certain plants. Consider a hypothetical population consisting exclusively of *Aa* heterozygotes. With self-fertilization, each plant would produce offspring in the proportions 1/4 *AA*, 1/2 *Aa*, and 1/4 *aa*, which means that one generation of self-fertilization reduces the proportion of heterozygotes from 1 to 1/2. In the second generation, only the heterozygous plants can again produce heterozygous offspring, and only half of their offspring will again be heterozygous. Heterozygosity is therefore reduced to 1/4 of what it was originally. Three generations of self-fertilization reduce the heterozygosity to

1/4 × 1/2 = 1/8, and so on. The remainder of this section demonstrates how the reduction in heterozygosity because of inbreeding can be expressed in quantitative terms.

The Inbreeding Coefficient

Repeated self-fertilization is a particularly intense form of inbreeding, but weaker forms of inbreeding are qualitatively similar in that they also lead to a reduction in heterozygosity. A convenient measure of the effect of inbreeding is based on the reduction in heterozygosity. Suppose that H_I is the frequency of heterozygous genotypes in a population of inbred individuals. The most widely used measure of inbreeding is called the **inbreeding coefficient,** symbolized F, which is defined as the proportionate reduction in H_I compared with the value of $2pq$ that would be expected with random mating; in symbols,

$$F = (2pq - H_I)/2pq \qquad (4)$$

Equation (4) can be rearranged as

$$H_I = 2pq(1 - F)$$

The homozygous genotypes increase in frequency as the heterozygotes decrease in frequency, and overall, the genotype frequencies in the inbred population are

$$\begin{aligned} &AA\text{:} \quad p^2(1 - F) + pF \\ &Aa\text{:} \quad 2pq(1 - F) \\ &aa\text{:} \quad q^2(1 - F) + qF \qquad (5) \end{aligned}$$

Equation (5) is a modification of the Hardy-Weinberg principle made to take inbreeding into account. When $F = 0$ (no inbreeding), the genotype frequencies are the same as those given in the Hardy-Weinberg principle in Equation (1): p^2, $2pq$, and q^2. At the other extreme, when $F = 1$ (complete inbreeding), the inbred population consists entirely of *AA* and *aa* genotypes in the frequencies p and q, respectively.

As an illustration of the use of Equation (5), consider again the *Pgm* gene in *Phlox cuspidata*, the genotype frequencies of which are not at all in agreement with the Hardy-Weinberg principle (Section 15.2). In this species, about 78 percent of the plants undergo self-fertilization. It can be shown that the inbreeding coefficient corresponding to this amount of self-fertilization is $F = 0.64$, and the expected genotype frequencies must be modified to take this inbreeding into account. The frequencies of alleles Pgm^a and Pgm^b are 0.51 and 0.49, respectively, so by Equation (5) for inbreeding, the expected genotype frequencies are

$$\begin{aligned} &Pgm^a\,Pgm^a\text{:} \quad (0.51)^2(1 - 0.64) + (0.51)(0.64) = 0.420 \\ &Pgm^a\,Pgm^b\text{:} \quad 2(0.51)(0.49)(1 - 0.64) = 0.180 \\ &Pgm^b\,Pgm^b\text{:} \quad (0.49)^2(1 - 0.64) + (0.49)(0.64) = 0.400 \end{aligned}$$

In a sample size of 35, the observed and expected numbers of the three genotypes were

$$\begin{aligned} &Pgm^a\,Pgm^a\text{:} \quad 15, \text{ versus } 0.420 \times 35 = 14.7 \\ &Pgm^a\,Pgm^b\text{:} \quad 6, \text{ versus } 0.180 \times 35 = 6.3 \\ &Pgm^b\,Pgm^b\text{:} \quad 14, \text{ versus } 0.400 \times 35 = 14 \end{aligned}$$

In this example, the agreement with the inbreeding model is excellent.

Allelic Identity by Descent

The approach to inbreeding based on genotype frequencies is useful for considering natural populations. However, in human genetics, as well as in animal and plant breeding, a geneticist may wish to calculate the inbreeding coefficient of a particular individual whose pedigree is known in order to determine the probability of each possible genotype for the individual. This section introduces a concept that is particularly suited to pedigree calculations.

The diagnostic criterion of inbreeding is a closed loop in the pedigree. Such a closed loop is often called a **path.** Figure 15.13A is an inbreeding pedigree in which individual I is the product of a mating between half sibs, and the path (closed loop) from I through the common ancestor A is evident. A streamlined version of this pedigree useful for inbreeding calculations is depicted in Figure 15.13B; individuals of both sexes are shown as circles, diagonal lines trace alleles transmitted from parent to offspring, and ancestors that did not contribute to the inbreeding of I (ancestors B and C) are left out altogether. The contrasting dots inside the circles represent particular alleles present in the corresponding individuals, and the depicted pattern of inheritance of the alleles represents the principal consequence of inbreeding and illustrates why closed loops are important. Specifically, individuals D and E both received the blue allele from A and received a white allele from B and C, respectively. The blue alleles in D and E have a very special relationship; they both originate by replication of the blue allele in A and therefore are said to be **identical by descent.** In the interval of the relatively few generations involved in most pedigrees, alleles that are identical by descent can be assumed to be identical in nucleotide sequence, because mutation can usually be ignored in so few generations. Individual I is shown as having received the blue allele from both D and E, and because the alleles in question are identical by descent, this individual must be homozygous for the allele inherited from the common ancestor. Without the closed loop in the pedigree, identity by descent of the alleles in I would be much less probable. The concept of identity by descent provides an alternative definition of the inbreeding coefficient, which can be shown to be completely equivalent to the definition in terms of heterozygosity given earlier. Specifically,

> The inbreeding coefficient of an individual is the probability that the individual carries alleles that are identical by descent.

Calculation of the Inbreeding Coefficient from Pedigrees

Figure 15.13C illustrates how the probability of identity by descent can be calculated in a pedigree, and the alleles are drawn as dots on the diagonal lines to emphasize their pattern of transmission. The double-headed arrows indicate which pairs of alleles must be identical by descent in order for I to have alleles that are identical by descent. The values of 1/2 represent the probability that a diploid genotype will transmit the allele inherited from a particular parent; this value must be 1/2 because of Mendelian segregation. The term(1/2) $(1 + F_A)$ can be explained by

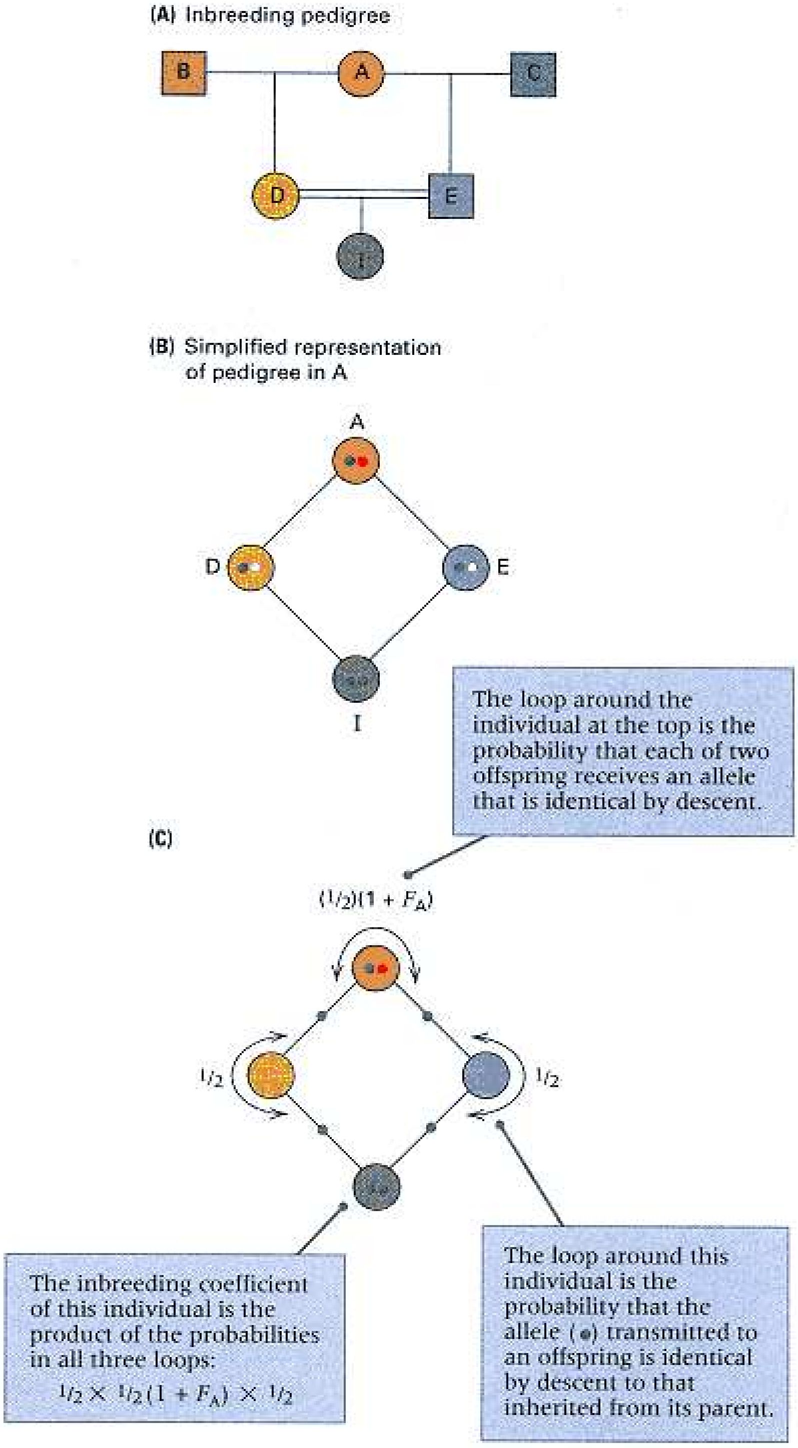


Figure 15.13
(A) An inbreeding pedigree in which individual I results from a mating between half siblings. (B) The same pedigree redrawn to show gametes in the closed loop leading to individual I. The colored dots represent alleles of a particular gene. (C) Calculation of the probability that the alleles indicated are identical by descent.

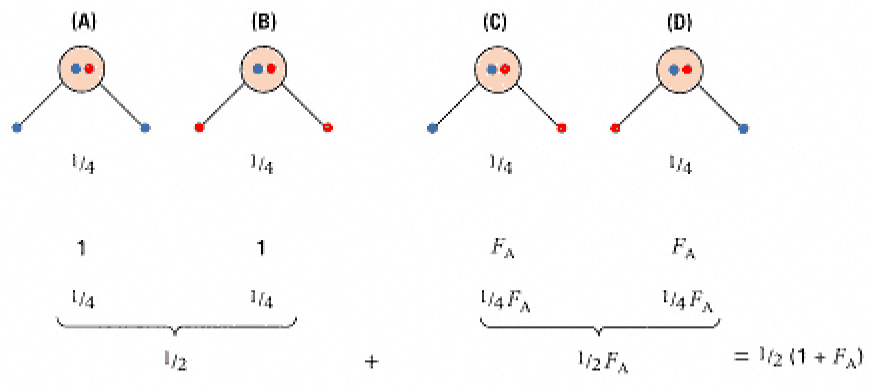


Figure 15.14
A diagram illustrating the reason why (1/2) (1 + F_A) is the probability that two different gametes from the same individual. A, carry alleles that are identical by descent.

looking at the situation in Figure 15.14, which shows four possible pairs of gametes that A can transmit, each equally likely. In cases A and B, the transmitted alleles are certainly identical by descent because they are products of DNA replication of the same allele (blue or red). In cases C and D, the transmitted alleles have contrasting colors, but they can still be identical by descent if A is already inbred. However, the probability that A carries alleles that are identical by descent is, by definition, the inbreeding coefficient of A, and this is designated F_A. Altogether, the probability that A transmits two alleles that are identical by descent is the sum of the probabilities of the four eventualities in Figure 15.14, and this sum equals (1/2) (1 + F_A). Matters are somewhat simpler when A is not itself inbred, because in this case F = 0, and the corresponding probability is simply 1/2.

Returning now to Figure 15.13C, the alleles in I can be identical by descent only if all three identities indicated by the loops are true, and this probability is the product of the three numbers shown. That is, the inbreeding coefficient of I, symbolized F_I, is

$$\begin{aligned} F_I &= (1/2) \times (1/2)(1 + F_A) \times (1/2) \\ &= (1/2)^3(1 + F_A) \end{aligned}$$

For a more remote common ancestor, the formula becomes

$$F_I = (1/2)^n(1 + F_A) \qquad (6)$$

in which n is the number of individuals in the path traced from one of the parents of I back through the common ancestor and forward again to the other parent of I.

Most pedigrees of interest are more complex than the one in Figure 15.13, but the inbreeding coefficient of any individual can still be calculated by using Equation (6). Application to complex pedigrees requires several additional rules:

1. If there are several distinct paths that can be traced through a common ancestor, then each path provides a contribution to the inbreeding coefficient calculated as in Equation (6).

2. If there are several common ancestors, then each is considered in turn in order to calculate its separate contribution to the inbreeding.

3. The overall inbreeding coefficient is then calculated as the sum of all these separate contributions of the common ancestors.

The extension of Equation (6) to complex pedigrees may be written as

$$F_I = \Sigma(1/2)^n(1 + F_A) \qquad (7)$$

in which σ denotes the summation of all paths through all common ancestors.

Application of Equation (7) can be illustrated in the pedigree in Figure 15.15. Each path and its contribution to the inbreeding coefficient of I is shown below (the red letter indicates the common ancestor).

E**B**D	$(1/2)^3(1 + F_B)$
E**C**D	$(1/2)^3(1 + F_C)$
EC**A**BD	$(1/2)^5(1 + F_A)$
EB**A**CD	$(1/2)^5(1 + F_A)$

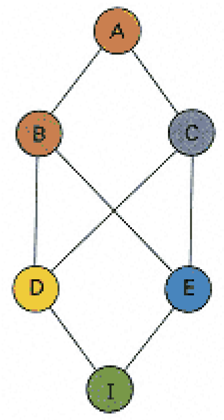


Figure 15.15
A complex pedigree in which the inbreeding of individual I results from several independent paths through three different common ancestors.

The inbreeding coefficient of I is the sum of these numbers. If none of the common ancestors is inbred (that is, if $F_A = F_B = F_C = 0$), then

$$F_I = (1/2)^3 + (1/2)^3 + (1/2)^5 + (1/2)^5 = 5/16$$

The value 5/16 is to be interpreted to mean that the probability that individual I will be heterozygous at any particular locus is smaller, by the proportion 5/16, than the probability that a noninbred individual will be heterozygous at the same locus. Equivalently, the 5/16 could be interpreted to mean that individual I is heterozygous at 5/16 fewer loci in the whole genome than a noninbred individual.

Effects of Inbreeding

The effects of inbreeding differ according to the normal mating system of an organism. At one extreme, in regularly self-fertilizing plants, inbreeding is already so intense and the organisms are so highly homozygous that additional inbreeding has virtually no effect. However,

> In most species, inbreeding is harmful, and much of the effect is due to rare recessive alleles that would not otherwise become homozygous.

Among human beings, inbreeding is usually uncommon because of social conventions and laws, although in small isolated populations (isolated villages, religious communities, aboriginal groups) it does occur, mainly through matings between relatives more distant than second cousins ("remote relatives"). The most common type of close inbreeding is between first cousins. The effect is always an increase in the frequency of genotypes that are homozygous for rare, usually harmful recessives. For example, among American whites, the frequency of albinism among offspring of matings between nonrelatives is approximately 1 in 20,000; among offspring of first-cousin matings, the frequency is approximately 1 in 2000. The reason for the increased risk may be understood by comparing the genotype frequencies of homozygous recessives in Equation (5) for inbreeding and Equation (1) for random mating. Among the offspring of a mating between first cousins, $F = 1/16$, or 0.062. Therefore, the frequency of homozygous recessives produced by first-cousin mating will be

$$q^2(1 - 0.062) + q(0.062)$$

whereas among the offspring of nonrelatives, the frequency of homozygous recessives is q^2. For albinism, $q = 0.007$ (approximately), and the calculated frequencies are 5×10^{-4} for the offspring of first cousins and 5×10^{-5} for the offspring of nonrelatives. The increased frequency with first-cousin mating is the principal consequence of inbreeding.

15.5— Genetics and Evolution

Evolution refers to changes in the gene pool resulting in the progressive adaptation of populations to their environment. Evolution occurs because genetic variation exists in populations and because there is a natural selection favoring organisms that are best adapted to the environment. Genetic variation and natural selection are population phenomena, so they are conveniently discussed in terms of allele frequencies.

Four processes account for most of the changes in allele frequency in populations. They form the basis of cumulative change in the genetic characteristics of populations, leading to the descent with modification that characterizes the process of evolution. Although the point has yet to be proved, most evolutionary biologists believe that these same processes, when carried out continuously over geological time, can also account for the formation of new species and higher taxonomic categories. These process are

1. Mutation, the origin of new genetic capabilities in populations by means of spontaneous heritable changes in genes.

2. Migration, the movement of individuals among subpopulations within a larger population.

3. Natural selection, which results from the differing abilities of individuals to survive and reproduce in their environment.

4. Random genetic drift, the random, undirected changes in allele frequency that occur by chance in all populations, but particularly in small ones.

Some of the implications of these processes for population genetics are considered in the following sections.

15.6—
Mutation and Migration

Mutation is the ultimate source of genetic variation. It is an essential process in evolution, but it is a relatively weak force for changing allele frequencies, primarily because typical mutation rates are too low. Moreover, most newly arising mutations are harmful to the organism. Although some mutations may be **selectively neutral,** which means they do not affect the ability of the organism to survive and reproduce, only a very few mutations are favorable for the organism and contribute to evolution. The low mutation rates that are observed are thought to have evolved as a compromise: The mutation rate is high enough to generate the favorable mutations that a species requires to continue evolving, but it is not so high that the species suffers excessive genetic damage from the preponderance of harmful mutations.

Migration is similar to mutation in that new alleles can be introduced into a local population, although the alleles derive from another subpopulation rather than from new mutations. In the absence of migration, the allele frequencies in each local population can change independently, so local populations can undergo considerable genetic differentiation. Genetic differentiation among subpopulations means that there are differing frequencies of common alleles among the local populations or that some local populations possess certain rare alleles not found in others. The accumulation of genetic differences among subpopulations can be minimized if the subpopulations exchange individuals (undergo migration). In fact, only a relatively small amount of migration among subpopulations (on the order of just a few migrant individuals in each local population in each generation) is usually sufficient to prevent the accumulation of high levels of genetic differentiation. On the other hand, genetic differentiation can occur in spite of migration if other evolutionary forces, such as natural selection for adaptation to the local environments, are sufficiently strong.

Irreversible Mutation

To appreciate the nature of the force of mutation, consider an allele *A* and a completely equivalent mutant allele *a*. (The mutational change in *a* could be, for example, a nucleotide substitution at a nonessential site in an intron or a change from one synonymous codon to another, so the protein product is unchanged.) Suppose allele *A* undergoes mutation to *a* at a rate of μ per *A* allele per generation. What this rate means is that in any generation, the probability that a particular *A* allele mutates to *a* is μ. Equivalently, we could say that in any generation, the proportion of *A* alleles that do *not* mutate to *a* equals $1 - \mu$. If we let *p* and *q* represent the allele frequencies of *A* and *a*, respectively, then the allele frequency of *A* changes according to the rule that the allele frequency of *A* in any generation equals the allele frequency of *A* in the previous generation times the proportion of *A* alleles that failed to mutate in that generation. (Here we are assuming that reverse mutation of *a* to *A* can be neglected.) In symbols, the rule implies that

$$p_n = p_{n-1}(1-\mu)$$

where the subscripts *n* and *n* - 1 denote any generation *n* and the previous generation, respectively. Hence the allele frequency of *A* in generations 0, 1, 2, 3, and so on changes according to

$$p_1 = p_0(1-\mu)$$
$$p_2 = p_1(1-\mu)$$
$$p_3 = p_2(1-\mu)$$

Substituting p_1 from the first equation into the second, we obtain

$$p_2 = p_1(1-\mu) = p_0(1-\mu)^2$$

Substituting this into the equation for p_3 yields

$$p_3 = p_2(1-\mu) = p_0(1-\mu)^3$$

The general rule follows by consecutive substitutions:

$$p_n = p_0(1-\mu)^n \qquad (8)$$

where p_n represents the allele frequency of *A* in the *nth* generation.

If we assume that allele *A* is fixed in the population at time *n* = 0, then $_0$ = 1, and Equation (8) becomes

$$p_n = (1-\mu)^n \qquad (9)$$

For realistic values of the mutation rate, Equation (9) implies a very slow rate of change in allele frequency. The pattern of change is illustrated by the lower curve in Figure 15.16, which assumes a mutation rate of $\mu = 1 \times 10^{-5}$ per generation. With this value of μ the allele frequency of *A* decreases by half its present value every 69,314 generations. To put this rate in human terms, assuming 20 years per generation, it would require 69,314 × 20 = 1,368,280 years, or approximately 1.4 million years, to reduce the allele frequency of *A* by half. For example, if a human allele *A* were fixed at the time of *Homo erectus*, 1.4 million years ago, and *A* underwent mutation to *a* at a rate of 1×10^{-5} per generation, then 50 percent of the alleles present in the modern human population would still be of type *A*.

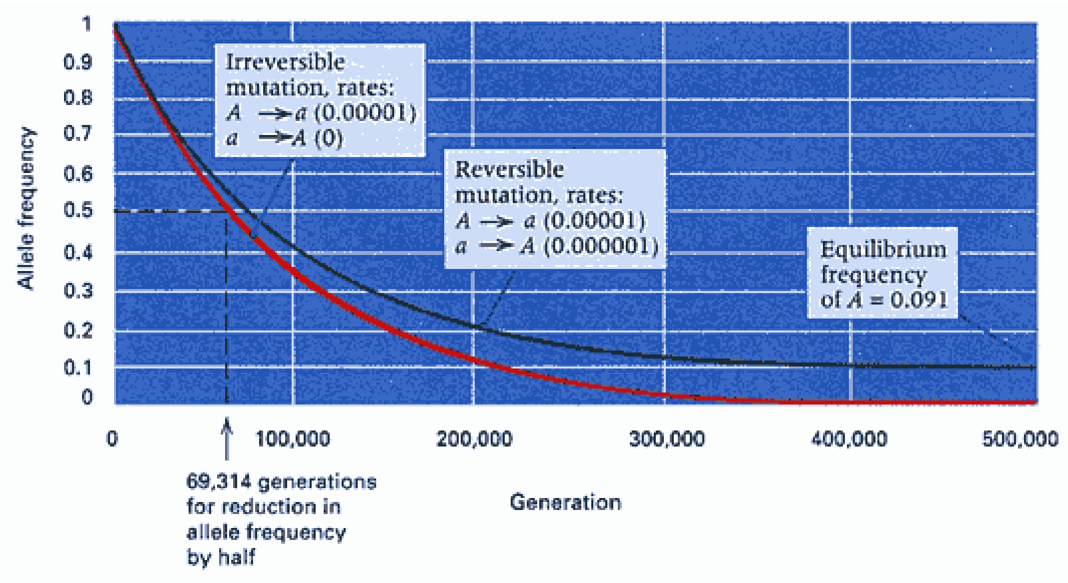


Figure 15.16
Effects of irreversible mutation (bottom curve) and reversible mutation (top curve) on allele frequency. In this example, the forward mutation rate is $\mu = 1 \times 10^{-5}$ and the reverse mutation rate is $\mu = 1 \times 10^{-6}$.

Reversible Mutation

When the possibility of reverse mutation is taken into account, then **forward mutation** of *A* to *a* is offset by **reverse mutation** of *a* to *A.* Eventually, an **equilibrium** of allele frequency is reached at which the allele frequencies become stable and no longer change, even though there is continuous mutation. The reason for the equilibrium is that at this point, each new *A* allele that is lost by forward mutation is replaced by a new *A* allele created by reverse mutation.

It is quite straightforward to deduce the equilibrium frequencies when mutation is reversible. As before, let *A* and *a* have allele frequencies *p* and *q,* respectively. Suppose that the rate of forward mutation ($A \rightarrow a$) is μ and that of reverse mutation ($a \rightarrow A$) is *v.* In any generation, the number of *A* alleles lost to forward mutation is proportional to *p* μ, and the number of new *A* alleles created by reverse mutation is proportional to *qv.* At equilibrium, the loss of *A* must equal the gain of *a;* hence *p*μ = *qv.* Substituting *q* = 1 - *p,* we obtain

$$p\mu = (1 - p)\nu = \nu - p\nu$$

and therefore, at equilibrium,

$$\hat{p} = \frac{\nu}{\mu + \nu} \qquad (10)$$

where the symbol $\hat{p}$ is used, by convention, to mean the equilibrium value of *p.* As a check of the correctness of Equation (10), note that $\hat{q}$ = μ (μ + *v)* and so $\hat{p}$ μ = *qv* as required.

In Figure 15.16, the upper curve gives the change in allele frequencies for reversible mutation with μ = 1×10^{-5} and *v* = 1×10^{-6}. The equilibrium in this example is

$$\hat{p} = \frac{1 \times 10^{-6}}{\qquad\qquad} = \frac{1}{11}$$

The time required to reach equilibrium is again very long. When μ = 1×10^{-6}. and *v* = 1×10^{-6}, it requires 63,013 generations for any specified values of *p* and *q* to go halfway to their equilibrium values. (The half-time of 69,314 generations derived earlier is different because of the effect of reverse mutation.)

15.7—
Natural Selection

The driving force of adaptive evolution is *natural selection,* which is a consequence of hereditary differences among organisms in their ability to survive and reproduce in the prevailing environment. Since it was first proposed by Charles Darwin in 1859, the concept of natural selection has been revised and extended, most notably by the incorporation of genetic concepts. In its modern formulation, the occurrence of natural selection rests on three premises.

1. In all organisms, more offspring are produced than can possibly survive and reproduce. (This part of the theory Darwin borrowed from Thomas Malthus.)

2. Organisms differ in their ability to survive and reproduce, and some of these differences are due to genotype.

3. In every generation, the genotypes that promote survival in the prevailing environment (favored genotypes) are present in excess among individuals of reproductive age, and hence the favored genotypes contribute disproportionately to the offspring of the next generation.

By means of this process, the alleles that enhance survival and reproduction increase in frequency from generation to generation, and the population becomes progressively better able to survive and reproduce in the environment. This progressive genetic improvement in populations constitutes the process of evolutionary **adaptation.**

Selection in a Laboratory Experiment

Selection is easily studied experimentally in bacterial populations because of the short generation time (about 30 minutes). Figure 15.17 shows the result of competition between two bacterial genotypes, *A* and *B.* Genotype *A* is the superior competitor under the particular conditions. In the experiment, the competition was allowed to continue for 290 generations, in which time the proportion of *A* genotypes (*p)* increased from 0.60 to 0.9995 and that of *B* genotypes decreased from 0.40 to 0.0005. The data points give a satisfactory fit to an equation of the form

$$\frac{p_n}{q_n} = \left(\frac{p_0}{q_0}\right)\left(\frac{1}{w}\right)^n \qquad (11)$$

in which p_0 and q_0 are the initial frequencies of *A* and *B* (in this case 0.6 and 0.4, respectively), p_n and q_n are the frequencies after *n* generations of competition, and *w* is a measure of the competitive ability of *B* when competing against *A* under the conditions of the experiment. Equation (11) can be derived in a similar manner as Equation (8) for mutation. Because the relative rates of survival and reproduction of strains *A* and *B* are in the ratio 1 : *w*, the ratio of frequencies in any generation, p_n q_n, must equal the ratio of frequencies in the previous generation, p_{n-1}/q_{n-1}, times 1*w.* Therefore,

$$(p_1/q_1) = (p_0/q_0) \times (1/w)$$
$$(p_2/q_2) = (p_1/q_1) \times (1/w)$$
$$(p_3/q_3) = (p_2/q_2) \times (1/w)$$

and so on. Equation (11) is derived by repeated substitution of each equation into the next.

The value of *w* is called the **relative fitness** of the *B* genotype relative to the *A* genotype under these particular conditions. The data in Figure 15.17 fit Equation (11) for a value of *w* = 0.958, which generates the smooth curve. Relative fitness measures the comparative contribution of each parental genotype to the pool of offspring genotypes produced in each generation. In this example, for each offspring cell produced by an *A* genotype, a *B* genotype produces an average of 0.958 offspring cell.

In population genetics, relative fitnesses are usually calculated with the most favored genotype (*A* in this case) taken as the standard with a fitness of 1.0. However, the selective disadvantage of a genotype is often of greater interest than its relative fitness, because some of the formulas are simplified. The selective disadvantage of a disfavored genotype is called the **selection coefficient** against the genotype, and it is calculated as the difference between the fit-

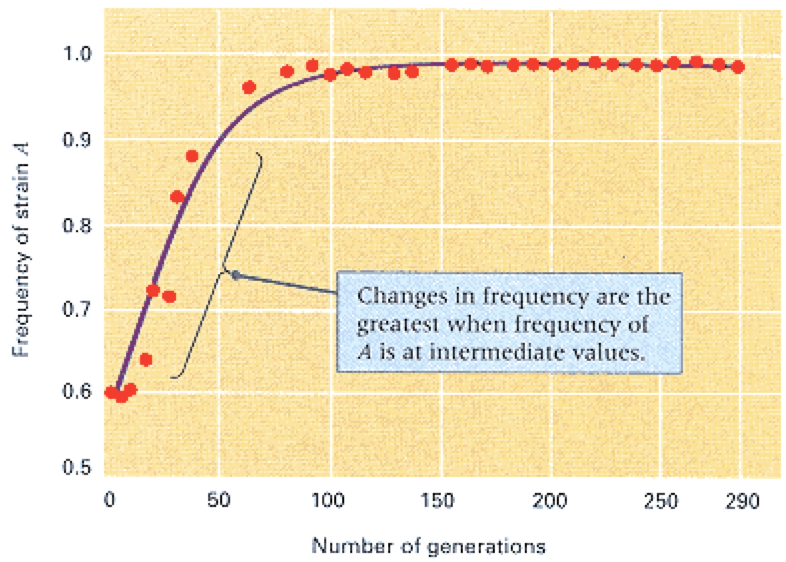


Figure 15.17
Increase in frequency of a favored strain of *E. coli* resulting from selection in a continuously growing population. The *y*-axis is the number of *A* cells at any time divided by the total number of cells (*A* + *B*). Note that the changes in frequency are the greatest when the frequency of the favored strain. *A* is at intermediate values.

ness of the standard (taken as 1.0) and the relative fitness of the genotype in question. In the case at hand, the selection coefficient against *B*, denoted *s*, is

$$s = 1.000 - 0.958 = 0.042 \qquad (12)$$

In words, the selective disadvantage of strain *B* is 4.2 percent per generation. When the fitnesses are known. Equation (11) also enables us to predict the allele frequencies in any future generation, given the original frequencies. Alternatively, it can be used to calculate the number of generations required for selection to change the allele frequencies from any specified initial values to any later ones. For example, from the relative fitnesses of *A* and *B* just estimated, we can calculate the number of generations required to change the frequency of *A* from 0.1 to 0.8. In this example, p_0/q_0 = 0.1/0.9, p_n/q_n= 0.8/0.2, and w = 0.958. A little manipulation of Equation (11) gives

$$\begin{aligned} n &= [\log(0.1/0.9) - \log(0.8/0.2)]/\log(0.958) \\ &= 83.5 \text{ generations} \end{aligned}$$

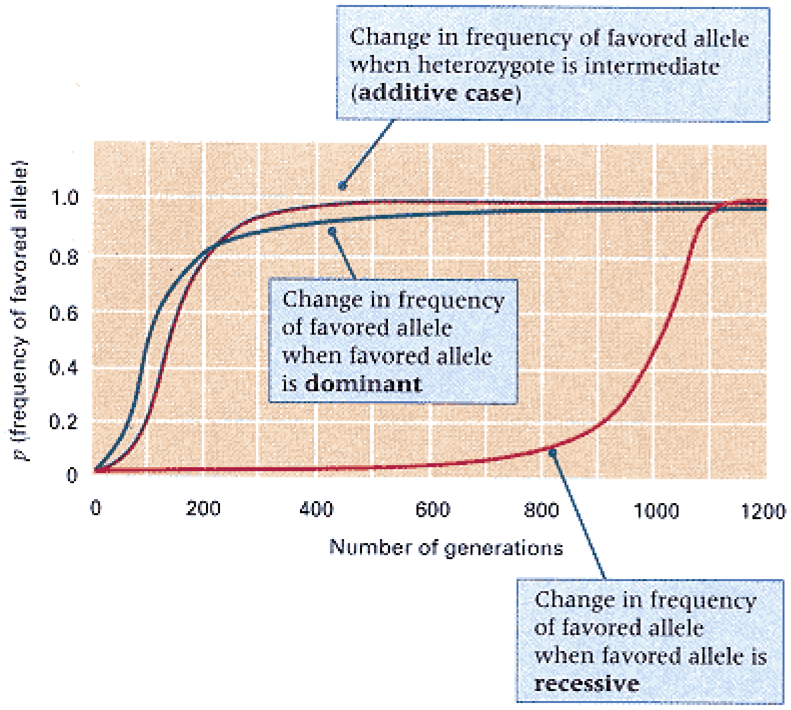


Figure 15.18
Theoretically expected change in frequency of an allele favored by selection in a diploid organism undergoing random mating when the favored allele is dominant, when the favored allele is recessive, and when the heterozygote is exactly intermediate in fitness (additive). In each case, the selection coefficient against the least fit homozygous genotype is 5 percent. The data are plotted in terms of *p*, the allele frequency of the beneficial allele. These curves demonstrate that selection for or against a recessive allele is very inefficient when the recessive allele is rare.

Selection in Diploids

Selection in diploids is analogous to that in haploids, but dominance and recessiveness create additional complications. Figure 15.18 shows the change in allele frequencies for both a favored dominant and a favored recessive in a random-mating diploid population. The striking feature of the figure is that the frequency of the favored dominant allele changes very slowly when the allele is common, and the frequency of the favored recessive allele changes very slowly when the allele is rare. The reason is that, with random mating, rare alleles are found much more frequently in heterozygotes than in homozygotes. With a favored dominant at high frequency, most of the recessive alleles are present in heterozygotes, and the heterozygotes are not exposed to selection and hence do not contribute to change in allele frequency. Conversely, with a favored recessive at low frequency, most of the favored alleles are in heterozygotes, and again the heterozygotes are not exposed to selection and do not contribute to change in allele frequency. The principle is quite general:

> Selection for or against a recessive allele is very inefficient when the recessive allele is rare.

The inefficiency of selection against rare recessive alleles has an important practical implication. There is a widely held belief that medical treatment to save the lives of persons with rare recessive disorders will cause a deterioration of the human gene pool because of the reproduction of persons who carry the harmful genes. This belief is unfounded. With rare alleles, the proportion of homozygous genotypes is so small that reproduction of the homozygous genotypes contributes a negligible amount to change in allele frequency. Considering their low frequency, it matters little whether homozygous genotypes reproduce or not. Similar reasoning applies to eugenic proposals to "cleanse" the human gene pool of harmful recessives by preventing the reproduction of affected persons. People with severe genetic disorders rarely reproduce anyway, and even when they do, they have essentially no effect on allele frequency. *The largest reservoir of harmful recessive alleles is in the genomes of phenotypically normal carrier heterozygotes.*

Components of Fitness

Data like those in Figure 15.17 are essentially unobtainable in higher organisms because of their long generation time. Observations extending over 290 generations would take 12 years with *Drosophila* and 6000 years with humans. How, then, is one to estimate fitness? The usual approach is to divide fitness into its component parts and estimate these separately. The two major components of the fitness of a genotype are usually considered to be

• The **viability,** which means the probability that a newly formed zygote of the specified genotype survives to reproductive age

• The **fertility,** which means the average number of offspring produced by an individual of the specified genotype during the reproductive period

Together, viability and fertility measure the contribution of each genotype to the offspring of the next generation.

Table 15.3 is an example of experiments carried out to estimate the viability of two genotypes coding for allozymes of the enzyme alcohol dehydrogenase in *Drosophila.* The genotypes are homozygous for either the *F* or the *S* allele. The tabulated numbers are averages obtained in seven separate experiments in which larvae were placed together in competition in culture vials. The ratio of surviving larvae to those introduced originally is the probability of survival of each genotype. As noted at the bottom of the table, these numbers can be converted into relative viabilities in two ways, using either *SS* or *FF* as the standard. Because of the convention that the larger of the two relative values should be 1.0, the situation can be summarized either by saying that the viability of *FF* relative to *SS* is 0.94 or that the selective disadvantage of *FF* relative to *SS* with respect to viability is 0.06.

Estimates of overall fitness based on viability and fertility components may be used for predicting changes in allele frequency that would be expected in experimental or natural populations. However, the fitness of a genotype may depend on the environment and on other factors such as population density and the frequencies of other genotypes. Consequently, fitnesses estimated in a particular laboratory situation may have little relevance to predicting allele frequency changes in natural populations unless other factors are comparable.

Selection-Mutation Balance

Natural selection usually acts to minimize the frequency of harmful alleles in a population. However, the harmful alleles can never be totally eliminated, because mutation of wildtype alleles continually creates new harmful mutations. With the forces of selection and mutation acting in opposite directions, the population eventually attains a state of equilibrium at which new mutations exactly offset selective eliminations. In determining the allele frequencies at equilibrium, it makes a great deal of difference whether the harmful allele is completely recessive or has, instead, a small effect (partial dominance) on the fitness of the heterozygous carriers. To deduce the equilibrium frequencies, we will symbolize the wildtype allele as *A* and the harmful allele as *a* and represent the allele frequencies as *p* and *q,* respectively.

Table 15.3 Relative viability of alcohol dehydrogenase genotypes in *Drosophila*

Experimental data	**Genotype**	
and relative viability	***FF***	***SS***
Experimental data		
Number of larvae introduced per vial	200	100
Average number of survivors per vial	132.72	70.43
Probability of survival	0.6636	0.7043
Relative viability		
With *SS* as standard	0.6636/0.7043 = 0.94	0.7043/0.7043 = 1.0
With *FF* as standard	0.6636/0.6636 = 1.0	0.7043/0.6636 = 1.06

In the case of a complete recessive, the fitnesses of *AA*, *Aa*, and *aa* genotypes can be written as 1 : 1 : 1 - *s*, where *s* represents the selection coefficient against the *aa* homozygote and measures the fraction of *aa* genotypes that fail to survive or reproduce. For a recessive lethal, for example, $s = 1$. When q is very small (as it will be near equilibrium), the number of *a* alleles eliminated by selection will be proportional to q^2s, assuming Hardy-Weinberg proportions. At the same time, the number of new *a* alleles introduced by mutation will be proportional to μ. At equilibrium, the selective eliminations must balance the new mutations, and so $q^2s = \mu$ or

$$\hat{q} = \sqrt{\frac{\mu}{s}} \qquad (13)$$

As before, $\hat{q}$ means the equilibrium value. To apply Equation (13) to a specific example, consider the recessive allele for Tay-Sachs disease, which in most non-Jewish populations has a frequency of $q = 0.001$. Because the condition is lethal, $s = 1$. Assuming that $q = 0.001$ represents the equilibrium frequency and that the allele has no effect on the fitness of the heterozygous carriers, the mutation rate required to account for the observed frequency would be

$$\mu = q^2s = (0.001)^2 \times 1.0 = 1 \times 10^{-6}$$

This example shows how considerations of selection-mutation balance can be used in estimating human mutation rates.

If a harmful allele shows partial dominance in having a small detrimental effect on the fitness of the heterozygous carriers, then the fitnesses of *AA*, *Aa*, and *aa* can be written as 1 : 1 - *hs:* 1 - *s*, in which *hs* is the selection coefficient against the heterozygous carriers. The parameter *h* is called the **degree of dominance.** When $h = 1$, the harmful allele is completely dominant; when $h = 0$, it is completely recessive; and when $h = 1/2$, the fitness of the heterozygous genotype is exactly the average of those of the homozygous genotypes. For most harmful alleles that show partial dominance, the value of *h* is considerably smaller than 1/2.

With random mating, when an allele is rare (as a harmful allele will be at equilibrium), the frequency of heterozygous genotypes far exceeds the frequency of those that are homozygous for the harmful recessive. This means that most of the alleles that are eliminated by selection are eliminated because of the selection against the heterozygous genotypes, because there are so many more of them. For each *Aa* genotype that fails to survive or reproduce, half of the alleles that are lost are *a*, so the number of alleles lost to selection in any generation is proportional to $2pq \times hs \times 1/2$. On the other hand, the number of new *a* alleles created by $A \rightarrow a$ mutation equals μ. At equilibrium, the loss due to selection must balance the gain from mutation, so $2pq \times hs \times 1/2 = \mu$, or

$$\hat{p}\hat{q}hs = \mu$$

Because a harmful allele will be rare at equilibrium, especially if it shows partial dominance, $\hat{p}$ will be so close to 1 that it can be replaced with 1 in the above expression. Therefore, the equilibrium value for the frequency of a partially dominant harmful allele is

$$\hat{q} = \frac{\mu}{hs} \qquad (14)$$

Note that Equation (14) does not include a square root, which makes a substantial difference in the allele frequency. This effect can be seen by comparing the curves in Figure 15.19 for a harmful allele that is lethal when homozygous ($s = 1$). The top curve is for a complete recessive ($h = 0$), and the other curves refer to various degrees of partial dominance; for example, $h = 1/2$ means that the heterozygous genotype has a fitness exactly intermediate between the homozygous genotypes. Even a small amount of partial dominance has a substantial effect on reducing the equilibrium allele frequency.

Heterozygote Superiority

So far we have considered examples of selection in which the fitness of the heterozygote is intermediate between those of the homozygotes (or possibly equal in fitness to one homozygote). In these cases, the allele associated with the most fit homozygote eventually becomes fixed, unless the selection is opposed by mutation. In this section we consider the possi-

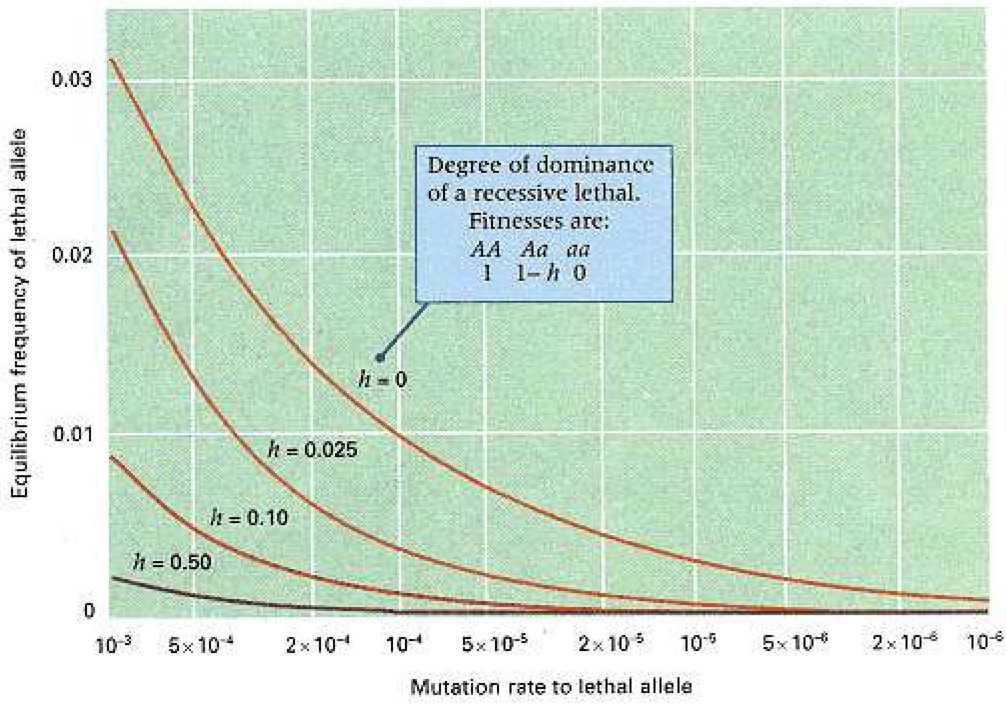


Figure 15.19
Effect of partial dominance on the equilibrium frequency of a harmful allele that is lethal when homozygous. The degree of dominance is measured by *h*. For example, a value of *h* = 0.025 means that the relative fitnesses of *AA* and *Aa* are 1 : 0.975.

bility that the fitness of the heterozygote is greater than that of both homozygotes; this case is called **overdominance** or **heterozygote superiority.** When there is overdominance, neither allele can be eliminated by selection, because in each generation, the heterozygotes produce more offspring than the homozygotes. The selection in favor of heterozygous genotypes keeps both alleles in the population. Eventually there is equilibrium in which the allele frequency no longer changes. With overdominance, the fitnesses of *AA*, *Aa* and *aa* may be written as $1 - s : 1 : 1 - t$, where *s* and *t* are the selection coefficients against *AA* and *aa*, respectively, relative to *Aa.* Assuming random mating, in any generation, the proportion of *A* alleles eliminated by selection is $p^2s/p = ps$, and the proportion of *a* alleles eliminated by selection is $q^2t/q = qt$. At equilibrium, the selective eliminations of *A* must balance the selective eliminations of *a*, and hence $ps = qt$, or

$$p = \frac{t}{s + t} \qquad (15)$$

With overdominance, the allele frequencies always go to equilibrium, but the rate of approach depends on the magnitudes of *s* and *t.* The equilibrium is attained most rapidly when there is strong selection against the homozygotes.

Overdominance does not appear to be a particularly common form of selection in natural populations. However, there are several well-established cases, the best known of which involves the sickle-cell hemoglobin mutation. The initial hint of a connection between sickle-cell anemia and falciparum malaria was the substantial overlap of the geographical distribution of sickle-cell anemia and the geographical distribution of malaria (Figure 15.20).

On the basis of the rates at which people of different genotypes are infected with malaria, and the mortality of infected people, the relative fitnesses of the genotypes *AA*, *AS* and *SS* have been estimated as 0.86 : 1 : 0. (*S* represents the sickle-cell mutation.) In the absence of intensive medical treatment, the homozygous *SS* genotype is lethal as a result of severe anemia. On the other hand, there is overdominance because *AS* heterozygotes are more resistant to malarial infection than either of the homozygous genotypes.

From these fitnesses, we have $s = 0.24$ and $t = 1$, so the predicted equilibrium

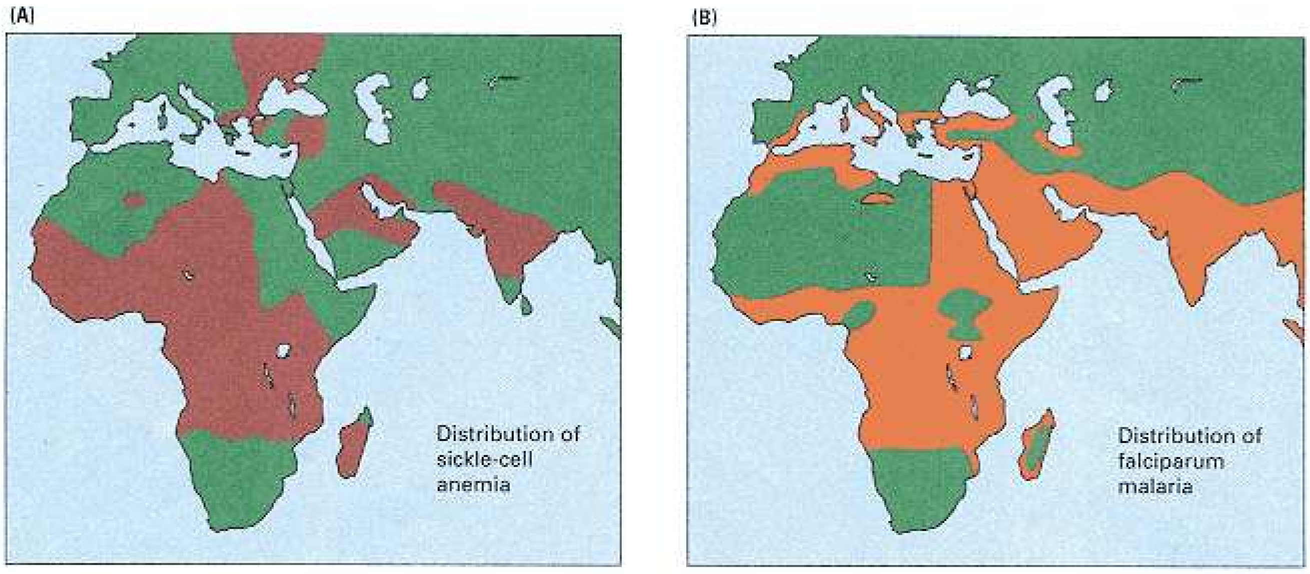


Figure 15.20
Geographic distribution of (A) sickle cell anemia and (B) falciparum malaria in the 1920s, before extensive malaria-control programs were launched. Shades of brown indicate the areas in question.

frequency of *A* from Equation (15) equals 1/(0.24 + 1) =0.81. The predicted equilibrium frequency of *S* is therefore 1 0.806 =0.19. This value is reasonably close to that observed in many parts of West Africa.

15.8—
Random Genetic Drift

The process of **random genetic drift** comes about because populations are not infinitely large (as we have been assuming all along) but rather are finite, or limited in size. The breeding individuals of any one generation produce a potentially infinite pool of gametes. In the absence of fertility differences, the allele frequencies among gametes will equal the allele frequencies among adults. However, because of the finite size of the population, only a few of these gametes will participate in fertilization and be represented among zygotes of the next generation. In other words, there is a process of *random sampling* that takes place in going from one generation to the next. Because there is variation among samples due to chance, the allele frequencies among gametes and zygotes may differ. Changes in allele frequency that come about because of the generation-to-generation sampling in finite populations are the cause of random genetic drift.

Consider a population consisting of exactly N diploid individuals in each generation. At each autosomal locus, there will be exactly $2N$ alleles. Suppose that i of these are of type *A* and the remaining $2N - i$ are of type *a.* (The allele frequency of *A*, designated p, therefore equals $i/2N$.) The number of *A* alleles in the next generation cannot be specified with certainty, because it will differ from case to case as a result of random genetic drift. On the other hand, it is possible to specify a *probability* for each possible number of *A* alleles in the next generation. This probability is given by the terms of the binomial distribution (Chapter 3). The probability $P(k|i)$ that there will be exactly k copies of *A* in the next generation, given that there are exactly i copies among the parents, is given by

$$P(k|i) = \frac{(2N)!}{k!\,(2N-k)!}p^k q^{2N-k} \qquad (16)$$

In Equation (16), $p = i/2N$, $q = 1 - p$, and k can equal 0, 1, 2, and so on, up to and including $2N$. The part of the equation with factorials is the number of possible orders in which the k copies of allele *A* can be chosen. (The *A* alleles could be the first k chosen, the last k chosen, alternating with *a*, and so on.) The powers of p and q represents the probability that any particular order will be realized. For any specified values of N and i, one can use Equation (16) along with a table of random numbers (or, better yet, a personal computer) to calculate an actual value for k.

A concrete example illustrates the essential features of random genetic drift. Table 15.4 shows the result of random genetic drift in 12 subpopulations (labeled *a* through *l*), each initially consisting of eight diploid individuals and containing eight copies of allele *A* and eight copies of allele *a*. Within each subpopulation, mating

Table 15.4 Effects of random genetic drift. Each entry is the number of *A* alleles in a subpopulation of size 16.

	Population designation												Average
Generation	***a***	***b***	***c***	***d***	***e***	***f***	***g***	***h***	***i***	***j***	***k***	***l***	$\bar{p}$
0	8	8	8	8	8	8	8	8	8	8	8	8	0.500
1	11	7	9	8	8	6	8	7	8	6	11	10	0.516
2	10	9	10	8	8	8	6	7	6	7	13	11	0.536
3	7	11	6	5	12	5	8	5	8	7	14	9	0.505
4	8	11	7	4	12	5	8	8	7	4	15	6	0.495
5	8	8	5	3	13	2	8	12	9	5	15	6	0.490
6	11	5	3	1	13	3	10	13	6	7	15	3	0.469
7	11	8	4	3	11	1	8	14	3	7	15	2	0.453
8	14	7	4	3	10	1	9	14	3	9	16	1	0.474
9	15	5	3	5	7	0	12	14	2	11	16	0	0.496
10	16	6	5	9	8	0	9	13	3	10	16	0	0.495
11	16	1	5	11	6	0	10	13	3	10	16	0	0.474
12	16	0	5	12	6	0	9	13	2	9	16	0	0.458
13	16	0	3	12	7	0	9	13	1	11	16	0	0.458
14	16	0	5	15	7	0	9	11	1	12	16	0	0.479
15	16	0	3	14	7	0	8	12	3	13	16	0	0.479
16	16	0	2	14	9	0	11	14	2	14	16	0	0.510
17	16	0	1	15	6	0	12	14	2	13	16	0	0.495
18	16	0	1	15	4	0	13	15	5	13	16	0	0.510
19	16	0	1	16	2	0	14	16	6	14	16	0	0.526
20	16	0	1	16	2	0	15	16	9	16	16	0	0.557
21	16	0	2	16	3	0	15	16	10	16	16	0	0.573
	A	*a*		*A*		*a*		*A*		*A*	*A*	*a*	
	fixed	fixed		fixed		fixed		fixed		fixed	fixed	fixed	

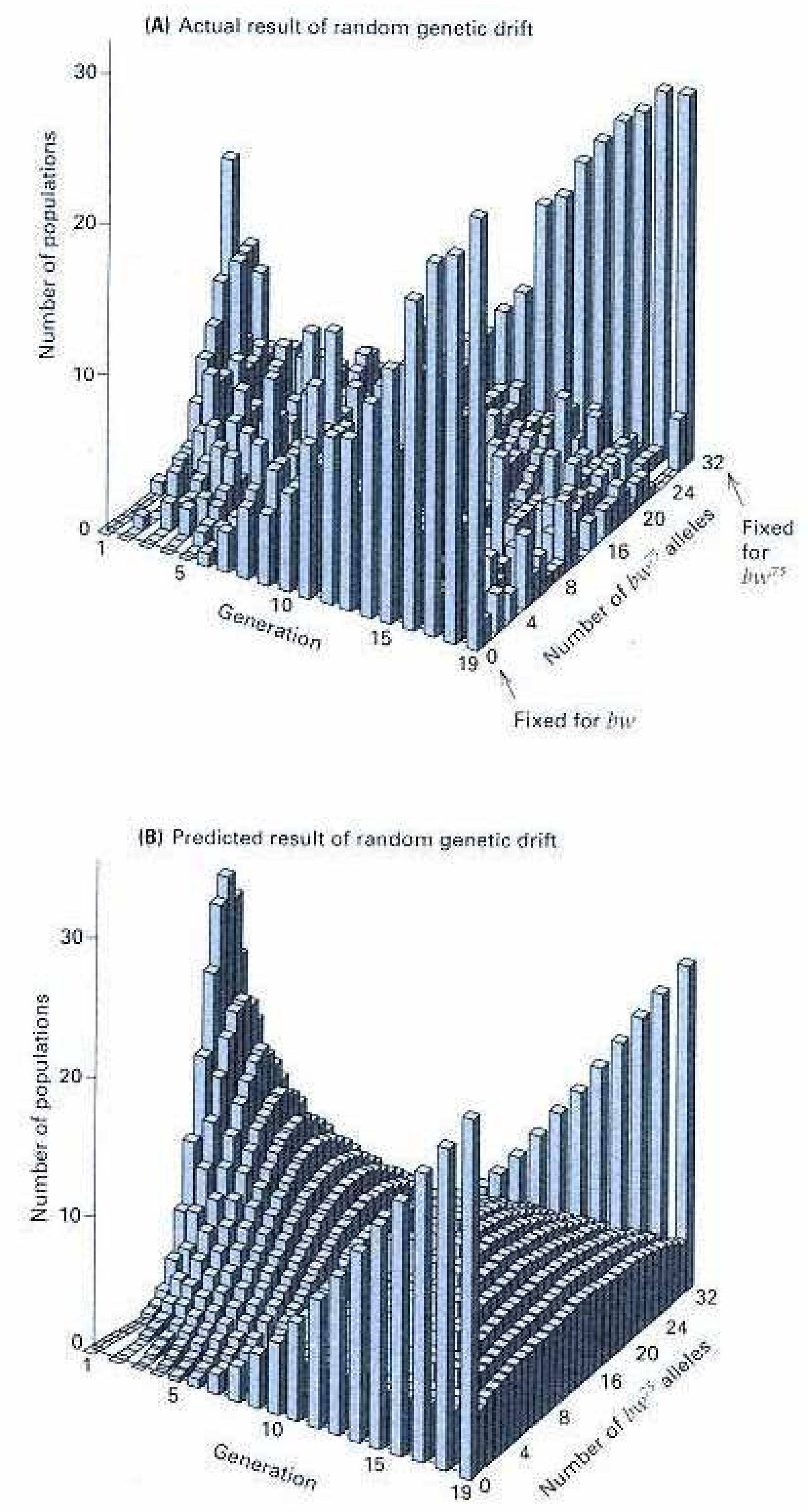


Figure 15.21
(A) Random genetic drift in 107 experimental populations of *Drosophila melanogaster*, each consisting of 8 females and 8 males. (B) Theoretical expectation of the same situation, calculated from the binomial distribution.
[Data in part A from P. Buri, 1956, Evolution 10:367. Graphs from L. Hartl and A. G. Clark, 1989, *Principles of Population Genetics, Sunderland,* MA: Sinauer Associates.]

is random. A computer program generating values of k with Equation (16) was used to calculate the number of A alleles in each subpopulation in each successive generation. These numbers are shown in the vertical columns, and the dispersion of allele frequencies resulting from random genetic drift is apparent. These changes in allele frequency would be less pronounced, and would require more generations, in larger populations than in the very small populations illustrated here, but the overall effect would be the same. Because of the prominent role of $2N$ in Equation (16), the dispersion of allele frequency resulting from random genetic drift depends on population size; the smaller the population, the greater the dispersion and the more rapidly it takes place.

In Table 15.4, the principal effect of random genetic drift is evident in the first generation: The allele frequencies have begun to spread out over a wider range. By the seventh generation, the spreading is extreme, and the number of *A* alleles ranges from 1 to 15. This speading out means that the allele frequencies among the subpopulations become progressively more different. In general:

> Random genetic drift causes differences in allele frequency among subpopulations; it is a major cause of genetic differentiation among subpopulations.

Although allele frequencies among subpopulations spread out over a wide range because of random genetic drift, the *average* allele frequency among subpopulations remains approximately constant. This point is illustrated by the column headed *p* in Table 15.4. The average allele frequency stays close to 0.5, its initial value. If an infinite number of subpopulations were being considered instead of the 12 subpopulatons in Table 15.4, then the average allele frequency would be exactly 0.5 in every generation. This principle implies that in spite of the random drift of allele frequency in individual subpopulations, the average allele frequency among a large number of subpopulations remains constant and equal to the average allele frequency among the original subpopulations.

After a sufficient number of generations of random genetic drift, some of the subpopulations become fixed for *A* and others for *a.* Because we are excluding the occurrence of mutation, a population that becomes fixed for an allele remains fixed thereafter. After 21 generations in Table 15.4, only four of the populations (*c, e, g,* and *i)* are still segregating; eventually, these too will become fixed. Because the average allele frequency of *A* remains constant, it follows that a fraction p_0 of the populations will ultimately become fixed for *A* and *a* fraction 1 - p_0 will become fixed for *a.* (The symbol p_0 represents the allele frequency of *A* in the initial generation.) Therefore,

> With random genetic drift, the probability of ultimate fixation of a particular allele is equal to the frequency of the allele in the original population.

In Table 15.4, five of the fixed populations are fixed for *A* and three for *a,* which is not very different from the equal numbers expected theoretically with an infinite number of subpopulations.

An example of random genetic drift in small experimental populations of *Drosophila* exhibiting the characteristics pointed out in connection with Table 15.4 is illustrated in Figure 15.21A. The figure is based on 107 subpopulations, each initiated with eight bw^{75}/bw females (*bw* = brown eyes) and eight bw^{75}/bw males and maintained at a constant size of 16 by randomly choosing eight males and eight females from among the progeny of each generation. Note how the allele frequencies among subpopulations spread out because of random genetic drift and how subpopulations soon begin to be fixed for either bw^{75} or *bw.* Although the data are somewhat rough because there are only 107 subpopulations, the overall pattern of genetic differentiation has a reasonable resemblance to that expected from the theory based on the binomial distribution (Figure 15.21B).

If random genetic drift were the only force at work, then all alleles would become either fixed or lost and there would be no polymorphism. On the other hand, many factors can act to retard or prevent the effects of random genetic drift, of which the following are the most important: (1) large population size; (2) mutation and migration, which impede fixation because alleles lost by random genetic drift can be reintroduced by either process; and (3) natural selection, particularly those modes of selection that tend to maintain genetic diversity, such as heterozygote superiority.

Chapter Summary

Population genetics is the application of Mendel's laws and other principles of genetics to populations of organisms. A subpopulation, or local population, is a group of organisms of the same species living within a geographical region of such size that most matings are between members of the group. In most natural populations, many genes are polymorphic in that they have two or more common alleles. One of the goals of population genetics is to determine the nature and causes of genetic variation in natural populations.

The relationship between the relative proportions of particular alleles (allele frequencies) and genotypes (genotype frequencies) is determined in part by the frequencies with which particular genotypes form mating pairs. In random mating, mating pairs are independent of genotype. When a population undergoes random mating for an autosomal gene with two alleles, the frequencies of the genotypes are given by the Hardy-Weinberg principle. If the alleles of the gene are *A* and *a,* and their allele frequencies are *p* and *q,* respectively, then the Hardy-Weinberg principle states that the genotype frequencies with random mating are *AA,* p^2; *Aa,* 2*pq;* and *aa,* q^2. These are often good approximations for genotype frequencies within subpopulations. An important implication of the Hardy-Weinberg principle is that rare alleles are found much more frequently in heterozygotes than in homozygotes (2*pq* versus q^2.

Inbreeding means mating between relatives, and the extent of inbreeding is measured by the inbreeding coefficient, *F.* The main consequence of inbreeding is that an allele present in a common ancestor may be transmitted to both parents of an inbred individual in a later generation and become homozygous in the inbred offspring. The replicas of an ancestral allele are said to be identical by descent. The inbreeding coefficient of an inbred organism can be deduced directly from the pedigree of inbreeding by calculating the probability of identity by descent along every possible path of descent through each common ancestor. Among inbred individuals, the frequency of heterozygous genotypes is smaller, and that of homozygous genotypes greater, than it would be with random mating.

Evolution is the progressive increase in the degree to which a species becomes genetically adapted to its environment. A principal mechanism of evolution is natural selection, in which individuals superior in survival or reproductive ability in the prevailing environment contribute a disproportionate share of genes to future generations, thereby gradually increasing the frequency of the favorable alleles in the whole population. However, at least three other processes also can change allele frequency: mutation (heritable change in a gene), migration (movement of individuals among subpopulations), and random genetic drift (resulting from restricted population size). Spontaneous mutation rates are generally so low that the effect of mutation on changing allele frequency is minor, except for rare alleles. Migration can have significant effects on allele frequency because migration rates may be very large. The main effect of migration is the tendency to equalize allele frequencies among the local populations that exchange migrants. Selection occurs through differences in viability (the probability of survival of a genotype) and in fertility (the probability of successful reproduction).

Populations maintain harmful alleles at low frequencies as a result of a balance between selection, which tends to eliminate the alleles, and mutation, which tends to increase their frequencies. When there is selection-mutation balance, the allele frequency at equilibrium is usually greater if the allele is completely recessive than if it is partially dominant. This difference arises because selection is quite ineffective in affecting the frequency of a completely recessive allele when the allele is rare, owing to the almost exclusive appearance of the allele in heterozygotes.

A few examples are known in which the heterozygous genotype has a greater fitness than either of the homozygous genotypes (heterozygote superiority). Heterozygote superiority results in an equilibrium in which both alleles are maintained in the population. An example is sickle-cell anemia in regions of the world where falciparum malaria is endemic. Heterozygous persons have an increased resistance to malaria and only a mild anemia, a circumstance that results in greater fitness than that of either homozygote.

Random genetic drift is a statistical process of change in allele frequency in small populations, resulting from the inability of every individual to contribute equally to the offspring of successive generations. In a subdivided population, random genetic drift results in differences in allele frequency among the subpopulations. In an isolated population, barring mutation, an allele will ultimately become either fixed or lost because of random genetic drift. The probability of ultimate fixation of an allele as a result of random genetic drift is equal to the frequency of the allele in the initial population.

Key Terms

adaptation	heterozygote superiority	random mating
allele frequency	identity by descent	relative fitness
allozyme	inbreeding	reverse mutation
assortative mating (positive and negative	inbreeding coefficient local population	selection coefficient selection-mutation balance
consanguineous mating	lost allele	selectively neutral mutation
degree of dominance	mating system	subpopulation
DNA typing	migration	viability
electrophoresis	mutation	VNTR (variable number of tandem repeats)
equilibrium	natural selection	
evolution	overdominance	
fertility	path in a pedigree	

fixed allele	polymorphic gene
forward mutation	population
gene pool	population genetics
genotype frequency	population substructure
Hardy-Weinberg principle	random genetic drift

Review the Basics

• With regard to allele frequency, what does it mean to say that an allele is fixed? That an allele is lost?

• What is the Hardy-Weinberg principle and how does it apply to multiple alleles?

• Name four evolutionary processes that can change allele frequencies in natural populations. How do allele frequencies change in the absence of these processes?

• Distinguish between inbreeding and assortative mating.

• What is the role of mutation in the evolutionary process? Why is mutation a weak force for changing allele frequencies?

• What is the meaning of the term *fitness* in population genetics? Give an example of an organism with high fitness and an example of an organism with low fitness.

Chapter 15 Genetics on the web

GeNETcs on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises blow, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis***and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes u of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. This **Hardy-Weinberg** site has a clever demonstration of the principle in a game that makes use of several decks o playing cards. If assigned to do so, write a one-paragraph descripition of the game, explaining why it works and why there have to be at least 25 players.

2. A colorful introduction to **DNA typing** is located at this site. The *how-is-it-done* link illustrates the procedure step step. If assigned to do so, use the information at this site to write a one-page report on the applications of DNA typin and identify some of the problmes in interpretation.

3. Search this keyword site for **Pingelap disease** to identify a human population in which from 4 to 9 percent of the population is blind from infancy. If assigned to do so, write on paragraph describing how random genetic drift might explain the high incidence of the condition in this population.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of gene resources available on the World Wide Web. Select the **Mutable Site** for Chapter 15, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 15.

• What is the fitness of an organism that dies before the age of reproduction? What is the fitness of an organism that is sterile?

• Why, in a random-mating population, does the allele frequency of a rare recessive allele change slowly under selecti

• What is a mutation-selection balance? How does the equilibrium allele frequency of a harmful allele depend on the degree of dominance?

• What is random genetic drift and why does it occur? In the absence of any counteracting forces, what is the ultimate effect of random genetic drift on allele frequency? Are random changes in allele frequency from one generation to the next greater in small or large populations, and why?

Guide to Problem Solving

Problem 1: A sample of 300 plants from a population is examined for the electrophoretic mobility of an enzyme that varies according to the genotype determined by two alleles, *F* and *S*, of a single gene. The results are 7 plants with genotype *FF*, 106 with genotype *FS*, and 187 with genotype *SS*. What are the allele frequencies of *F* and *S*? What are the expected numbers of the three genotypes, assuming random mating?

Answer: The allele frequencies are determined by counting the alleles. The *7 FF* plants represent 14 *F* alleles, the 106 *FS* plants represent 106 *F* and 106 *S* alleles, and the 187 *SS* plants represent 374 *S* alleles, for a total of 300 × 2 = 600 alleles altogether. The allele frequency *p* of *F* is (14 + 106)/600 = 0.2, and the allele frequency *q* of *S* is (106 + 374)/600 = 0.8. As a check on the calculations, note that the allele frequencies sum to unity, as they should. For the second part of the question, the expected genotype frequencies with random mating are p^2 *FF*,2*pq* *FS*, andq^2 *SS*, so the expected numbers are

$$FF\colon\ (0.2)^2 \times 300 = 12$$
$$FS\colon\ 2(0.2)(0.8) \times 300 = 96$$
$$SS\colon\ (0.8)^2 \times 300 = 192$$

As a check on these calculations, note that 12 + 96 + 192 = 300.

Problem 2: Excessive secretion of male sex hormones results in premature sexual maturation in males and masculization of the sex characters in females. This disorder is called the adrenogenital syndrome, and in Switzerland there is an autosomal recessive form of the disease that affects about one in 5000 newborns.

(a) Assuming random mating, what is the allele frequency of the recessive?

(b) What is the frequency of heterozygous carriers?

(c) What is the expected frequency of the condition among the offspring of first cousins?

Answer:

(a) Set q^2 = 1/5000, which implies that

(a) Set $q^2 = 1/5000$, which implies that

$$q = (1/5000)^{1/2} = 0.014.$$

(b) The frequency of heterozygotes is I2 *pq*, in which *p* = 1 - 0.014 = 0.986. Thus

(b) The frequency of heterozygotes is $2pq$, in which $p = 1 - 0.014 = 0.986$. Thus

$$2pq = 2 \times 0.014 \times 0.986 = 0.028 = 1/36$$

that is, almost 3 percent of the population are carriers, even though only about 0.02 percent of the population are affected.

(c) With first-cousin mating, the inbreeding coefficient *F* = 1 16 = 0.062, and the expected proportion of affected people is

$$q^2(1 - F) + qF = (0.014)^2(0.938) + (0.014)(0.062)$$
$$= 0.001$$

In other words, there is about a five-fold greater risk of homozygosity for this allele among the offspring of first cousins.

Problem 3: Warfarin is a rat killer that acts by hindering blood coagulation. Continued use of the poison has resulted in the evolution of resistance in most rat populations because of selection favoring a resistance mutation *R*. The normal, sensitive allele may be denoted *S*. In the absence of warfarin, the relative fitnesses of *SS*, *RS*, and *RR* are in the ratio 1.00 : 0.77 : 0.46, and in the presence of warfarin, the ratio of fitnesses is 0.68 : 1.00 : 0.37.

(a) With regard to persistence or loss of the *R* and *S* alleles, what is the expected result of the continued use of the chemical?

(b) What would happen if warfarin were no longer used?

(c) Under what circumstances might the *R* allele become fixed?

Answer:

(a) In the presence of warfarin, there is heterozygote superiority, so continued use would be expected to result in a stable equilibrium in which both *R* and *S* persisted in the population.

(b) In the absence of warfarin, the genotype *SS* is favored and *RS* is intermediate in fitness, so allele *S* would become fixed and *R* would become lost.

(c) The allele *R* could become fixed in a small population where, because of chance fluctuations in allele frequency, the *S* allele might become lost. (However, small populations would also increase the chance that the *R* allele might be lost.)

Analysis and Applications

15.1 If the genotype *AA* is an embryonic lethal and the genotype *aa* is fully viable but sterile, what genotype frequencies would be found in adults in an equilibrium population containing the *A* and *a* alleles? Is it necessary to assume random mating?

15.2 For an X-linked recessive allele maintained by mutation-selection balance, would you expect the equilibrium frequency of the allele to be greater or smaller than that of an autosomal recessive allele, assuming that the relative fitnesses are the same in both cases? Why?

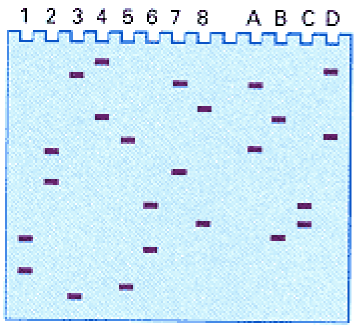
15.3 How many *A* and *a* alleles are present in a sample of organisms that consists of 10 *AA*, 15 *Aa*, and 4 *aa* individuals? What are the allele frequencies in this sample?

15.4 Allozyme phenotypes of alcohol dehydrogenase in the flowering plant *Phlox drummondii* are determined by codominant alleles of a single gene. In one sample of 35 plants, the following data were obtained:

Genotype	*AA*	*AB*	*BB*	*BC*	*CC*	*AC*
Number	2	5	12	10	5	1

What are the frequencies of the alleles A, B, and C in this sample?

15.5 The accompanying illustration shows the gel patterns observed with a probe for a VNTR (variable number of tandem repeats) locus among four pairs of parents (1–8) and four children (A–D). One child comes from each pair of parents.



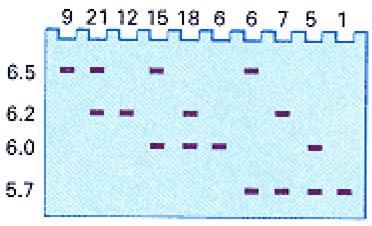
(a) Why does each person have two bands?

(b) How would it be possible for a person to have only one band?

(c) What type of dominance is illustrated by the VNTR alleles?

(d) Which pairs of people who have the DNA in lanes 1–8 are the parents of each child?

15.6 DNA from 100 unrelated people was digested with the restriction enzyme *Hin*dIII, and the resulting fragments were separated and probed with a sequence for a particular gene. Four fragment lengths that hybridized with the probe were observed-namely 5.7, 6.0, 6.2, and 6.5 kb-where each fragment defines a different restriction-fragment allele. The accompanying illustration shows the gel patterns observed;the number of individuals with each gel pattern is shown across the top. Estimate the allele frequencies of the four restriction-fragment alleles.



15.7 Which of the following genotype frequencies of *AA*, *Aa*, and *aa*, respectively, satisfy the Hardy-Weinberg principle?

(a) 0.25,0.50,0.25

(b) 0.36,0.55,0.09

(c) 0.49,0.42, 0.09

(d) 0.64,0.27,0.09

(e) 0.29,0.42,0.29

15.8 If the frequency of a homozygous dominant genotype in a randomly mating population is 0.09, what is the frequency of the dominant allele? What is the combined frequency of all the other alleles of this gene?

15.9 Hartnup disease is an autosomal-recessive disorder of intestinal and renal transport of amino acids. The frequency of affected newborn infants is about 1 in 14,000. Assuming random mating, what is the frequency of heterozygotes?

15.10 A randomly mating population of dairy cattle contains an autosomal-recessive allele causing dwarfism. If the frequency of dwarf calves is 10 percent, what is the frequency of heterozygous carriers of the allele in the entire herd? What is the frequency of heterozygotes among nondwarf cattle?

15.11 In certain grasses, the ability to grow in soil contaminated with the toxic metal nickel is determined by a dominant allele.

(a) If 60 percent of the seeds in a randomly mating population are able to germinate in contaminated soil, what is the frequency of the resistance allele?

(b) Among plants that germinate, what proportion are homozygous?

15.12 In Caucasians, the *M*-shaped hairline that recedes with age (sometimes to nearly complete baldness) is due to an allele that is dominant in males but recessive in females. The frequency of the baldness allele is approximately 0.3. Assuming random mating, what frequencies of the bald and nonbald phenotypes are expected in males and females?

15.13 In a Pygmy group in Central Africa, the frequencies of alleles determining the ABO blood groups were estimated as 0.74 for I^0, 0.16 for I^A, and 0.10 for I^B. Assuming random mating, what are the expected frequencies of ABO genotypes and phenotypes?

15.14 Among 35 individuals of the flowering plant *Phlox roemariana*, the following genotypes were observed for a gene determining electrophoretic forms of the enzyme phosphoglucose isomerase: 2 *AA*, 13 *AB*, 20 *BB.*

(a) What are the frequencies of the alleles *A* and *B*?

(b) Assuming random mating, what are the expected numbers of the genotypes?

15.15 If an X-linked recessive trait is present in 2 percent of the males in a population with random mating, what is the frequency of the trait in females? What is the frequency of carrier females?

15.16 In a population of *Drosophila,* an X-linked recessive allele causing yellow body color is present in genotypes at frequencies typical of random mating; the frequency of the recessive allele is 0.20. Among 1000 females and 1000 males, what are the expected numbers of the yellow and wildtype phenotypes in each sex?

15.17 How does the frequency of heterozygotes in an inbred population compare with that in a randomly mating population with the same allele frequencies?

15.18 Which of the following genotype frequencies of *AA, Aa,* and *aa,* respectively, are suggestive of inbreeding?

(a) 0.25,0.50,0.25

(b) 0.36,0.55,0.09

(c) 0.49,0.42,0.09

(d) 0.64,0.27,0.09

(e) 0.29,0.42,0.29

15.19 Galactosemia is an autosomal-recessive condition associated with liver enlargement, cataracts, and mental retardation. Among the offspring of unrelated individuals, the frequency of galactosemia is 8.5×10^{-6}. What is the expected frequency among the offspring of first cousins (F = 1/16) and among the offspring of second cousins (F = 1/64)?

15.20 Self-fertilization in the annual plant *Phlox cuspidata* results in an average inbreeding coefficient of $F = 0.66$.

(a) What frequencies of the genotypes for the enzyme phosphoglucose isomerase would be expected in a population with alleles *A* and *B* at frequencies 0.43 and 0.57, respectively?

(b) What frequencies of the genotypes would be expected with random mating?

15.21 Two strains of *Escherichia coli, A* and *B,* are inoculated into a chemostat in equal frequencies and undergo competition. After 40 generations, the frequency of the *B* strain is 35 percent. What is the fitness of strain *B* relative to strain *A*, under the particular experimental conditions, and what is the selection coefficient against strain *B*?

15.22 Determine the ultimate fate of the *A* allele with random mating when the relative fitnesses of *AA, Aa,* and *aa* are

(a) 1.00,0.98,0.96

(b) 1.00, 1.02, 1.02

(c) 1.00, 1.02, 0.98

(d) 1.00, 1.01, 1.02

15.23 For an autosomal gene with two alleles in a randomly mating population, what is the probability that a heterozygous female will have a heterozygous offspring?

Challenge Problems

15.24 In a randomly mating population that contains n equally frequent alleles, what is the expected frequency of heterozygotes?

15.25 Human genetic polymorphisms in which there are a variable number of tandem repeats (VNTR) are often used in DNA typing. The accompanying table shows the alleles of four VNTR loci and their estimated average frequencies in genetically heterogeneous Caucasian populations. (The allele 9.3 of the *HUMTH01* locus has a partial copy of one of the tandem repeats.) Using Equation (3), estimate

(a) the probability that an individual will be of genotype (6, 9.3) for *HUMTH01*, (10, 11) for *HUMFES*, (6, 6) for *D12S67*, and (18, 24) for *D1S80*.

(b) The probability that an individual will be of genotype (8, 10) for *HUMTH01*, (8, 13) for *HUMFES*, (1, 2) for *D12S67*, and (19, 19) for *D1S80*.

VNTR locus							
HUMTH01		**HUMFES**		**D12S67**		**DIS80**	
Repeats	**Freq.**	**Repeats**	**Freq.**	**Repeats**	**Freq.**	**Repeats**	**Freq.**
6	0.230	8	0.007	1	0.015	18	0.293
7	0.160	10	0.321	2	0.005	19	0.011
8	0.105	11	0.373	3	0.058	20	0.021
9	0.193	12	0.233	4	0.078	21	0.032
9.3	0.310	13	0.066	5	0.118	22	0.043
10	0.002			6	0.324	23	0.016
				7	0.196	24	0.335
				8	0.127	25	0.037
				9	0.059	26	0.016
				10	0.020	28	0.059
						29	0.059
						30	0.016
						31	0.043

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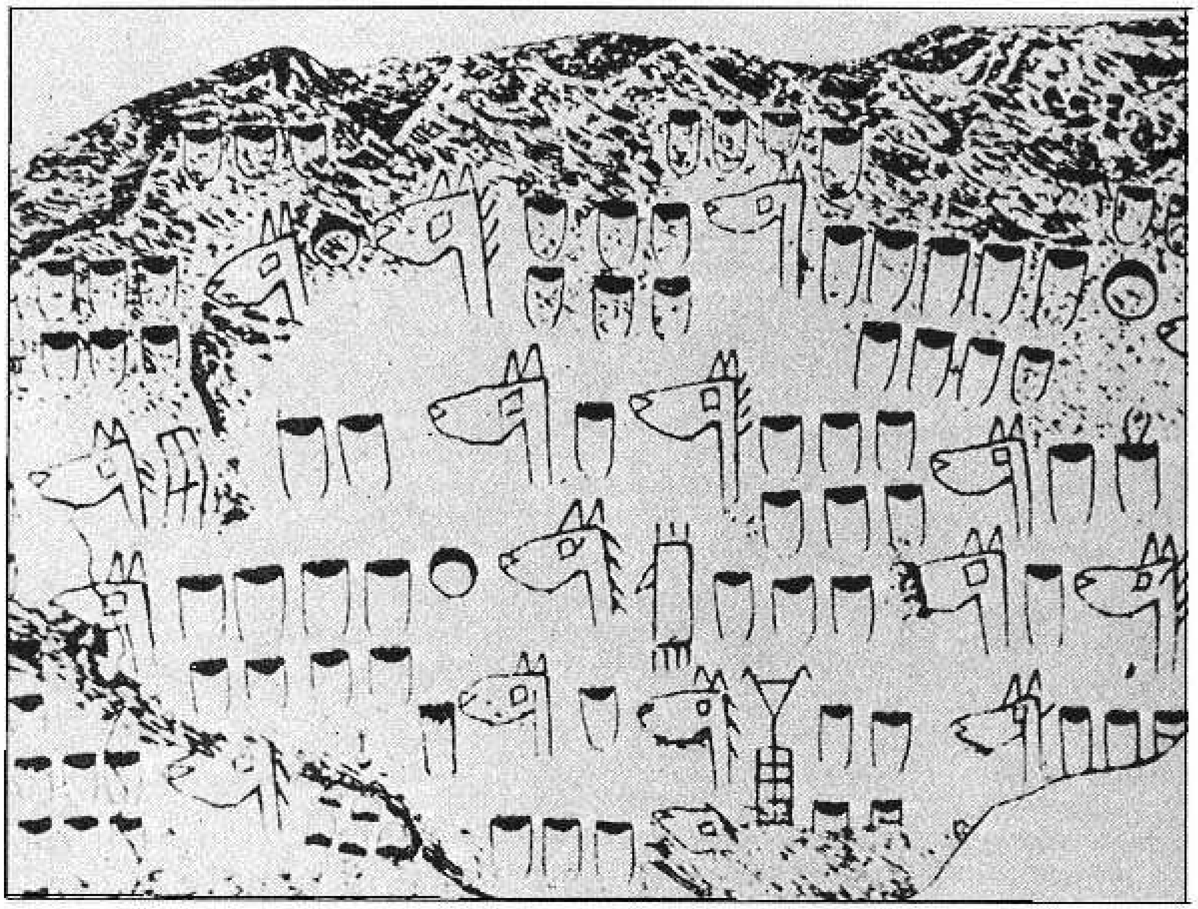

A stone carving found in Asia, about 4000 years old, showing a herd of domesticated horses. The markings indicate that the breeders kept records of desirable traits in their horses and may have used these traits as the basis of artificial selection.
[Courtesy of Dorsey Stuart.]

Chapter 16—
Quantitative Genetics and Multifactorial Inheritance

CHAPTER OUTLINE

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Chapter Summary

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GeNETics on the web

PRINCIPLES

• Multifactorial traits are determined by multiple genetic and environmental factors acting together.

• The relative contributions of genotype and environment to a trait are measured by the variance due to genotype (genotypic variance) and the variance due to environment (environmental variance).

• Correlations between relatives are used to estimate various components of variation, such as genotypic variance, additive variance, and dominance variance.

• Additive variance accounts for the parent-offspring correlation; dominance variance accounts for the sib-sib correlation over and above that expected from the additive variance.

• Narrow-sense heritability is the ratio of additive (transmissible) variance to the total phenotypic variance;

it is widely used in plant and animal breeding.

• Genes that affect quantitative traits can be identified and genetically mapped using genetic polymorphisms, especially highly polymorphic DNA sequences.

CONNECTIONS

CONNECTION: The Supreme Law of Unreason

Francis Galton 1989
Natural Inheritance

CONNECTION: Human Gene Map

Jeffrey C. Murry and 26 other investigators 1994
A comprehensive human linkage map with centimorgan density

Earlier chapters have emphasized traits in which differences in phenotype result from alternative genotypes of a single gene. Examples include green versus yellow peas, red eyes versus white eyes in *Drosophila*, normal versus sickle-cell hemoglobin, and the ABO blood groups. These traits are particularly suited for genetic analysis through the study of pedigrees because of the small number of genotypes and phenotypes and because of the simple correspondence between genotype and phenotype. However, many traits of importance in plant breeding, animal breeding, and medical genetics are influenced by *multiple* genes as well as by the effects of environment. These are known as **multifactorial traits** because of the multiple genetic and environmental factors implicated in their causation. With a multifactorial trait, a single genotype can have many possible phenotypes (depending on the environment), and a single phenotype can include many possible genotypes. Genetic analysis of such complex traits requires special concepts and methods, which are introduced in this chapter.

16.1— Quantitative Inheritance

Multifactorial traits are often called **quantitative traits** because the phenotypes in a population differ in the quantity of a characteristic rather than in type. Height is a quantitative trait. Heights are not found in discrete categories but differ merely in quantity from one person to the next. The opposite of a quantitative trait is a *discrete* trait, in which the phenotypes differ in kind—for example, brown eyes versus blue eyes.

Quantitative traits are typically influenced not only by the alleles of two or more genes but also by the effects of environment. Therefore, with a quantitative trait, the phenotype of an organism is potentially influenced by

• **Genetic factors** in the form of alternative genotypes of one or more genes.

• **Environmental factors** in the form of conditions that are favorable or unfavorable for the development of the trait. Examples include the effect of nutrition on the growth rate of animals, and the effects of fertilizer, rainfall, and planting density on the yield of crop plants.

With some quantitative traits, differences in phenotype result largely from differences in genotype, and the environment plays a minor role. With other quantitative traits, differences in phenotype result largely from the effects of environment, and genetic factors play a minor role. However, most quantitative traits fall between these extremes, and both genotype and environment must be taken into account in their analysis.

In a genetically heterogeneous population, many genotypes are formed by the processes of segregation and recombination. Variation in genotype can be eliminated by studying inbred lines, which are homozygous for most genes, or the F_1 progeny from a cross of inbred lines, which are uniformly heterozygous for all genes in which the parental inbreds differ. In contrast, complete elimination of environmental variation is impossible, no matter how hard the experimenter may try to render the environment identical for all members of a population. With plants, for example, small variations in soil quality or exposure to the sun will produce slightly different environments, sometimes even for adjacent plants. Similarly, highly inbred *Drosophila* still show variation in phenotype (for example, in body size) brought about by environmental differences among animals within the same culture bottle. Therefore, traits that are susceptible to small environmental effects will never be uniform, even in inbred lines.

Most traits that are important in plant and animal breeding are quantitative traits. In agricultural production, one economically important quantitative trait is yield—for example, the size of the harvest of corn, tomatoes, soybeans, or grapes per unit area. In domestic animals, important quantitative traits include meat quality, milk production per cow, egg laying per hen, fleece weight per sheep, and litter size per sow. Important quantitative traits in human genetics include infant growth rate, adult weight, blood pressure, serum cholesterol, and length of life. In evolutionary studies, fitness is the preeminent quantitative trait.

Most quantitative traits cannot be studied by means of the usual pedigree methods because the effects of segregation of alleles of one gene may be concealed by effects of other genes, and environmental effects may cause identical genotypes to have different phenotypes. Therefore, individual pedigrees of quantitative traits do not fit any simple pattern of dominance, recessiveness, or X linkage. Nevertheless, genetic effects on quantitative traits can be assessed by comparing the phenotypes of relatives who, because of their familial relationship, must have a certain proportion of their genes in common. Such studies utilize many of the concepts of population genetics discussed in Chapter 15.

Three categories of traits are frequently found to have quantitative inheritance. They are described in the following section.

Continuous, Meristic, and Threshold Traits

Most phenotypic variation in populations is not manifested in a few easily distinguished categories. Instead, the traits vary continuously from one phenotypic extreme to the other, with no clear-cut breaks in between. Height is again a prime example of such a trait. Other examples include milk production in cattle, growth rate in poultry, yield in corn, and blood pressure in human beings. Such traits are called **continuous traits** because there is a continuous gradation from one phenotype to the next, from minimum to maximum, with no clear categories. Weight is an example of such a trait, because the weight of an organism can fall anywhere along a continuous scale of weights. The distinguishing characteristic of continuous traits is that the phenotype of an individual organism can fall anywhere on a continuous scale of measurement, and so the number of possible phenotypes is virtually unlimited.

Two other types of quantitative traits are not continuous:

Meristic traits are traits in which the phenotype is determined by counting. Some examples are number of skin ridges forming the fingerprints, number of kernels on an ear of corn, number of eggs laid by a hen, number of bristles on the abdomen of a fly, and number of puppies in a litter. Another example of a meristic trait is the number of ears on a stalk of corn, which typically has the value 1, 2, 3, or 4 ears on a given stalk.

Threshold traits are traits that have only two, or a few, phenotypic classes, but their inheritance is determined by the effects of multiple genes together with the environment. Examples of threshold traits are twinning in cattle and parthenogenesis (development of unfertilized eggs) in turkeys. In a threshold trait, each organism has an underlying and not directly observable predisposition to express the trait, such as a predisposition for a cow to give birth to twins. If the underlying predisposition is high enough (above a "threshold"), the cow will actually give birth to twins; otherwise she will give birth to a single calf. In many threshold-trait disorders, the phenotypic classes are "affected" versus "not affected." Examples of threshold-trait disorders in human beings include adultonset diabetes, schizophrenia, and many congenital abnormalities, such as spina bifida. Threshold traits can be interpreted as continuous traits by imagining that each individual has an underlying risk or *liability* toward manifestation of the condition. A liability above a certain cutoff, or *threshold,* results in expression of the condition; a liability below the threshold results in normality. The liability of an individual to expressing a threshold trait cannot be observed directly, but inferences about liability can be drawn from the incidence of the condition among individuals and their relatives. The manner in which this is done is discussed in Section 16.6.

Distributions

The **distribution** of a trait in a population is a description of the population in terms of the proportion of individuals that have each of the possible phenotypes. Characterizing the distribution of some traits is straightforward because the number of phenotypic classes is small. For example, the distribution of progeny in a certain pea cross may consist of 3/4 green seeds and 1/4 yellow seeds, and the distribution of ABO blood groups among Greeks consists of 42 percent O, 39 percent A, 14 percent B, and 5 percent AB. However, with continuous traits, the large number of possible phenotypes makes such summaries impractical. Often, it is convenient to reduce the number of phenotypic classes by grouping similar phenotypes together. Data for an example pertaining to the distribution of height among 4995 British women are given in Table 16.1 and in Figure 16.1. You can imagine each bar in the graph in Figure 16.1 being built, step by step, as each of the women is measured, by placing a small square along the *x*-axis at the location corresponding to the height of that woman. As sampling proceeds, the squares begin to pile up in certain places, leading ultimately to the bar graph shown.

Displaying a distribution completely, in either tabular form (Table 16.1) or graphical form (Figure 16.1), is always helpful but often unnecessary; frequently, a description of the distribution in terms of two major features—the *mean* and the *variance*—is sufficient. To discuss the mean and the variance in quantitative terms, we will need some symbols. In Table 16.1, the height intervals are numbered from 1 (53–55 inches) to 11 (73–75 inches). The

Table 16.1 Distribution of height among British women

Interval number (i)	Height interval (inches)	Midpoint (x_i)	Number of women (f_i)
1	53–55	54	5
2	55–57	56	33
3	57–59	58	254
4	59–61	60	813
5	61–63	62	1340
6	63–65	64	1454
7	65–67	66	750
8	67–69	68	275
9	69–71	70	56
10	71–73	72	11
11	73–75	74	4
		Total N =	995

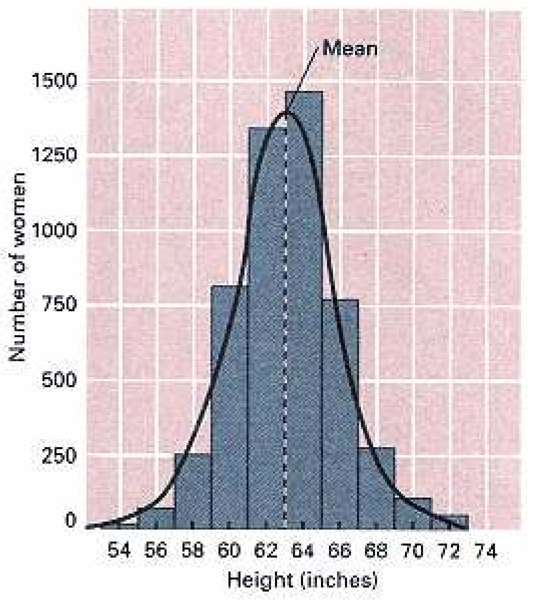


Figure 16.1
Distribution of height among 4995 British women and the smooth normal distribution that approximates the data.

symbol x_i designates the midpoint of the height interval numbered *i;* for example, $x_1 = 54$ inches, $x_2 = 56$ inches, and so on. The number of women in height interval *i* is designated f_i; for example, $f_1 = 5$ women, $f_2 = 33$ women, and so forth. The total size of the sample, in this case 4995, is denoted as *N*. The mean and variance serve to characterize the distribution of height among these women as well as the distribution of many other quantitative traits.

• The **mean,** or average, is the peak of the distribution. The mean of a population is estimated from a sample of individuals from the population, as follows:

$$\bar{x} = \frac{\Sigma f_i x_i}{N} \qquad (1)$$

in which *x* is the estimate of the mean and Σ symbolizes summation over all classes of data (in this example, summation over all 11 height intervals). In Table 16.1, the mean height in the sample of women is 63.1 inches.

• The **variance** is a measure of the spread of the distribution and is estimated in terms of the squared *deviation* (difference) of each observation from the mean. The variance is estimated from a sample of individuals as follows:

$$s^2 = \frac{\Sigma f_i (x_i - \bar{x})^2}{N - 1} \qquad (2)$$

in which s^2 is the estimated variance and x^i, f^i and N are as in Table 16.1. Note that $(x_i - \bar{x})$ is the difference from the mean of each height category and that the denominator is the total number of individuals minus 1. The variance describes the extent to which the phenotypes are clustered around the mean, as shown in Figure 16.2. A large value implies that the distribution is spread out, and a small value implies that it is clustered near the mean. From the data in Table 16.1, the variance of the population of British women is estimated as $s^2 = 7.24$ in^2.

A quantity closely related to the variance—the **standard deviation** of the distribution—is defined as the square root of the variance. For the data in Table 16.1, the estimated standard deviation s is obtained from Equation 2 as

$$s = (s^2)^{1/2} = (7.24 \text{ in}^2)^{1/2} = 2.29 \text{ inches}$$

The standard deviation has the useful feature of having the same units of dimension as the mean—in this example, inches.

When the data are symmetrical, or approximately symmetrical, the distribution of a trait can often be approximated by a smooth arching curve of the type shown in Figure 16.1. The arch-shaped curve is called the **normal distribution.** Because the normal curve is symmetrical, half of its area is determined by points that have values greater than the mean and half by points with values less than the mean, and thus the proportion of phenotypes that exceed the mean is 1/2. One important characteristic of the normal distribution is that it is completely determined by the value of the mean and the variance.

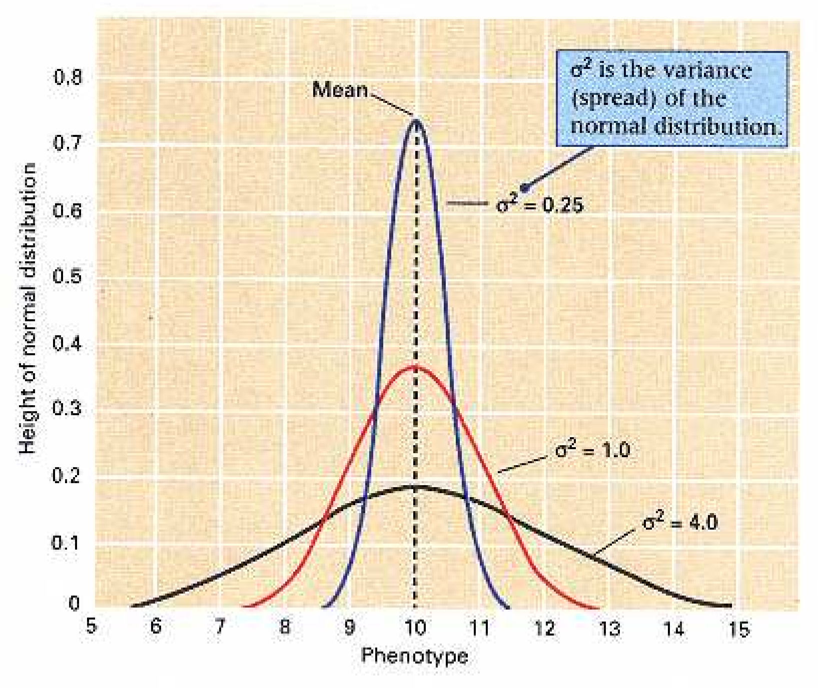


Figure 16.2
Graphs showing that the variance of a distribution measures the spread of the distribution around the mean. The area under each curve covering any range of phenotypes equals the proportion of individuals having phenotypes within that range.

Connection The Supreme Law of Unreason

Francis Galton 1889

42 Rutland Gate, South Kensington,
London, England

Francis Galton pioneered the application of statistics to biological problems. He was captivated by the properties of the normal distribution. This form of distribution is also known by several other names, including the "bell curve," the "Gaussian distribution," and the "law of the frequency of errors." The name Gaussian comes from Carl Friedrich Gauss, a German mathematician who studied the distribution in the early 1800s. The distribution was popularized by the Belgian mathematician and astronomer Adolphe Quetelet (1796–1874). He showed that various human measurements conformed to the normal distribution, including the heights of French draftees and the chest girths of Scottish soldiers. From this conformity, Quetelet developed the concept of the homme moyen, or "average person," as the mean of the distribution. Deviations from the homme moyen were regarded as "errors" of the same sort that arise in astronomy when the magnitude of any celestial quantity is estimated. In this concept, variation around the mean is a regrettable fact of an imperfect world. It was Galton who realized that in biology, variation is an intrinsic result of differences in genotype and environment among individuals. He regarded biological variation as a subject demanding study in its own right, and to him we owe such fundamental concepts as correlation and regression. Why is the normal distribution so often found in practice? A theorem important in statistics, called the Central Limit Theorem, states that under quite general conditions, the sum of a large number of independent random quantities is bound to converge to a normal distribu-

The law would have been personified by the Greeks if they had known of it.

tion. In quantitative genetics, the independent quantities are multiple genes and multiple environmental factors, which often do interact to yield a normal distribution of a trait.

It is difficult to understand why statisticians commonly limit their inquiries to averages, and do not revel in more comprehensive views. Their souls seem dull to the charm of variety . . .. An average is but a solitary fact, whereas if a single other fact [the standard deviation] be added to it, an entire normal distribution, which nearly corresponds to the observed one, starts potentially into existence . . .. Some people hate the very name of statistics, but it find it full of beauty and interest. Measurements, whenever they are not brutalized, but delicately handled, and are warily interpreted, their power of dealing with complicated phenomena is extraordinary. They are the only tools by which an opening can be cut through the formidable thicket of difficulties that bar the path of those who pursue the science of humanity . . .. I know of scarcely anything so apt to impress the imagination as the wonderful form of cosmic order expressed by the "law of frequency of error" [the normal distribution]. Whenever a large sample of chaotic elements is taken in hand and marshaled in the order of their magnitude, this unexpected and most beautiful form of regularity proves to have been latent all along. The law would have been personified by the Greeks if they had known of it. It reigns with serenity and complete self-effacement amidst the wildest confusion. The larger the mob and the greater the apparent anarchy, the more perfect is its sway. It is the supreme law of unreason.

Source: Natural Inheritance. **Macmillan Publishers, London**

The mean and standard deviation (square root of the variance) of a normal distribution provide a great deal of information about the distribution of phenotypes in a population, as is illustrated in Figure 16.3. Specifically, for a normal distribution,

1. Approximately 68 percent of the population have a phenotype within 1 standard deviation of the mean (in the symbols of Figure 16.3, between $\mu - \sigma$ and $\mu + \sigma$).

2. Approximately 95 percent lie within 2 standard deviations of the mean (between $\mu - 2\sigma$ and $\mu + 2\sigma$).

3. Approximately 99.7 percent lie within 3 standard deviations of the mean (between $\mu - 3\sigma$ and $\mu + 3\sigma$).

Applying these rules to the data in Figure 16.1, in which the mean andstandard deviation are 63.1 and 2.69 inches, reveals that approximately 68 percent of the women are expected to have heights in the range

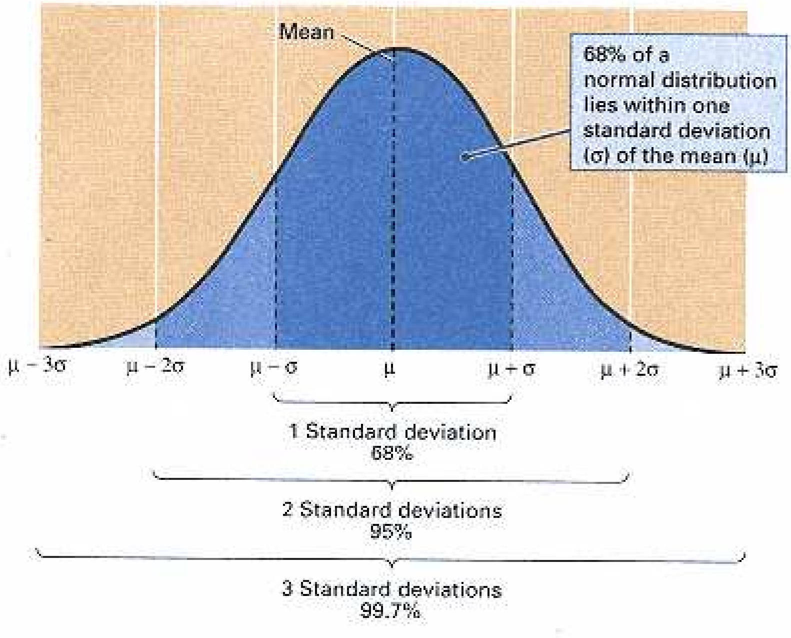


Figure 16.3
Features of a normal distribution. The proportions of individuals lying within 1, 2, and 3 standard deviations of the mean are approximately 68 percent, 95 percent, and 99.7 percent, respectively. In this normal distribution, the mean is symbolized μ and the standard deviation σ.

from 63.1 - 2.69 inches to 63.1 + 2.69 inches (that is, 60.4–65.8), and approximately 95 percent are expected to have heights in the range from 63.1 - 2 × 2.69 inches to 63.1 + 2 × 2.69 inches (that is, 57.7–68.5).

Real data frequently conform to the normal distribution. Normal distributions are usually the rule when the phenotype is determined by the cumulative effect of many individually small independent factors. This is the case for many multifactorial traits.

16.2—
Causes of Variation

In considering the genetics of multifactorial traits, an important objective is to assess the relative importance of genotype and environment. In some cases in experimental organisms, it is possible to separate genotype and environment with respect to their effects on the mean. For example, a plant breeder may study the yield of a series of inbred lines grown in a group of environments that differ in planting density or amount of fertilizer. It would then be possible (1) to compare yields of the same genotype grown in different environments and thereby rank the *environments* relative to their effects on yield, or (2) to compare yields of different genotypes grown in the same environment and thereby rank the *genotypes* relative to their effects on yield.

Such a fine discrimination between genetic and environmental effects is not usually possible, particularly in human quantitative genetics. For example, with regard to the height of the women in Figure 16.1, environment could be considered favorable or unfavorable for tall stature only in comparison with the mean height of a genetically identical population reared in a different environment. This reference population does not exist. Likewise, the genetic composition of the population could be judged as favorable or unfavorable for tall stature only in comparison with the mean of a genetically different population reared in an identical environment. This reference population does not exist either.

Without such standards of comparison, it is impossible to distinguish genetic from environmental effects on the mean. However, it is still possible to assess genetic versus environmental contributions to the *variance*, because instead of comparing the means of two or more populations, we can compare the phenotypes of individuals within the *same* population. Some of the differences in phenotype result from differences in genotype and others from differences in environment, and it is often possible to separate these effects.

In any distribution of phenotypes, such as the one in Figure 16.1, four sources contribute to phenotypic variation:

1. Genotypic variation.

2. Environmental variation.

3. Variation due to genotype-environment interaction.

4. Variation due to genotype-environment association.

Each of these sources of variation is discussed in the following sections.

Genotypic Variance

The variation in phenotype caused by differences in genotype among individuals is termed **genotypic variance.** Figure 16.4 illustrates the genetic variation expected among the F_2 generation from a cross of two inbred lines that differ in genotype for three unlinked genes. The alleles of the three genes are represented as *Aa, Bb,* and *Cc,* and the genetic variation in the F_2 caused by segregation and recombination is

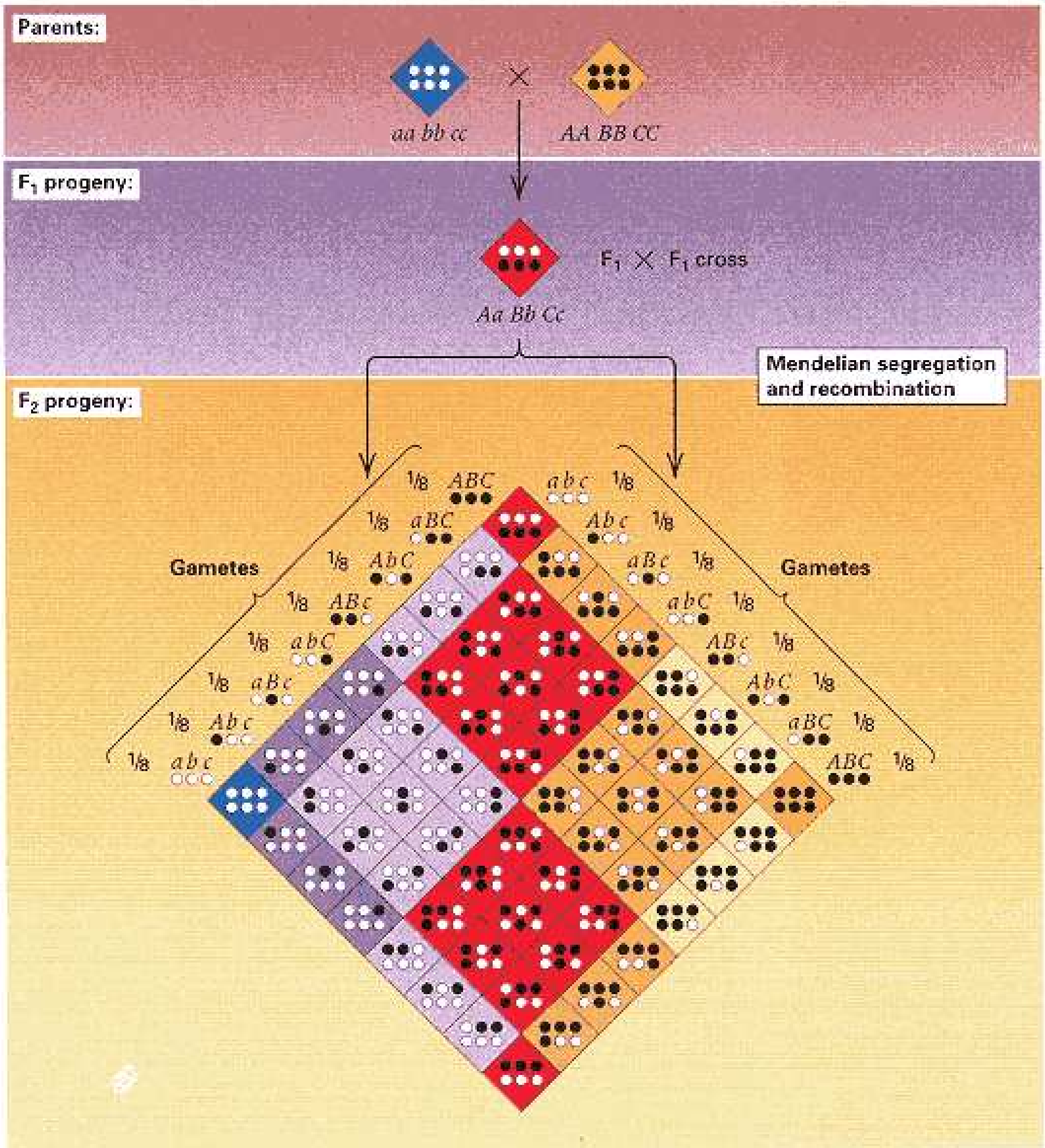


Figure 16.4
Segregation of three independent genes affecting a quantitative trait. Each uppercase allele in a genotype contributes one unit to the phenotype.

evident in the differences in color. Relative to a meristic trait (a trait whose phenotype is determined by counting, such as ears per stalk in corn), if we assume that each uppercase allele is favorable for the expression of the trait and adds one unit to the phenotype, whereas each lowercase allele is without effect, then the *aa bb cc* genotype has a phenotype of 0, and the *AA BB CC* genotype has a phenotype of 6. There are seven possible phenotypes (0–6) in the F_2 generation. The distribution of phenotypes in the F_2 generation is shown in the colored bar graph in Figure 16.5. The normal distribution approximating the data has a mean of 3 and a variance of 1.5. In this case, we are assuming that *all* of the variation in phenotype in the population results from differences in genotype among the individuals.

Figure 16.5 also includes a bar graph with diagonal lines that represent the theoretical distribution when the trait is determined by 30 unlinked genes segregating in a randomly mating population, grouped into the same number of phenotypic classes as the 3-gene case. We assume that 15 of the genes are nearly fixed for the favorable allele and 15 are nearly fixed for the unfavorable allele. The contribution of each favorable allele to the phenotype has been chosen to make the mean of the distribution equal to 3 and the variance equal to 1.5. Note that the distribution with 30 genes is virtually identical to that with 3 genes and that both are approximated by the same normal curve. If such distributions were encountered in actual research, the experimenter would not be able to distinguish between them. The key point is that

> Even in the absence of environmental variation, the distribution of phenotypes, by itself, provides no information about the number of genes influencing a trait and no information about the dominance relations of the alleles.

However, the number of genes influencing a quantitative trait is important in determining the potential for long-term genetic improvement of a population by means of artificial selection. For example, in the 3-gene case in Figure 16.5, the best possible genotype would have a phenotype of 6, but in the 30-gene case, the best possible

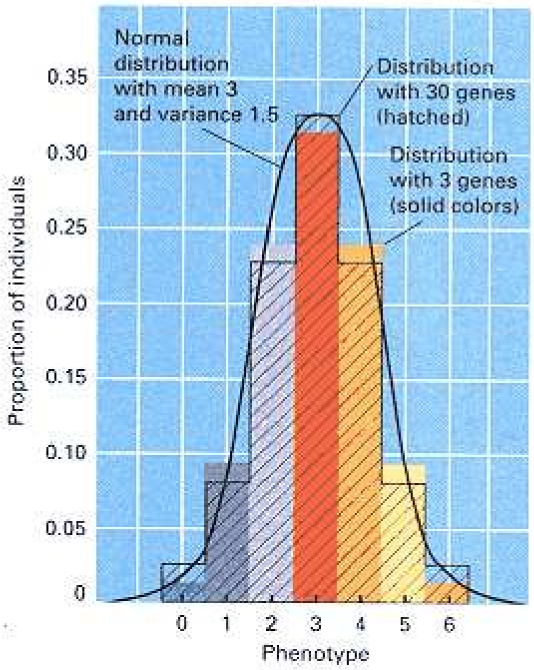


Figure 16.5
The colored bar graph is the distribution of phenotypes determined by the segregation of the 3 genes illustrated in Figure 16.4. The bar graph with the diagonal lines is the theoretical distribution expected from the segregation of 30 independent genes. Both distributions are approximated by the same normal distribution (black curve).

genotype (homozygous for the favorable allele of all 30 genes) would have a phenotype of 30.

Later in this chapter, some methods for estimating how many genes affect a quantitative trait will be presented. All the methods depend on comparing the phenotypic distributions in the F_1 and F_2 generations of crosses between nearly or completely homozygous lines.

Environmental Variance

The variation in phenotype caused by differences in environment among individuals is termed **environmental variance.** Figure 16.6 is an example showing the distribution of seed weight in edible beans. The mean of the distribution is 500 mg, and the standard deviation is 95 mg. All of the beans in this population are genetically identical and homozygous because they are highly inbred. Therefore, in this population, *all* of the phenotypic variation in seed

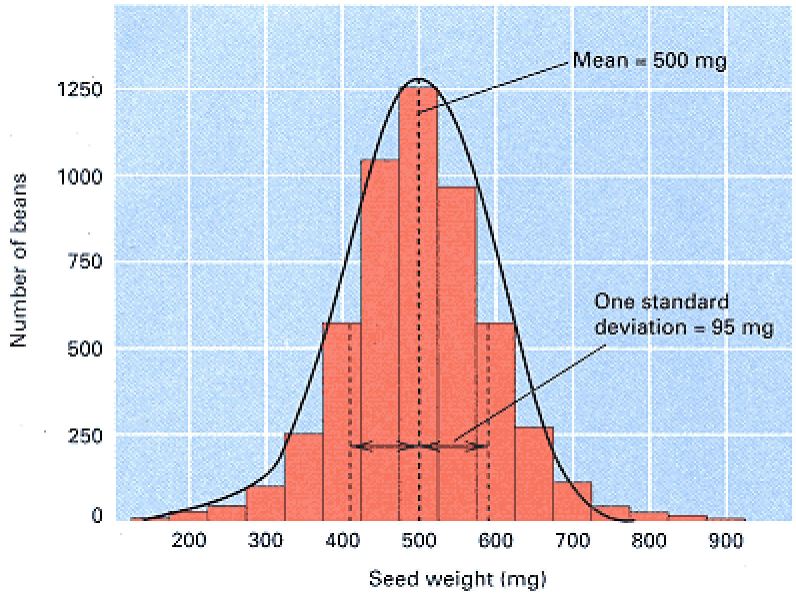


Figure 16.6
Distribution of seed weight in a homozygous line of edible beans. All variation in phenotype among individuals results from environmental differences.

weight results from environmental variance. A comparison of Figure 16.5 and 16.6 demonstrates the following principle:

> The distribution of a trait in a population provides no information about the relative importance of genotype and environment. Variation in the trait can be entirely genetic, it can be entirely environmental, or it can reflect a combination of both influences.

Genotypic and environmental variance are seldom separated as clearly as in Figures 16.5 and 16.6, because usually they work together. Their combined effects are illustrated for a simple hypothetical case in Figure 16.7. Figure 16.7A is the distribution of phenotypes for three genotypes assumed not to be influenced by environment. As depicted, the trait can have one of three distinct and nonoverlapping phenotypes determined by the effects of two additive alleles. The genotypes are in random-mating proportions for an allele frequency of 1/2, and the distribution of phenotypes has mean 5 and variance 2. Because it results solely from differences in genotype, this variance is *genotypic variance*, which is symbolized σ_g^2. Figure 16.7B is the distribution of phenotypes in the presence of environmental variation, illustrated for each genotype separately. Each distribution corresponds to the one in Figure 16.6, and the variance is 1. Because this variance results solely from differences in environment, it is *environmental variance*, which is symbolized σ_e^2. When the effects of genotype and environment are combined in the same population, each genotype is affected by environmental variation, and the distribution shown in Figure 16.7C of the figure results. The variance of this distribution is the **total variance** in phenotype, which is symbolized σ_t^2. Because we are assuming that genotype and environment have separate and independent effects on phenotype, we expect σ_t^2 to be greater than either σ_g^2 or σ_e^2 alone. In fact,

$$\sigma_t^2 = \sigma_g^2 + \sigma_e^2 \qquad (3)$$

In words, Equation (3) states that

> When genetic and environmental effects contribute independently to phenotype, the total variance equals the sum of the genotypic variance and the environmental variance.

Equation (3) is one of the most important relations in quantitative genetics. How it can be used to analyze data will be explained shortly. Although the equation serves as an excellent approximation in very many cases, it is valid in an exact sense only when genotype and environment are independent in their effects on phenotype. The two most important departures from independence are discussed in the next section.

Genotype-Environment Interaction and Genotype-Environment Association

In the simplest cases, environmental effects on phenotype are additive, and each environment adds or detracts the same amount from the phenotype, independent of the genotype. When this is not true, environmental effects on phenotype differ according to genotype, and a **genotype-environment interaction (G-E interaction)** is said to be present. In some cases,

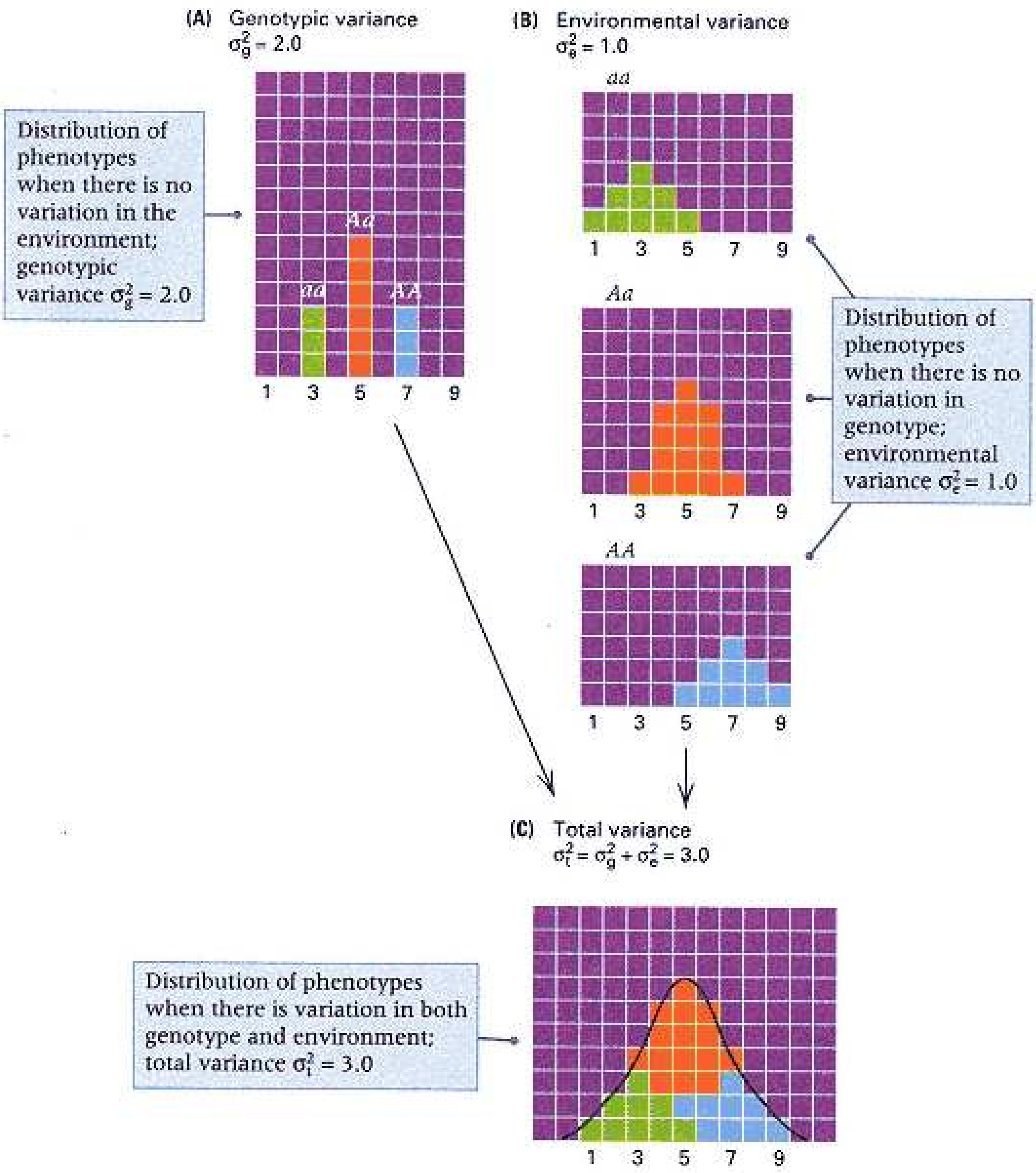


Figure 16.7
The combined effects of genotypic and environmental variance. (A) Population affected only by genotypic variance, σ_g^2 (B) Populations of each genotype affected only by environmental variance, σ_e^2. (C) Population affected by both genotypic and environmental variance, in which the total phenotypic variance, σ_t^2 equals the sum of σ_g^2 and σ_e^2

G-E interaction can even change the relative ranking of the genotypes, so a genotype that is superior in one environment may become inferior in another.

An example of genotype-environment interaction in maize is illustrated in Figure 16.8. The two strains of corn are hybrids formed by crossing different pairs of inbred lines, and their overall means, averaged across all of the environments, are approximately the same. However, the strain designated A clearly outperforms B in the stressful (negative) environments, whereas the performance is reversed when the environment is of high quality. (Environmental quality is judged on the basis of soil fertility, moisture, and other factors.) In some organisms, particularly plants, experiments like those illustrated in Figure 16.8 can be carried out to determine the contribution of G-E interaction to the total observed variation in phenotype. In other organisms, particularly human beings, the effect cannot be evaluated separately.

Interaction of genotype and environment is common and is very important in both plants and animals. Because of interaction, no one plant variety will

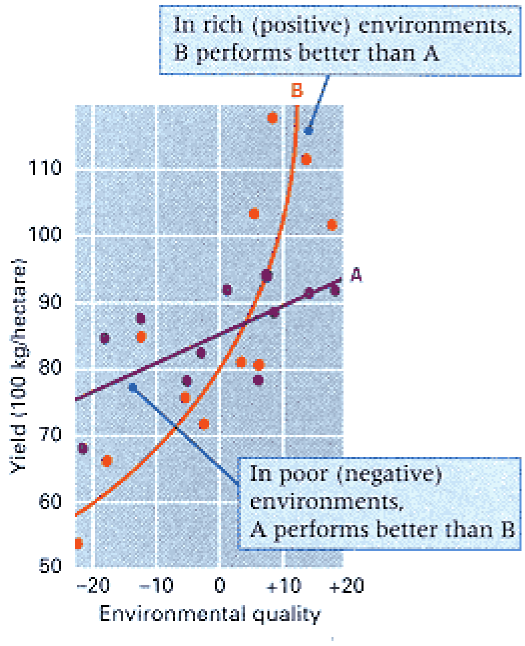


Figure 16.8
Genotype-environment interaction in maize. Strain A is superior when environmental quality is low (negative numbers), but strain B is superior when environmental quality is high. [Data from W. A. Russell. 1974. *Annual Corn * Sorghum Research Conference,* 29: 81.]

outperform all others in all types of soil and climate, and therefore plant breeders must develop special varieties that are suited to each growing area.

When the different genotypes are not distributed at random in all the possible environments, there is **genotype-environment association (G-E association).** In these circumstances, certain genotypes are preferentially associated with certain environments, which may either increase or decrease the phenotype of these genotypes compared with what would result in the absence of G-E association. An example of deliberate genotype-environment association can be found in dairy husbandry, in which some farmers feed each of their cows in proportion to its level of milk production. Because of this practice, cows with superior genotypes with respect to milk production also have a superior environment in that they receive more feed. In plant breeding, genotype-environment association can often be eliminated or minimized by appropriate randomization of genotypes within the experimental plots. In other cases, human genetics again being a prime example, the possibility of G-E association cannot usually be controlled.

16.3— Analysis of Quantitative Traits

Equation (3) can be used to separate the effects of genotype and environment on the total phenotypic variance. Two types of data are required: (1) the phenotypic variance of a genetically uniform population, which provides an estimate of σ_e^2 because a genetically uniform population has a value of σ_g^2 = 0, and (2) the phenotypic variance of a genetically heterogeneous population, which provides an estimate of $\sigma_g^2 + \sigma_e^2$. An example of a genetically uniform population is the F_1 generation from a cross between two highly homozygous strains, such as inbred lines. An example of a genetically heterogeneous population is the F_2 generation from the same cross. If the environments of both populations are the same, and if there is no G-E interaction, then the estimates may be combined to deduce the value of σ_g^2.

To take a specific numerical illustration, consider variation in the size of the eyes in the cave-dwelling fish, *Astyanax*, all individuals being reared in the same environment. The variances in eye diameter in the F_1 and F_2 generations from a cross of two highly homozygous strains were estimated as 0.057 and 0.563, respectively. Written in terms of the components of variance, these are

$$F_2: \quad \sigma_t^2 = \sigma_g^2 + \sigma_e^2 = 0.563$$
$$F_1: \quad \sigma_e^2 = 0.057$$

The estimate of genotypic variance σ_g^2 is obtained by subtracting the second equation from the first; that is,

$$\sigma_g^2 = 0.563 - 0.057 = 0.506$$

because

$$(\sigma_g^2 + \sigma_e^2) - \sigma_e^2 = \sigma_g^2$$

Hence, the estimate of σ_g^2 is 0.506, whereas that of σ_e^2 is 0.057. In this example, the genotypic variance is much greater than the environmental variance, but this is not always the case.

The next section shows what information can be obtained from an estimate of the genotypic variance.

The Number of Genes Affecting a Quantitative Trait

When the number of genes influencing a quantitative trait is not too large, knowledge of the genotypic variance can be used to estimate the number of genes. All we need are the means and variances of two phenotypically divergent strains and their F_1, F_2, and backcrosses. In ideal cases, the data appear as in Figure 16.9, in which P_1 and P_2 represent the divergent parental strains (for example, inbred lines). The points lie on a triangle, with increasing variance according to the increasing genetic heterogeneity (genotypic variance) of the populations. If the F_1 and backcross means lie exactly between their parental means, then these means will lie at the midpoints along the sides of the triangle, as shown in Figure 16.9. This finding implies that the alleles affecting the trait are *additive;* that is, for each gene, the phenotype of the heterozygote is the average of the phenotypes of the corresponding homozygotes. In such a simple situation, it can be shown that the number, *n*, of genes contributing to the trait is

$$n = \frac{D^2}{8\sigma_g^2} \qquad (4)$$

in which *D* represents the difference between the means of the original parental strains, P_1 and P_2. This equation can be verified by applying it to the ideal case in Figure 16.7. In this case, the parental strains would be the homozygous genotypes *AA*, with a mean phenotype of 7, and *aa*, with a mean phenotype of 3. Consequently, $D = 7 - 3 = 4$. The genotypic variance is given in Figure 16.7 as $\sigma_g^2 = 2$. Substituting *D* and σ_g^2 into Equation 4, we obtain $n = 16/(8 \times 2) = 1$, which is correct because there is only one gene, with alleles *A* and *a*, that affects the trait.

Applied to actual data, Equation (4) requires several assumptions that are not necessarily valid. In addition to the assumption that all generations are reared in the same environment, the theory also makes the genetic assumptions that (1) the alleles of each gene are additive, (2) the genes contribute equally to the trait, (3) the genes are unlinked, and (4) the original parental strains are homozygous for alternative alleles of each gene. However, when the assumptions are invalid, then the calculated *n* is smaller than the actual number of genes affecting the trait. The estimated number is a minimum because almost any departure from the genetic assumptions leads to a smaller genotypic variance in the F_2 generation and so, for the same value of *D*, would yield a larger value of *n* in Equation (4). This is why the estimated *n* is the *minimum* number of genes that can account for the data. For the cave-dwelling *Astyanax* fish discussed in the preceding section, the parental strains had average phenotypes of 7.05 and 2.10, giving $D = 4.95$. The estimated value of $\sigma_g^2 = 0.506$, and so the minimum number of genes affecting eye diameter is $n = (4.95)^2/(8 \times 0.506) = 6.0$. Therefore, at least six different genes affect the diameter of the eye of the fish.

The number of genes that affect a quantitative trait is important because it

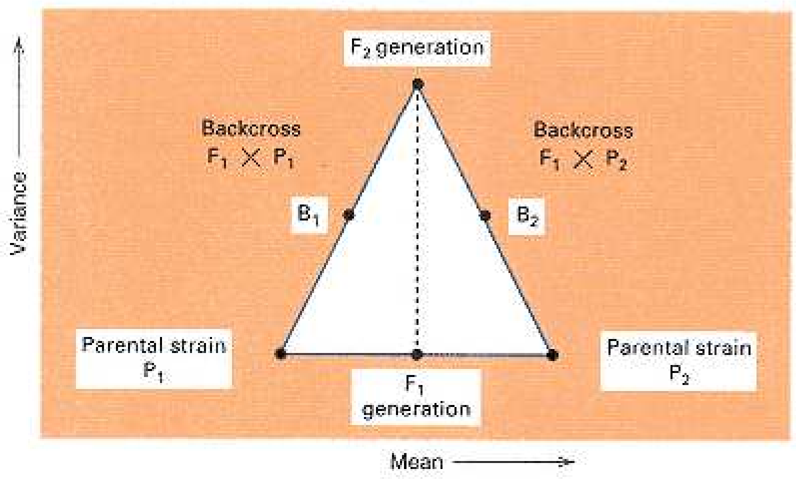


Figure 16.9
Means and variances of parents (P), backcross (B), and hybrid (F) progeny of inbred lines for an ideal quantitative trait affected by unlinked and completely additive genes.
[After R. Lande. 1981. *Genetics*, 99: 541.]

influences the amount by which a population can be genetically improved by selective breeding. With traits determined by a small number of genes, the potential for change in a trait is relatively small, and a population consisting of the best possible genotypes may have a mean value that is only 2 or 3 standard deviations above the mean of the original population. However, traits determined by a large number of genes have a large potential for improvement. For example, after a population of the flour beetle *Tribolium* was bred for increased pupa weight, the mean value for pupa weight was found to be 17 standard deviations above the mean of the original population. Determination of traits by a large number of genes implies that

> Selective breeding can create an improved population in which the value of *every* individual greatly exceeds that of the *best* individuals that existed in the original population.

This principle at first seems paradoxical, because in a large enough population, every possible genotype should be created at some low frequency. The resolution of the paradox is that real populations subjected to selective breeding typically consist of a few hundred organisms (at most), and hence many of the theoretically possible genotypes are never formed because their frequencies are much too rare. As selection takes place, and the allele frequencies change, these genotypes become more common and allow the selection of superior organisms in future generations.

Broad-Sense Heritability

Estimates of the number of genes that determine quantitative traits are frequently unavailable because the necessary experiments are impractical or have not been carried out. Another attribute of quantitative traits, which requires less data to evaluate, makes use of the ratio of the genotypic variance to the total phenotypic variance. This ratio of σ_g^2 to σ_t^2 is called the **broad-sense heritability,** symbolized as H^2, and it measures the importance of genetic variation, relative to environmental variation,in causing variation in the phenotype of a trait of interest. Broad-sense heritability is a ratio of variances, specifically

$$H^2 = \frac{\sigma_g^2}{\sigma_t^2} = \frac{\sigma_g^2}{\sigma_g^2 + \sigma_e^2} \qquad (5)$$

Substitution of the data for eye diameter in *Astyanax*, in which $\sigma_g^2 = 0.506$ and $\sigma_g^2 + \sigma_e^2 = 0.563$, into Equation (5) yields $H^2 = 0.506/0.563 = 0.90$ for the estimate of broad-sense heritability. This value implies that 90 percent of the variation in eye diameter in this population results from differences in genotype among the fish.

Knowledge of heritability is useful in the context of plant and animal breeding, because heritability can be used to predict the magnitude and speed of population improvement. The broad-sense heritability defined in Equation (5) is used in predicting the outcome of selection practiced among clones, inbred lines, or varieties. Analogous predictions for randomly bred populations utilize another type of heritability, different from H^2, which we will discuss shortly. Broad-sense heritability measures how much of the total variance in phenotype results from differences in genotype. For this reason, $H^2$is often of interest in regard to human quantitative traits.

Twin Studies

In human beings, twins would seem to be ideal subjects for separating genotypic and environmental variance, because **identical twins,** which arise from the splitting of a single fertilized egg, are genetically identical and are often strikingly similar in such traits as facial features and body build. **Fraternal twins,** which arise from two fertilized eggs, have the same genetic relationship as ordinary siblings, and therefore only half of the genes in either twin are identical with those in the other. Theoretically, the variance between members of an identical-twin pair would be equivalent to σ_e^2 , because the twins are genetically identical; whereas the variance between members of a fraternal-twin pair would include not only σ_e^2 but also part of the genotypic variance (approximately

σ_g^2 /2, because of the identity of half of the genes in fraternal twins). Consequently, both σ_g^2 and σ_e^2 could be estimated from twin data and combined as in Equation (5) to estimate H^2. Table 16.2 summarizes estimates of H^2 based on twin studies of several traits.

Unfortunately, twin studies are subject to several important sources of error, most of which increase the similarity of identical twins, so the numbers in Table 16.2 should be considered very approximate and probably too high. Four of the potential sources of error are

1. Genotype-environment interaction, which increases the variance in fraternal twins but not in identical twins.

2. Frequent sharing of embryonic membranes between identical twins, resulting in more similar intrauterine environments than those of fraternal twins.

3. Greater similarity in the treatment of identical twins by parents, teachers, and peers, resulting in a decreased environmental variance in identical twins.

4. Different sexes in half of the pairs of fraternal twins, in contrast with the same sex of identical twins.

These pitfalls and others imply that data from human twin studies should be interpreted with caution and reservation.

16.4— Artificial Selection

The practice among breeders of choosing a select group of organisms from a population to become the parents of the next generation is termed **artificial selection.** When artificial selection is carried out either by choosing the best organisms in a species that reproduces asexually or by choosing the best subpopulation among a series of subpopulations, each propagated by close inbreeding (such as self-fertilization), the broad-sense heritability makes possible an assessment of how rapidly progress can be achieved. Broad-sense heritability is important in this context because, with clones, inbred lines, or varieties, superior genotypes can be perpetuated without disruption of favorable gene combinations by Mendelian segregation. An example is the selection of superior varieties of plants that are propagated asexually by means of cuttings or grafts. Because there is no sexual reproduction, each plant has exactly the same genotype as its parent.

In sexually reproducing populations that are genetically heterogeneous, broadsense heritability is not relevant in predicting progress resulting from artificial selection, because superior genotypes must necessarily be broken up by the processes of segregation and recombination. For example, if the best genotype is heterozygous for

Table 16.2 Broad-sense heritability, in percent, based on twin studies

Trait	Heritability (H^2)	Trait	Heritability (H^2)
Longevity	29	Verbal ability	63
Height	85	Numerical ability	76
Weight	63	Memory	47
Amino acid excretion	72	Sociability index	66
Serum lipid levels	44	Masculinity index	12
Maximum blood lactate	34	Temperament index	58
Maximum heart rate	84		

each of two unlinked loci, *Aa Bb*, then because of segregation and independent assortment, among the progeny of a cross between parents with the best genotypes—*Aa Bb* × *Aa Bb*—only 1/4 will have the same favorable *Aa Bb* genotype as the parents. The rest of the progeny will be genetically inferior to the parents. For this reason, to the extent that high genetic merit may depend on particular combinations of alleles, each generation of artificial selection results in a slight setback in that the offspring of superior parents are generally not quite so good as the parents themselves. Progress under selection can still be predicted, but the prediction must make use of another type of heritability, the narrow-sense heritability, which is discussed in the next section.

Narrow-Sense Heritability

Figure 16.10 illustrates a typical form of artificial selection and its result. The organism is *Nicotiana longiflora* (tobacco), and the trait is the length of the corolla tube (the corolla is a collective term for all the petals of a flower). Figure 16.10A shows the distribution of phenotypes in the parental generation, and Figure 16.10B shows the distribution of phenotypes in the offspring generation. The parental generation is the population from which the parents were chosen for breeding. The type of selection is called **individual selection,** because each member of the population to be selected is evaluated according to its own individual phenotype. The selection is practiced by choosing some arbitrary level of phenotype—called the **truncation point**—that determines which individuals will be saved for breeding purposes. All individuals with a phenotype above the threshold are randomly mated among themselves to produce the next generation.

Phenotypic Change with Selection: A Prediction Equation

In evaluating progress through individual selection, three distinct phenotypic means are important. In Figure 16.10, these means are symbolized as M, M^* and M' and are defined as follows:

1. M is the mean phenotype of the entire population in the parental generation, including both the selected and the nonselected individuals.

2. M^* is the mean phenotype among those individuals selected as parents (those with a phenotype above the truncation point).

3. M' is the mean phenotype among the progeny of selected parents.

The relationship among these three means is given by

$$M' = M + h^2(M^* - M) \qquad (6)$$

where the symbol h^2 is the **narrow-sense heritability** of the trait in question.

Later in this chapter, a method for estimating narrow-sense heritability from the similarity in phenotype among relatives will be explained. In Figure 16.10, h^2 is the only unknown quantity, so it can be estimated from the data themselves. Rearranging Equation (6) and substituting the values for the means from Figure 16.10, we find that

$$h^2 = \frac{M' - M}{M^* - M} = \frac{77 - 70}{81 - 70} = 0.64$$

In a manner analogous to the way in which total phenotypic variance can be split into the sum of the genotypic variance and the environmental variance (Equation (3), the genotypic variance can be split into parts resulting from the additive effects of alleles, dominance effects, and effects of interaction between alleles of different genes. The difference between the broad-sense heritability, H^2, and the narrow-sense heritability, h^2, is that H^2 includes all of these genetic contributions to variation, whereas h^2 includes only the additive effects of alleles. From the standpoint of animal or plant improvement, h^2 is the heritability of interest, because

> The narrow sense heritability, h^2, is the proportion of the variance in phenotype that can be used to predict changes in population mean with individual selection according to Equation (6).

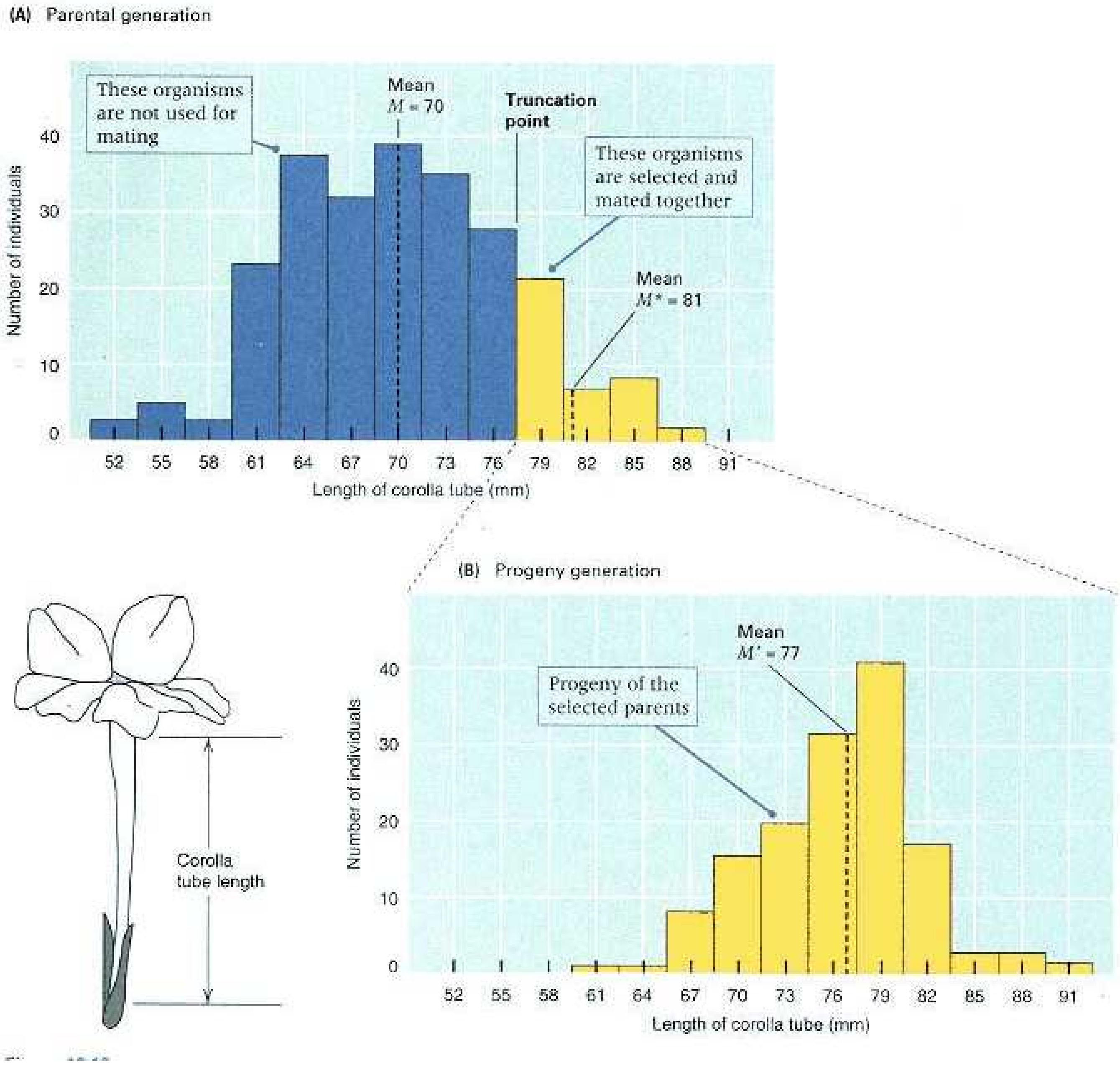


Figure 16.10
Selection for increased length of corolla tube in tobacco. (A) Distribution of phenotypes in the parental generation. The symbol *M* denotes the mean phenotype of the entire population, and *M** denotes the mean phenotype of the organisms chosen for breeding (organisms with a phenotype that exceeds the truncation point). (B) Distribution of phenotypes among the offspring bred from the selected parents. The symbol *M'* denotes the mean.

The distinction between the broad-sense heritability and the narrow-sense heritability can be appreciated intuitively by considering a population in which there is a rare recessive gene. In such a case, most homozygous recessive genotypes come from matings between heterozygous carriers. Some such kindreds have more than one affected offspring. Hence, affected siblings can resemble each other more than they resemble their parents. For example, if *a* is a recessive allele, the mating *Aa* × *Aa,* in which both parents show the dominant phenotype, may yield two offspring that are *aa*, which both show the recessive phenotype; in this case, the offspring are more similar to each other than to either of the parents. It is the dominance of the wildtype allele that causes this paradox, contributing to the broad-sense heritability of the trait but not to the narrow-sense heritability. The narrow-sense heritability includes only those genetic effects that contribute to the resemblance between parents and their offspring, because narrow-sense heritability measures how similar offspring are to their parents.

In general, the narrow-sense heritability of a trait is smaller than the broad-sense heritability. For example, in the parental generation in Figure 16.10, the broad-sense heritability of corolla tube length is $H^2 = 0.82$. The two types of heritability are equal only when the alleles affecting the trait are additive in their effects; with additive effects, each heterozygous genotype shows a phenotype that is exactly intermediate between the phenotypes of the respective homozygous genotypes.

Equation (6) is of fundamental importance in quantitative genetics because of its predictive value. This can be seen in the following example. The selection in Figure 16.10 was carried out for several generations. After two generations, the mean of the population was 83, and parents having a mean of 90 were selected. By use of the estimate $h^2 = 0.64$, the mean in the next generation can be predicted. The information provided is that $M = 83$ and $M^* = 90$. Therefore, Equation (6) implies that the predicted mean is

$$M' = 83 + (0.64)(90 - 83) = 87.5$$

This value is in good agreement with the observed value of 87.9.

Long-Term Artificial Selection

Artificial selection is analogous to natural selection in that both types of selection cause an increase in the frequency of alleles that improve the selected trait (or traits). The principles of natural selection discussed in Chapter 15 also apply to artificial selection. For example, artificial selection is most effective in changing the frequency of alleles that are in an intermediate range of frequency ($0.2 < p < 0.8$). Selection is less effective for alleles with frequencies outside this range and is least effective for rare recessive alleles. For quantitative traits, including fitness, the total selection is shared among all the genes that influence the trait, and the selection coefficient affecting each allele is determined by (1) the magnitude of the effect of the allele, (2) the frequency of the allele, (3) the total number of genes affecting the trait, (4) the narrow-sense heritability of the trait, and (5) the proportion of the population that is selected for breeding.

The value of heritability is determined both by the magnitude of effects and by the frequency of alleles. If all favorable alleles were fixed ($p = 1$) or lost ($p = 0$), then the heritability of the trait would be 0. Therefore, the heritability of a quantitative trait is expected to decrease over many generations of artificial selection as a result of favorable alleles becoming nearly fixed. For example, ten generations of selection for less fat in a population of Duroc pigs decreased the heritability of fatness from 73 to 30 percent because of changes in allele frequency that resulted from the selection.

Population improvement by means of artificial selection cannot continue indefinitely. A population may respond to selection until its mean is many standard deviations different from the mean of the original population, but eventually the population reaches a **selection limit** at which successive generations show no further improvement. Progress may stop because all alleles affecting the trait are either fixed or lost, and so the narrow-sense heritability of the trait becomes 0. However, it is more common for a selection limit to be reached because natural selection counteracts artificial selection. Many genes that respond to artificial selection as a result of their favorable effect on a selected trait also have indirect harmful effects on fitness. For example, selection for increased size of eggs in poultry results in a decrease in the number of eggs, and selection for extreme body size (large or small) in most animals results in a decrease in fertility. When one trait (for example, number of eggs) changes in the course of selection for a different trait (for example, size of eggs), then the unselected trait is said to have undergone a **correlated response** to selection. Correlated response of fitness is typical in long-term artificial selection. Each increment of progress in the selected trait is partially offset by a decrease in fitness because of correlated response; eventually, artificial selection for the trait of interest is exactly balanced by natural selection against the trait. Thus a selection limit is reached, and no further progress is possible without changing the strategy of selection.

Inbreeding Depression and Heterosis

Inbreeding can have harmful effects on economically important traits such as yield of grain and egg production. This decline in performance is called **inbreeding depression,** and it results principally from rare harmful recessive alleles becoming homozygous because of inbreeding (Chapter 15). The degree of inbreeding is measured by the inbreeding coefficient *F* discussed in Chapter 15. Figure 16.11 is an example of inbreeding depression in yield of corn, in which the yield decreases linearly as the inbreeding coefficient increases.

Most highly inbred strains suffer from many genetic defects, as might be expected from the uncovering of deleterious recessive alleles. One would also expect that if two different inbred strains were crossed, then the F_1 would show improved features, because a harmful recessive allele inherited from one parent would be likely to be covered up by a normal dominant allele from the other parent. In many organisms, the F_1 generation of a cross between inbred lines is superior to the parental lines, and the phenomenon is called **heterosis,** or **hybrid vigor.** The phenomenon, which is widely used in the production of corn and other agricultural products, yields genetically identical hybrid plants with traits that are sometimes more favorable than those of the ancestral plants from which the inbreds were derived. The features that most commonly distinguish hybrid plants from their inbred parents are their rapid growth, larger size, and greater yield.

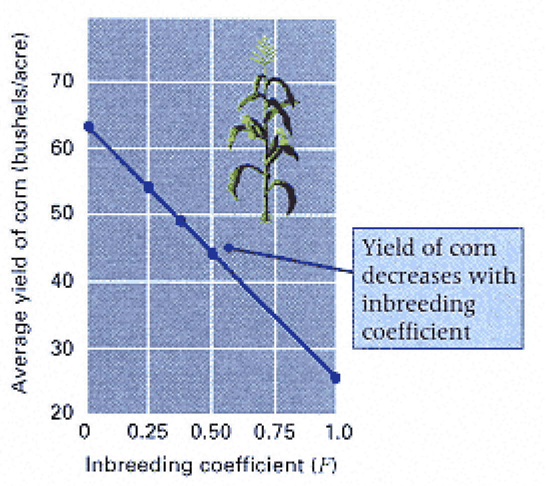


Figure 16.11
Inbreeding depression for yield in corn.
[Data from N. Neal. 1935. *J. Amer. Soc. Agronomy*, 27:666]

Furthermore, the F_1 plants have a fairly uniform phenotype (because $\sigma_g^2 = 0$). Genetically heterogeneous crops with high yields or certain other desirable features can also be produced by traditional plant-breeding programs, but growers often prefer hybrid plants because of their uniformity. For example, uniform height and time of maturity facilitate machine harvesting, and plants that all bear fruit at the same time accommodate picking and shipping schedules.

Hybrid varieties of corn are used almost exclusively in the United States for commercial crops. A farmer cannot plant the seeds from his crop because the F_2 generation consists of a variety of genotypes, most of which do not show hybrid vigor. The production of hybrid seeds is a major industry in corn-growing sections of the United States.

16.5—
Correlation Between Relatives

Quantitative genetics relies extensively on similarity among relatives to assess the importance of genetic factors. Particularly in the study of complex behavioral traits in human beings, genetic interpretation of familial resemblance is not always straightforward because of the possibility of nongenetic, but nevertheless familial, sources of resemblance. However, the situation is usually less complex in plant and animal breeding, because genotypes and environments are under experimental control.

Covariance and Correlation

Genetic data about families are frequently pairs of numbers: pairs of parents, pairs of twins, or pairs consisting of a single parent and a single offspring. An important issue in quantitative genetics is the degree to which the numbers in each pair are associated. The usual way to measure the asso

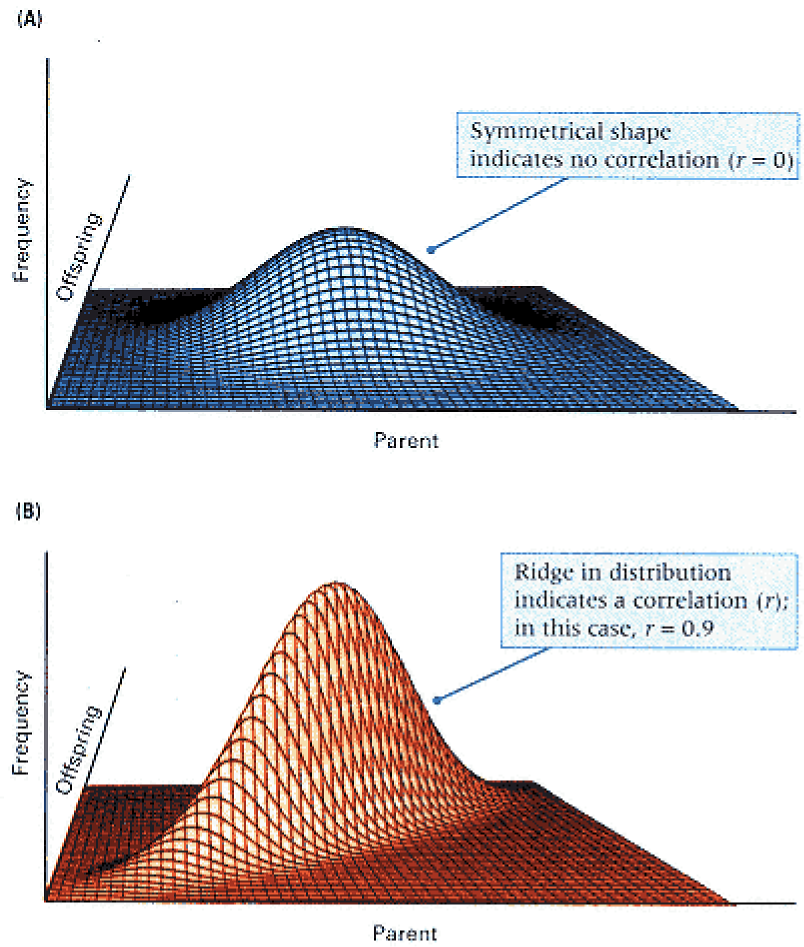


Figure 16.12
Distribution of a quantitative trait in parents and offspring
when there is no correlation (A) or a high correlation (B) between them.
[From R. R. Sokal and F. J. Rohlf. 1969. Biometry. San Francisco: Freeman.]

ciation is to calculate a statistical quantity called the **correlation coefficient** between the members of each pair.

The correlation coefficient among relatives is based on the covariance in phenotype among them. Much as the variance describes the tendency of a set of measurements to vary (Equation 2), the covariance describes the tendency of pairs of numbers to vary together (co-vary). Calculation of the covariance is similar to calculation of the variance in Equation (2) except that the squared deviation term $(x_i - \bar{x})^2$ is replaced with the product of the deviations of the pairs of measurements from their respective means---that is, $(x_i - \bar{x})(y_i - \bar{y})$.

For example, $(x_i - \bar{x})$ could be the deviation of a particular father's height from the overall father mean, and $(y_i - \bar{y})$ could be the deviation of his son's height from the overall son mean. In symbols, let f_i be the number of pairs of relatives with phenotypic measurements x_i and y_i. Then the estimated **covariance *Cov)*** of the trait among the relatives is

$$Cov(x,y) = \frac{\Sigma f_i(x_i - \bar{x})(y_i - \bar{y})}{N - 1} \qquad (7)$$

where N is the total number of pairs of relatives studied.

The **correlation coefficient**(r) of the trait between the relatives is calculated from the covariance as follows:

$$r = \frac{Cov(x,y)}{s_x s_y} \qquad (8)$$

where s_x and s_y are the standard deviations of the measurements, estimated from Equation 2. The correlation coefficient can range from -1.0 to + 1.0. A value of + 1.0 means perfect association. When $r = 0$, x and y are not associated.

The Geometrical Meaning of a Correlation

Figure 16.12 shows a geometrical interpretation of the correlation coefficient. Just as the distribution of a single variable can be built up, step by step, by placing small squares corresponding to each individual along the x-axis, so too can the joint distribution of two variables (for example, height of fathers and height of sons) be built up, step by step, by placing small cubes on the *x*–*y* plane at a position determined by each pair of measurements. As these cubes pile up, the joint distribution assumes the form of an inverted bowl, cross sections of which are themselves normal distributions. When there is no correlation between the variables, the surface is completely symmetrical (Figure 16.12A), which means that the distribution of *y* is completely independent of that of *x.* However, when the distributions are correlated, the distribution of *y* varies according to the particular value of *x,* and the joint distribution becomes asymmetrical with a ridge built up in the *x*–*y* plane (Figure 16.12B).

Normally, there are not enough data to form a full three-dimensional joint distribution of *x* and *y* as depicted in Figure 16.12. Instead, the data are in the form of individual points, sparse enough to be plotted on a scatter diagram like those in Figure 16.13. In these diagrams, the clustering of points around each line corresponds to the ridge of points in Figure 16.13. The higher the ridge, the more clustered the points are around the line.

Estimation of Narrow-Sense Heritability

Covariance and correlation are important in quantitative genetics, because the correlation coefficient of a trait between individuals with various degrees of genetic relationship is related fairly simply to the narrow-sense or broad-sense heritability, as

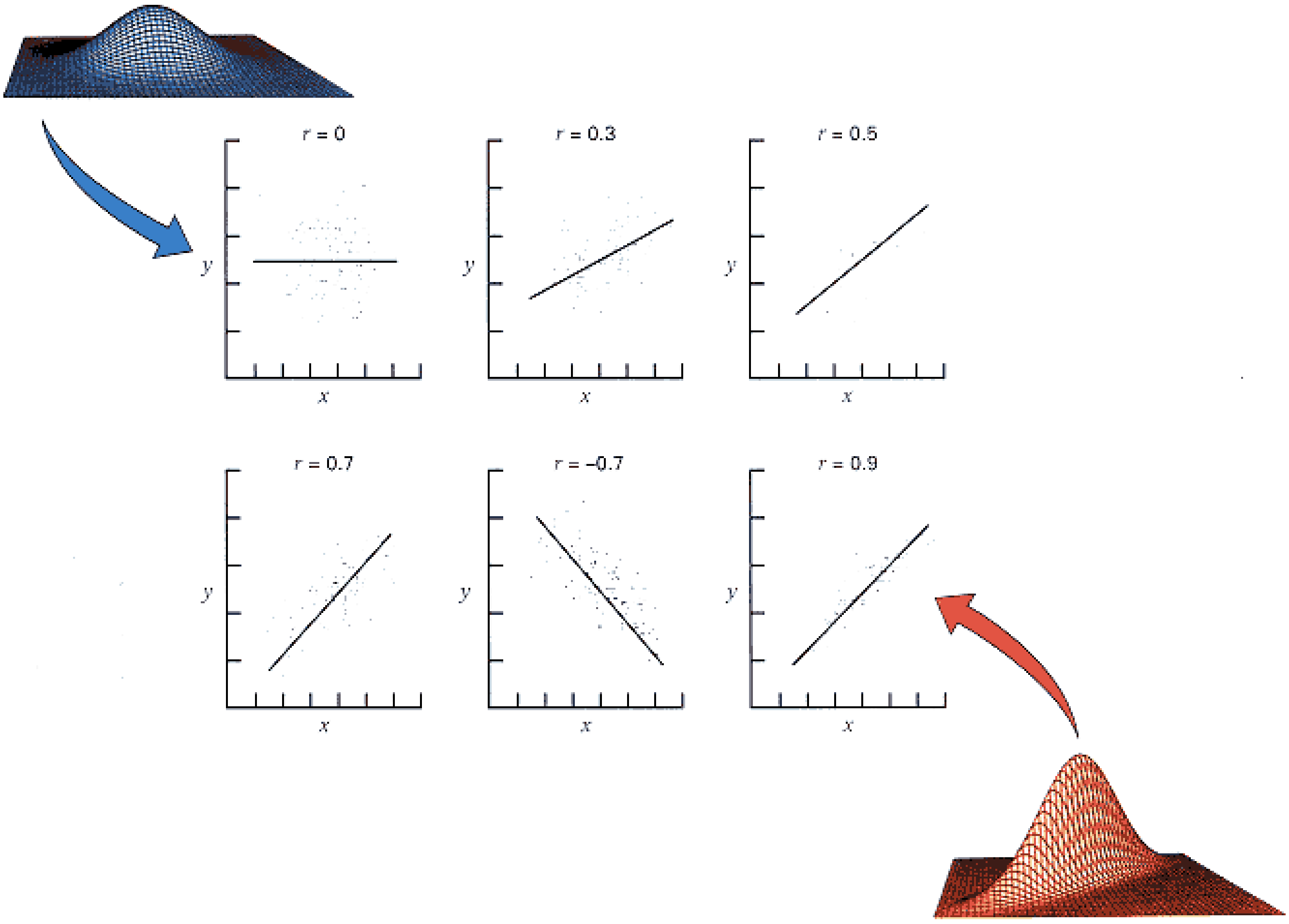


Figure 16.13
Scatter diagrams of x and y variables with various degrees of correlation r.
[From R. R. Sokal and F. J. Rohlf. 1969. *Biometry*. San Francisco: Freeman.]

Table 16.3 Theoretical correlation coefficient in phenotype between relatives

Degree of relationship	Correlation coefficeint*
Offspring and one parent	$h^2/2$
Offspring and average of parents	$h^2/2$
Half siblings	$h^2/4$
First cousins	$h^2/8$
Monozygotic twins	H^2
Full siblings	$\sim H^2/2$

*Contributions from interactions among alleles of different genes have been ignored. For this and other reasons, H^2 correlations are approximate

shown in Table 16.3. The table gives the theoretical values of the correlation coefficient for various pairs of relatives; h^2 represents the narrow-sense heritability and H^2 the broad-sense heritability. For parent-offspring, half-sibling, or first-cousin pairs, narrow-sense heritability can be estimated directly by multiplication. Specifically, h^2 can be estimated as twice the parent-offspring correlation, four times the half-sibling correlation, or eight times the first-cousin correlation.

With full siblings, identical twins, and double first cousins, the correlation coefficient is related to the broad-sense heritability, H^2, because phenotypic resemblance depends not only on additive effects but also on dominance. In these relatives, dominance contributes to resemblance because the relatives can share *both* of their alleles as a result of their common ancestry, whereas parents and offspring, half siblings, and first cousins can share at most a single allele of any gene because of common ancestry. Therefore, to the extent that phenotype depends on dominance effects, full siblings can resemble each other more than they resemble their parents.

The potentially greater resemblance between siblings than between parents and offspring may be understood by considering an autosomal recessive trait caused by a rare recessive allele. In a randomly mating population, the probability that an offspring will be affected, if the mother is affected, is q (the allele frequency), which corresponds to the random-mating probability that the sperm giving rise to the offspring carries the recessive allele. In contrast, the probability that both of two siblings will be affected, if one of them is affected, is 1/4, because most of the matings that produce affected individuals will be between heterozygous parents. When the trait is rare, the parent-offspring resemblance will be very small (as q approaches 0), whereas the sibling-sibling resemblance will be substantial. This discrepancy is entirely a result of dominance, and it arises only because, at any particular locus, full siblings may have the same genotype.

16.6— Heritabilities of Threshold Traits

Application of the concepts of individual selection can be used to understand the meaning of heritability in the context of threshold traits. The analogy is illustrated in Figure 16.14. Threshold traits depend on an underlying quantitative trait, called the **liability,** which refers to the degree of genetic risk or predisposition to the trait. Only those individuals with a liability greater than a certain threshold develop the trait. Figure 16.14A shows the distribution of liability in a population, where liability is measured on an arbitrary scale so that the mean liability (*M*) equals 0 and the variance in liability equals 1. Affected individuals are in the right-hand area of the distribution. Using various characteristics of the normal distribution, we can use the observed proportion of affected individuals to calculate the position of the threshold and the value of M^*, which in this context is the mean liability of affected individuals.

The distribution of liability among offspring of affected individuals is shown in Figure 16.14B. It is shifted slightly to the right, which implies that, relative to the population as a whole, a greater proportion of individuals with affected parents have liabilities above the threshold and so will be affected. The observed proportion of affected individuals in the offspring generation can be used to calculate M'. The important point is that *M** and *M*' can be obtained from the observed proportions corresponding to the shaded red areas in

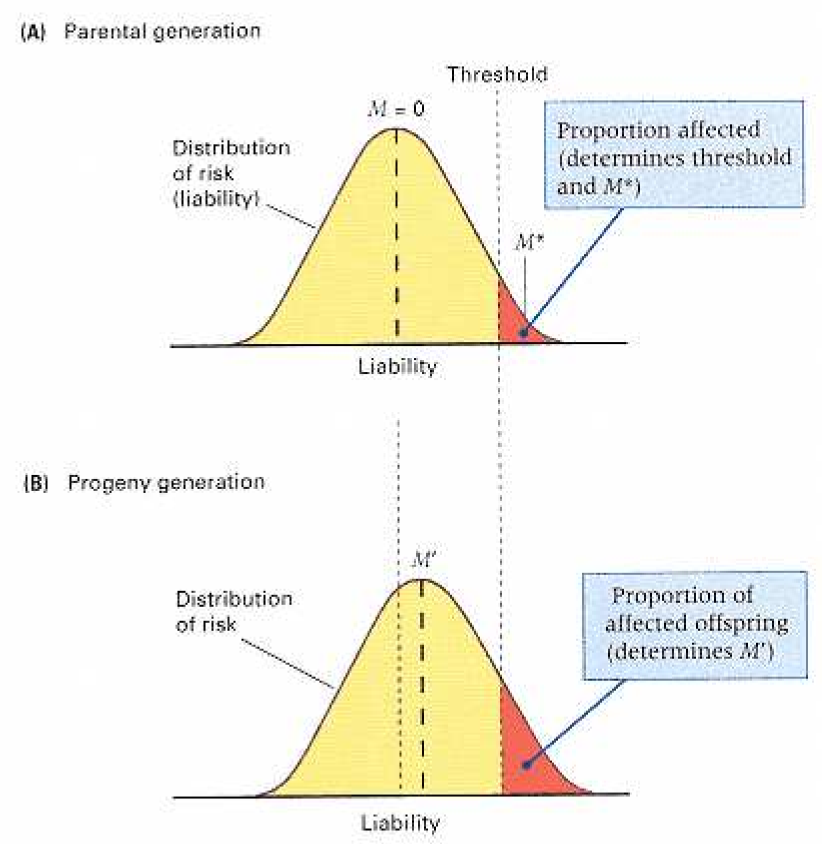


Figure 16.14
Interpretation of a threshold trait in terms of an underlying continuous distribution of liability. Individuals with a liability greater than the threshold are affected. *M* (arbitrarily set equal to 0), *M**, and *M'* are the mean liabilities of the parental generation, of affected individuals, and of the offspring of affected individuals, respectively.

Figure 16.14. With *M* = 0, Equation (6) can be used to calculate the narrow-sense heritability of liability of the trait in question.

Examples of narrow-sense heritability of the liability to important congenital abnormalities are given in Table 16.4. How these heritabilities translate into risk is illustrated in Figure 16.15, along with observed data for the most common congenital abnormalities in Caucasians. The theoretical curves correspond to the relationships among the incidence of the trait in the general population, the type of inheritance, and the risk among first-degree relatives of affected individuals. (First-degree relatives of an affected individual include parents, offspring, and siblings.) Simple Mendelian inheritance produces the highest risks, but these traits, as a group, tend to be rare. The most common traits are threshold traits, and with a few exceptions, their risks tend to be moderate or low, corresponding to the heritabilities indicated.

Table 16.4 Narrow-sense heritability of liability for congenital abnormalities

Trait	Description	Population frequency (%)	Narrow-sense Heritability (%)
Cleft lip	Upper lip not completely formed	0.10	76
Spina bifida	Exposed spinal cord	0.50	60
Club foot	Abnormally formed foot	0.10	68
Pyloric stenosis	Obstructed stomach	0.30	75
Dislocation of hip	Hip joint mispositioned	0.06	70
Hydrocephalus	Fluid accumulation around brain	0.05	76
Celiac disease	Inability to digest fats, starches, and sugars	0.05	45

Hypospadias	Abnormally formed penis	0.30	75
Atrial septal defect	Hole between upper chambers of hearts	0.07	70
Patent ductus arteriosus	Hole between aorta and pulmonary artery; blood bypasses lungs	0.05	60

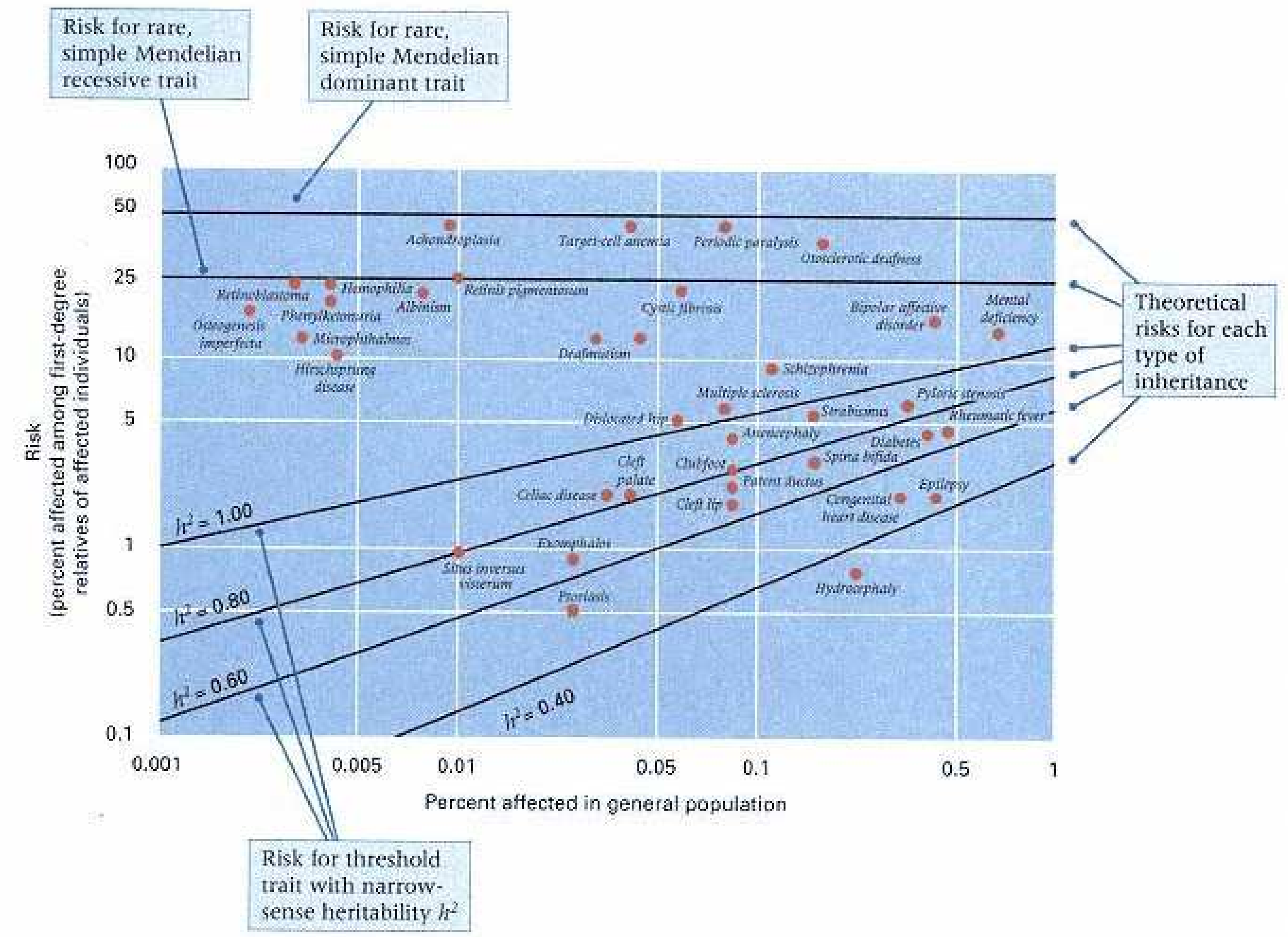


Figure 16.15
Recurrence risks of common abnormalities. Diagonal lines are the theoretical risks for threshold traits with the indicated values of the narrow-sense heritability of liability. Horizontal lines are the theoretical risks for simple dominant or recessive traits.

16.7— Linkage Analysis of Quantitative-Trait Loci

Genes that affect quantitative traits cannot usually be identified in pedigrees, because their individual effects are obscured by the segregation of other genes and by environmental variation. Even so, genes that affect quantitative traits can be localized if they are genetically linked with genetic markers, such as simple tandem repeat polymorphisms (STRPs), discussed in Chapter 4, because the effects of the genotype affecting the quantitative trait will be correlated with the genotype of the linked STRP. A gene that affects a quantitative trait is a **quantitative-trait locus (QTL).** Locating QTLs in the genome is important to the manipulation of genes in breeding programs and to the cloning and study of genes in order to identify their functions.

STRPs result from variation in number, *n*, of repeating units of a simple-sequence repeat, such as $(CA)_n$. The number of repeats can be determined by PCR amplification of the STRP using oligonucleotide primers to flanking unique-sequence DNA. The size of the amplified DNA fragment is a function of the number of repeats in the STRP. Such STRP markers are abundant, are distributed throughout the genome,

Connection Human Gene Map

Jeffrey C. Murry and 26 other investigators 1994

University of Iowa and 9 other research institutions
A Comprehensive Human Linkage Map with Centimorgan Density

Assessed by the distance between genetic markers, the human genetic map is one of the most dense genetic maps of any organism. At present, the human map includes more than 5000 genetic markers. The assembly of the human genetic map represents a major achievement in international science. It was pieced together by hundreds of investigators working in many countries throughout the world. This paper is one example of a progress report illustrating several important features. First, the human genetic map was made possible by researchers studying a single, very extensive set of families assembled by CEPH (Centre d'Étude du Polymorphisme Humain, Paris) so that their data could be collated. Second, a very large proportion of the markers in the human map are detected by means of the polymerase chain reaction (PCR) to amplify short tandem repeats (STRPs) that are abundant and highly polymorphic in the human genome. Third, the data are so extensive that they are not presented in the paper itself but rather in a large wall chart and (more conveniently) at a site on the Internet. Fourth, the authors point out the utility of the genetic map in determining the chromosomal locations of mutations that cause disease, using the methods of QTL mapping, as a first step in identifying the genes themselves. Finally, as is clear from the closing passages of the excerpt, the authors are well aware of the implications of modern human genetics for ethics, law, and social policy.

For the first time, humans have been presented with the capability of understanding their own genetic makeup and how it contributes to morbidity of the individual and the species. Rapid scientific advances have made this possible, and developments in molecular biology, genetics, and computing, coupled with a cooperative and interactive biomedical community, have accelerated the progress of investigation into human inherited disorders. A primary

It should be emphasized that the new opportunities and challenges for biomedical research provided by these marker-dense maps also create an urgency to face the parallel challenges in the areas of ethics, law, and social policy.

engine driving these advances has been the development and use of human gene maps that allow the rapid positional assignment of an inherited disorder as a starting point for gene identification and characterization . . .. The genetic maps were based on genotypes generated from DNA samples obtained from the CEPH reference pedigree set. Since this material is publicly available, individual research groups can add their own genetic markers in the future . . .. Although only markers genotyped on CEPH reference families are included in the maps, the list is extensive. The CEPH database contains blood group markers and protein polymorphisms as well as a variety of DNA-based markers including RFLPs and short tandem repeat polymorphisms (STRPs). There has been an emphasis on PCR-based markers . . .. The 3617 STRP markers alone provide about one marker every 1×10^6 base pairs (bp) throughout the genome. Once an initial localization [of a new gene] is identified by means of a set of genome-wide genetic maps, subsequent steps can be undertaken to increasingly narrow the region to be searched and eventually to select a specific interval on which studies of physical reagents, such as cloned DNA fragments, can be carried out . . .. Once the physical reagents are in hand, gene and mutation searches are possible with a number of strategies, . . . but considerable problems and difficulties still remain . . .. Public access to databases is an especially important feature of the Human Genome Project. They are easy to use and facilitate rapid communication of new findings and can be updated efficiently. We have made primary data available electronically on the World Wide Web at http://gdbwww.gdb.org/ . . . Finally, it should be emphasized that the new opportunities and challenges for biomedical research provided by these marker-dense maps also create an urgency to face the parallel challenges in the areas of ethics, law, and social policy. Our ability to distinguish individuals for forensic purposes, identify genetic predispositions for rare and common inherited disorders, and to characterize, if present, the underlying nature of the genetic components of normal trait variability such as height, intelligence, sexual preference, or personality type, has never been greater. Although technically feasible, whether these maps should be used for these ends should be resolved after open dialogue to review those implications and devise policy to deal with the as yet unpredictable outcomes.

Source; Science 265:2049-2054

and often have multiple codominant alleles; thus they are ideally suited for linkage studies of quantitative traits. In STRP studies, as many widely scattered STRPs as possible are monitored, along with the quantitative trait, in successive generations of a genetically heterogeneous population. Statistical studies are then carried out to identify which STRP alleles are the best predictors of phenotype of the quantitative trait because their presence in a genotype is consistently accompanied by superior

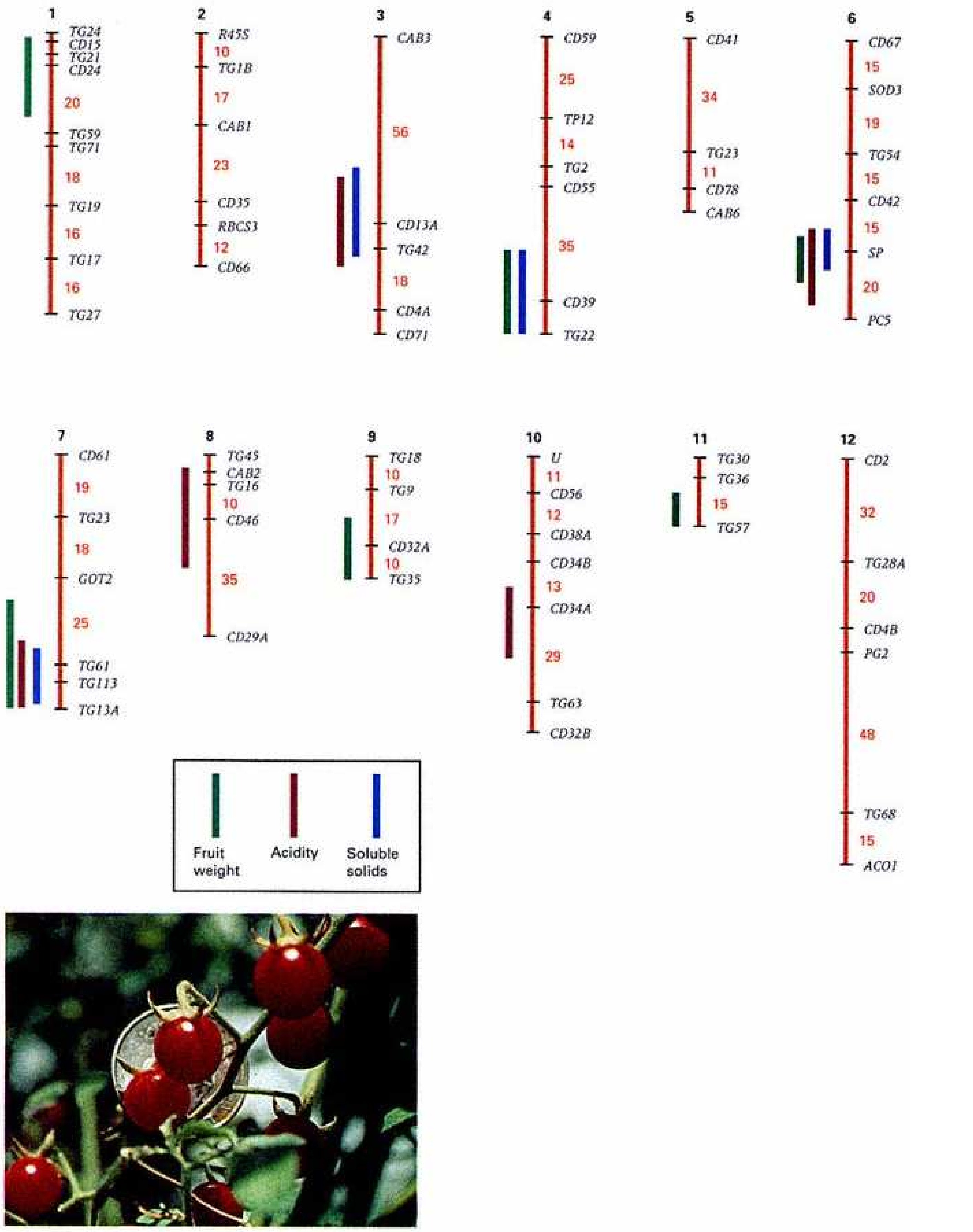


Figure 16.16
Location of QTLs for several quantitative traits in the tomato genome. The genetic markers are shown for each of the12 chromosomes. The numbers in red are distances in map units between adjacent markers, but only map distances of 10 or greater are indicated.The regions in which the QTLs are located are indicated by the bars—green bars, QTLs for fruit weight; blue bars, QTLs for content of soluble solids; and dark red bars, QTLs for acidity (pH). The data are from crosses between the domestic tomato (Lycopersicon esculentum) and a wild South American relative with small, green fruit (Lycopersicon chmielewskii). The photograph is of fruits of wild tomato. Note the small size in comparison with the coin (a U.S. quarter). The F_1 generation was backcrossed with the domestic tomato, and fruits from the progeny were assayed for the genetic markers and each of the

quantitative traits.
[Data from A. H. Paterson, et al. 1988. Nature, 335: 721. Photograph courtesy of Steven D. Tanksley.]

performance with respect to the quantitative trait. These STRPs identify regions of the genome that contain one or more genes that have important effects on the quantitative trait, and the STRPs can be used to trace the segregation of the important regions in breeding programs and even as starting points for cloning genes with particularly large effects.

An example of genetic mapping of quantitative-trait loci for several quantitative traits in tomatoes is illustrated in Figure 16.16. More than 300 highly polymorphic genetic markers have been mapped in the tomato genome, with an average spacing between markers of 5 map units. The chromosome maps in Figure 16.16 show a subset of 67 markers that were segregating in crosses between the domestic tomato and a wild South American relative. The average spacing between the markers is 20 map units. Backcross progeny of the cross F_1 × domestic tomato were tested for the genetic markers, and the fruits of the backcross progeny were assayed for three quantitative traits—fruit weight, content of soluble solids, and acidity. Statistical analysis of the data was carried out in order to detect marker alleles that were associated with phenotypic differences in any of the traits; a significant association indicates genetic linkage between the marker gene and one or more QTLs affecting the trait. A total of six QTLs affecting fruit weight were detected (green bars), as well as four QTLs affecting soluble solids (blue bars) and five QTLs affecting acidity (dark red bars). Although additional QTLs with smaller effects undoubtedly remained undetected in these types of experiments, the effects of the mapped QTLs are substantial: The mapped QTLs account for 58 percent of the total phenotypic variance in fruit weight, 44 percent of the phenotypic variance in soluble solids, and 48 percent of the phenotypic variance in acidity. The genetic markers linked to the QTLs with substantial effects make it possible to trace the transmission of the QTLs in pedigrees and to manipulate the QTLs inbreeding programs by following the transmission of the linked marker genes. Figure 16.16 also indicates a number of chromosomal regions containing QTLs for two or more of the traits—for example, the QTL regions on chromosomes 6 and 7, which affect all three traits.Because the locations of the QTLs can be specified only to within 20 to 30 map units, it is unclear whether the coincidences result from (1) pleiotropy, in which a single gene affects several traits simultaneously, or(2) independent effects of multiple, tightly linked genes.However, the locations of QTLs for different traits coincide frequently enough so that, in many cases, pleiotropy is likely to be the explanation.

Chapter Summary

Many traits that are important in agriculture and human genetics are determined by the effects of multiple genes and by the environment. Such traits are multifactorial traits (also known as quantitative traits), and their analysis is known as quantitative genetics. There are three types of multifactorial traits: continuous, meristic, and threshold traits. Continuous traits are expressed according to a continuous scale of measurement, such as height.Meristic traits are traits that are expressed in whole numbers, such as the number of grains on an ear of corn.Threshold traits have an underlying risk and are either expressed or not expressed in each individual;an example is diabetes. The genes affecting quantitative traits are no different from those affecting simple Mendelian traits, and the genes can have multiple alleles,partial dominance, and so forth. When several genes affect a trait, the pattern of genetic transmission need not fit a simple Mendelian pattern, because the effects of one gene can be obscured by other genes or the environment.However, the number of genes can be estimated and many of them can be mapped using linkage to simple tandem repeat polymorphisms.

Many quantitative and meristic traits have a distribution that approximates the bell-shaped curve of a normal distribution. A normal distribution can be completely described by two quantities: the mean and the variance.The standard deviation of a distribution is the square root of the variance. In a normal distribution, approximately 68 percent of the individuals have a phenotype within 1 standard deviation of the mean, and approximately 95 percent of the individuals have a phenotype within 2 standard deviations of the mean.

Variation in phenotype of multifactorial traits among individuals in a population derives from four principal sources: (1) variation in genotype, which is measured by the genotypic variance; (2) variation in environment, which is measured by the environmental variance; (3) variation resulting from the interaction between genotype and environment (G-E interaction); and (4) variation resulting from nonrandom association of genotypes and environments (G-E association). The ratio of genotypic variance to the total phenotypic variance of a trait is called the broad-sense heritability; this quantity is useful in predicting the outcome of artificial selection among clones, inbred lines, or varieties. When artificial selection is carried out in a randomly mating population, the narrow-sense heritability is used for prediction. The value of the narrow-sense heritability can be determined from the correlation in phenotype among groups of relatives.

One common type of artificial selection is called truncation selection, in which only those individuals that have a phenotype above a certain value (the truncation point) are saved for breeding the next generation. Artificial selection usually results in improvement of the selected population. The general principle is that the deviation of the progeny mean from the mean of the previous generation equals the narrow-sense heritability times the deviation of the parental mean from the mean of the parental generation: $M' M = h^2(M^* - M)$. When selection is carried out for many generations, progress often slows or ceases because (1) some of the favorable genes become nearly fixed in the population, which decreases the narrow-sense heritability, and (2) natural selection may counteract the artificial selection.

A gene that affects a quantitative trait is a quantitativetrait locus, or QTL. The locations of QTLs in the genome can be determined by genetic mapping with respect to simple tandem repeat polymorphisms or other types of genetic markers. The mapping of QTLs aids in the manipulation of genes in breeding programs and in the identification of the genes.

Key Terms

artificial selection	genotype-environment interaction	multifactorial trait
broad-sense heritability	genotypic variance	narrow-sense heritability
continuous trait	heterosis	normal distribution
correlated response	hybrid vigor	quantitative trait
correlation coefficient	identical twins	quantitative-trait locus (QTL)
covariance	inbreeding depression	selection limit
distribution	individual selection	standard deviation
environmental variance	liability	threshold trait
fraternal twins	mean	truncation point
genotype-environment association	meristic trait	variance

Review the Basics

• What is a quantitative trait? How do a continuous trait, a meristic trait, and a threshold trait differ?

• How can a trait be affected both by genetic factors and by environmental factors? Give an example.

• What is the genotypic variance of a quantitative trait? What is the environmental variance?

• In terms of genotypic variance, what is special about an inbred line or about the F_1 progeny of a cross between two inbred lines?

• Why are correlations between relatives of interest to the quantitative geneticist?

• Distinguish between broad-sense heritability and narrow-sense heritability. Which type of heritability is relevant to individual truncation selection?

• Distinguish between the variance due to genotype-environment interaction and the variance due to genotype environment association. Which type of variance is more easily controlled by the experimentalist?

• How are the variance and the standard deviation related? When does the variance of a distribution of phenotypes equal zero?

Guide to Problem Solving

Problem 1: The following data are the values of a quantitative trait measured in a sample of 100 individuals taken at random from a population.

(a) Estimate the mean, variance and standard deviation in the population.

(b) Assuming a normal distribution, what proportion of individuals in the entire population would be expected to have a phenotype in the range of 93 to 107? Above 114? (For purposes of this problem, round off the value of the standard deviation to the nearest whole number.)

Phenotype	**Number of individuals**
86–90	10
91–95	16
96–100	26
101–105	29
106–110	10
111–115	9

Answer: The data are tabulated in ranges, but for purposes of calculation it is better to retabulate the data as the midpoints of the ranges, as is shown in the next table.

Chapter 16 GeNETics on the web

GeNETics on the web will introduce you to some of the most important sites for finding genetic information on the Internet. To complete the exercises below, visit the Jones and Bartlett home page at

http://www.jbpub.com/genetics

Select the link to ***Genetics: Principles and Analysis*** and then choose the link to ***GeNETics on the web.*** You will be presented with a chapter-by-chapter list of highlighted keywords.

GeNETics EXERCISES

Select the highlighted keyword in any of the exercises below, and you will be linked to a web site containing the genetic information necessary to complete the exercise. Each exercise suggests a specific, written report that makes use of the information available at the site. This report, or an alternative, may be assigned by your instructor.

1. This keyword will open a link to a table frequencies of the most common types of **birth defects.** The overall frequency of these conditions is about one in 25 births. Most of the genetically determined conditions are multifactorial. If assigned to do so, make a bar graph showing the relative incidence of each of the types of abnormality.

2. The trait of **parthogenesis** in turkeys is an example of a quantitative threshold trait influenced by genetic and environmental factors. At this site you can learn about parthenogenesis and its role in commercial turkey production. If assigned to do so, prepare a table listing the genetic and environmental factors affecting parthenogenesis that have so far been identified.

3. One of the founders of evolutionary genetics was **Sewall Wright,**who related his strong interest in animal breeding to the principles of quantitative genetics. His paper on *the relation of live-stock breeding to theories of evolution* is good summary of his thoughts and is well worth reading. If assigned to do so, write a 250-word paper describing his shifting-balance theory of evolution and the four very diverse research projects that led him to it.

MUTABLE SITE EXERCISES

The Mutable Site Exercise changes frequently. Each new update includes a different exercise that makes use of genetics resources available on the World Wide Web. Select the **Mutable Site** for Chapter 16, and you will be linked to the current exercise that relates to the material presented in this chapter.

PIC SITE

The Pic Site showcases some of the most visually appealing genetics sites on the World Wide Web. To visit the showcase genetics site, select the **Pic Site** for Chapter 16.

Phenotype	Number of individuals
88	10
93	16
98	26
103	29
108	10
113	9

(a) The mean is estimated as (88 × 10 + 93 × 16 + 98 × 26 + 103 × 29 + 108 × 10 + 113 × 9)/100 = 10,000/100 = 100. The variance is estimated as $[(88 - 100)^2 \times 10 + (93 - 100)^2 \times 16 + (98 - 100)^2 \times 26 + (103 - 100)^2 \times 29 + (108 - 100)^2 \times 10 + (113 - 100)^2 \times 9]/99$ = 4750/99 = 47.98. The standard deviation is then estimated as $(47.98)^{1/2}$=6.93.

(b) Rounding off the standard deviation to 7, the range of 93 to 107 equals the mean plus or minus one standard deviation (100 ± 7), and approximately 68 percent of the population is expected to be within this range. To estimate the proportion with a phenotype above 114, note that this is two standard deviations above the mean. Because approximately 95 percent of the population have phenotypes within the range 100 ± 14 (two standard deviations), the remaining 5 percent will have phenotypes either greater than 114 or less than 86. Because the normal distribution is symmetrical, half of the 5 percent (or 2.5 percent) will be expected to have phenotypes greater than 114.

Problem 2: A genetically heterogeneous population of wheat has a variance in the number of days to maturation of 40, whereas two inbred populations derived from it have a variance in the number of days to maturation of 10.

(a) What is the genotypic variance, σ^2_g, the environmental variance, σ^2_e, and the broad-sense heritability, H^2, of days to maturation in this population?

(b) If the inbred lines were crossed, what would be the predicted variance in days to maturation of the F_1 generation?

Answer:

(a) The total variance, σ^2_t, in the genetically heterogeneous population is the sum $\sigma^2_g + \sigma^2_e = 40$. The variance in the inbred lines equals the environmental variance (because $\sigma^2_g = 0$ in a genetically homogeneous population); hence $\sigma^2_g = 10$. Therefore, σ^2_g in the heterogeneous population equals 40 - 10 = 30. The broad-sense heritability H^2 equals $\sigma^2_g/(\sigma^2_g + \sigma^2_e = 30/40 = 75$ percent.

(b) If the inbred lines were crossed, the F_1 generation would be genetically uniform, and consequently the predicted variance would be $\sigma^2_e = 10$.

Problem 3: A breeder aims to increase the height of maize plants by artificial selection. In a population with an average plant height of 150 cm, plants averaging 180 cm are selected and mated at random to give the next generation. If the narrow-sense heritability of plant height in this population is 40 percent, what is the expected average plant height among the progeny generation?

Answer: Use Equation (6) with $M = 150$, $M^* = 180$, and $h^2 = 0.40$. Then the predicted mean is calculated as $M' = 150 + 0.40 \times (180 - 150) = 162$ cm.

Problem 4: In maize, data for the logarithm of the percent oil content in the kernels lie approximately on a triangle like that depicted in Figure 16.9. Two inbred lines with average values of 0.513 and 1.122 are crossed. The variance of the trait in the F_1 generation is 0.0003. The F_1 plants are self-fertilized, and the variance of the trait in the F_2 generation is 0.0030. Estimate the minimum number of genes affecting oil content.

Answer:The minimum number of genes is estimated as $n = D^2/8\sigma^2_g$, where D is the difference between inbred lines and σ^2_g is the genotypic variance. In this problem, $D = 1.122 - 0.513 = 0.609$. The variance of the F_2 generation equals $\sigma^2_g + \sigma^2_e$ and that of the F_1 generation equals σ^2_e (because the F_1 generation is genetically uniform). Therefore, $\sigma^2_g = 0.0030 - 0.0003 = 0.0027$. The minimum number of genes affecting the oil content of kernels is estimated as $(0.609)^2/[8 \times 0.0027] = 17.2$.

Analysis and Applications

16.1 Two varieties of corn, A and B, are field-tested in Indiana and North Carolina. Strain A is more productive in Indiana, but strain B is more productive in North Carolina. What phenomenon in quantitative genetics does this example illustrate?

16.2 A distribution has the feature that the standard deviation is equal to the variance. What are the possible values for the variance?

16.3 The following questions pertain to a normal distribution.

(a) What term applies to the value along the x-axis that corresponds to the peak of the distribution?

(b) If two normal distributions have the same mean but different variances, which is the broader?

(c) What proportion of the population is expected to lie within one standard deviation of the mean? Within two standard deviations?

16.4 Distinguish between the broad-sense heritability of a quantitative trait and the narrow-sense heritability. If a population is fixed for all genes that affect a particular quantitative trait, what are the values of the narrow-sense and broad-sense heritabilities?

16.5 When we compare a quantitative trait in the F_1 and F_2 generations obtained by crossing two highly inbred strains, which set of progeny provides an estimate of the environmental variance? What determines the variance of the other set of progeny?

16.6 For the difference between the domestic tomato, *Lycopersicon esculentum*, and the wild South American relative, *Lycopersicon chmielewskii*, the environmental variance σ^{2e} accounts for 13 percent of the total phenotypic variance $\sigma 2_t$ of fruit weight, 9 percent of $\sigma 2_t$ of soluble-solid content, and 11 percent of $\sigma 2_t$ of acidity. What are the broad-sense heritabilities of these traits?

16.7 Ten female mice had the following numbers of live born offspring in their first litters: 11, 9, 13, 10, 9, 8, 10, 11, 10, 13. Considering these females as representative of the total population from which they came, estimate the mean, variance, and standard deviation of size of the first litter in the entire population.

16.8 Values of IQ score are distributed approximately according to a normal distribution with mean 100 and standard deviation 15. What proportion of the population has a value above 130? Below 85? Above 85?

16.9 The data in the adjoining table pertain to milk production over an 8-month lactation among 304 two-year-old Jersey cows.

Pounds of milk produced	**Number of cows**
1000–1500	1
1500–2000	0
2000–2500	5
2500–3000	23
3000–3500	60
3500–4000	58
4000–4500	67
4500–5000	54
5000–5500	23
5500–6000	11
6000–6500	2

The data have been grouped by intervals, but for purposes of computation, each cow may be treated as though the milk production were equal to the midpoint of the interval (for example, 1250 for cows in the 1000–1500 interval).

(a) Estimate the mean, variance, and standard deviation in milk yield.

(b) What range of yield would be expected to include 68 percent of the cows?

(c) Round the limits of this range to the nearest 500 pounds, and compare the observed with the expected numbers of animals.

(d) Do the same calculations as in parts (b) and (c) for the range expected to include 95 percent of the animals.

16.10 In the F_2 generation of a cross of two cultivated varieties of tobacco, the number of leaves per plant was distributed according to a normal distribution with mean 18 and standard deviation 3. What proportion of the population is expected to have the following phenotypes?

(a) between 15 and 21 leaves

(b) between 12 and 24 leaves

(c) fewer than 15 leaves

(d) more than 24 leaves

(e) between 21 and 24 leaves

16.11 Two highly homozygous inbred strains of mice are crossed and the 6-week weight of the F_1 progeny determined.

(a) What is the magnitude of the genotypic variance in the F_1 generation?

(b) If all alleles affecting 6-week weight were additive, what would be the expected mean phenotype of the F_1 generation compared with the average 6-week weight of the original inbred strains?

16.12 In a cross between two cultivated inbred varieties of tobacco, the variance in leaf number per plant is 1.46 in the F_1 generation and 5.97 in the F_2 generation. What are the genotypic and environmental variances? What is the broad sense heritability in leaf number?

16.13 Two inbred lines of *Drosophila* are crossed, and the F_1 population has a mean number of abdominal bristles of 20 and a standard deviation of 2. The F_2 generation has a mean of 20 and a standard deviation of 3. What are the environmental variance, the genetic variance, and the broad-sense heritability of bristle number in this population?

16.14 In an experiment with weight gain between ages 3 and 6 weeks in mice, the difference in mean phenotype between two strains was 17.6 grams and the genotypic variance was estimated as 0.88. Estimate the minimum number of genes affecting this trait.

16.15 Estimate the minimum number of genes affecting fruit weight in a population of the domestic tomato produced by crossing two inbred strains. When average fruit weight is expressed as the logarithm of fruit weight in grams, the inbred lines have average fruit weights of -0.137 and 1.689. The F_1 generation had a variance of 0.0144, and the F_2 generation had a variance of 0.0570.

16.16 A flock of broiler chickens has a mean weight gain of 700 g between ages 5 and 9 weeks, and the narrow-sense heritability of weight gain in this flock is 0.80. Selection for increased weight gain is carried out for five consecutive generations, and in each generation the average of the parents is 50 g greater than the average of the population from which the parents were derived. Assuming that the heritability of the trait remains constant at 80 percent, what is the expected mean weight gain after the five generations?

16.17 To estimate the heritability of maze-learning ability in rats, a selection experiment was carried out. From a population in which the average number of trials necessary to learn the maze was 10.8, with a variance of 4.0, animals were selected that managed to learn the maze in an average of 5.8 trials. Their offspring required an average of 8.8 trials to learn the maze. What is the estimated narrow-sense heritability of maze-learning ability in this population?

16.18 A replicate of the population in the preceding problem was reared in another laboratory under rather different conditions of handling and other stimulation. The mean number of trials required to learn the maze was still 10.8, but the variance was increased to 9.0. Animals with a mean learning time of 5.8 trials were again selected, and the mean learning time of the offspring was 9.9.

(a) What is the heritability of the trait under these conditions?

(b) Is this result consistent with that in Problem 16.17 and how can the difference be explained?

16.19 In terms of the narrow-sense heritability, what is the theoretical correlation coefficient in phenotype between first cousins who are the offspring of monozygotic twins?

16.20 If the correlation coefficient of a trait between first cousins is 0.09, what is the estimated narrow-sense heritability of the trait?

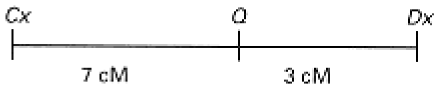
16.21 A representative sample of lamb weights at the time of weaning in a large flock is shown to the right. If the narrow sense heritability of weaning weight is 20 percent, and the half of the flock consisting of the heaviest lambs is saved for breeding for the next generation, what is the best estimate of the average weaning weight of the progeny? (*Note:* If a normal distribution has mean μ and standard deviation σ, then the mean of the upper half of the distribution is given by $\mu + 0.8\sigma$.)

81	81	83	101	86
65	68	77	66	92
94	85	105	60	90
94	90	81	63	58

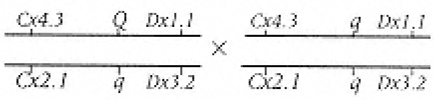
Challenge Problems

16.22 Consider a trait determined by *n* unlinked, additive genes with two alleles of each gene, in which each favorable allele contributes one unit to the phenotypic value and each unfavorable allele contributes nothing. What is the value of the genotypic variance?

16.23 Two varieties of chickens differ in the average number of eggs laid per hen per year. One variety yields an average of 300 eggs per year, the other an average of 270 eggs per year. The difference is due to a single quantitative-trait locus (QTL) located between two restriction fragment length polymorphisms. The genetic map is



In this diagram, *Cx* and *Dx* are the locations of the restriction fragment length polymorphisms, and *Q* is the location of the QTL affecting egg production. The map distances are given in centimorgans (cM). The *Cx* alleles are denoted *Cx4.3* and *Cx2.1* according to the size of the restriction fragment produced by digestion with a particular restriction enzyme, and the *Dx* alleles are denoted *Dx3.2* and *Dx1.1* The allele *Q* is associated with the higher egg production and *q* with the lower egg production, with the *Qq* genotype having an annual egg production equal to the average of the *QQ* (300 eggs) and *qq* (270 eggs) genotypes. A cross of the following type is made:



Female offspring are classified into the following four types:

(a) *Cx4.3 Dx1.1/Cx2.1 Dx3.2*

(b) *Cx4.3 Dx3.2/Cx2.1 Dx3.2*

(c) *Cx2.1 Dx1.1/Cx2.1 Dx3.2*

(d) *Cx2.1 Dx3.2/Cx2.1 Dx3.2*

What is the expected average egg production of each of these genotypes? You may assume complete chromosome interference across the region.

16.24 The pedigree shown to the right includes information about the alleles of three genes that are segregating.

• Gene 1 has four alleles denoted *A, B, C,* and *D.* Each allele is identified according to the electrophoretic mobility of a DNA restriction fragment that hybridizes with a probe for gene 1. The pattern of bands observed in each individual in the pedigree is shown in the diagram of the gel below the pedigre. When an allele is homozygous, the band is twice as thick as otherwise, because homozygous genotypes yield twice as many restriction fragments corresponding to the allele than do homozygotes. The small table at the upper right of the pedigree gives the frequencies of the gene 1 alleles in the population as a whole.

• Gene 2 has four alleles denoted *E, F, G,* and *H.* These also are identified by the electrophoretic mobility of restriction fragments. For each person in the pedigree, the genotype with respect to gene 2 is indicated. The population frequencies of the gene-2 alleles are also given in the table at the upper right.

• Gene 3 is associated with a rare genetic disorder. Affected people are indicated by the brown symbols in the pedigree. The genotype associated with the disease has complete penetrance. There are two alleles of gene *3,* which are denoted *Q* and *q.*

(a) What genetic hypothesis can explain the pattern of affected and non affected individuals in the pedigree?

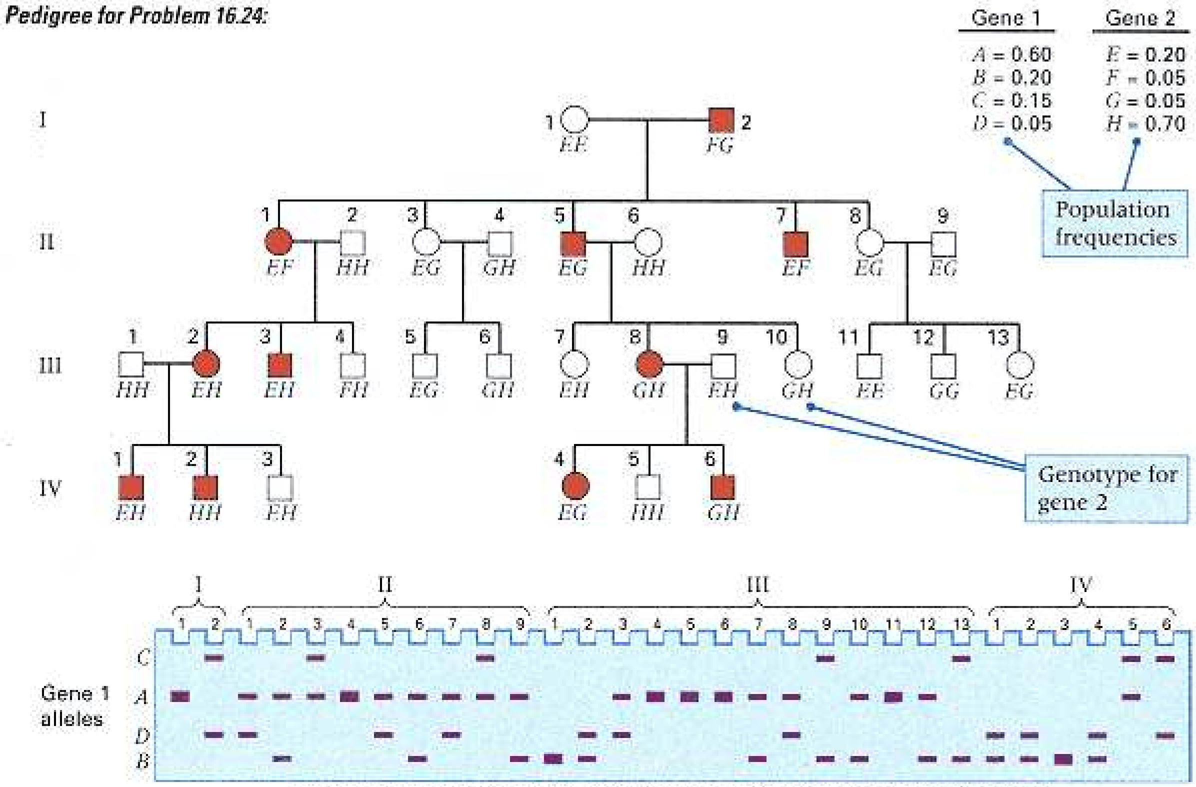
(b) If the male IV-2 fathers a daughter, what is the probability that the daughter will be affected?

(c) Assuming random mating, if the male IV-2 fathers a daughter, what is the probability that the daughter will have the genotype *HF* for gene 2?

(d) Identify the genotype of each person in the pedigree with respect to genes 1, 2, and 3. Wherever possible, specify the linkage phase (coupling or repulsion) of the alleles present in each individual.

(e) Examine the data for evidence of linkage. Are any of the pairs of genes linked? On the basis of this pedigree, what is the estimated recombination frequency between gene 1 and gene 2? Between gene 2 and gene 3?

Pedigree for Problem 16.24:



Electrophoresis pattern when DNA from each individual is probed for gene 1

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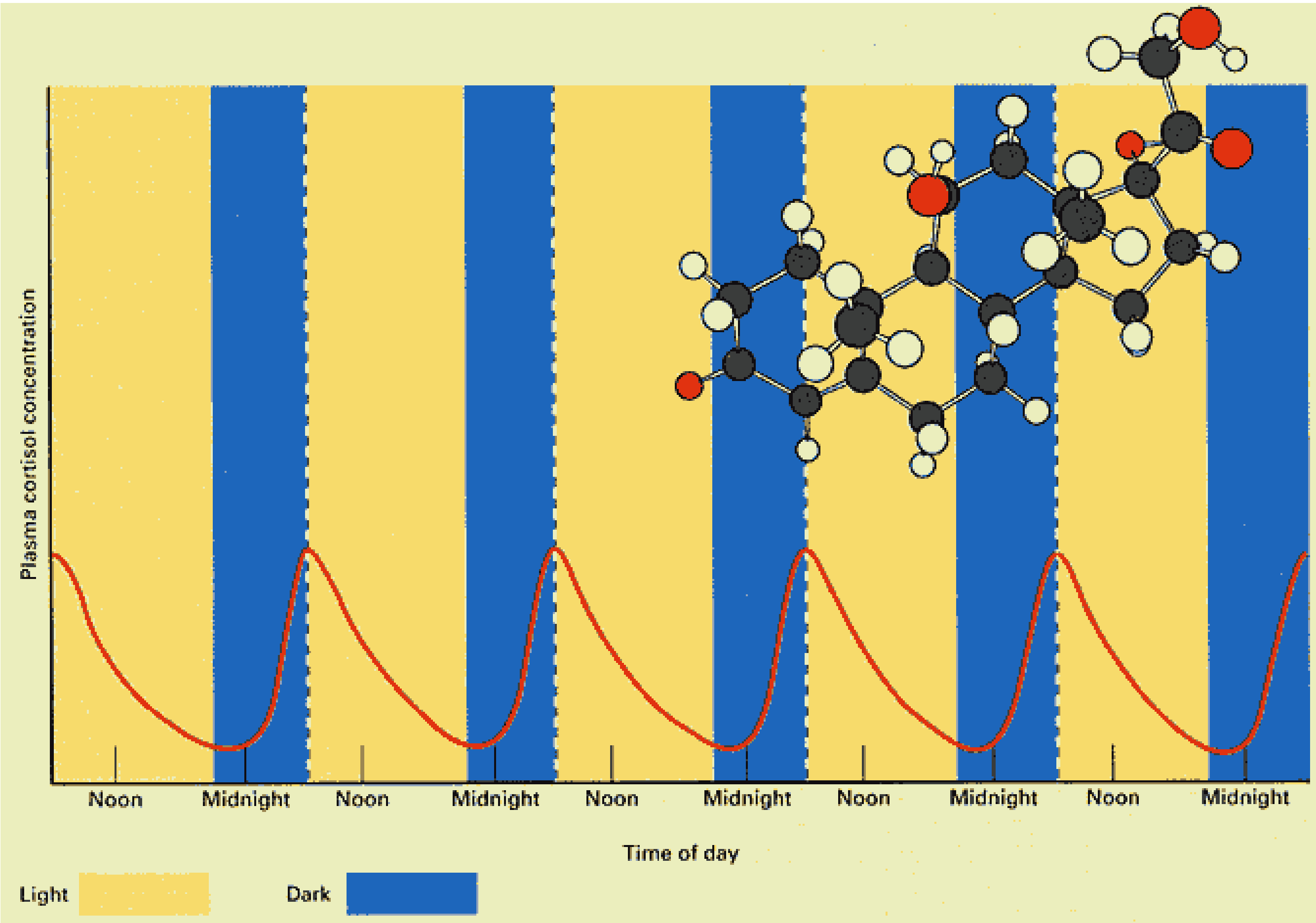

Circadian rhythms with a period of approximately 24 hours affect many physiological and neurological traits. This example shows a circadian rhythm in the production of cortisol in human beings. The highest levels occur at night, just before waking. N means noontime and M midnight.
[Adapted with permission from G. A. Hedge, H. D. Colby, and R. L. Goodman 1987.*Clinical Endocrine Physiology*. Philadelphia: Saunders:]

Chapter 17—
Genetics of Biorhythms and Behavior

CHAPTER OUTLINE

PRINCIPLES

• Behavior includes all changes that an organism undergoes in response to its environment.

• Some behaviors are completely determined genetically.

An example is chemotaxis in *E. coli*, which results from control of counterclockwise or clockwise rotation of the flagella by means of intracellular signals transmitted in response to attractants or repellents in the medium.

• Other behaviors are determined by innate biological rhythms, such as the circadian rhythm of roughly 24 hours. The molecular genetics of the *Drosophila* circadian rhythm involves cyclic transcriptional control of genes such as *per (period)* and *tim (timeless).*

• Learning in laboratory animals is affected by genetic as well as environmental factors. Genetic factors are evidenced by the success of artificial selection in increasing or decreasing maze-learning ability in laboratory rats. In this case, there is also a strong genotype-environment interaction.

• Some human behavioral differences are due to simple Mendelian factors, such as the ability to detect the chemical phenylthiocarbamide. More complex human behaviors are affected by genetic and environmental factors and the interactions between them.

• Heritability estimates of human behavioral traits are often inflated because of cultural transmission; these values, however estimated, cannot be used to infer the genetic or environmental causation of differences among human racial or ethnic groups.

CONNECTIONS

CONNECTION: A Trip to the Zoo

Joan Fisher Box 1978
R. A. Fisher: The Life of a Scientist

All organisms interact with their environments. The bacterium *E. coli* responds to the presence of lactose (milk sugar) in its environment by synthesizing new enzymes that allow the lactose to be brought into the cell and metabolized. Sunflowers respond to sunlight by tracking the sun across the sky. Canada geese respond to signs of autumn by beginning their long migrations. You respond to touching a hot stove by involuntarily jerking your hand away. **Behavior** in its broadest sense includes all changes that an organism undergoes in response to its environment. The underlying genetics and molecular biology of some seemingly complex behaviors may be relatively simple. For example, the enzymes for utilizing lactose in *E. coli* are induced by the straightforward molecular mechanism discussed in Chapter 11. At the other extreme of complexity, behaviors such as learning and memory in humans beings are currently beyond explanation by genetics and molecular biology. Between the bacterial cell's response to lactose and a human child's ability to learn a language are immeasurable gradations of behavioral complexity. The field of **behavior genetics** deals with the study of observable genetic differences that affect behavior.

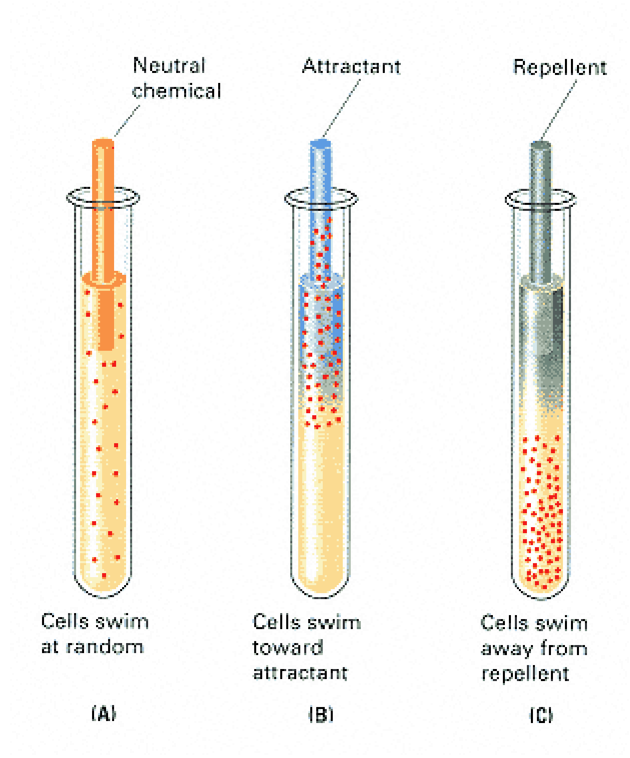

Figure 17.1
Chemotactic behavior of E. coli in a soft agar solution with an inserted capillary tube containing (A) a neutral chemical, (B) an attractant, or (C) a repellent. The cells swim toward the attractant, swim away from the repellent, and are indifferent to the neutral chemical.

Because behavior genetics is carried out in a wide variety of organisms, from bacteria to human beings, virtually all techniques of genetic analysis are applicable at one level or another. The induction and analysis of specific mutations affecting behavior has proved fruitful in behavioral studies in *E. coli* and *Drosophila.* In human behavior, the most common techniques of genetic analysis are pedigree studies and quantitative genetics. Quantitative genetics is essential in human behavior genetics, because most variation in human behavior-particularly variation within the range considered "normal"-is multifactorial in origin; most normal human behavior is influenced by both genetic and environmental factors.

This chapter summarizes examples of behavioral genetic analysis. Its emphasis is on the types of genetic approaches that have been developed for the study of behavior in different organisms. The examples start with *E. coli* and end with human beings.

17.1— Chemotaxis in Bacteria

Chemotaxis is the tendency of organisms to move toward certain chemical substances (attractants) and away from others (repellents). Such behaviors are widespread among living organisms. Among human beings, who is not attracted to the aroma of freshly baked bread? Some organisms rely on chemical attractants for finding mates; one example is the female gypsy moth, whose pheromones can attract males across long distances. Other organisms, such as the skunk, rely on chemical repellents for self-defense. Even relatively simple creatures, such as bacteria, have a chemotactic response. In *E. coli,* the chemotactic response can be observed by inserting a capillary tube filled with an attractant or repellent into a test tube containing bacterial cells in agar dilute enough that the bacteria can swim (Figure 17.1). If the capillary contains a neutral chemical that neither attracts nor repels, the bacterial cells are indifferent to it and remain uniformly dispersed throughout the medium (part A). If the capillary contains an attractant, the cells move toward the mouth of the capillary and eventually swim up into it (part B). If the capillary contains a repellent, the cells swim away from it (part C). Most attractants are nutritious substances such as sugars and amino acids, whereas most repellents are harmful substances or excretory products such as acids or alcohols. Hence, the chemotactic response is important in helping bacterial cells move toward favorable conditions or away from harmful ones.

Cells of *E. coli* propel themselves by means of hollow filaments, called **flagella,** that twirl individually like propellers on a motorboat. A typical cell has six flagella, each approximately 7 μm long (Figure 17.2), which is roughly three times the length of the cell itself. At the base of each flagellum is a molecular "motor" that rotates the flagellum either counterclockwise or clockwise. Counterclockwise rotation twists the flagella into a single bundle that propels the cell forward (Figure 17.3A), whereas clockwise rotation causes the flagella to splay out and work at cross

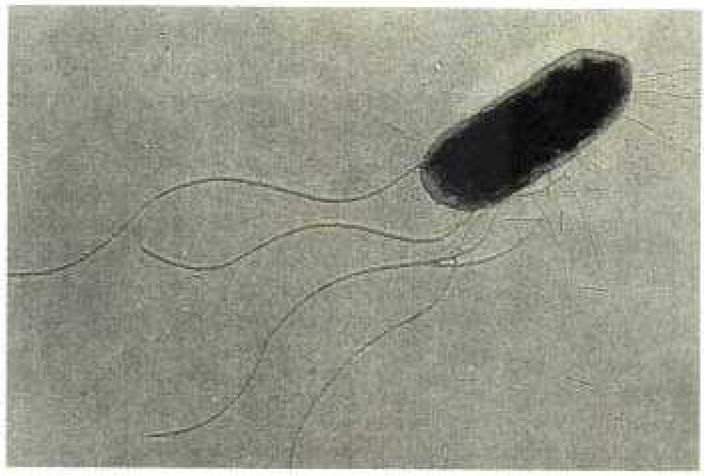

Figure 17.2
Flagella (long fibers) emanate from random places from the sides of an *E. coli* cell. When the flagella rotate counterclockwise, they form a bundle and propel the cell forward; when they rotate clockwise, they splay apart and cause the cell to tumble. The shorter, thinner objects projecting from the cell are a different type of fiber that aids in adherence to surfaces. Magnification × 27,600.
[Courtesy of Howard C. Berg.]

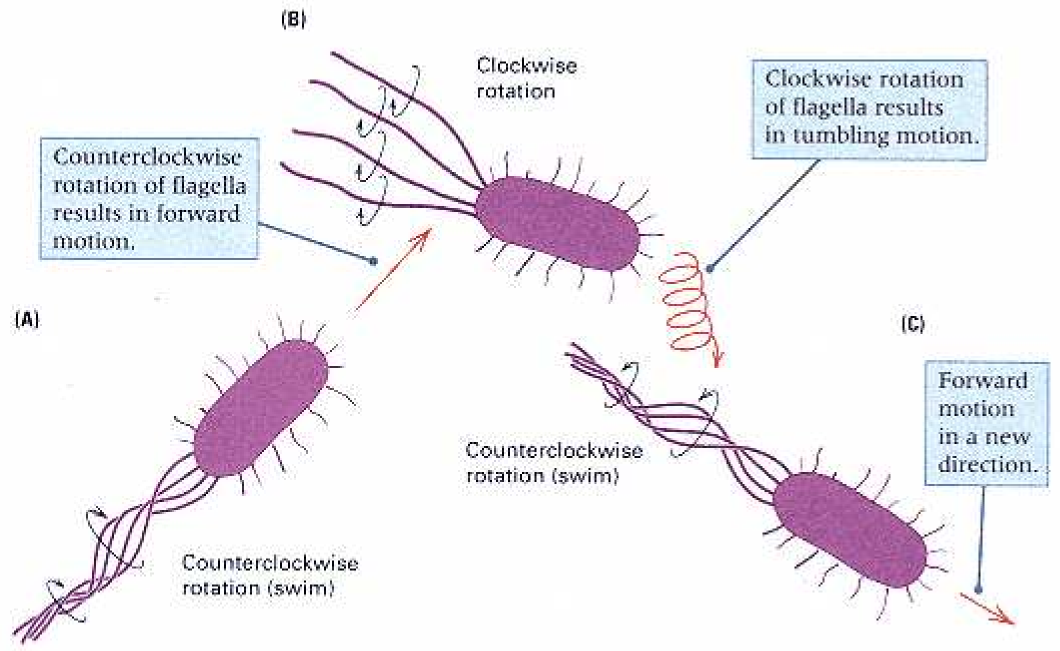

Figure 17.3
Each flagellum of E. coli is rotated by a molecular "motor" at the base. (A) If the flagella are rotated counterclockwise, they form a coherent bundle that propels the cell forward (swimming). (B) If the flagella are rotated clockwise, they spread apart and the cell tumbles, causing an unpredictable change in direction. (C) After tumbling, a return to counterclockwise rotation causes the cell to dart

off in a new direction.

purposes (Figure 17.3B). Consequently, a cell that switches flagellar rotation from counterclockwise to clockwise stops swimming forward and begins to tumble, thereby changing its orientation in the medium. A return to counterclockwise rotation of the flagella then sends the cell off swimming again in a new direction (Figure 17.3C). In the absence of attractants or repellents, cells trace a kind of random walk, darting forward in some direction for a second or two, then tumbling for a fraction of a second, and then darting off in perhaps a quite different direction (Figure 17.4). In the presence of an attractant, periods of swimming toward the attractant become longer, and periods of moving in the other direction become

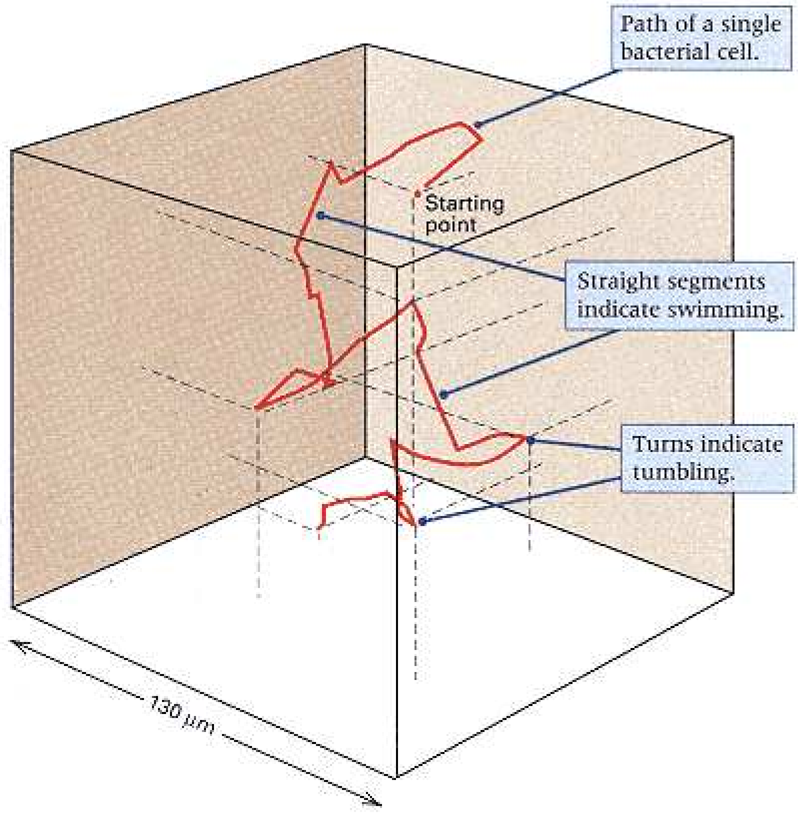

Figure 17.4
Three-dimensional path of a single *E. coli* cell followed automatically in a computerized tracking microscope. The track starts at the position of the red dot. In the 30-second tracking period, the cell made 26 swims (straight paths) and 26 tumbles (changes of direction). The swims were at a speed of about 20 μm per second. The cube has dimensions of 130 μm on each side.
[After H. C. Berg. 1975. *Scientific American* 233: 36.]

shorter. The random walk consequently becomes biased toward the direction of the attractant, and the cells eventually swim and tumble their way to the source of the attractant (Figure 17.1). Repellents act in the reverse manner, and the cells gradually disperse from them.

Mutations Affecting Chemotaxis

Mutations affecting chemotaxis have revealed the genetic components of the chemotactic response. The phenotypes of chemotaxis mutations fall into five very general categories, which are summarized in Table 17.1.

1. Specific nonchemotaxis means that mutant cells show no response to a particular attractant or repellent. For example, galactose is an attractant. A mutant phenotype that fails to respond to galactose, but that is otherwise completely wildtype, shows a nonchemotaxis that is specific for galactose. A mutation in a different gene results in a phenotype of specific nonchemotaxis in which mutant cells fail to be attracted to glucose.

2. Multiple nonchemotaxis refers to a phenotype with no chemotactic response to a group of chemically unrelated substances. For example, mutations in the gene designated *tsr* cause the mutants to fail to respond to the amino acid attractant serine or to the repellents indole, leucine, and acetate. Mutants that are multiply nonchemotactic fall into four complementation groups and thus define four genes, which are called *tsr, tar, trg,* and *tap.*

3. General nonchemotaxis is a failure to respond to any attractant or repellent. Although such mutants have normal flagella, as well as the apparatus necessary for switching flagellar rotation between clockwise and counterclockwise, they are unable to control the flagellar rotation in response to attractants or repellents. Mutations for general nonchemotaxis can be grouped into six complementation groups, each of

Table 17.1 Major categories of nonchemotactic mutants

Type of mutant	Genes	Phenotype
1. Specific nonchemotaxis	Galactose-binding protein, glucose transporter, etc.	No response to a particular attractant (e.g., galactose)
2. Multiply nonchemotactic	*tsr, tar, trg, tap*	No response to attractants that stimulate a particular receptor
3. Generally nonchemotactic	*cheA, cheW, cheR, cheB, cheZ, cheY*	No response to any attractant
4. Motor components	*fliG, fliM, fliN*	Continuous tumbling or continuous swimming
5. Flagellar components	Genes needed for synthesis of flagella	No motility

which defines a gene that is essential for any kind of chemotaxis. These genes are designated *cheA, cheB, cheR, cheW, cheY,* and *cheZ* (Table 17.1).

4. Mutations in the **motor components** of the flagella cause nonchemotactic behavior because of defects in the apparatus that controls switching between clockwise and counterclockwise flagellar rotation. These mutants continuously swim or continuously tumble, depending on the particular type of mutation. Figure 17.5 is the track of a mutant cell that swims continuously. Mutations in the motor components define three genes necessary for switching between swimming and tumbling, which are designated *fliG,fliM,* and *fliN.*

5. Nonmotile mutants are nonchemotactic for the trivial reason that they are unable either to swim or to tumble. Nonmotile mutations define a large number of different genes that are necessary for the synthesis or function of the flagella. Figure 17.6 is a diagram showing some of the gene products that are structural components of the flagellum.

Analysis of mutations has been critical to our understanding of the process of chemotaxis. For example, the correlation between the direction of rotation of the flagella and the swimming or tumbling behavior of the cells was determined by observation of the two types of mutations in the motor components of the flagella

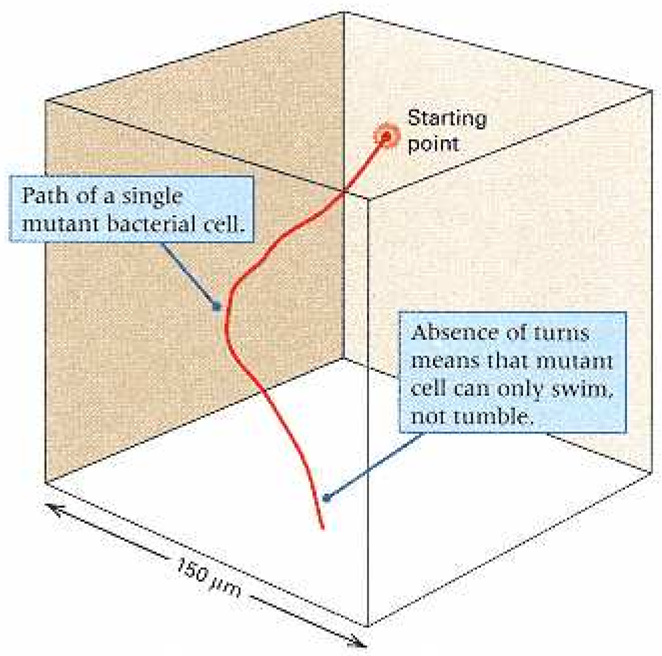

Figure 17.5
Swimming path of an *E. coli* cell with a mutation in one of the flagellar motor genes, making the cell unable to tumble. The track starts at the position of the red dot. Compare this pattern with that of the wildtype cell in Figure 17.4. In a 7-second tracking period, the cell swam 225 μm without a tumble. The square is 150 μm on each side.
[After H. C. Berg. 1972. *Nature* 239: 500.]

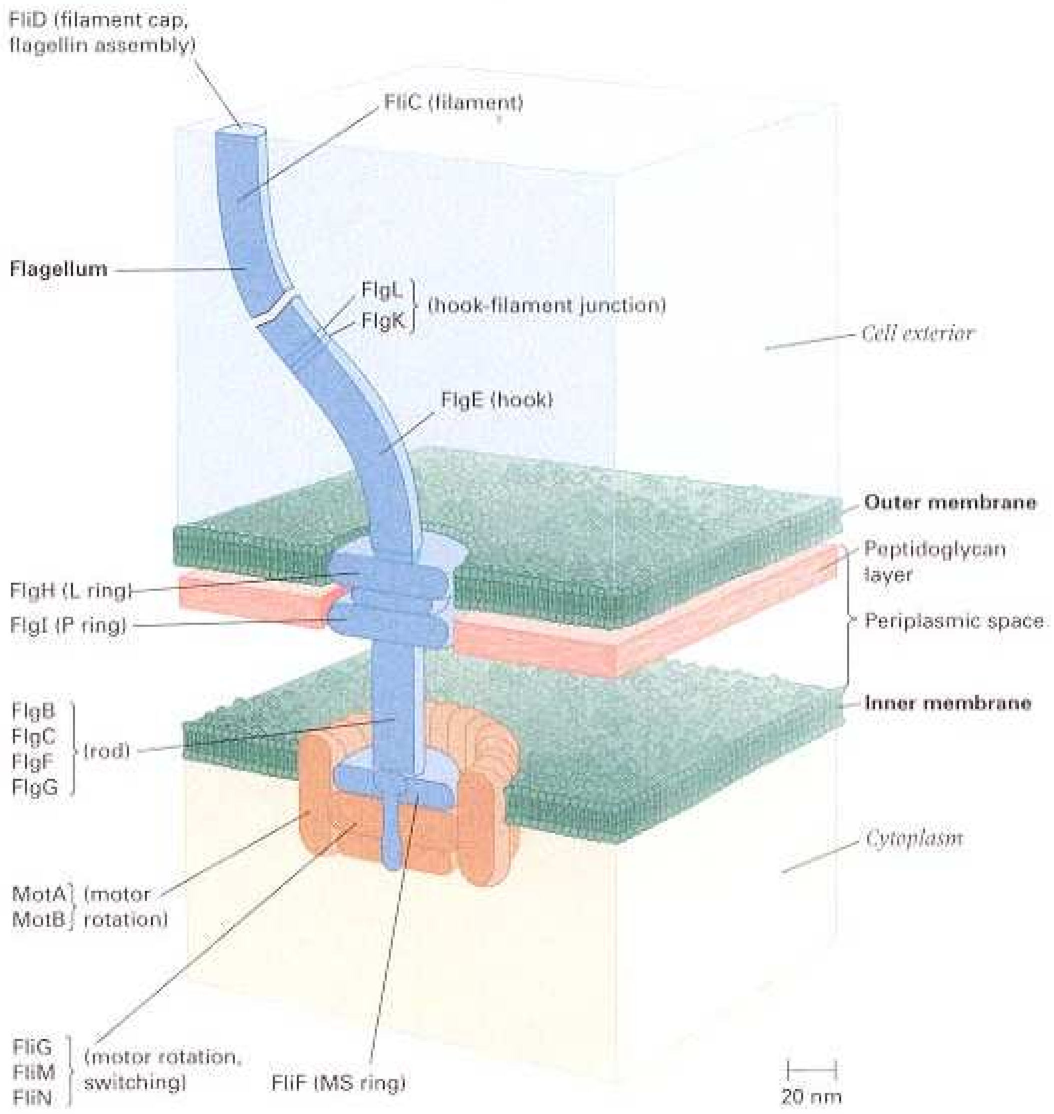

Figure 17.6
Protein components of the flagellum of *E. coli*. The products of more than 40 genes necessary for flagellar synthesis and function have been identified. The flagellum is anchored to the outer and inner cell membranes and is rotated by means of a motor apparatus at the base. The products of the genes *fliG*, *fliM*, and *fliN*. denoted FliG, FliM, and FliN, respectively, are necessary for switching between counterclockwise and clockwise rotation. Mutations in these genes result in motile but nonchemotactic cells. Mutations in the genes for most of the other flagellar components result in loss of motility.
[From R. M. Macnab and J. S. Parkinson. 1991. *Trends Genet*. 7: 196.]

(category 4 in Table 17.1). The flagella are too thin for their rotation to be observed directly in the light microscope, but they can be attached to glass slides coated with antiflagellar antibody proteins that combine tightly with the flagellar proteins and hold them to the slide. Under these conditions, the bacteria become tethered to the slide through their flagella, and the flagella remain fixed while the *body* of the cell rotates. The cell bodies are large enough that the direction of rotation can be observed directly, and the cell bodies rotate about the tethered flagella in the same direction as the flagella themselves would rotate if they were untethered (Figure

17.7). Observation of tethered cells has revealed that

> Mutants that swim continuously rotate counterclockwise continuously; mutants that tumble continuously rotate clockwise continuously.

All of the genes required for chemotaxis in categories 2 through 4 have been cloned and sequenced, and their gene products have been identified. The complex behavior of chemotaxis is a relatively simple consequence of the various products of these genes and their interactions. The cellular components of chemotaxis and the mechanism of the response are the subjects of the next section.

The Cellular Components of Chemotaxis

The cellular components of chemotaxis are illustrated in the diagram of an *E. coli* cell in Figure 17.8. A mutational defect in any of these components results in abnormal chemotaxis, and the phenotypes of the major classes of mutants listed in Table 17.1 are indicated in the blue boxes. Mutations can affect components in the system at any of the following levels.

Wildtype chemotaxis requires a set of proteins called **chemosensors,** each of which binds with a particular attractant or repellent. Examples of chemosensors include the galactose-binding protein and the glucose transporter. More than 20 different chemosensors are known; about half of them bind with attractants and half with repellents. Some chemosensors, such as the galactose-binding protein, are present in the periplasmic space between the cytoplasmic membrane and the outer membrane. Other chemosensors, such as the glucose transporter, are tightly associated with the cytoplasmic membrane. The chemosensors for attractants are typically components of the molecular systems used for transporting the attractant into the cell. Mutations in the chemosensors result in a phenotype of specific nonchemotaxis (Table 17.1).

The next step in chemotaxis is interaction between the chemosensor and any of

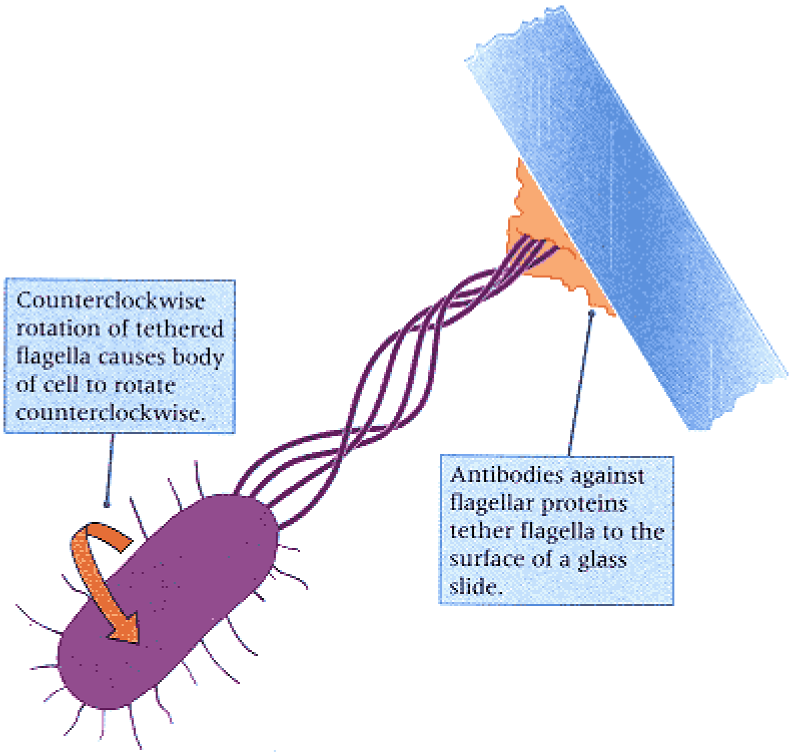

Figure 17.7
A molecular motor at the base of a cell that rotates the flagellum in any direction (in this case, counterclockwise) will rotate the cell body in the same direction (in this case, counterclockwise) when the flagellum is tethered and unable to turn.

four **chemoreceptors,** which are the products of the genes *tsr, tar, trg,* and *tap.* Each chemoreceptor interacts with a particular subset of chemosensors. For example, the chemoreceptor Tsr (Figure 17.9), encoded by the gene *tsr,* interacts with the chemosensor for the attractant serine (an amino acid) and with chemosensors for the repellents leucine, indole, and acetate. Similarly, the chemoreceptor Trg, corresponding to the *trg* gene, responds to several different chemosensors, including those for galactose and ribose. Mutations in any of the chemoreceptor genes result in loss of responsiveness to all of the chemosensors with which the chemoreceptor interacts. Hence chemoreceptor mutants have a phenotype of multiple nonchemotaxis (category 2 in Table 17.1).

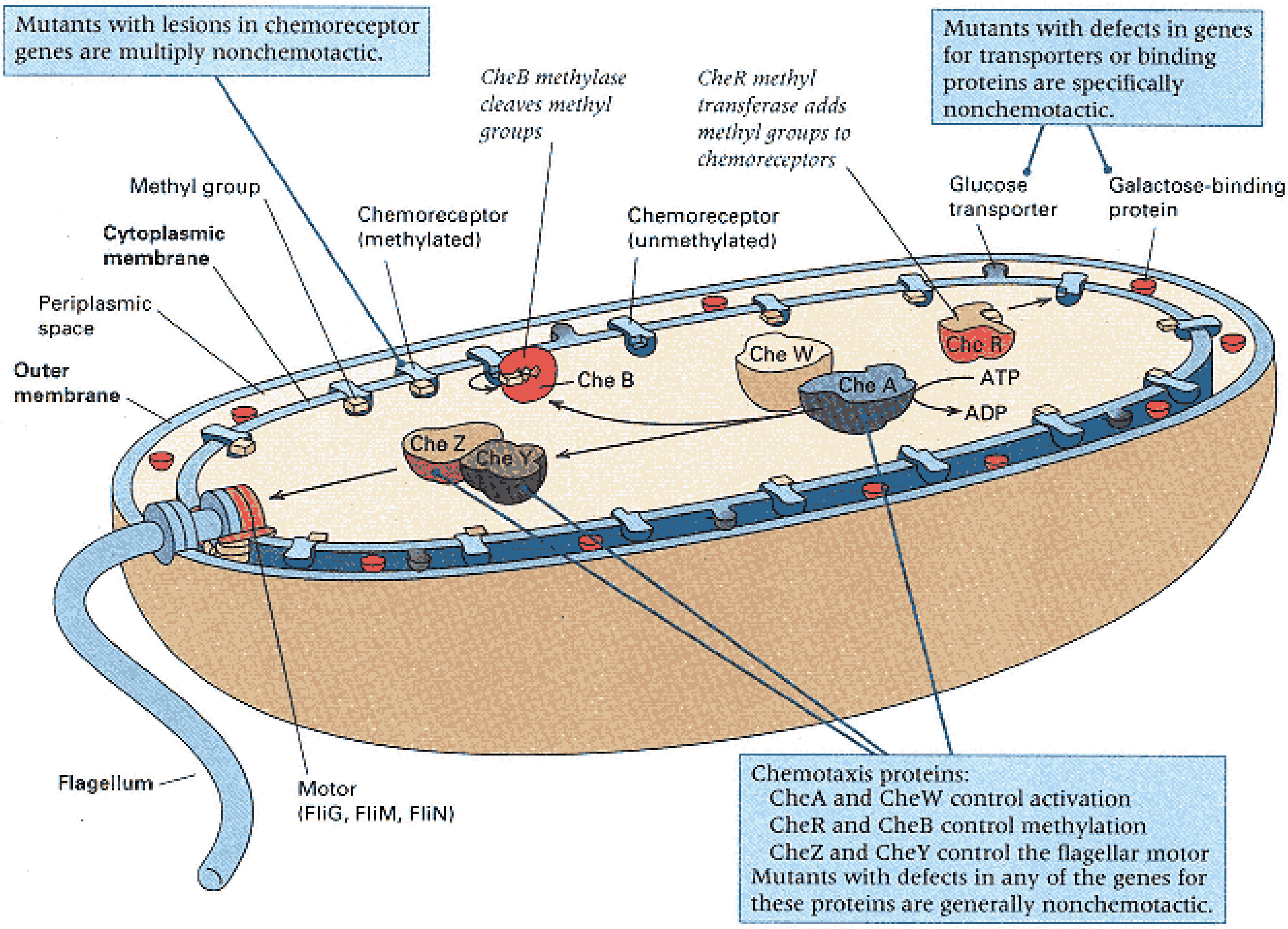

Figure 17.8
Diagram of a cell of *E. coli* showing the interactions of the major types of proteins implicated in chemotaxis. The phenotypes associated with the major classes of mutations are indicated in the blue boxes.

All of the chemoreceptors feed molecular signals into a common pathway containing the proteins denoted CheA, CheB, CheR, CheW, CheY, and CheZ, illustrated in Figure 17.8; these proteins are the products of the genes *cheA, cheB, cheR, cheW, cheY,* and *cheZ,* respectively. The molecular nature of the chemotactic signals is described in the next section. For the present, the important point is that all of the chemoreceptors use the same pathway, and this explains why mutations in any of the genes in the common pathway result in a phenotype of general nonchemotaxis (category 3 in Table 17.1).

The target of the common pathway is the molecular motor at the base of the flagella. Switching between clockwise rotation (tumbling) and counterclockwise rotation (swimming) is controlled by the products of the genes *fliG, fliM,* and *fliN.* Mutations in any of these genes result in continuous tumbling or continuous swimming (Table 17.1).

Once the chemotaxis signal is passed to the switching apparatus, successful attraction or repulsion requires functional flagella that are able to rotate. The flagella are composed of a number of essential components (Figure 17.6) in which mutational defects result in nonmotile cells (category 5 in Table 17.1).

Molecular Mechanisms in Chemotaxis

The molecular basis of the chemotactic response is mediated by chemical modification of some of the key components in the central pathway illustrated in Figure 17.8. For clarity, the chemoreceptors are shown distributed uniformly in the cytoplasmic

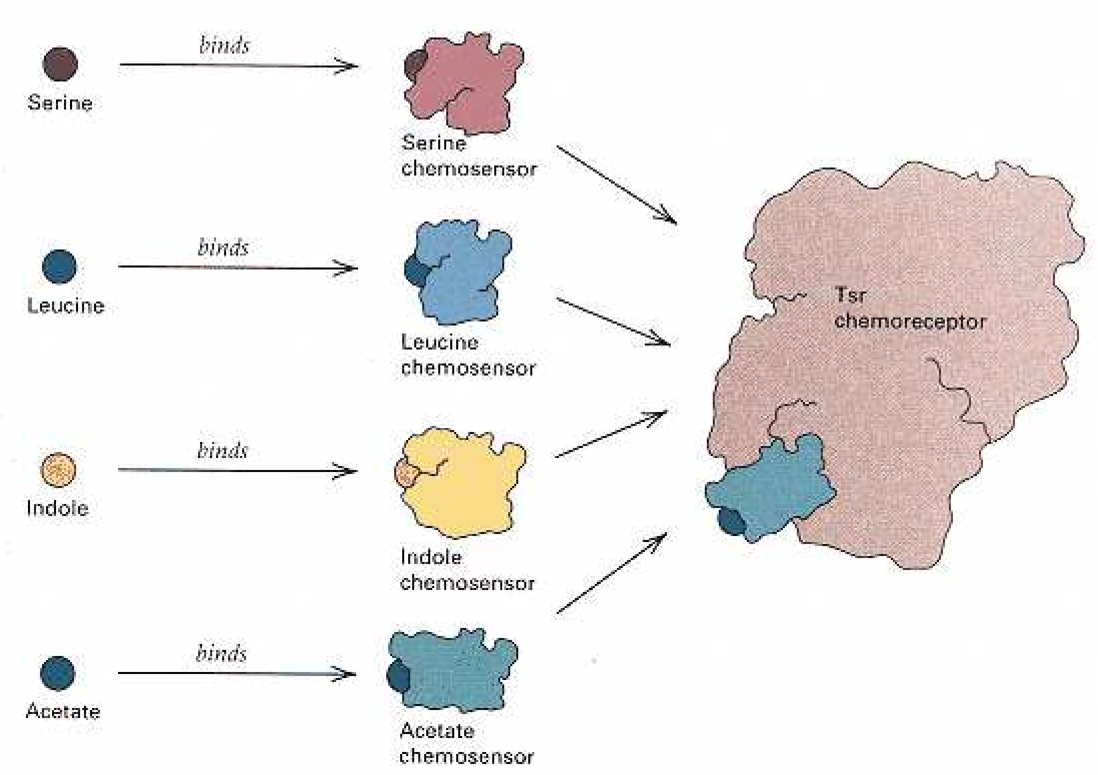

Figure 17.9
Pattern in which the Tsr chemoreceptor interacts with a number of different types of chemosensors.

membrane, and CheA and CheW in the cytoplasm. However, localization of key components of the chemotaxis apparatus by the use of fluorescent antibodies implies that they are concentrated at the poles of the cell (Figure 17.10). To understand how chemotaxis in *E. coli* works, let us first follow the normal course of events in the absence of attractants or repellents, as shown by the arrows in Figure 17.8.

The key player in regulating chemotaxis is CheA. This protein contains a region that catalyzes the transfer of a phosphate group from ATP to a specific histidine within the molecule (Figure 17.11A), yielding the phosphorylated form of CheA (denoted P-CheA). The formation of P-CheA is stimulated by the chemoreceptors acting in combination with CheW. Once transferred to P-CheA, the phosphoryl group on the histidine in P-CheA is readily transferred to either CheY or CheB, resulting in the phosphorylated forms P-CheY and P-CheB. The P-CheY form interacts directly with the switching apparatus that controls flagellar

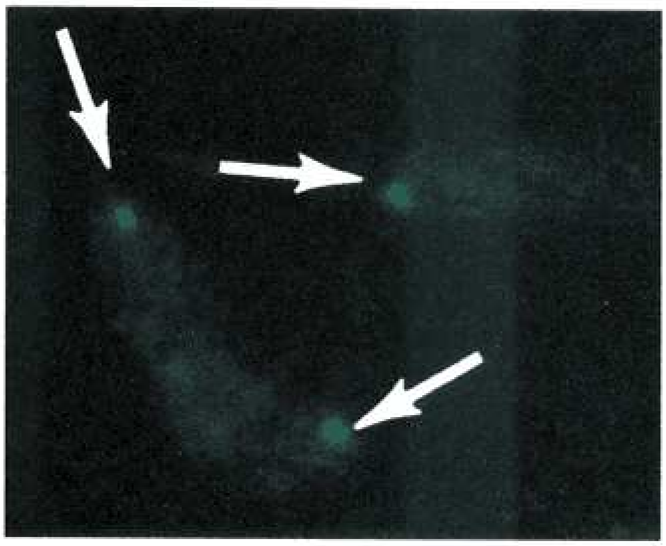

Figure 17.10
Evidence that the chemoreceptor proteins are preferentially located at the poles of the cell. In this photograph, the green color results from fluorescent antibodies to Tsr combining with the Tsr protein where it is localized in the cell-namely, at the poles. Similar polar localization is also observed for CheA

and CheW.
[Courtesy of L. Shapiro, from J. R. Maddock and L. Shapiro. 1993. *Science* 259: 1717.]

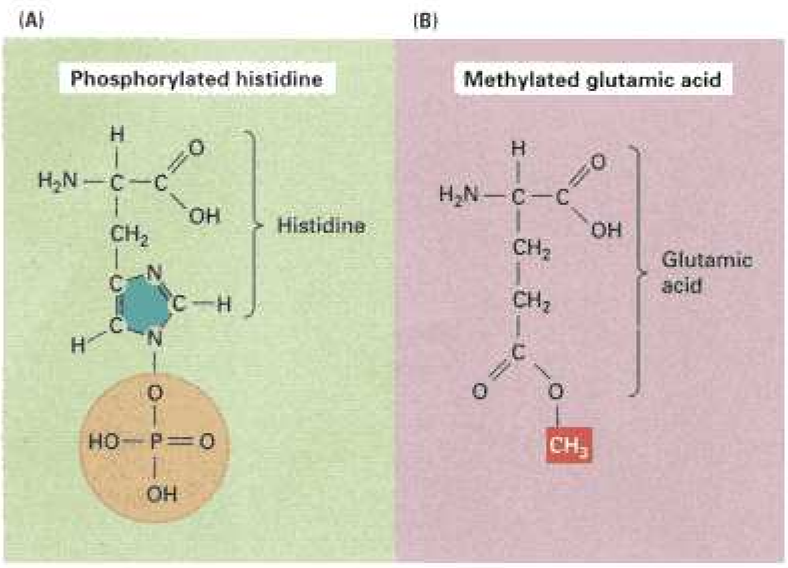

Figure 17.11
Protein modifications in the regulation of chemotaxis. (A) One particular histidine at amino acid position 48 in the CheA protein becomes phosphorylated (chemical group in yellow) at the position shown. (B) A number of glutamic acid residues in the chemoreceptor proteins become methylated (methyl group shown in red). Addition of the methyl groups is catalyzed by CheR, and removal is catalyzed by P-CheB.

rotation and causes the flagella to rotate clockwise (tumbling). As might be expected from this description, extreme mutations in *cheA*that cannot make P-CheA, and extreme mutations in *cheY* that cannot make P-CheY, result in a phenotype of continuous swimming because the clockwise rotation of the flagella necessary for tumbling cannot be stimulated.

If only CheA and CheY were at work, the cells could be expected to tumble continuously because of nearly constant stimulation of the rotational switch by P-CheY. However, continuous tumbling is prevented by P-CheB, which acts on the chemoreceptors to remove methyl groups from certain glutamic acid subunits that are methylated (Figure 17.11B). The methylated forms of the chemoreceptors stimulate production of P-CheA. Removal of the methyl groups by P-CheB inhibits the formation of P-CheA and thereby diminishes further tumbling. Continuous tumbling is also prevented by CheZ, which rapidly removes the phosphoryl group from P-CheY, so that each tumbling episode caused by P-CheY typically lasts for only a fraction of a second. As might be expected, extreme *cheZ* mutations result in a phenotype of continuous tumbling because of the increased levels of P-CheY. Extreme *cheB* mutations also result in a phenotype of continuous tumbling, because when the chemoreceptors remain methylated, they stimulate additional production of P-CheA and therefore P-CheY.

As noted, methylation of the chemoreceptors stimulates greater production of P-CheA and enhances the tendency of the cells to tumble. Addition of the methyl groups is the function of CheR. In the chemoreceptor Tsr, for example, as manyas four glutamic acid residues can be methylated. Methylation requires the amino acid methionine because a derivative of methionine (*S-adenosylmethionine)* is the donor of the methyl groups that CheR transfers. Because they accept methyl groups, the chemoreceptor proteins are often called **MCP** proteins, which stands for **methyl-accepting chemotaxis proteins.** Extreme mutations in *cheR* cannot methylate the receptors; therefore, the mutants swim continuously.

In the absence of any chemotactic stimulus, the basal level of phosphorylation of CheA results in enough P-CheY that the cells undergo repeated episodes of tumbling and reorientation in random directions; the tumbling episodes are terminated by the effects of CheZ and P-CheB, after which the cells swim for several seconds in the direction in which they happen to be oriented at the moment. The phosphorylation of CheA is inhibited by the binding of attractants to the chemoreceptors. Therefore, in the presence of an attractant, the episodes of tumbling become less frequent. Cells that, by chance, happen to be propelling themselves toward higher concentrations of the attractant undergo greater inhibition of tumbling, which results in longer runs of swimming toward the attractant. The upshot is that, in time, all the cells in the medium converge toward the attractant (Figure 17.1B). With repellents, the situation is the reverse: The repellent-excited chemoreceptors increase the rate of CheA phosphorylation, which results in more tumbling and, ultimately, divergence from the source of the repellent (Figure 17.1C).

Connection A Trip to the Zoo

Joan Fisher Box 1978
Madison, Wisconsin
Ronald A. Fisher combined extraordinary mathematical skills with a deep love of biological observation. Shortly after the discovery that the human population is polymorphic for the ability to taste phenylthiocarbamide (PTC)-the substance is harmless, but tasters report that it is very bitter-Fisher and a small group of other geneticists decided to find out whether chimpanzees are also polymorphic for the taster trait. If such a polymorphism persists over long periods of evolutionary time, even through events of species formation and divergence, then the persistence is evidence that the polymorphism may be maintained by some sort of balancing selection in which, on the average, the heterozygous genotype is favored. With this in mind, and solutions of PTC in hand, the researchers paid a visit to the chimpanzees at the Edinburgh Zoo. The episode is recalled by Joan Fisher Box, Fisher's daughter and biographer.
At the end of August 1939 the International Genetical Congress met in Edinburgh. It was an uneasy session in anticipation of the outbreak of World War II . . . ..Fisher and a number of others abandoned the conference one afternoon and made their way in high spirits to the cage of the chimpanzees [at the Edinburgh Zoo]. The eight chimpanzees were given mugs containing first a small quantity of slightly sweetened water (2% sucrose), a pleasant mixture for which they were expected to return for more. They drank it up and came back. This time they were given a sugared mixture containing 6 parts per million of phenylthiocarbamide; there were one or two dubious
Obviously,there were tasters among the chimpanzees.
expressions but they returned again, and received a mixture of 50 parts per million of phenylthiocarbamide. This time there was no doubt about their reaction: one chimpanzee turned his back emphatically on his visitors; another poured his drink on the ground with disgust and went to the opposite side of the cage; one sat gazing reproachfully at his audience while the fluid dribbled from a lower lip, a picture of misery; a more spirited beast spat it out in a strong jet straight at his persecutors. The geneticists were much diverted by these exhibitions. Obviously, there were tasters among the chimpanzees. There were also nontasters: two of the eight animals tested returned to receive and to drink a mixture containing 400 parts per million and still returned for more. This result was astonishing. The small sample was inconclusive as to the exact proportion of tasters, but it suggested not only that chimpanzees were polymorphic for the taste factor in the same way as people but in approximately the same proportions.
Source: R. A. Fisher: The Life of a Scientist.
John Wiley & Sons, New York (pp. 371–372)
It should be noted that the Animal Caretaker approved of these experiments and that the animals were not harmed. Today the experiments would need advance approval by an Animal Welfare Committee in response to a written application explaining the details of the experiment and making a case for its scientific value. Try having a discussion with a few fellow students by pretending that youare members of such a committee; the issue for debate is whether to approve an application for similar experiments with a different, but also harmless, chemical.

Sensory Adaptation

Methylation of the chemoreceptors by CheR is the basis of **sensory adaptation,** a process by which bacterial cells become accommodated to external stimuli. Sensory adaptation can be demonstrated by a simple experiment in which a solution of attractant is suddenly stirred into a suspension of cells. Because of the mixing, the concentration of the attractant is the same everywhere in the medium, but nevertheless tumbling is suppressed and each cell swims in whatever direction it happens to be oriented. After a few minutes, however, the cells adapt to the presence of the attractant and begin to tumble and swim in the normal, unstimulated manner. What happens is that the temporarily reduced level of P-CheB allows the activity of CheR to increase the overall level of methylation of the chemoreceptors. Methylation stimulates the production of P-CheA, which not only restores tumbling behavior through the production of P-CheY but also yields enough P-CheB methylase activity to result in an equilibrium level of methylation that is higher than it was before the attractant wasadded. The attractant still attracts the cells, but higher concentrations are necessary to be effective. For example, the Tsr chemoreceptor for serine contains four glutamic acid residues that are capable of being methylated. Each additional methyl group increases the level of serine required for excitation, so sensory adaptation to serine is graded in response to the existing level of serine. Should the level of an attractant suddenly decrease, the P-CheB methylase removes additional methyl groups and returns the chemoreceptor to its basal level of methylation.

The mixing experiment also demonstrates another important point about chemotaxis. Because bacterial cells swim toward increasing concentrations of an attractant, it may seem that the cells are comparing concentrations at various points in space-for example, between opposite ends of the cell. This would be very difficult because the bacterial cell is so small (approximately 1 × 2 µm). What the cells, in fact, do is compare the concentration that they experience at any time with the concentration of a moment ago. If the present concentration of an attractant is greater than that in the immediate past, the cell swims. If the present concentration is smaller, the cell tumbles. The comparison of concentrations at different points in *time,* rather than different points in *space,* explains why bacteria respond as they do to a substance suddenly dispersed uniformly in a liquid medium: If it is an attractant, each cell swims in a random direction; if it is a repellent, each cell tumbles.

17.2— Animal Behavior

The most complex behaviors are found in animals with well-developed nervous systems, such as insects, fish, reptiles, birds, and mammals. Although behaviors can differ among normal organisms in a species, the inheritance of these differences is usually through multiple genes. For example, inbred lines of mice differ in their nesting behavior, but the differences are due to multiple genes, no one of which has such a strong effect that it stands out above the rest. Complex behaviors are usually also influenced by environmental factors, such as learning and experience. Nevertheless, mutations in single genes can affect complex behaviors, so the genetic approach can be used to identify genes that are essential for the behavior. Because of the relative ease with which specific mutations can be obtained and analyzed in *D. melanogaster*, this organism is a favorite for research in behavior genetics. Behavioral mutants in *Drosophila* include mutations affecting the ability of the animal to walk or fly, the response to light, the ability to learn or to retain prior learning, sexual behavior or sexual responsiveness, and the length of cycle of the internal clock that regulates the animal's activities in synchrony with the cycle of day and night. The following sections illustrate the genetic approach to dissecting a particular complex behavior in *Drosophila.*

Circadian Rhythms

Many animal behaviors repeat at regular intervals. Biological rhythms synchronized with the 24-hour cycle of day and night caused by Earth's rotation are known as **circadian** rhythms. The natural rhythm of human sleep is an example of a circadian rhythm. The unpleasantness of jet lag after high-speed, long-distance air travel results from the tendency of the sleep rhythm to persist for several days even after the day-night cycle has undergone a major dislocation. It takes a traveler about one day to adjust for each time zone passed through, but it is usually easier to adjust to travel from East to West than from West to East. The jet-lag example, and many others, indicates that circadian rhythms are intrinsic to the metabolism of the organism, even though the rhythms eventually undergo adjustment to changes in the day-night cycle.

Among many behaviors in *Drosophila* that exhibit a circadian rhythm is **locomotor activity,** or movement from one place to another, with the peak period of activity during the night. An apparatus for automatic monitoring of locomotor activity is illustrated in Figure 17.12A. It consists of a narrow glass tube containing a single fly.

Each end of the tube is illuminated with red light of a wavelength invisible to the fly, and the intensity of the light transmitted through the tube at each end is measuredby a photocell. The photocells are connected to an event recorder in such a way that a single event is recorded whenever the light intensity on the two photo-

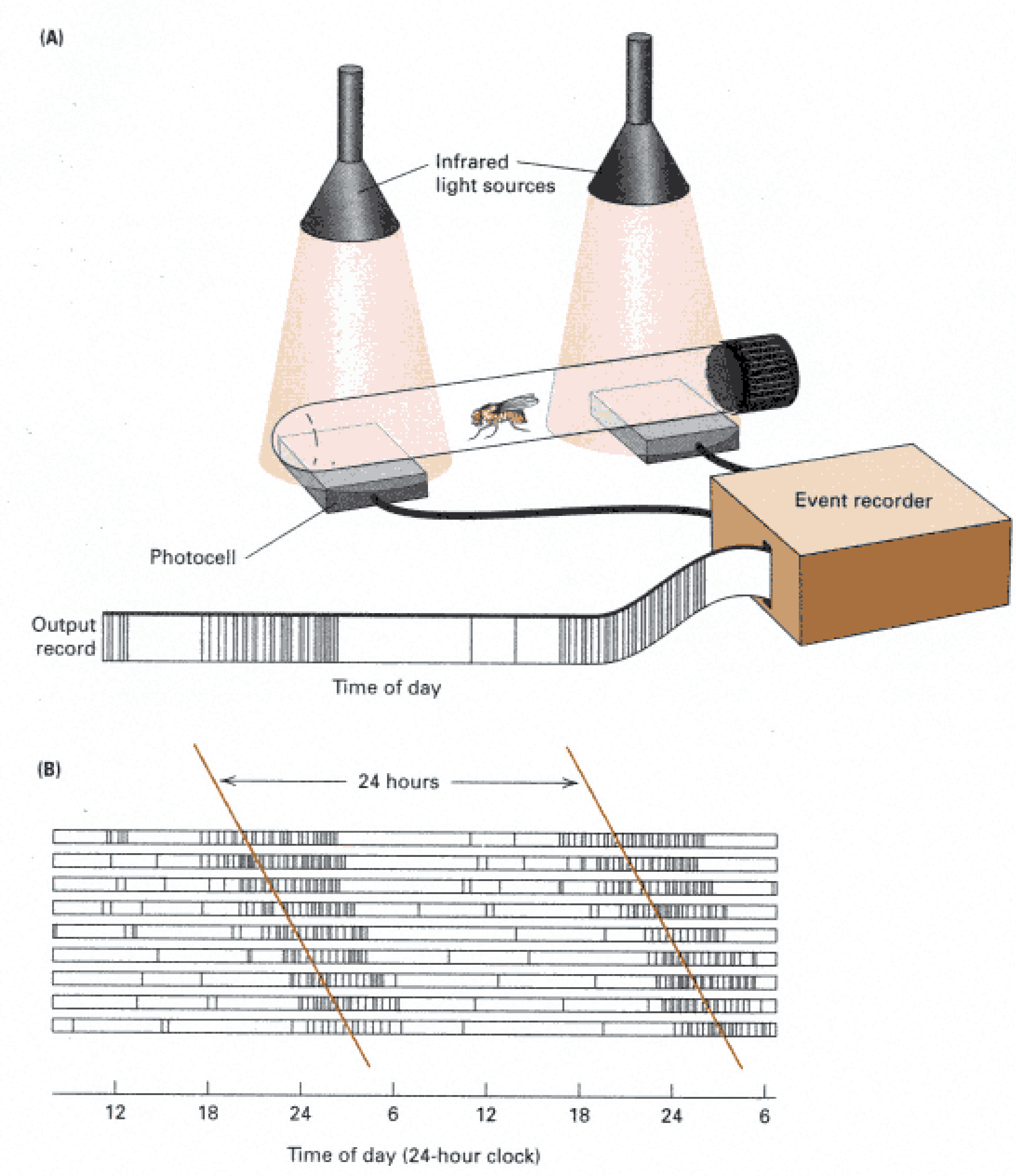

Figure 17.12
Locomotor activity in *Drosophila*. (A) The monitoring apparatus consists of a narrow tube illuminated with red light (invisible to the fly) at each end. The event recorder is adjusted to register an event whenever the light intensity on the two photocells becomes unequal. Hence, each vertical line in the output record indicates that the fly moved into one end of the tube or the other. (B) Output record of a single wildtype fly. Note that the periods of high activity (indicated by the dense vertical marks) have an approximately 24-hour periodicity.
[Data from Y. Citri, H. V. Colot, A. C. Jacquier, Q. Yu, J. C. Hall, D. Baltimore, and M. Rosbash. 1987. *Nature* 346: 82.]

cells becomes unequal. Hence, in the output record, each vertical line indicates that the fly moved into one end of the tube or the other. The high density of vertical lines in the middle of the output record indicates a period of high locomotor activity. Data from this apparatus are illustrated in Figure 17.12B, which shows the locomotor activity of a single wildtype fly over a period of 9 days. (Each horizontal line represents only 24 hours, because in order to maintain visual continuity, the activity pattern on

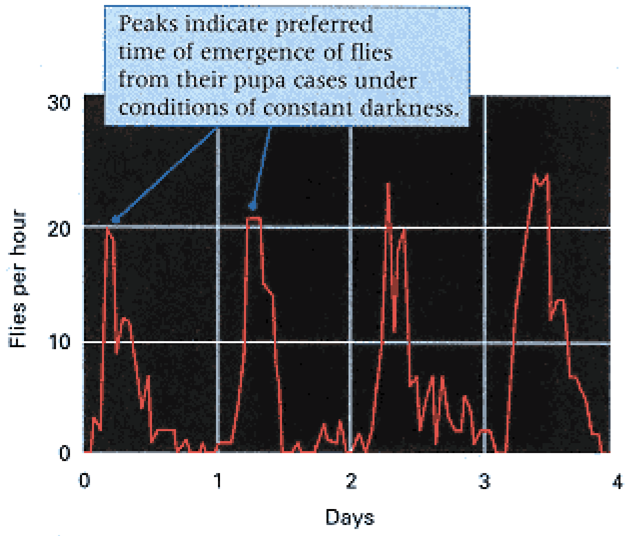

Figure 17.13
Circadian rhythm in time of emergence of adult Drosophila from their pupa cases. Wildtype flies were grown as larvae under a cycle of 12 hours light: 12 hours dark and then shifted to constant darkness for formation of pupae and metamorphosis. The graph shows the number of adult flies emerging during each hour of a 4-day period.
[Data from R. J. Konopka and S. Benzer. 1971. *Proc. Natl. Acad. Sci. USA* 68: 2112.]

the right-hand side of each line is duplicated on the left-hand side of the line below.) Before being monitored, the fly was maintained in a regular cycle of 12 hours light and 12 hours dark, but during the monitoring, there was no visible light. Nevertheless, a daily peak of locomotor activity is evident, and although the onset of activity shifts slightly from one day to the next, the periods of peak activity recur at regular intervals of approximately 24 hours.

Another example of circadian rhythm in *Drosophila* is the time of day at which the newly developed adults emerge from their pupa cases after undergoing metamorphosis. An example is shown in Figure 17.13, which plots the number of wildtype flies emerging each hour of a 4-day period. Periods of emergence peak at approximately 24-hour intervals, even though the pupae were in total darkness during metamorphosis. Abnormal emergence rhythms of a number of *Drosophila* mutants are shown in Figure 17.14. The mutant strains were obtained by exposing adult flies to a chemical mutagen and mating them in such a way as to produce a series of populations, within each of which the males carried a replica of a single mutagen-treated X chromosome. Each mutant strain was maintained in total darkness during metamorphosis, like the wildtype strain in Figure 17.13, and the pattern of emergence was examined to identify abnormal circadian rhythms. The three mutants in Figure 17.14 were obtained from tests of 2000 mutagen-treated X chromosomes.

All three mutations in Figure 17.14 proved to be alleles of the same X-linked gene, called *period (per)*. However, the mutant alleles have dramatically different circadian phenotypes, which suggests that the *per* gene is important in the timing of circadian rhythms. The allele per^{O} knocks out the circadian clock (Figure 17.14A). Males hemizygous for per^{O}, and females that are homozygous, are arrhythmic, so the adults are equally likely to emerge at any time of day. The alleles per^{S} and per^{L} result in phenotypes that still exhibit an emergence rhythm, but the period of the rhythm is altered. A short period of about 19 hours is characteristic of per^{S} (Figure 17.14B), and a long period of about 29 hours is characteristic of per^{L} (Figure 17.14C). In a manner paralleling their effects on adult emergence, the *per* alleles also affect other circadian rhythms. For example, in locomotor activity, the per^{O} mutation results in an arrhythmic phenotype, whereas per^{S} and per^{L} cause, respectively, phenotypes with short (19-hour) and long (29-hour) periods between successive peaks of locomotor activity.

Love-Song Rhythms in Drosophila

Fruit flies exhibit many types of behavioral rhythms that are not circadian, and it is remarkable that some of the noncircadian rhythms are also affected by mutations in *per.* An example is the *Drosophila* "love song," which consists of rhythmic patterns of wing vibrations that the male directs toward the female during courtship. The *Drosophila* courtship ritual is depicted in Figure 17.15. In part A, the male approaches the female from the front, ori-

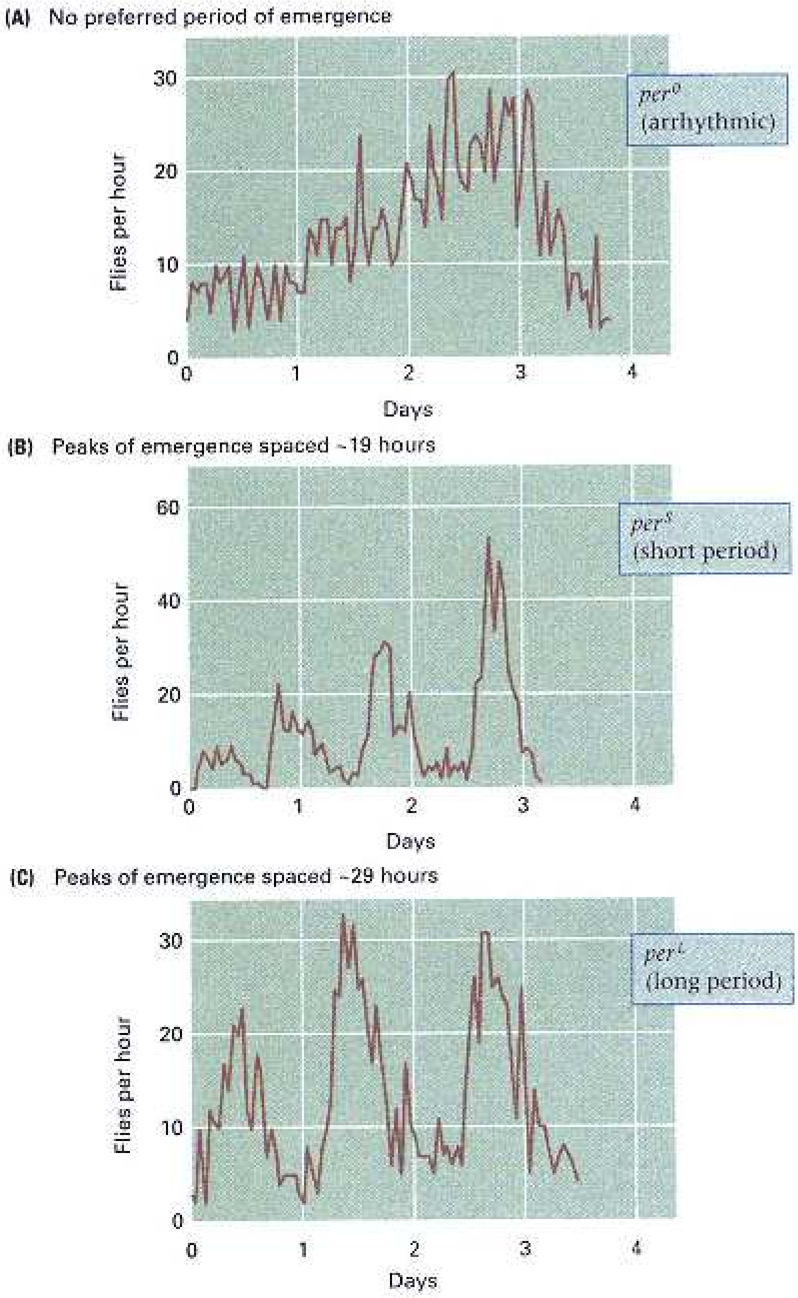

Figure 17.14
Mutant alleles of the per gene affecting the period of the adult emergence rhythm.
(A) In *per*^{O} flies, emergence is equally likely at any time of day, and because there is no circadian rhythm, the mutant is arrhythmic. (B) In *per*^{S} flies, there is an emergence rhythm, but the period is shortened to about 19 hours. (C) In *per*^{L} flies, there is an emergence rhythm, but the period is lengthened to about 29 hours.
[Data from R. J. Konopka and S. Benzer. 1971. *Proc.Natl. Acad. Sci*. USA 68: 2112.]

ented at an angle. In parts B and C, the male moves toward the rear, following the female if she moves, and repeatedly extends one of his wings, vibrating it rapidly up and down. In parts D and E, the male moves directly behind the female and touches her genitalia with his proboscis. In parts F and G, there is attempted copulation. More often than not, the attempt at copulation is unsuccessful, and the sequence of behaviors must be repeated several times until either copulation occurs or the encounter is broken off.

The *Drosophila* courtship song is defined by the pattern of wing vibrations in the stage of courtship depicted in Figure 17.15C. The wing vibrations are studied by the use of a small microphone connected to a device that converts the sound into electrical signals that can be displayed visually. A typical song burst is shown in Figure 17.16. The song burst is preceded by a

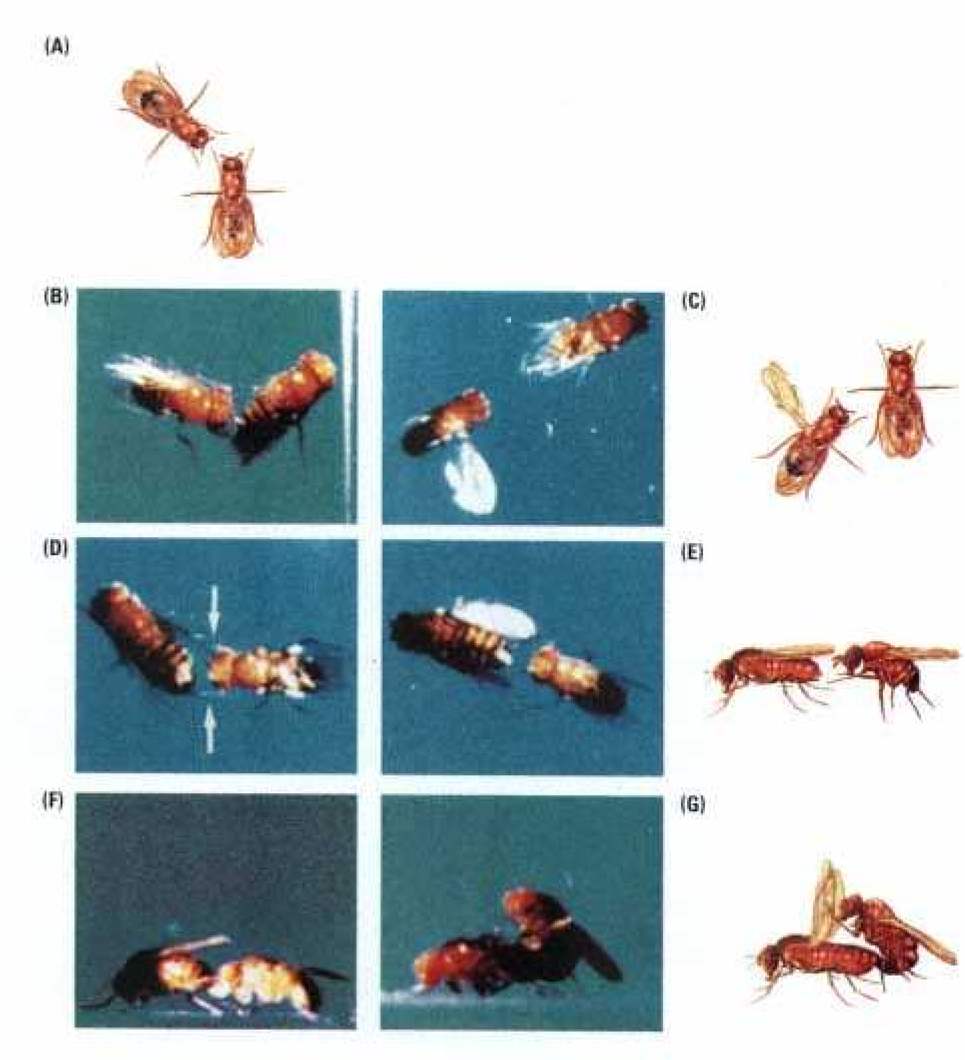

Figure 17.15
Courtship behavior in *Drosophila melanogaster*. (A) The male approaches the female and becomes oriented toward her, usually slightly to one side. (B) The male moves to a position slightly behind the female and extends the wing nearer her head (C), vibrating it rapidly up and down before returning the wing to its normal position (this process may be repeated several times). (D–F) The male moves to a position directly behind the female and touches her genitalia with his proboscis. (G) Attempted copulation. [Photographs courtesy of Jeffrey C. Hall; sketches from B. Burnet and K. Connolly, in *The Genetics of Behaviour*, edited by J. H. F. van Abeelen (Amsterdam: North-Holland, 1974), pp. 212–213.]

"hum" of extremely rapid vibrations, which serves to "warm up," or sensitize, the female. The "song" follows: a series of discrete pulses that vary in length from 15 to 100 milliseconds from one song burst to the next. A sequence of song bursts is diagrammed in Figure 17.17A. As the sequence progresses, the interpulse intervals become shorter, then longer, then shorter again. It is the rhythmic expansion and contraction of the interpulse intervals in the sequence of song bursts that defines the *Drosophila* courtship song. A plot of the interpulse intervals, as in Figure 17.17B, produces a smooth oscillating curve.

The effects of *per* alleles on the period of the courtship song are illustrated in Figure 17.18. The period observed with wildtype *D. melanogaster* is approximately 60 seconds. The per^S allele shortens the period to about 40 seconds, and the per^L allele lengthens the period to about 90 seconds.